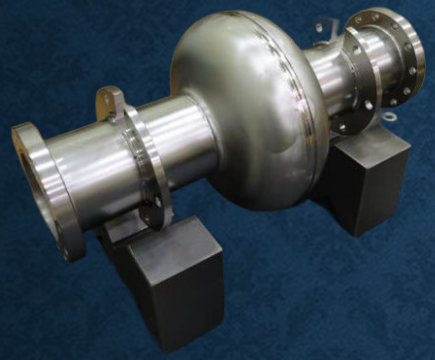
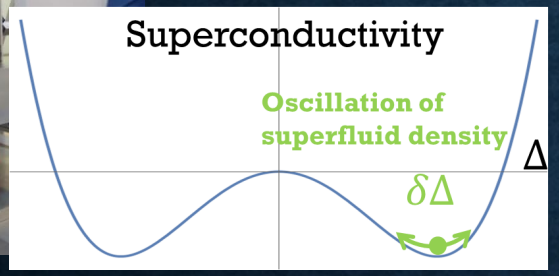
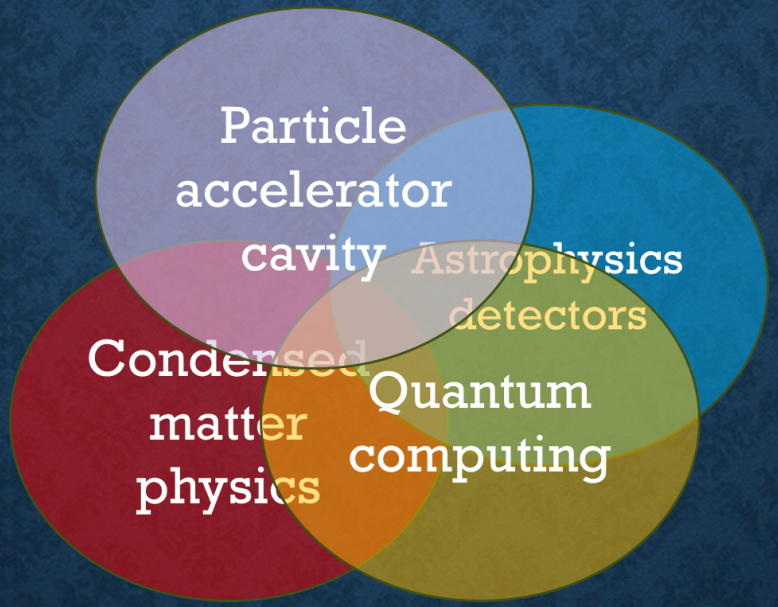
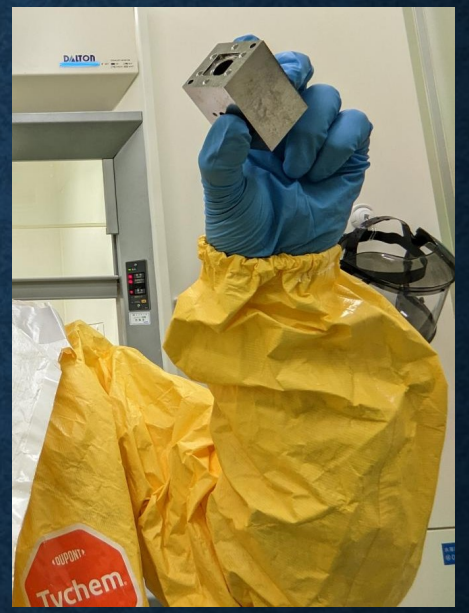
Superconducting Cavities and Their Physics:
From Particle Accelerators to Quantum Technologies









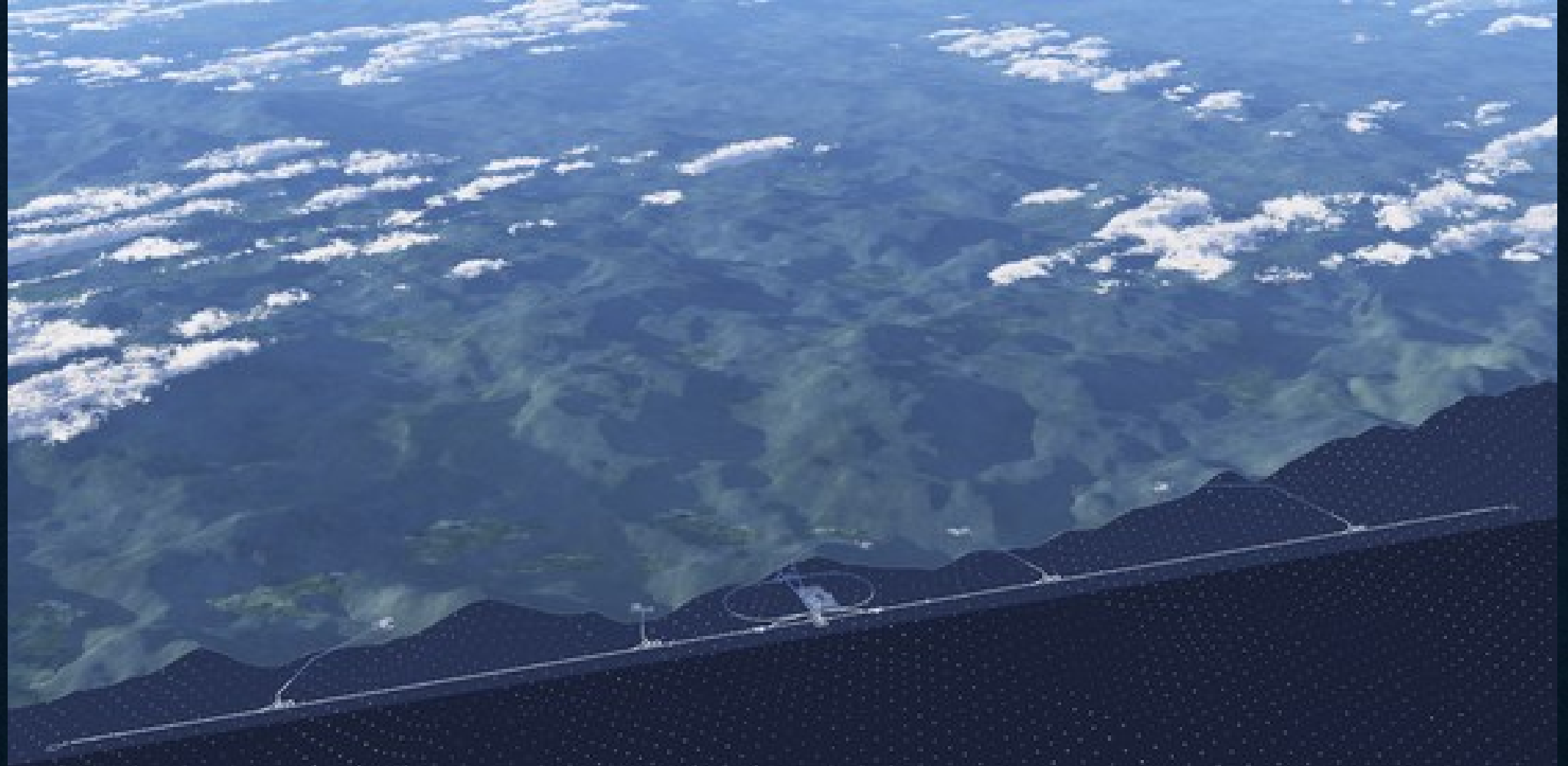




High energy Accelerator Research Organization (KEK)
Innovation Center for Applied Superconducting Accelerators (iCASA)

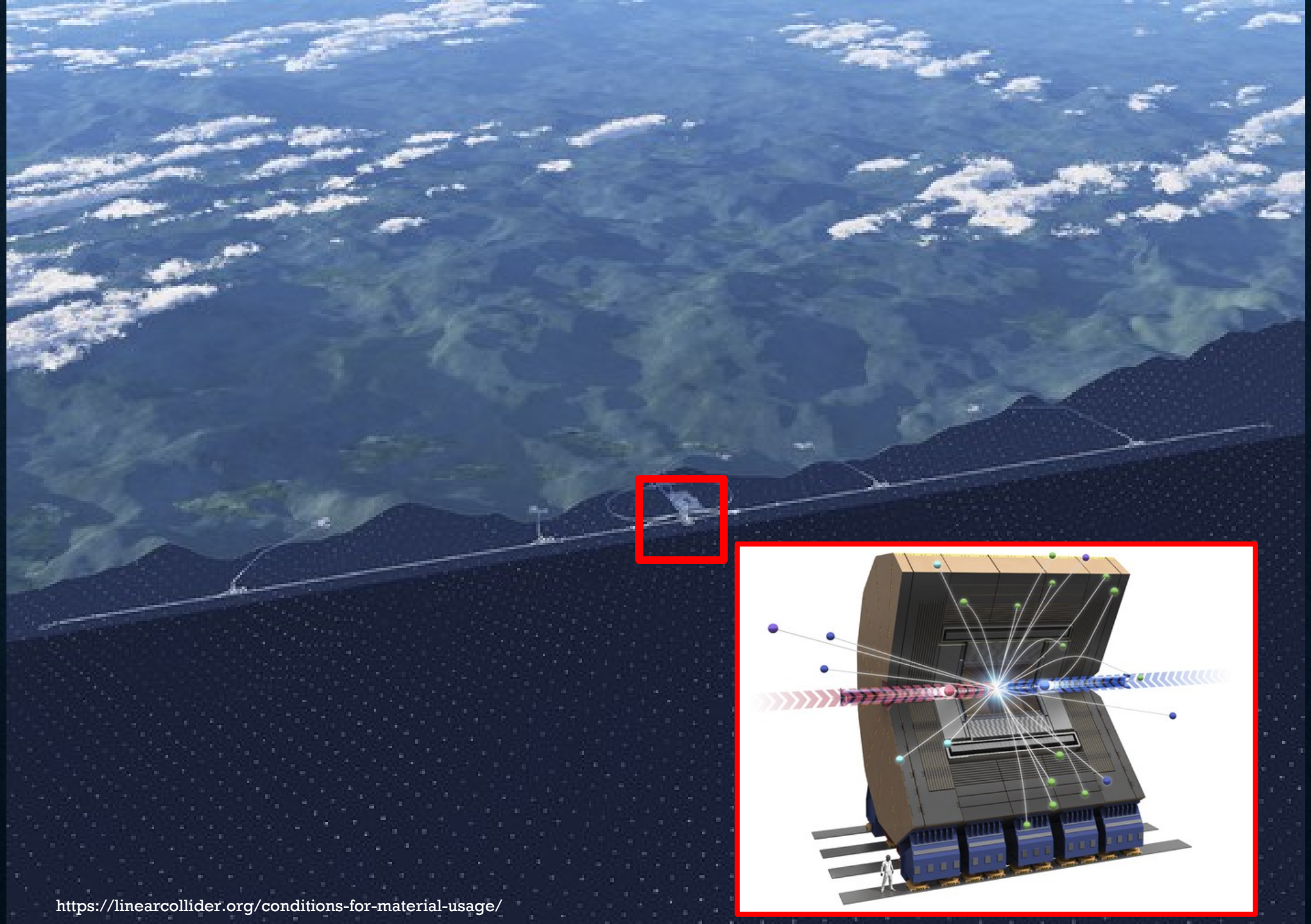
Takayuki Kubo

https://kbtkphys.github.io/
https://scholar.google.com/citations?user=MN4RQIsAAAAJ

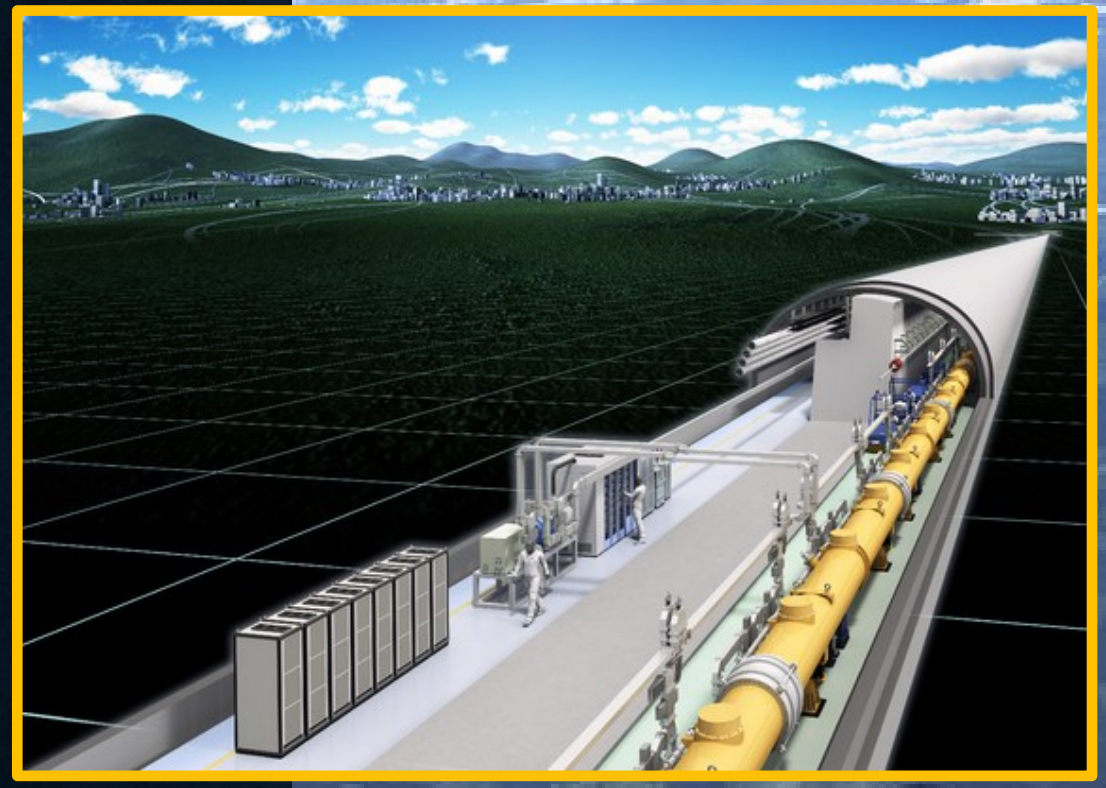


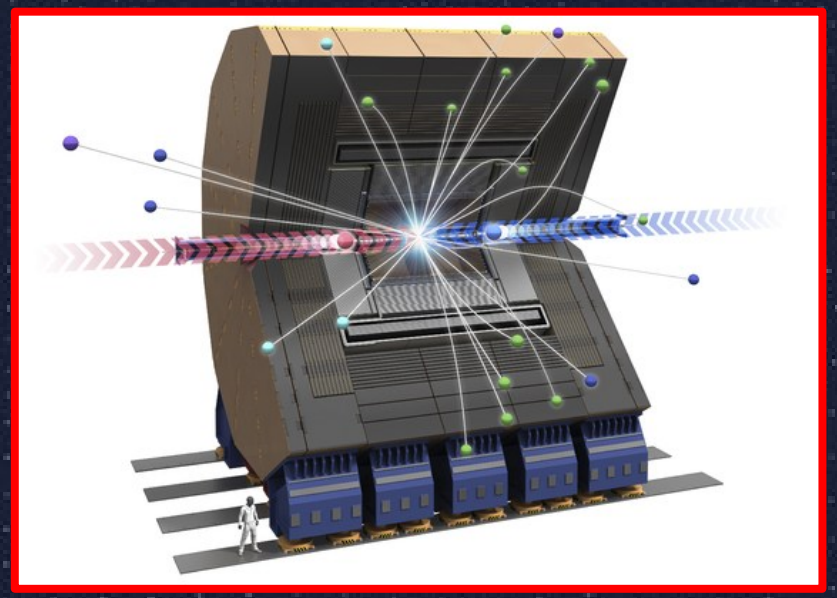
Here we use the International Linear Collider (ILC), an electron–positron Higgs factory, **as an example**, but the European X-ray Free-Electron Laser (European XFEL) in Germany and the Linac Coherent Light Source II (LCLS-II) at SLAC in California are also based on similar superconducting RF linac technology.

https://linearcollider.org/conditions-for-material-usage/

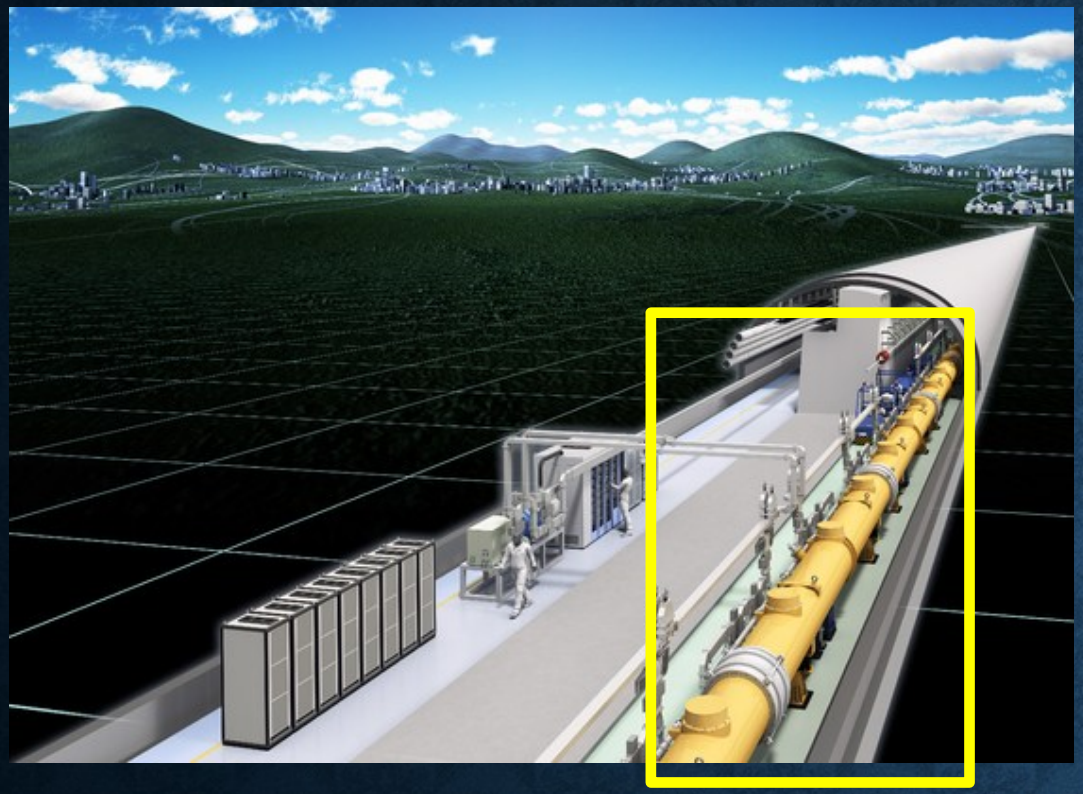


https://linearcollider.org/conditions-for-material-usage/





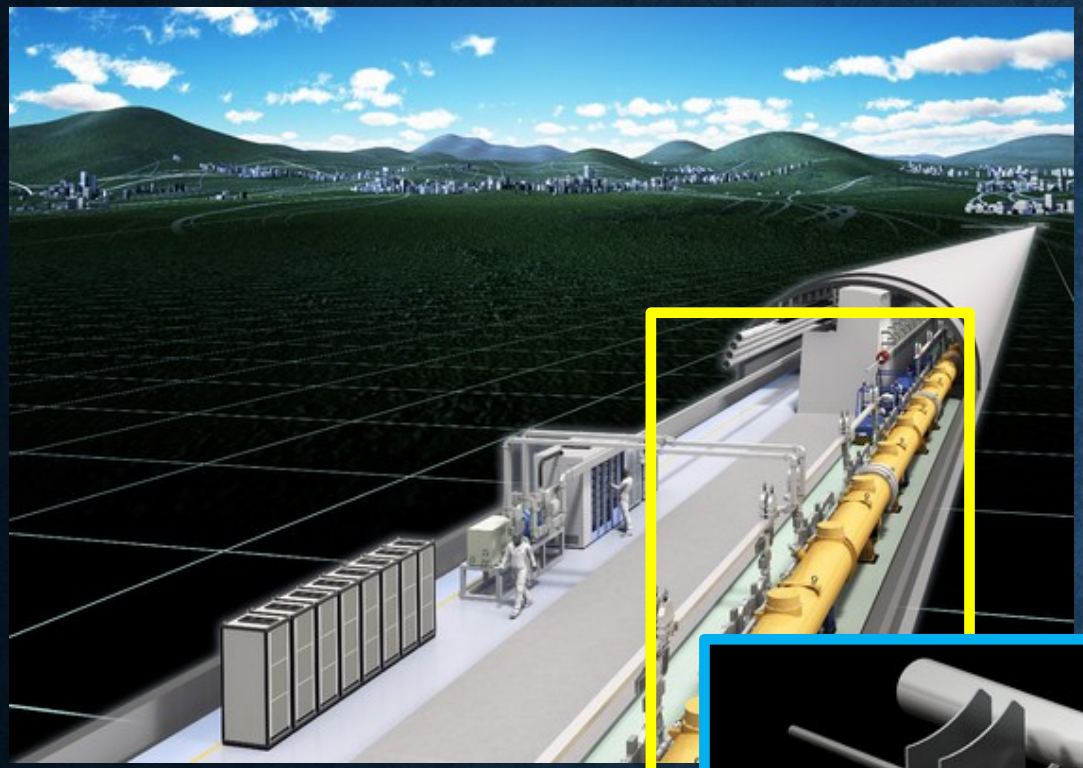
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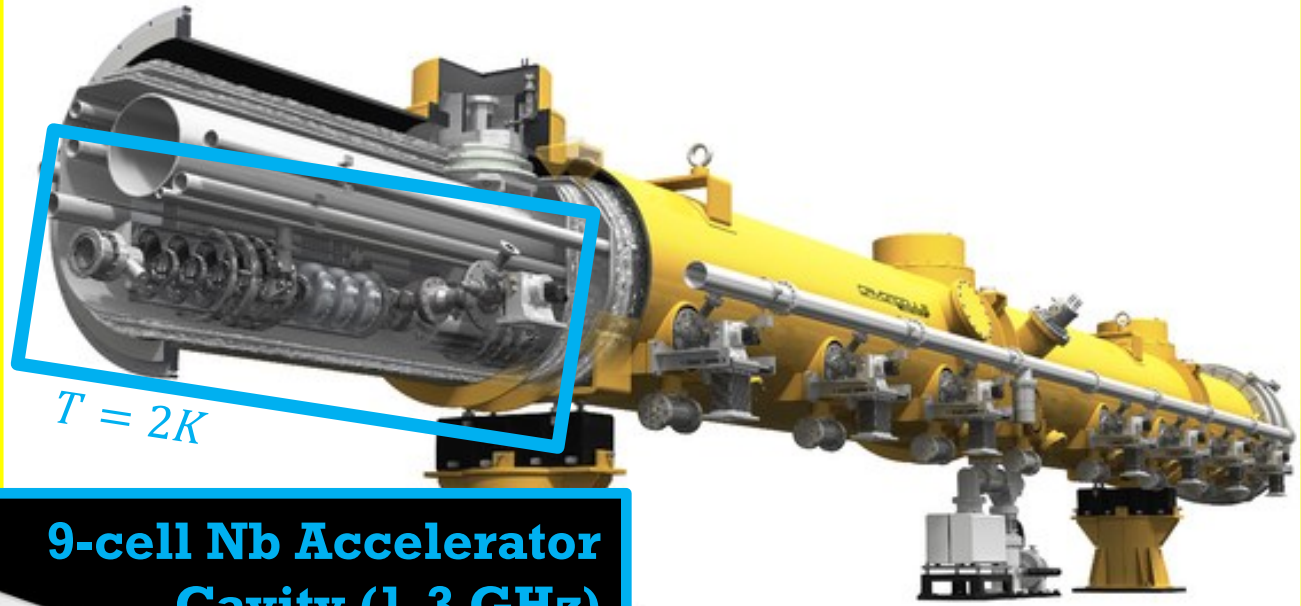
Cryomodule



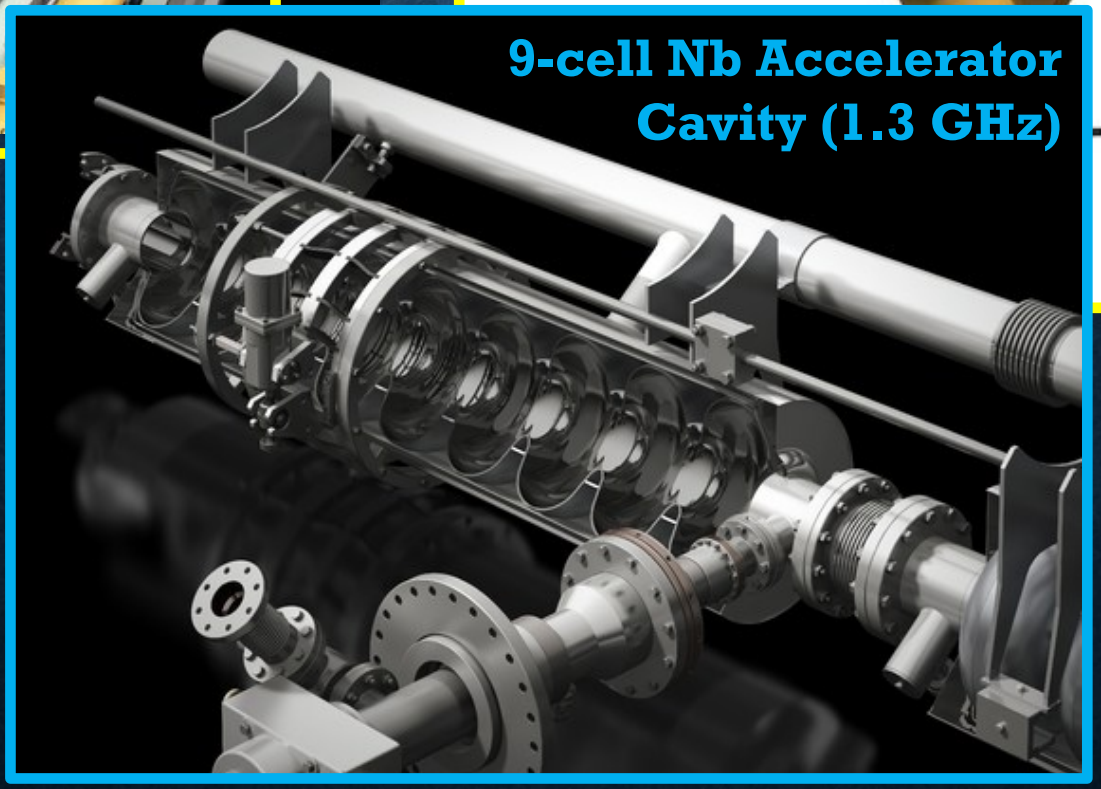
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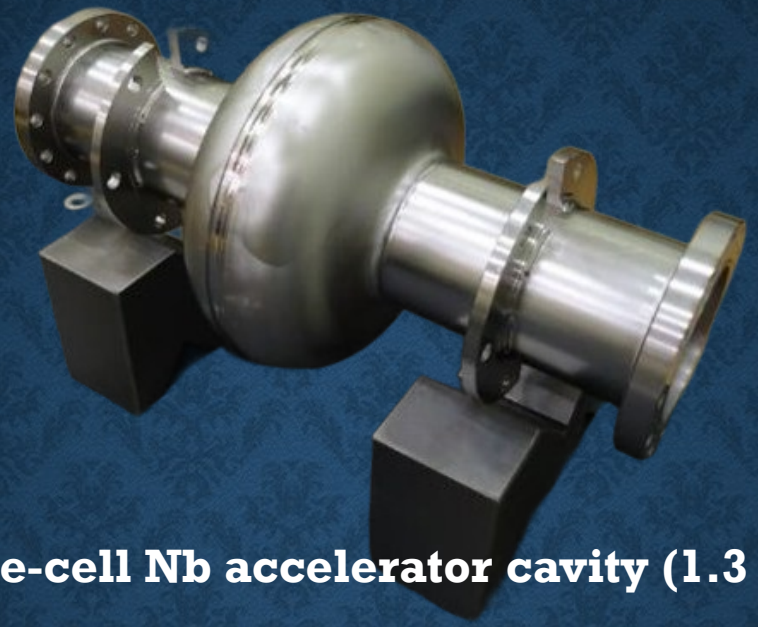
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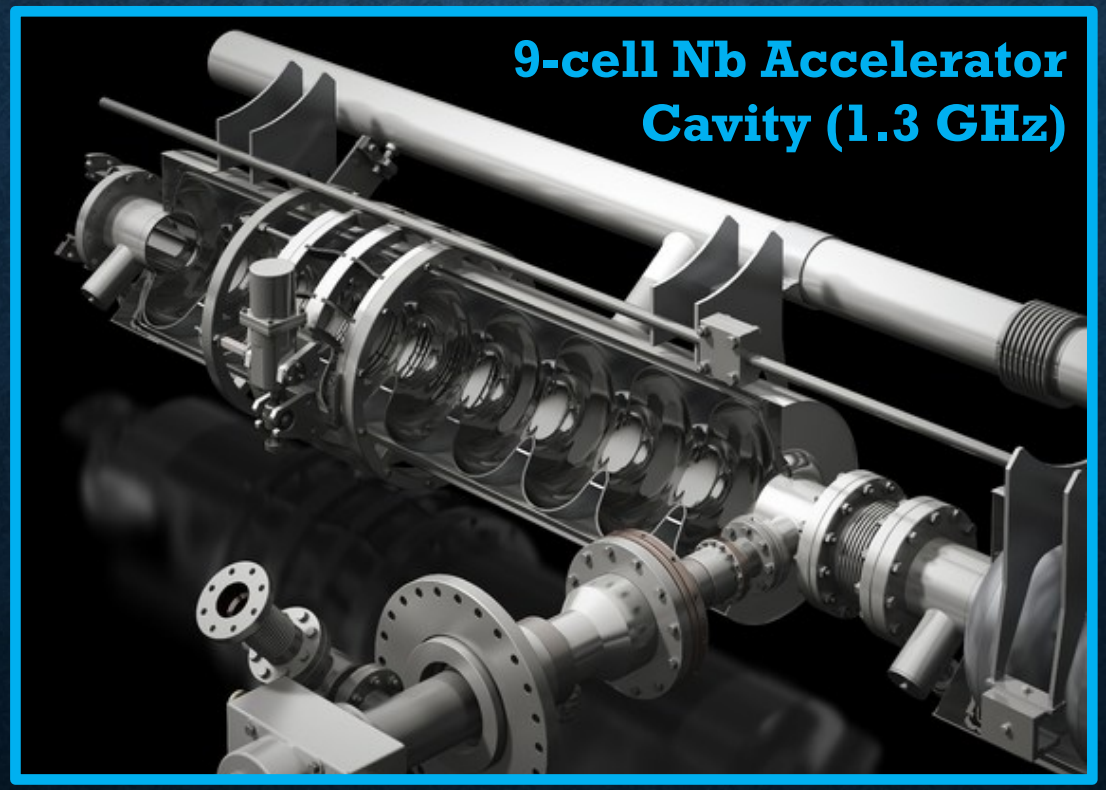
9-cell Nb Accelerator Cavity (1.3 GHz)

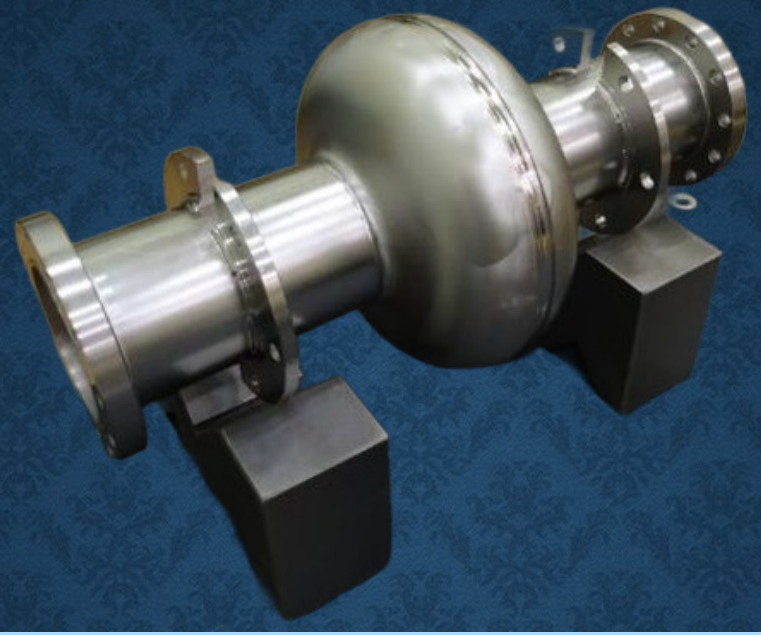


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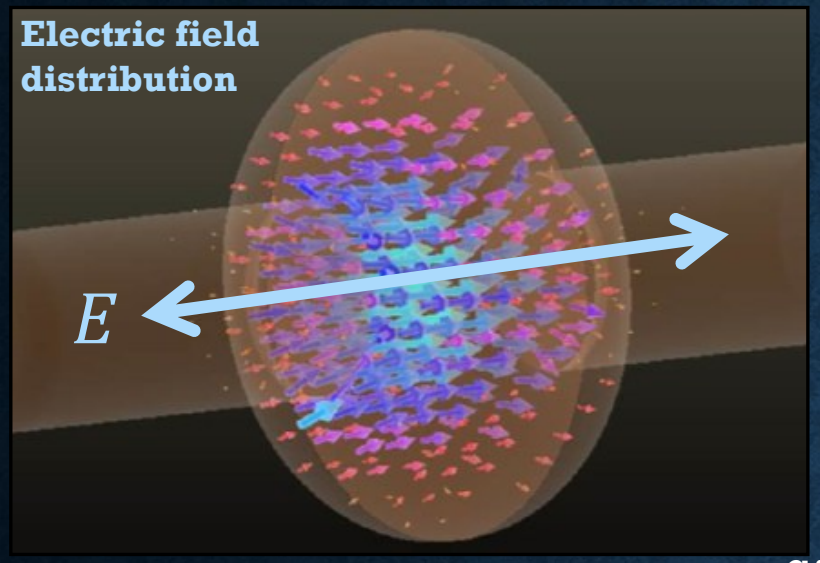


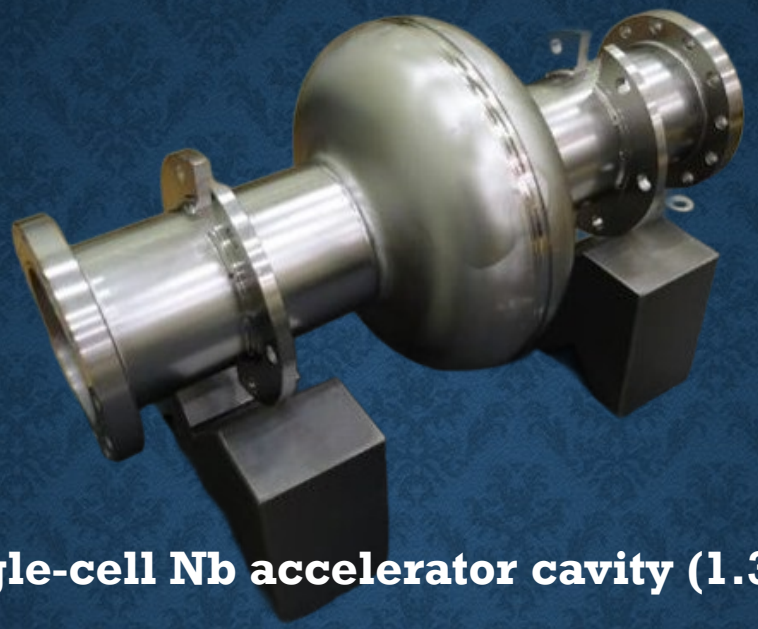
Single-cell Nb accelerator cavity (1.3 GHz)



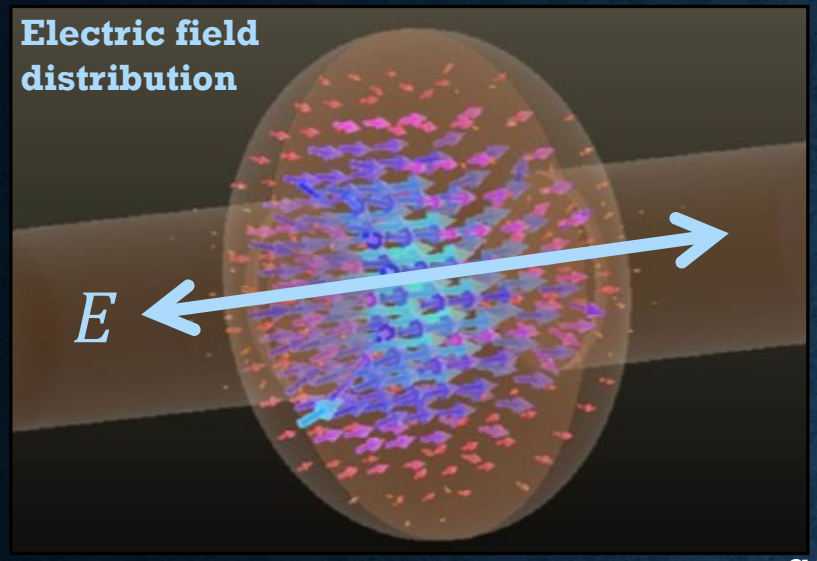


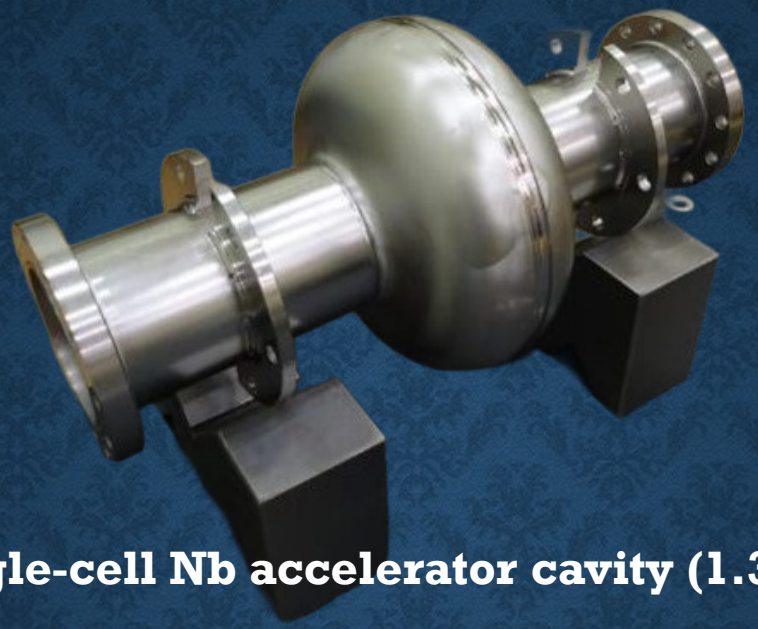
Overview

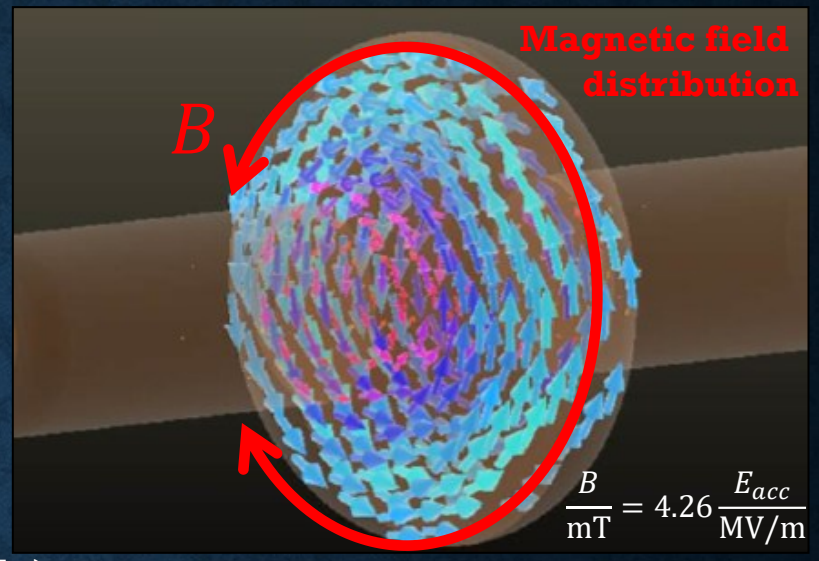




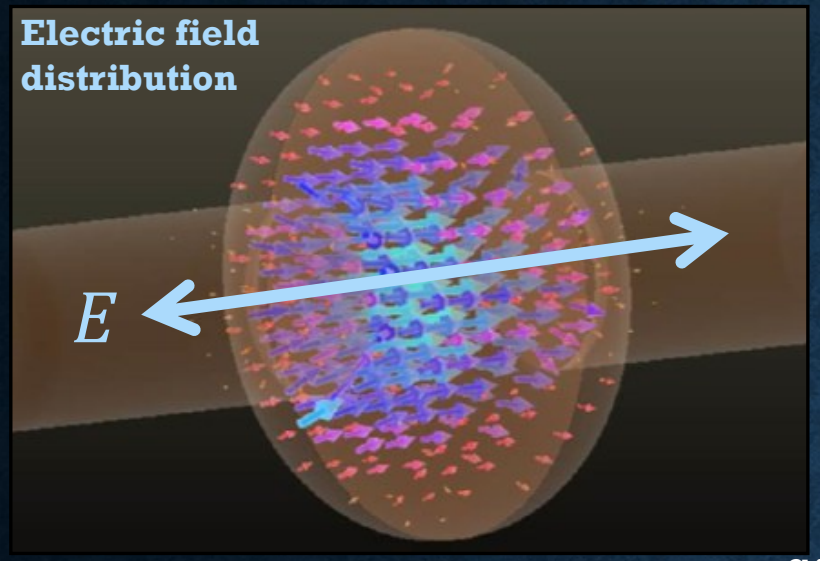
Single-cell Nb accelerator cavity (1.3 GHz)

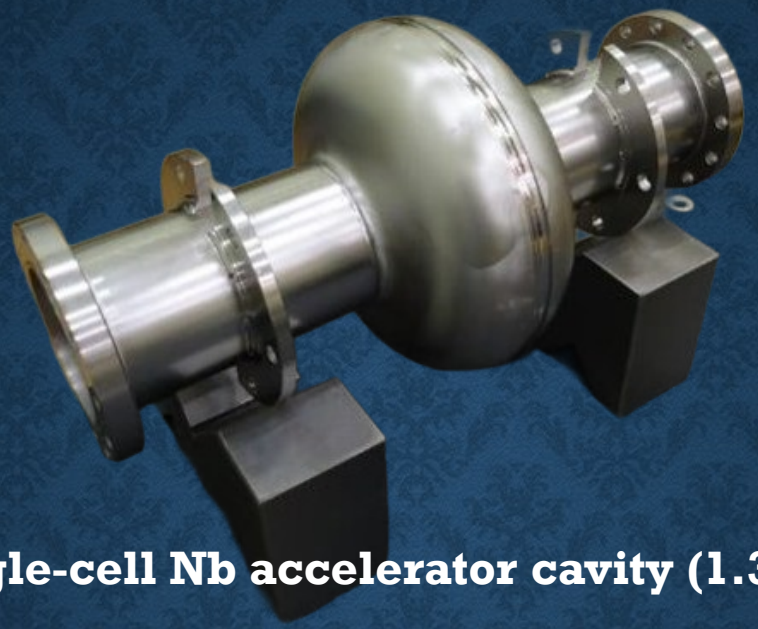


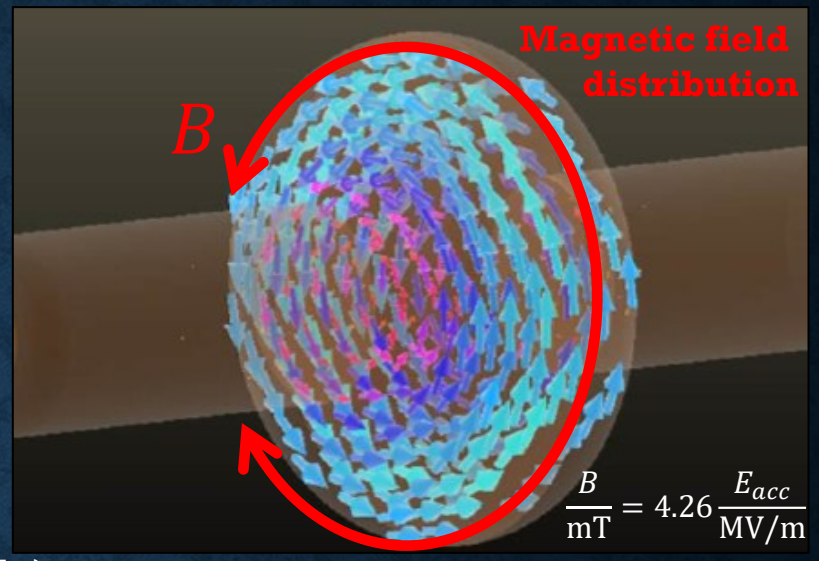




Single-cell Nb accelerator cavity (1.3 GHz)





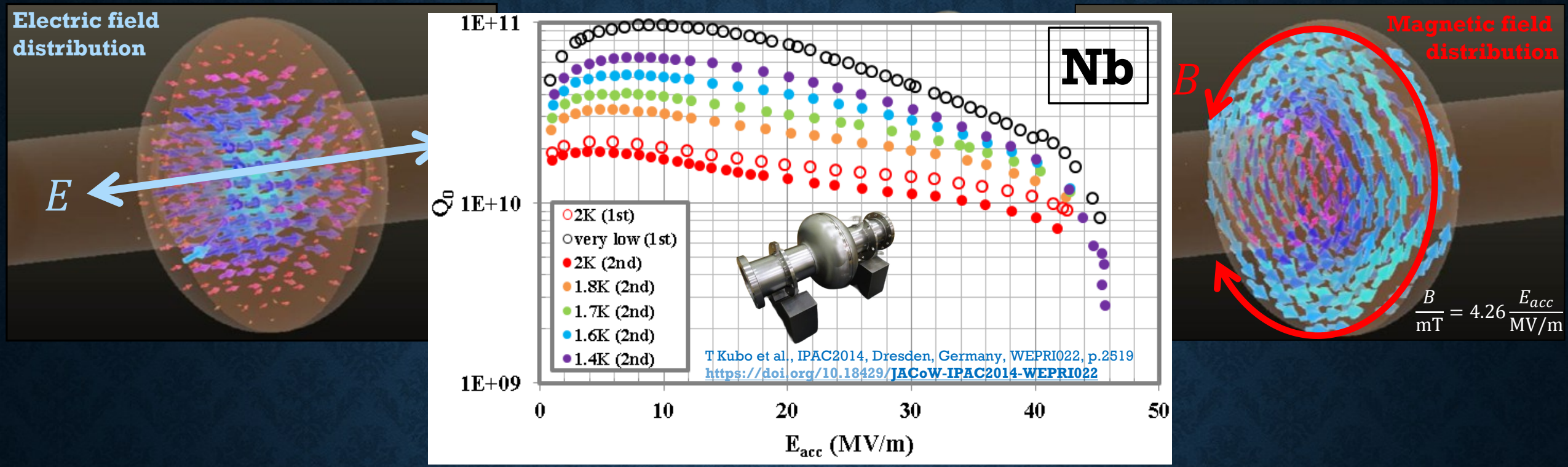


Single-cell Nb accelerator cavity (1.3 GHz)

There are two key performance metrics

(1) Maximum field (stability limit of the vortex-free state)

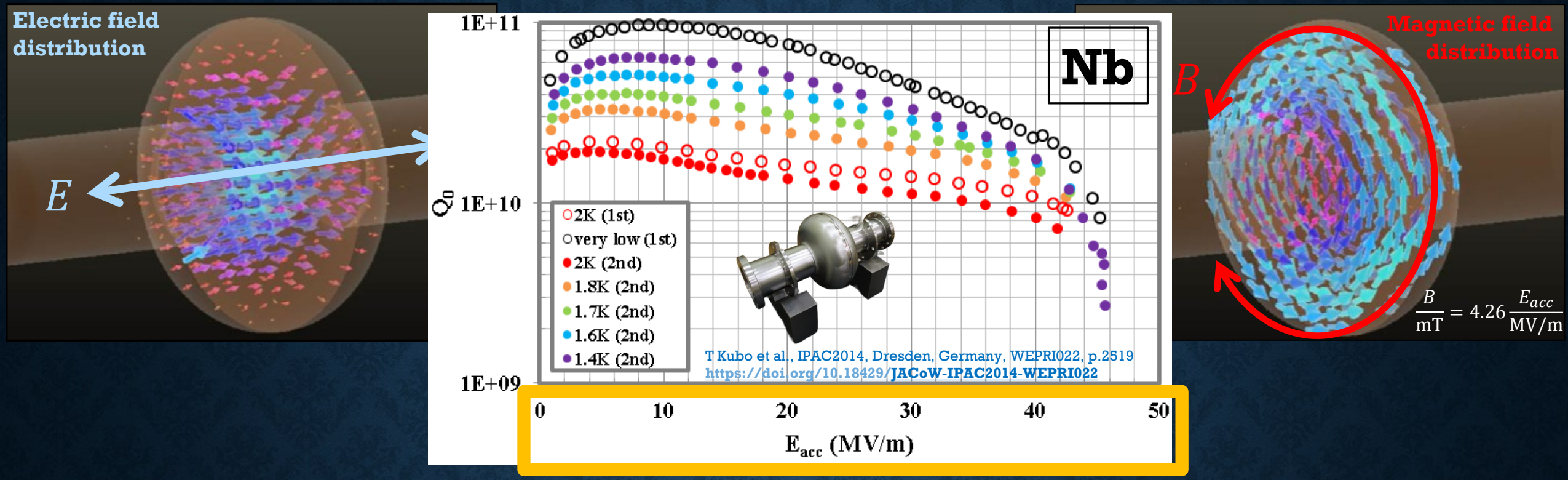
(2) Quality factor Q_0 (smallness of dissipation)



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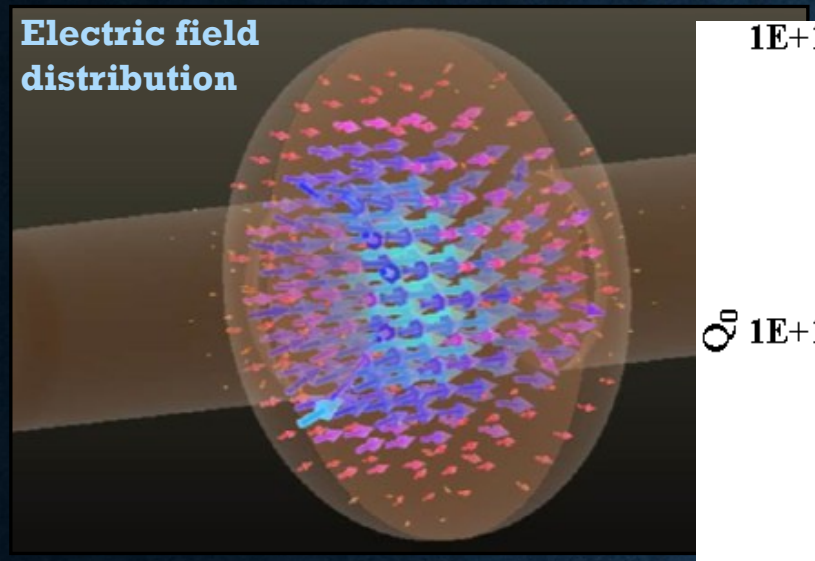
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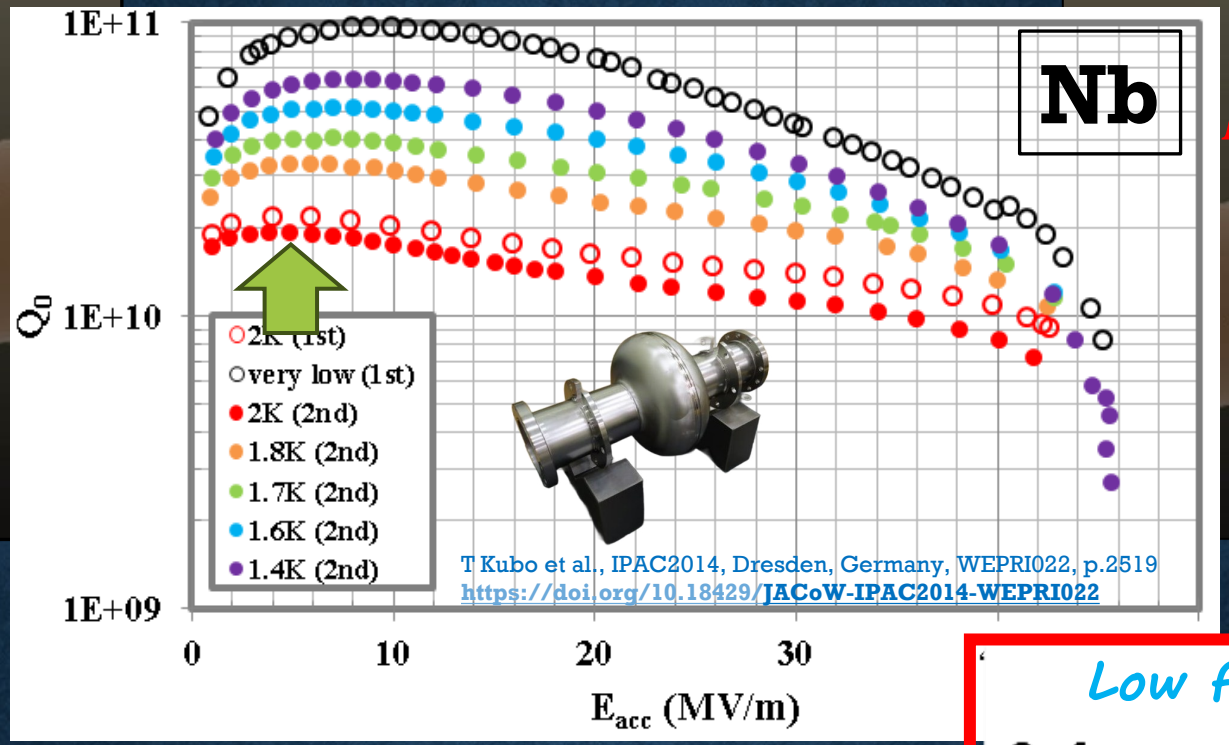


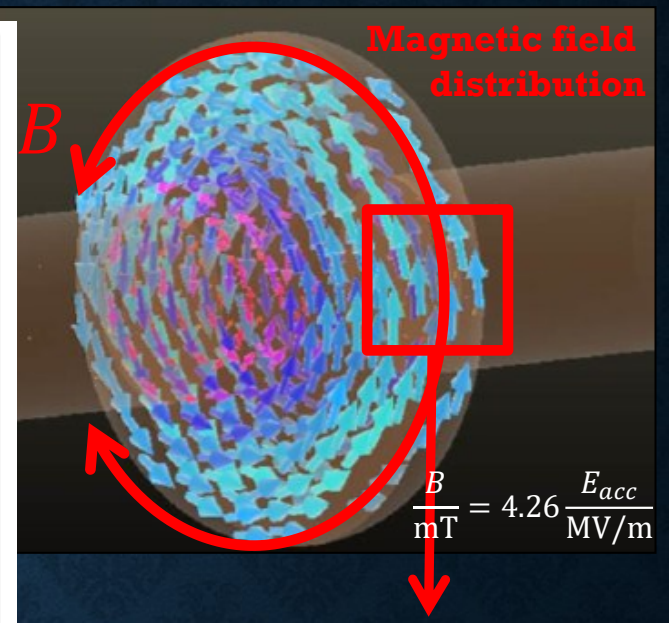
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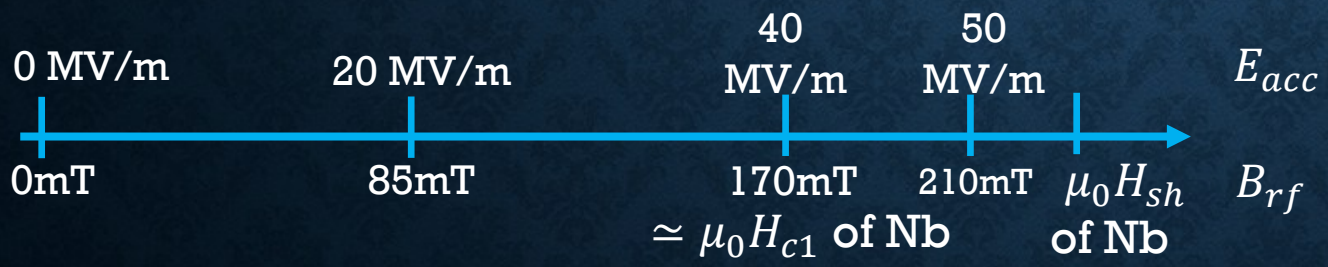
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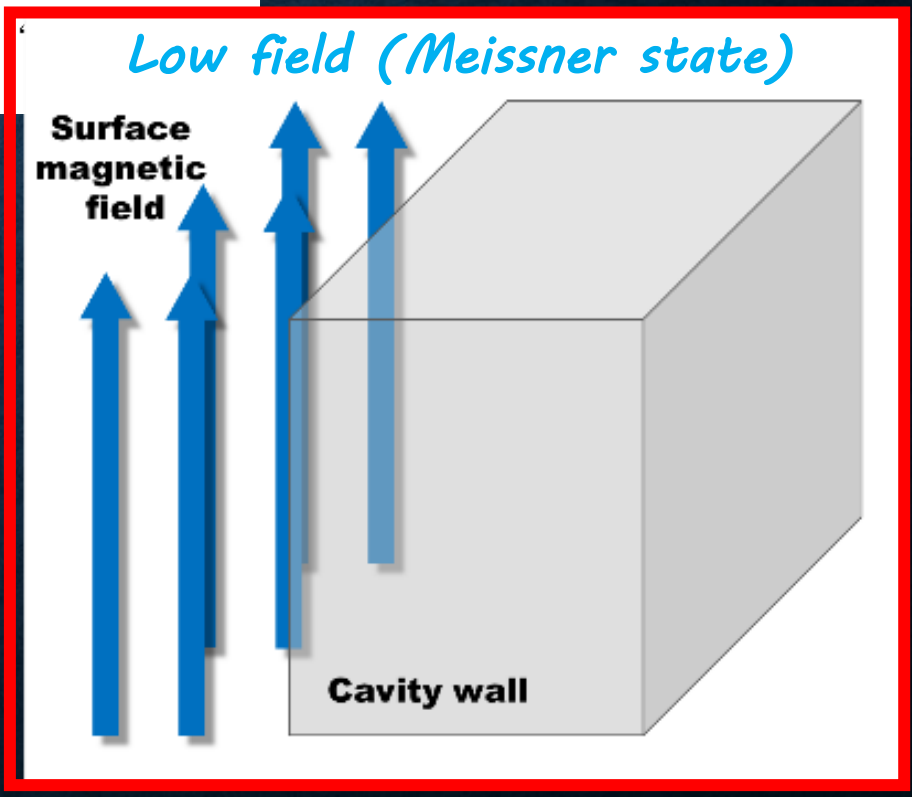


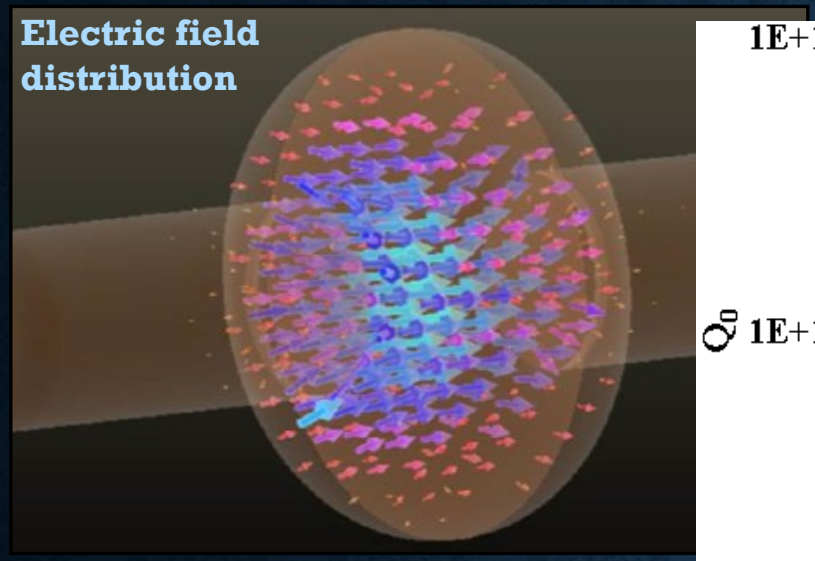


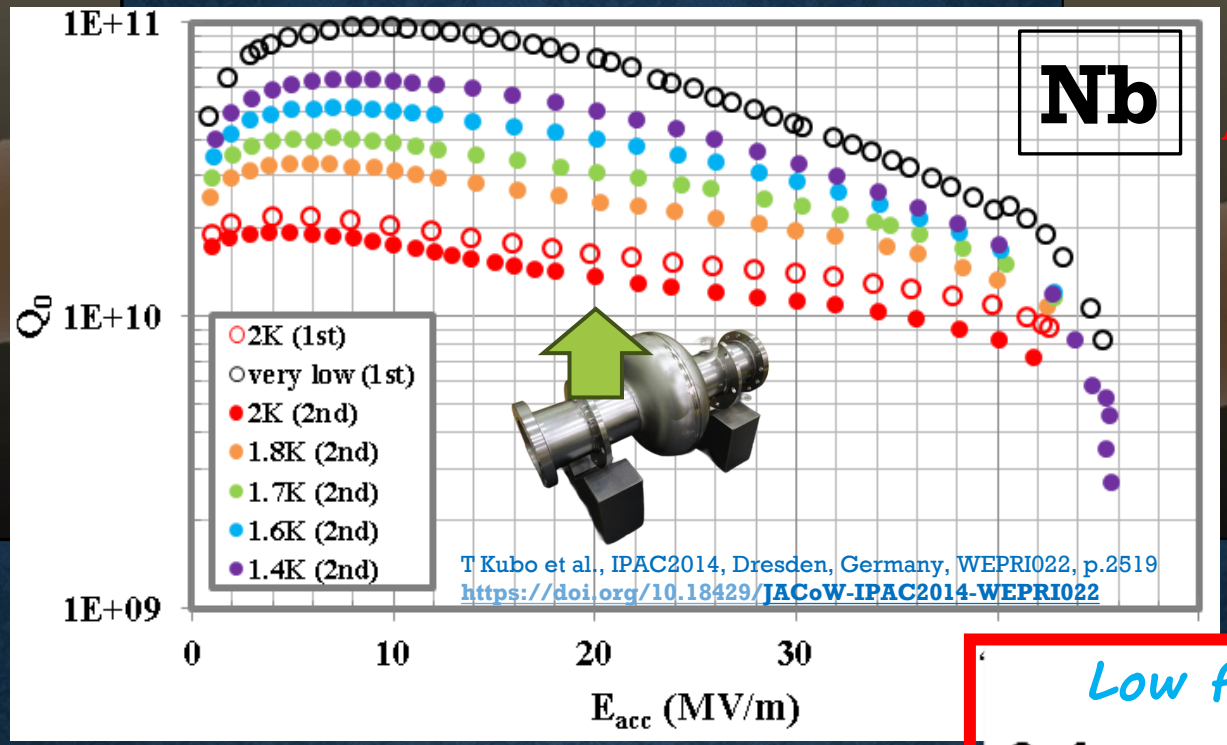
Note: Cavities are operated under the **Meissner state**

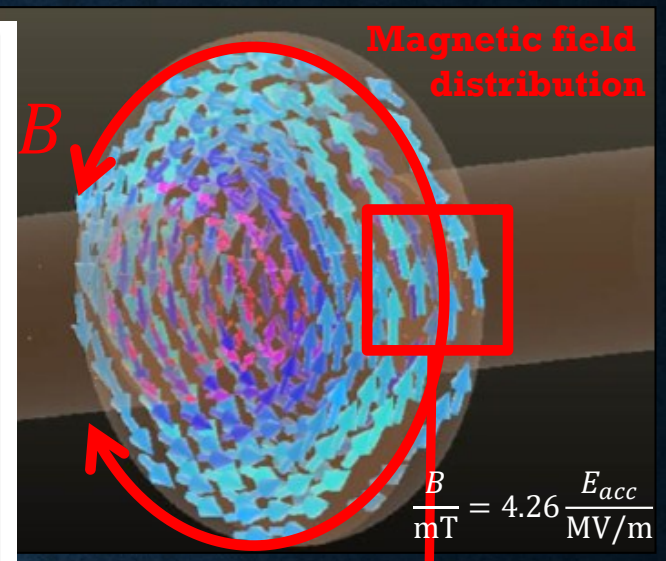


(⌐■_■) good

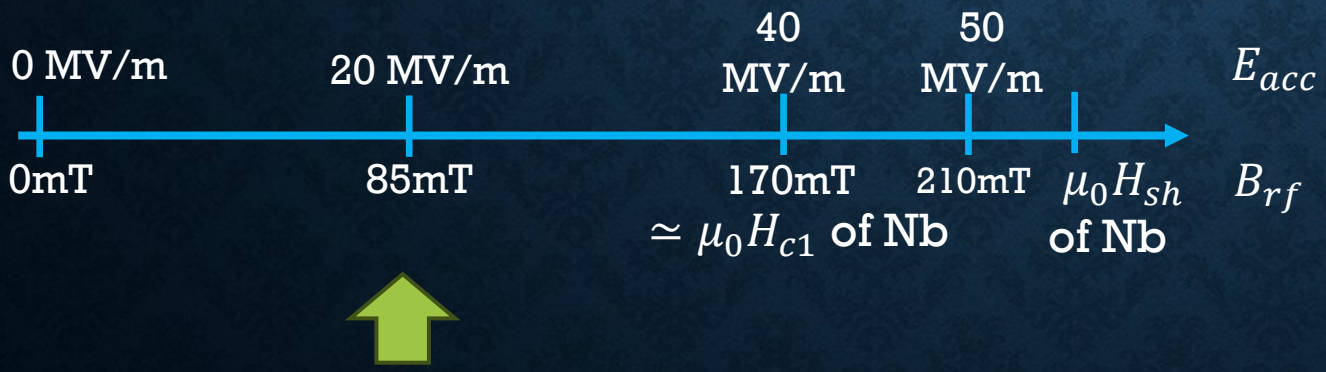




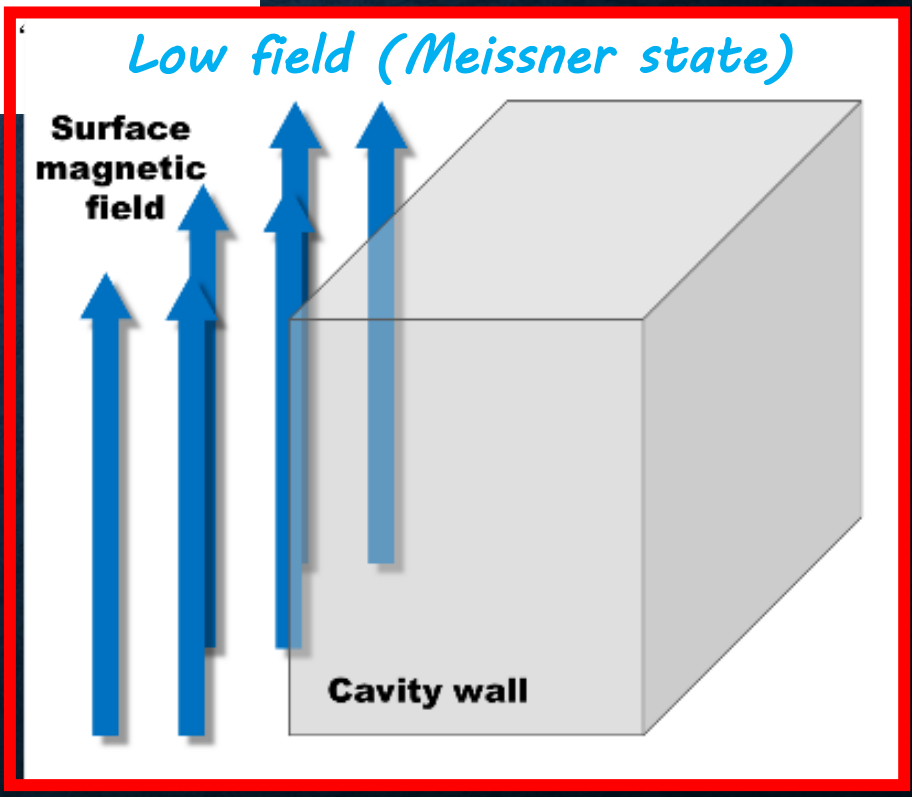


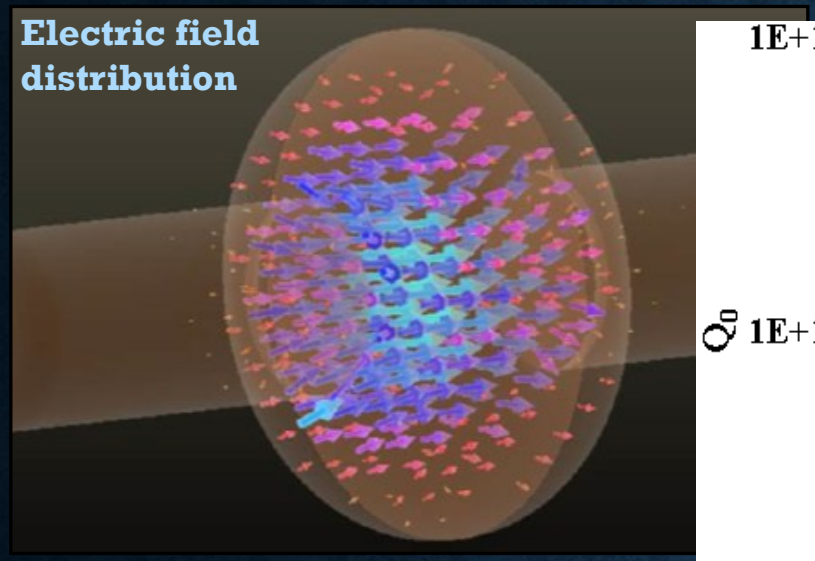


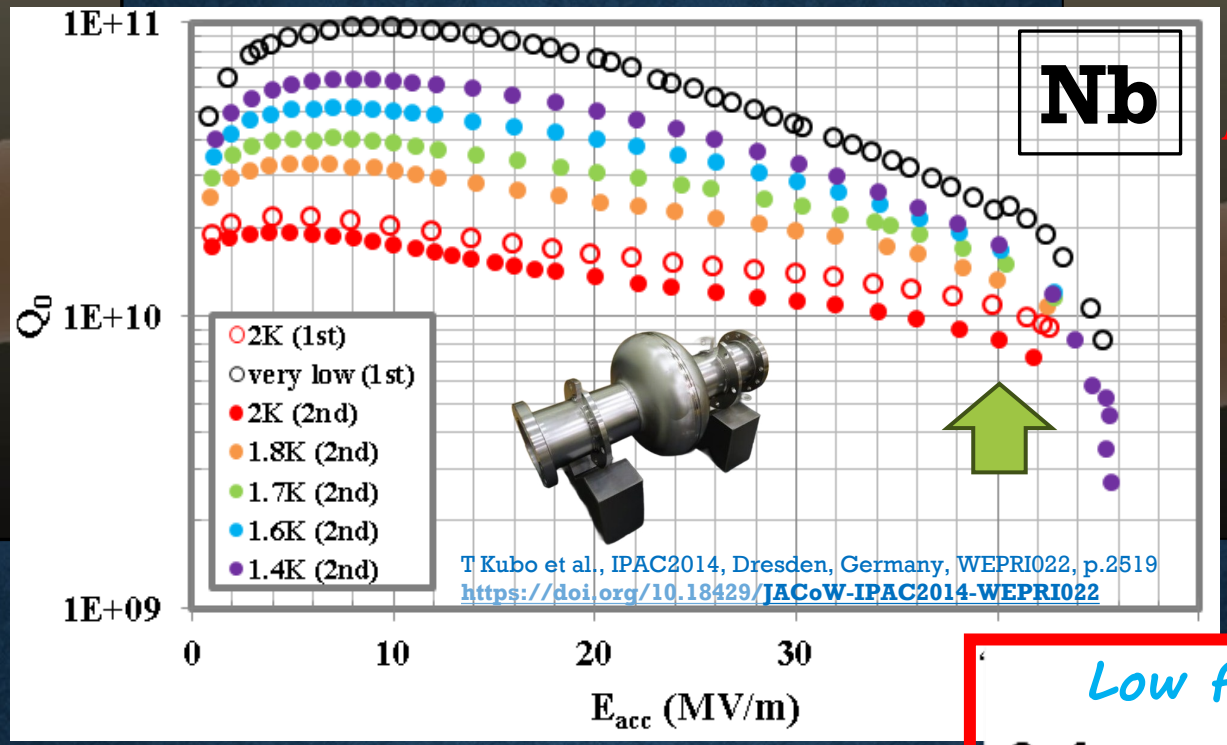
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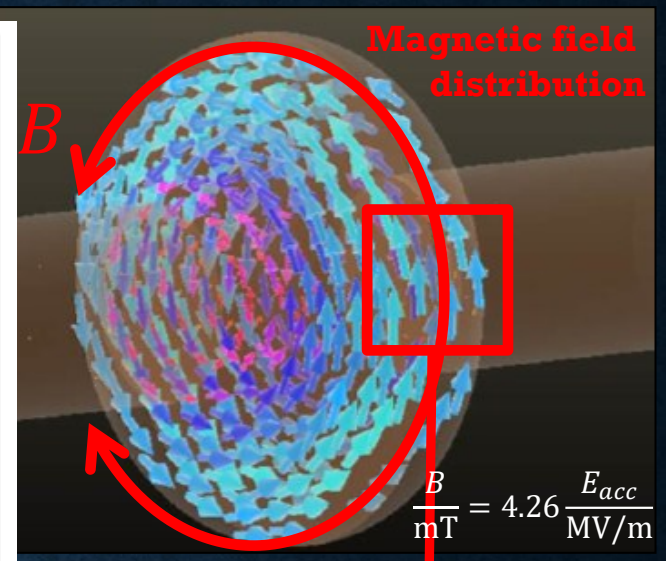


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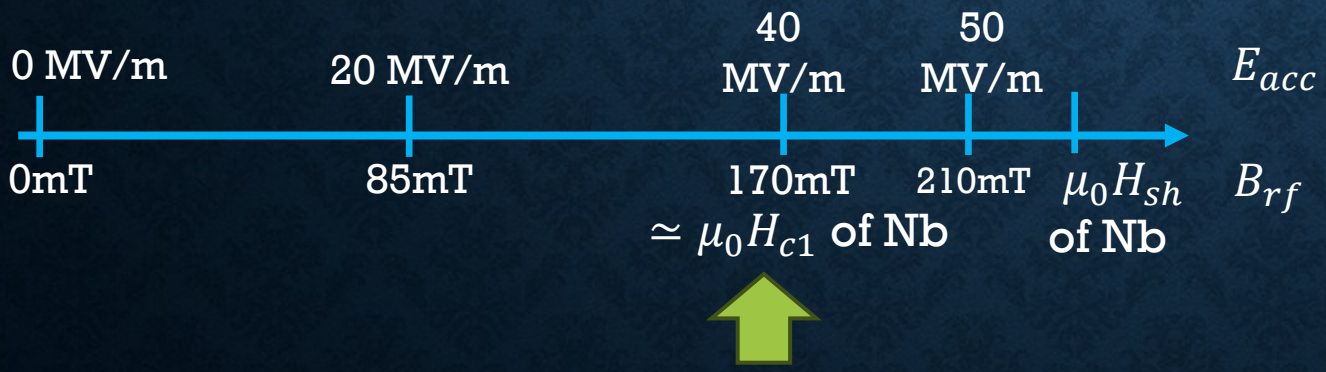




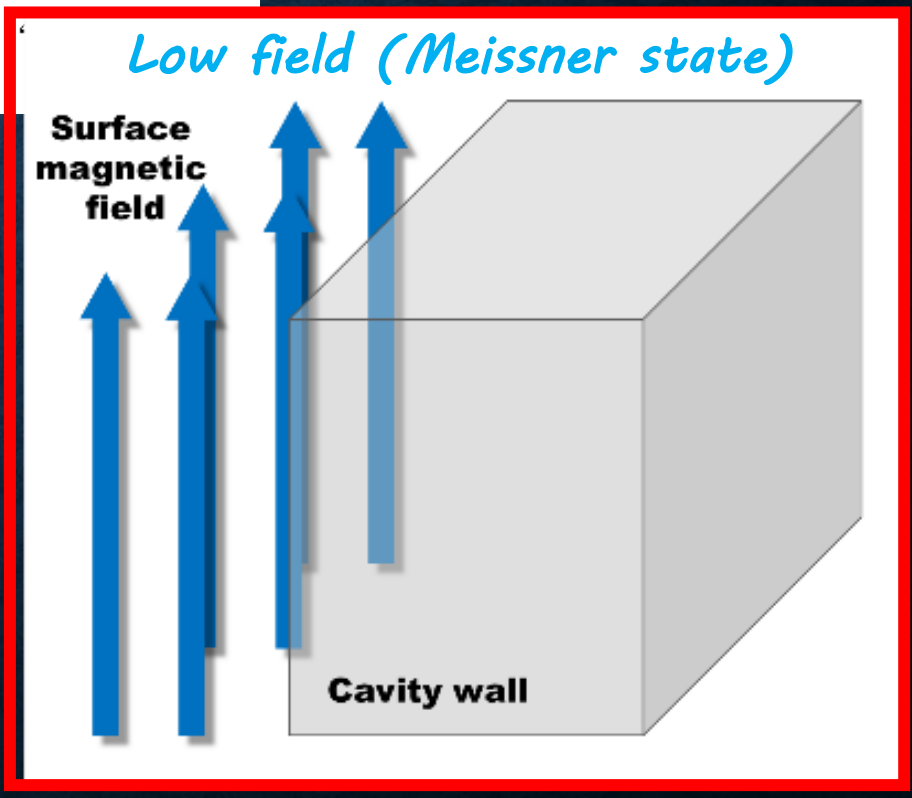


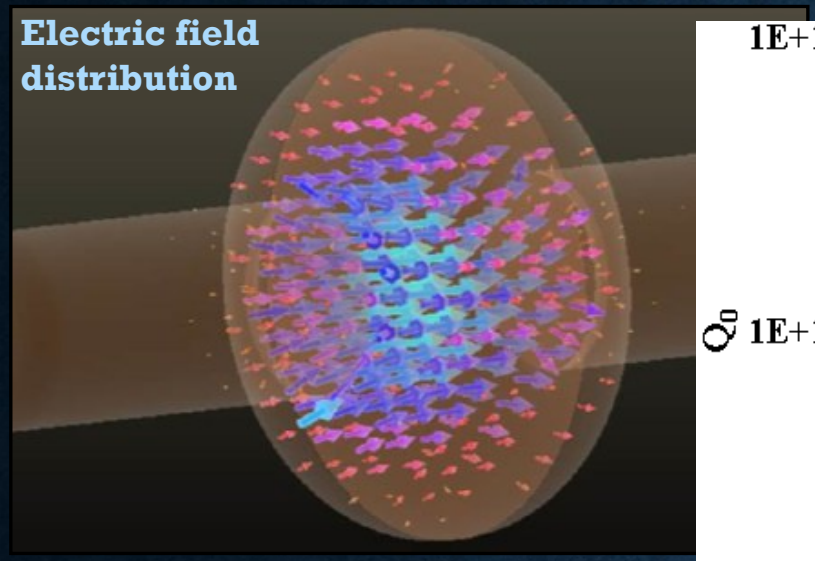


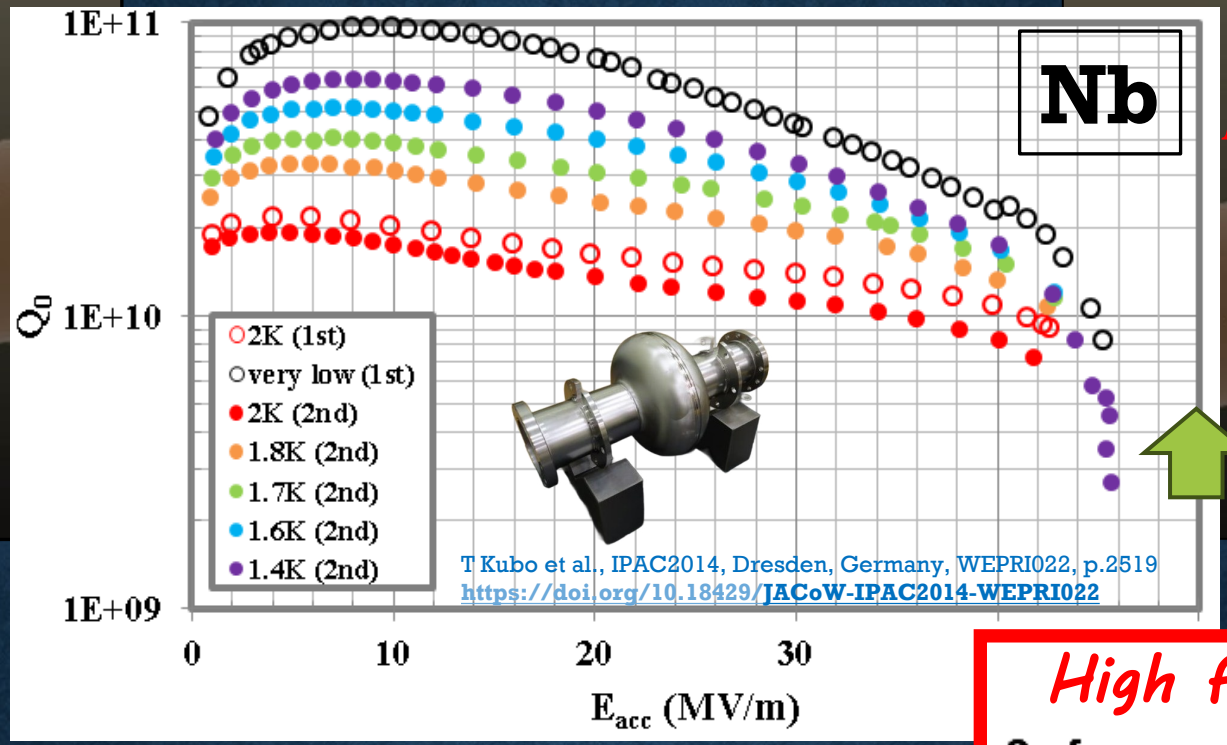
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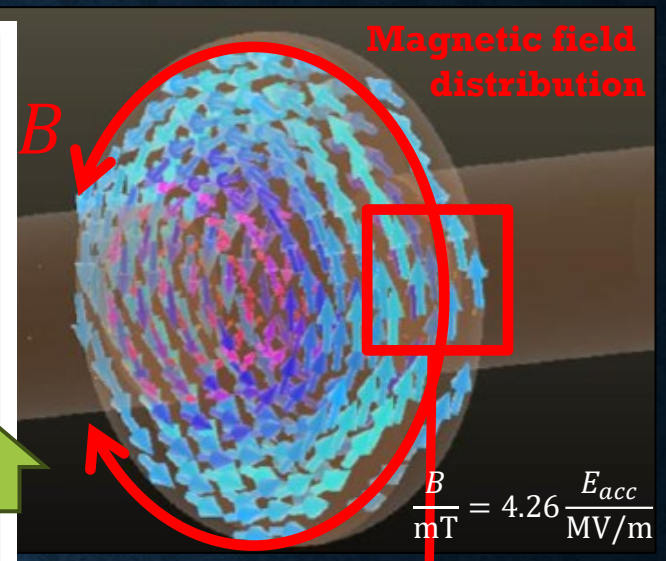
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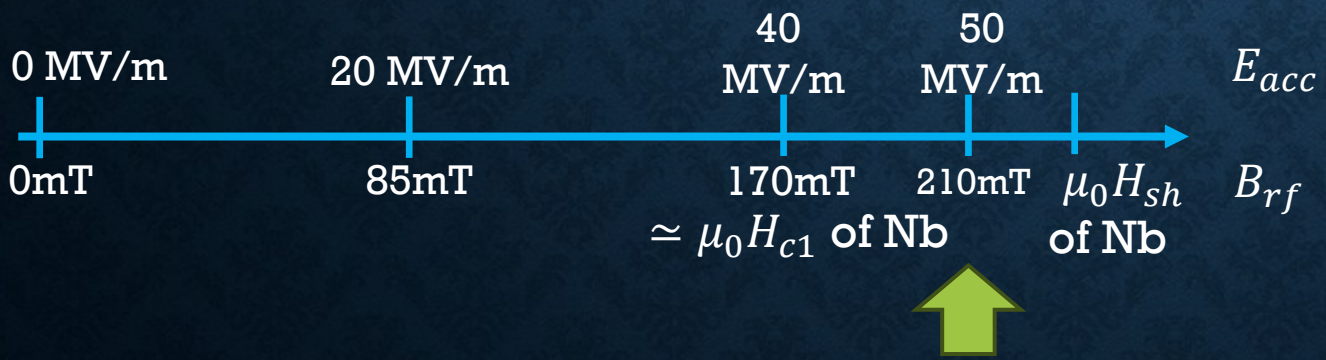




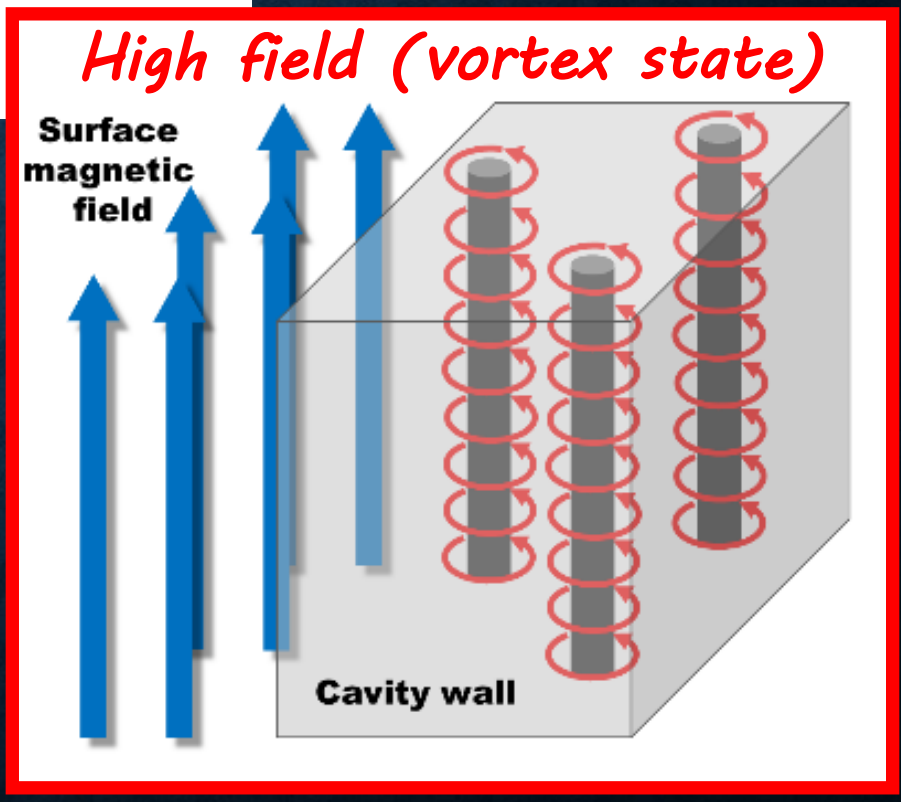
T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

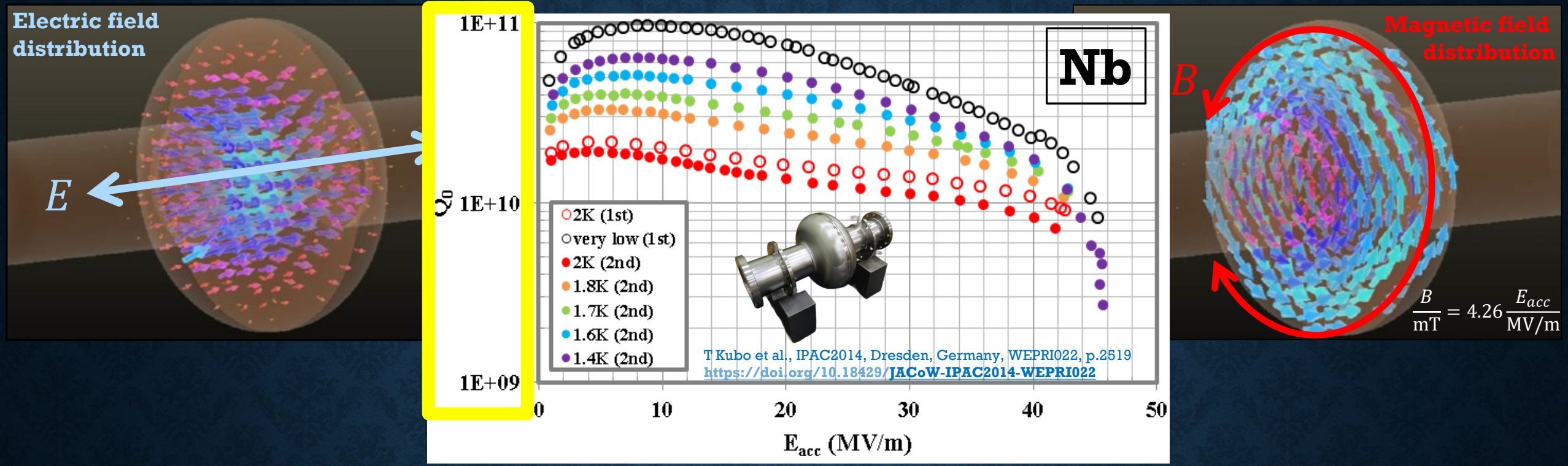


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I'm doomed





There are two key performance metrics

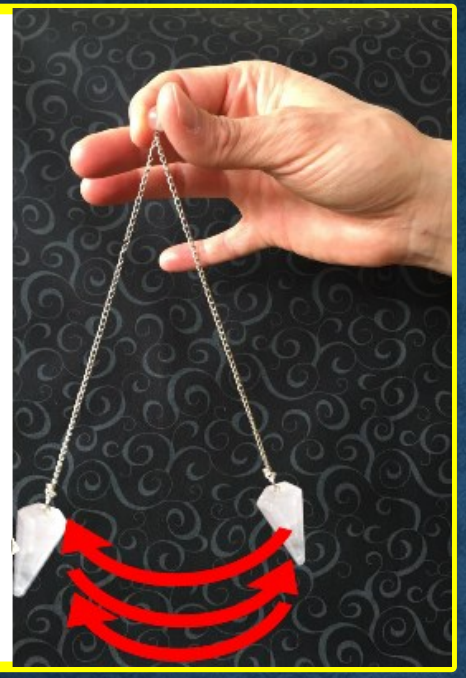
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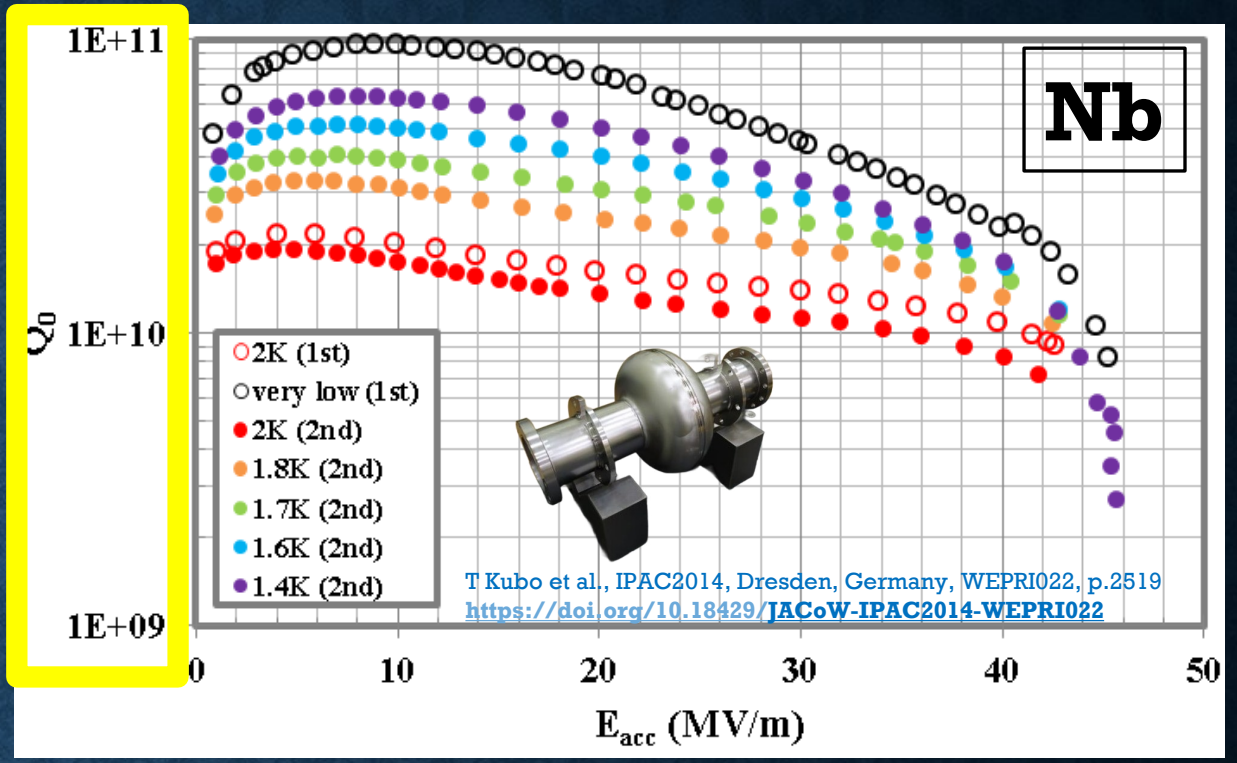
(2) Quality factor Q_0 (smallness of dissipation)

$$Q_0 = \frac{\omega U}{P} \sim \frac{U}{P_1} \propto \frac{1}{R_s}$$

P_1 : dissipation per RF cycle

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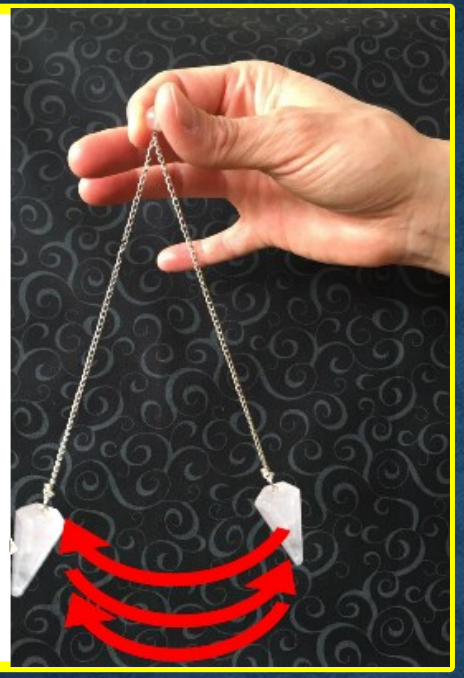
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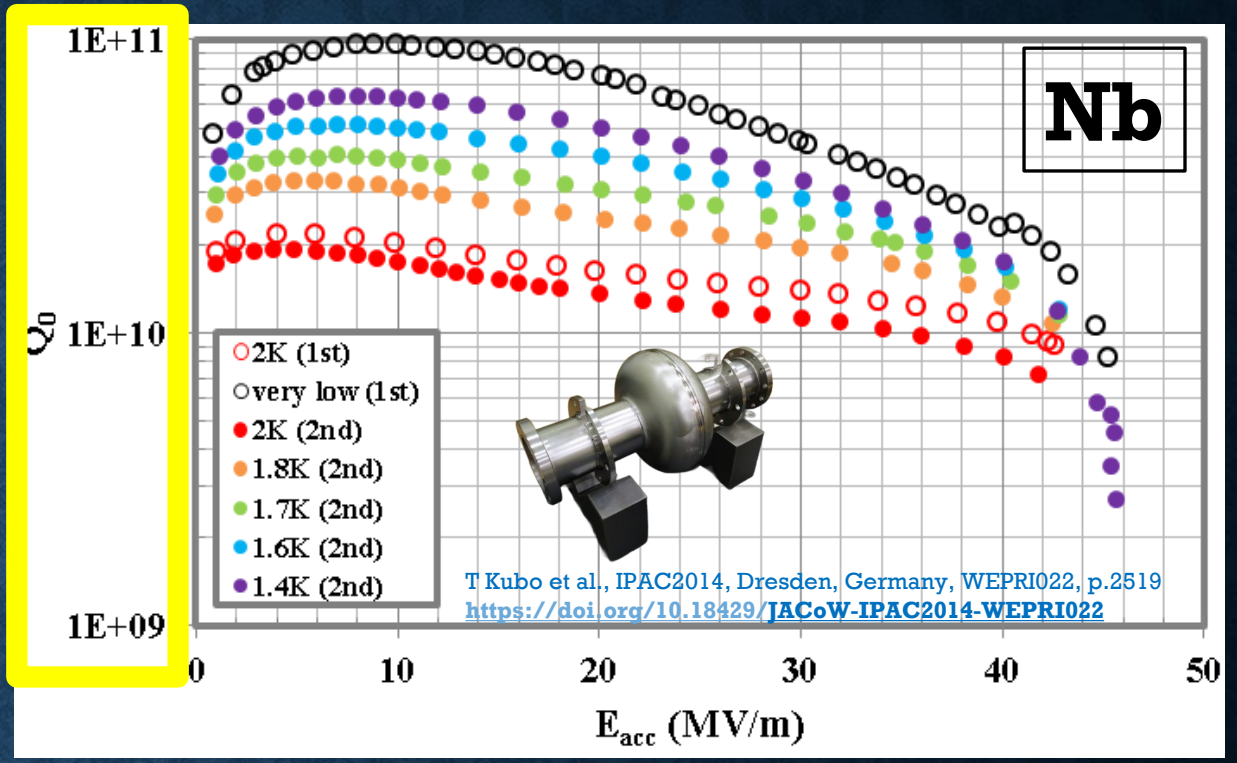
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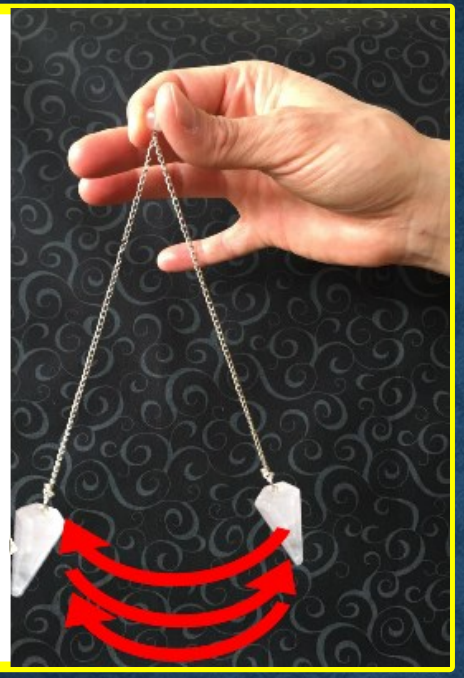
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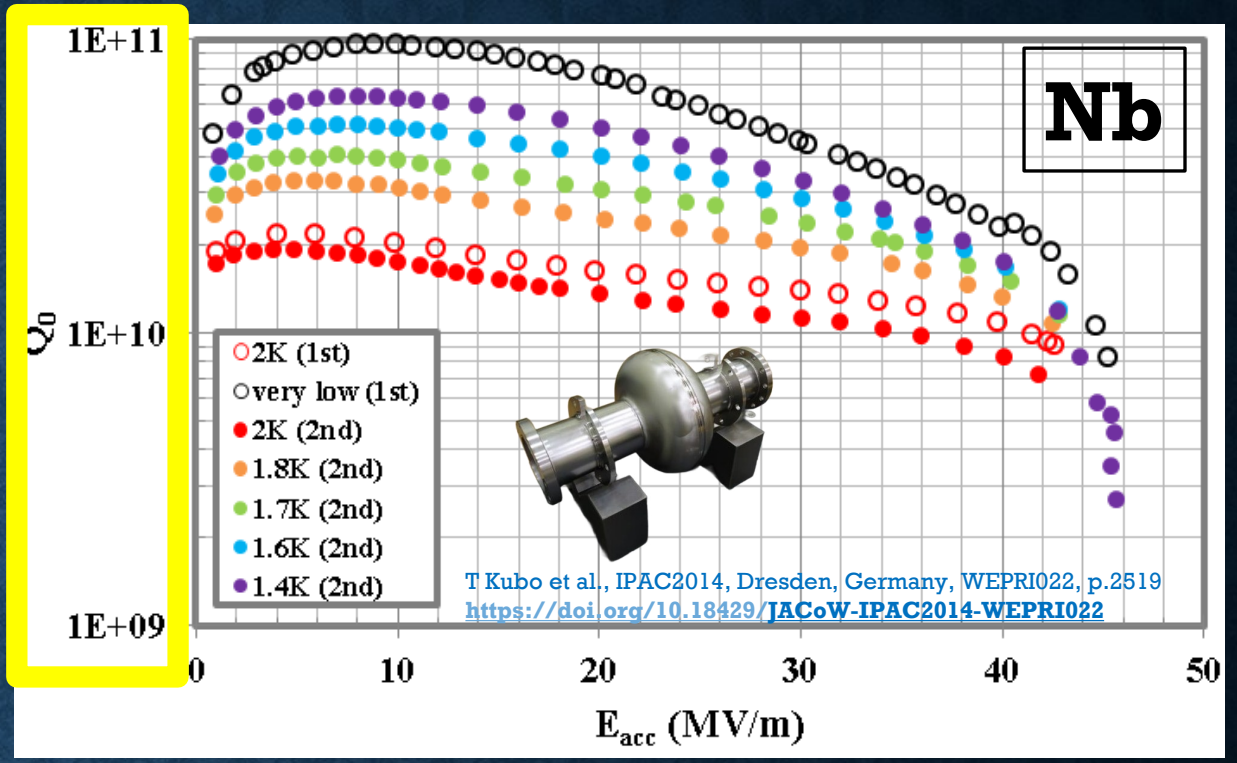
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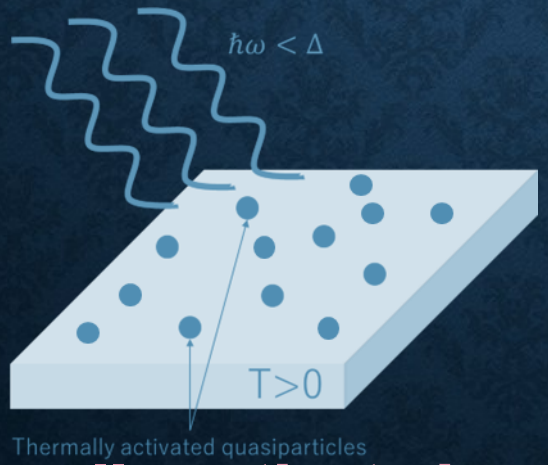




T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

There are two different temperature regimes

At a higher T ($T/T_c \gtrsim 0.2$)



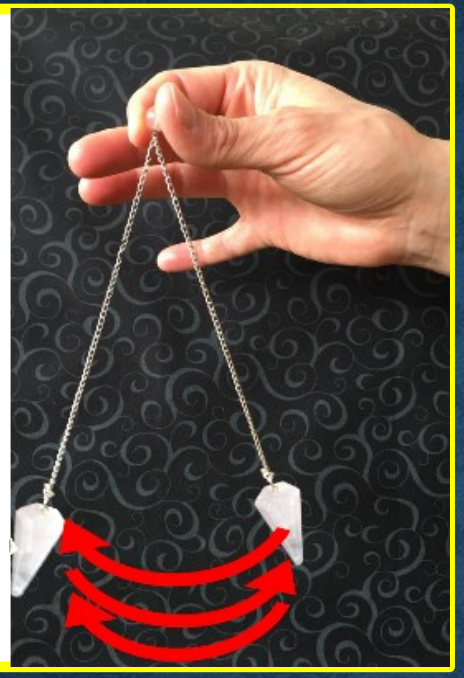
thermally activated quasiparticles

$$Q_0 = \frac{\omega U}{P} \sim \frac{U}{P_1} \propto \frac{1}{R_s}$$

P_1 : dissipation per RF cycle

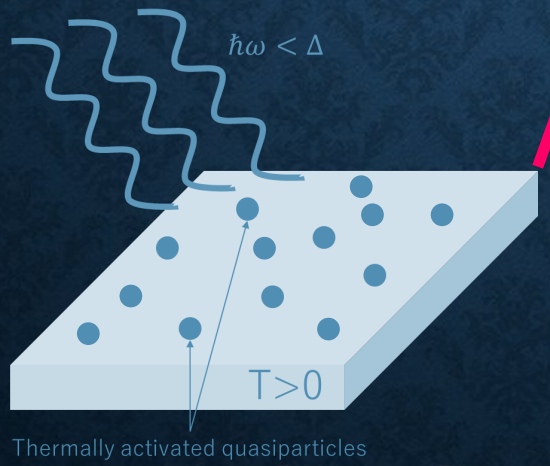
Q_0 : how many cycles it takes for the stored energy to completely dissipate

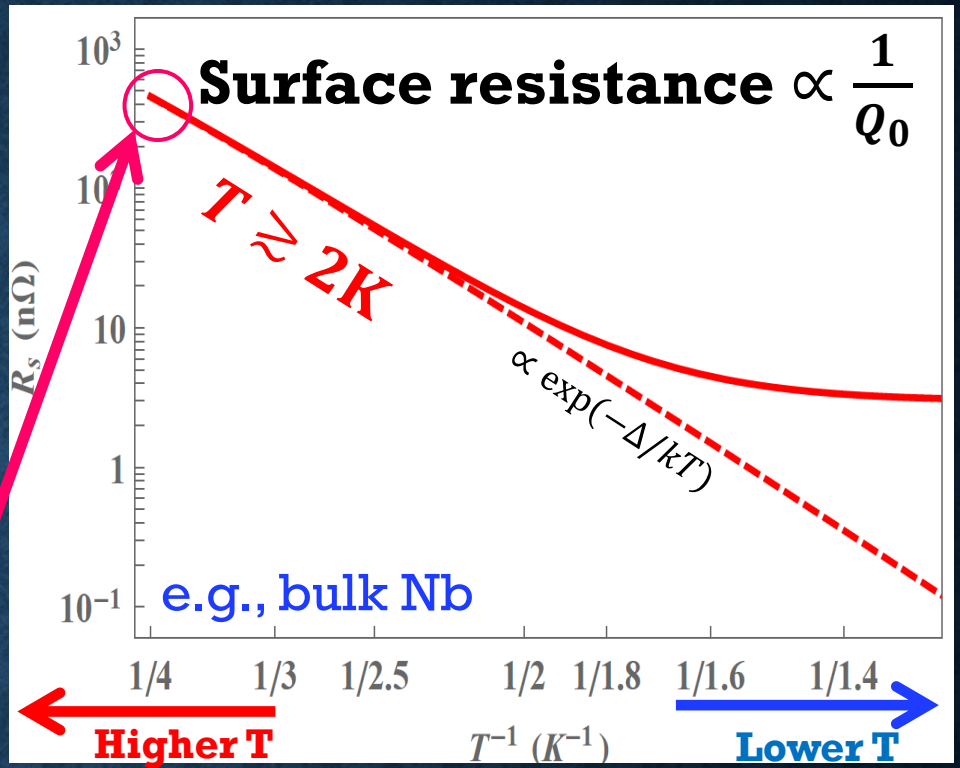
What are the causes of dissipation?



There are two different temperature regimes

At a higher T ($T/T_c \gtrsim 0.2$)





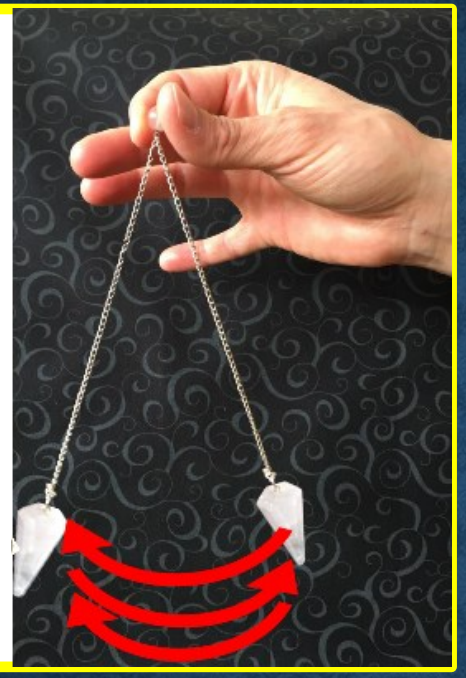
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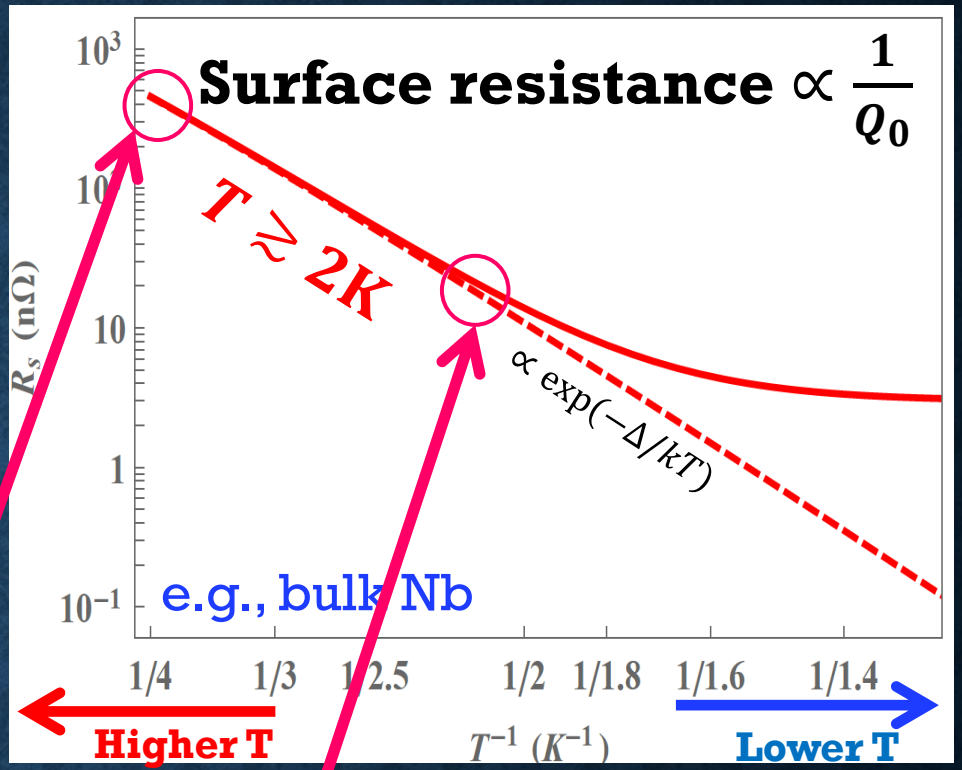
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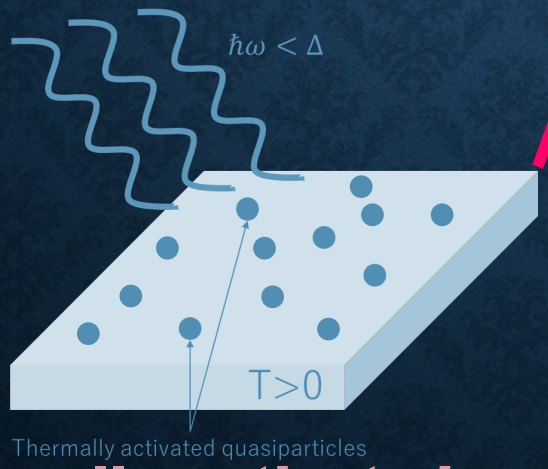
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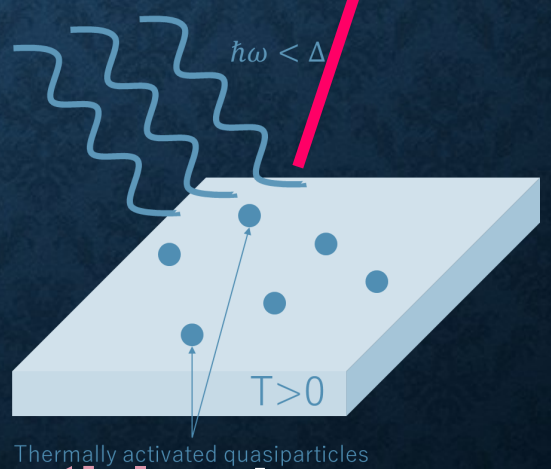
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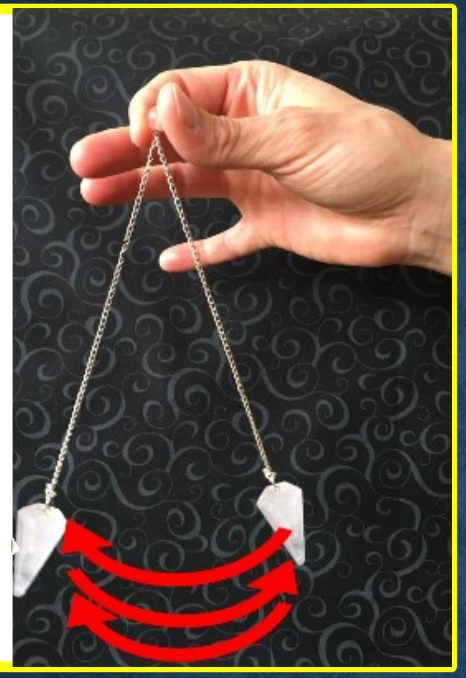
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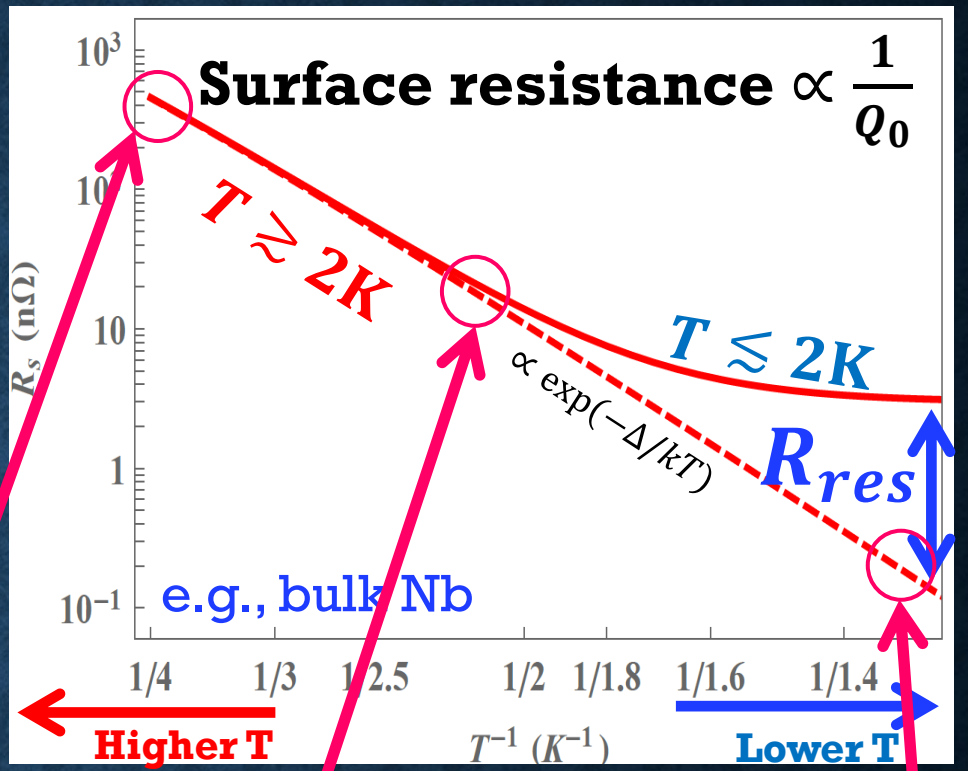
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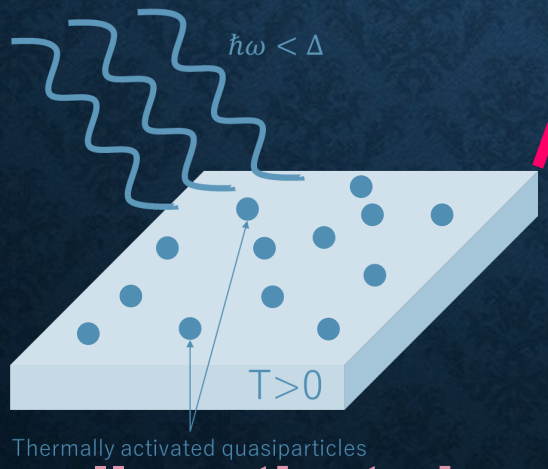
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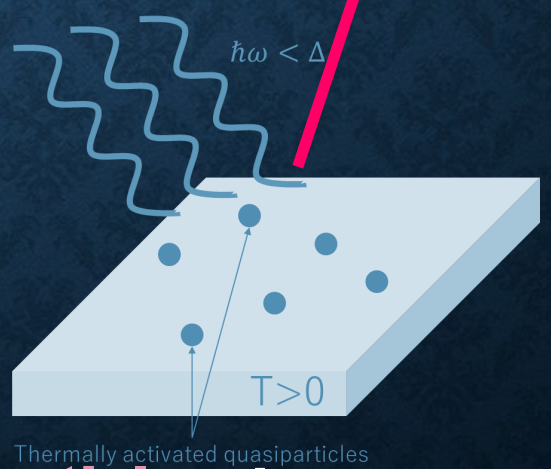
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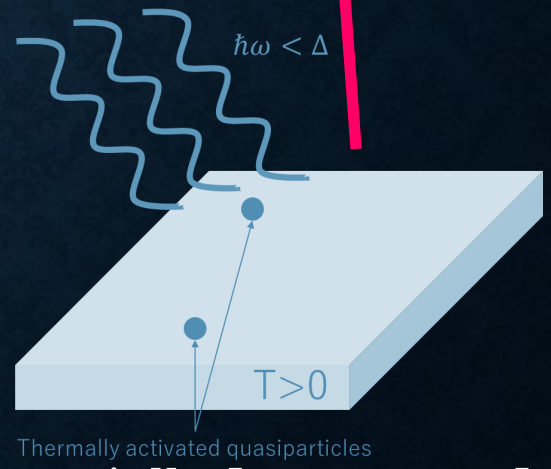


There are two different temperature regimes









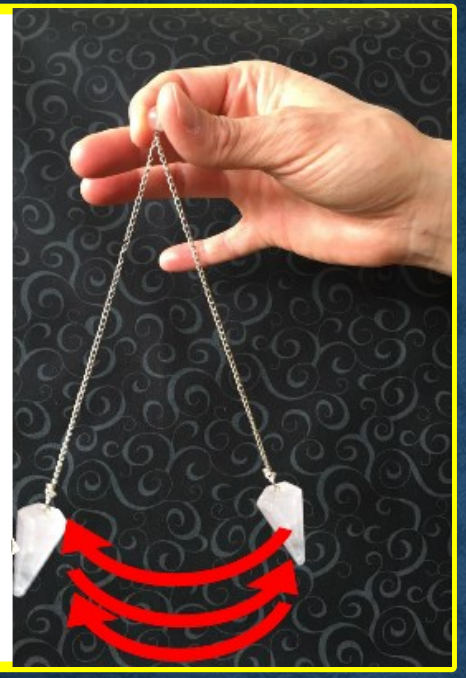
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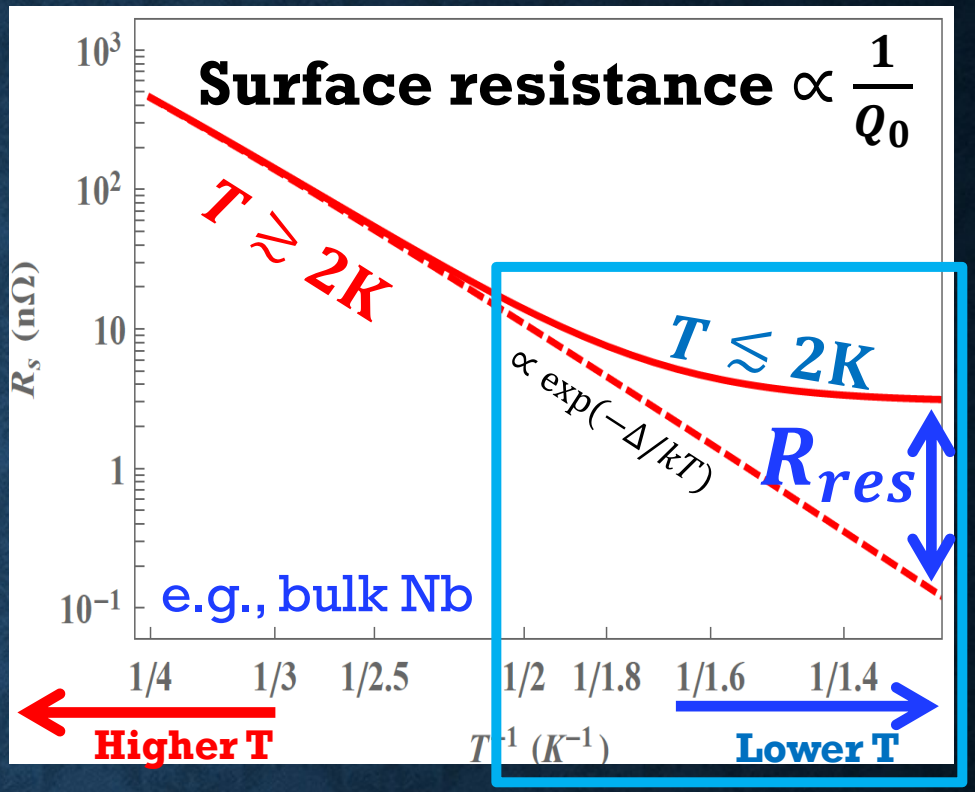
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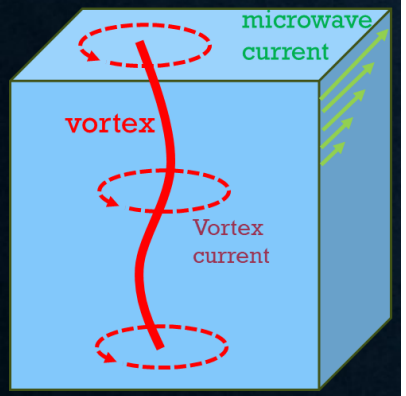
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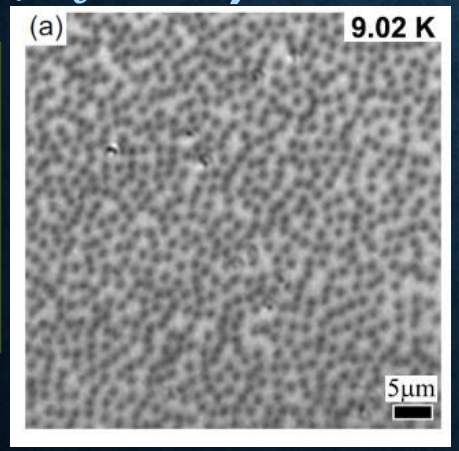




There are two different temperature regimes

At a lower T ($T/T_c \lesssim 0.2$)





Trapped ambient vortex

Microwave-driven vortex dynamics cause dissipation, which is only weakly dependent on temperature, leading to an elevated surface resistance: R_{res}

Figures adapted from Ooi et al., PRB **111**, 094519 (2025)
© American Physical Society.

Videos of vortex dynamics in **cavity-grade Nb (RRR>300)** can be found in Ooi et al., PRB **111**, 094519 (2025)

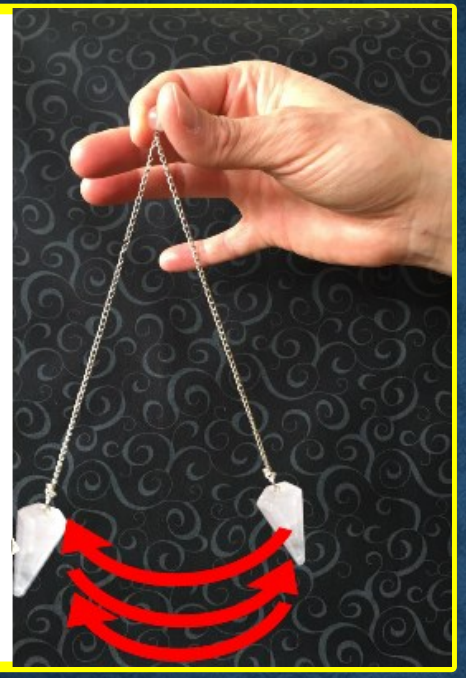
See also,
A. Gurevich and G. Ciovati, PRB **87**, 054502 (2013)
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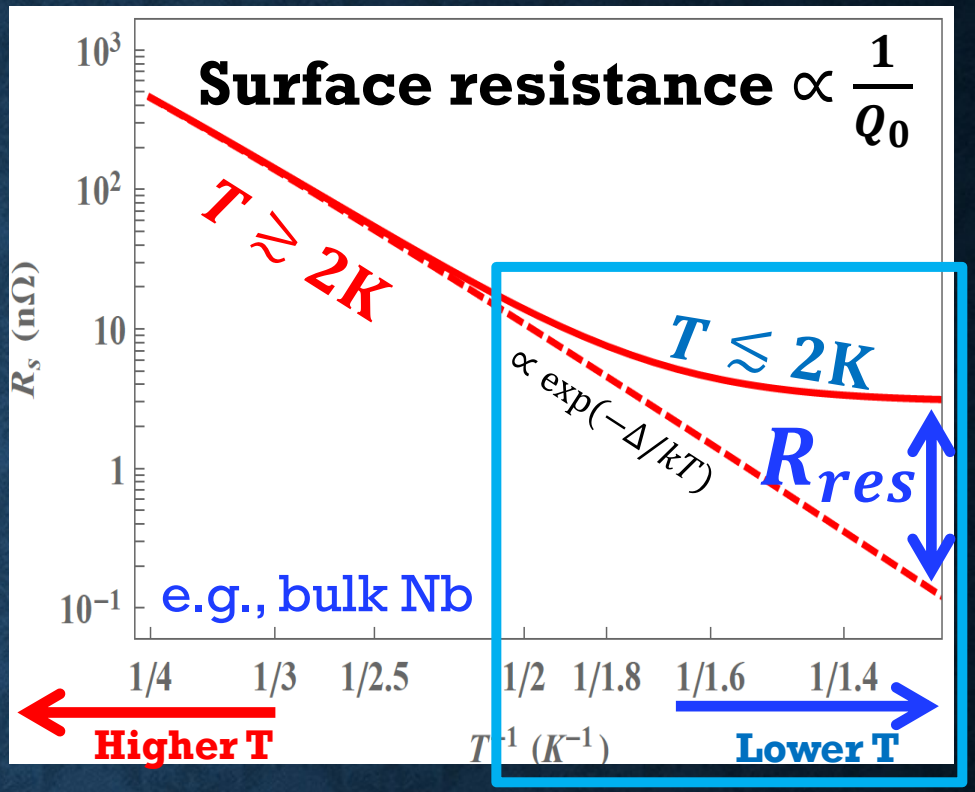
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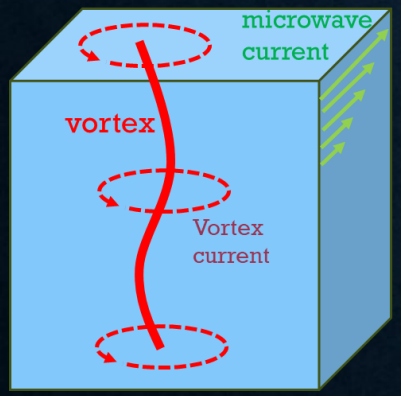
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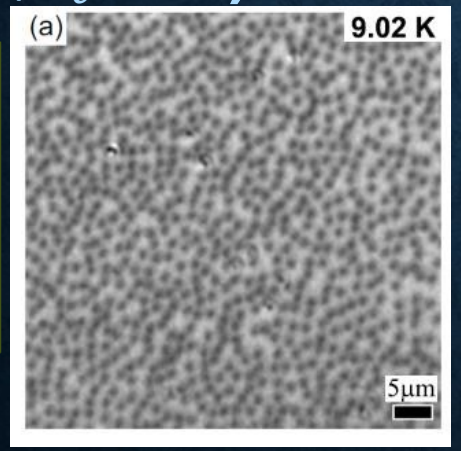




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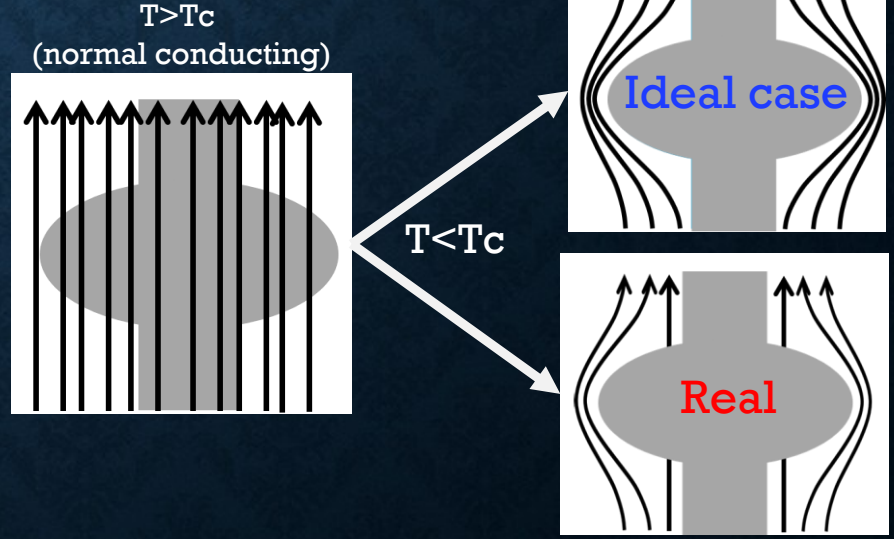


Figures adapted from Ooi et al., PRB **111**, 094519 (2025)
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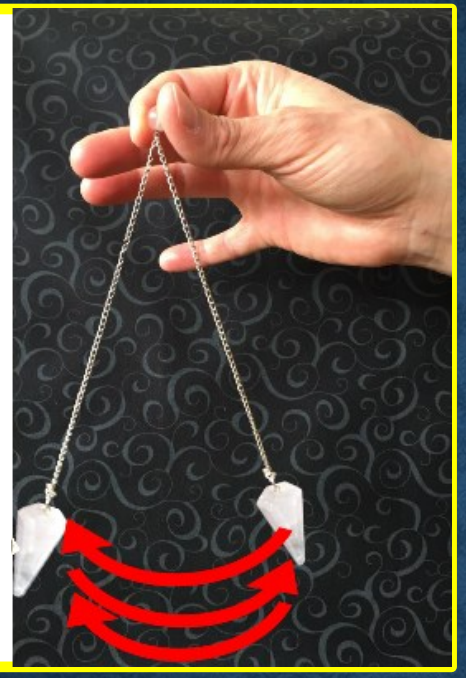
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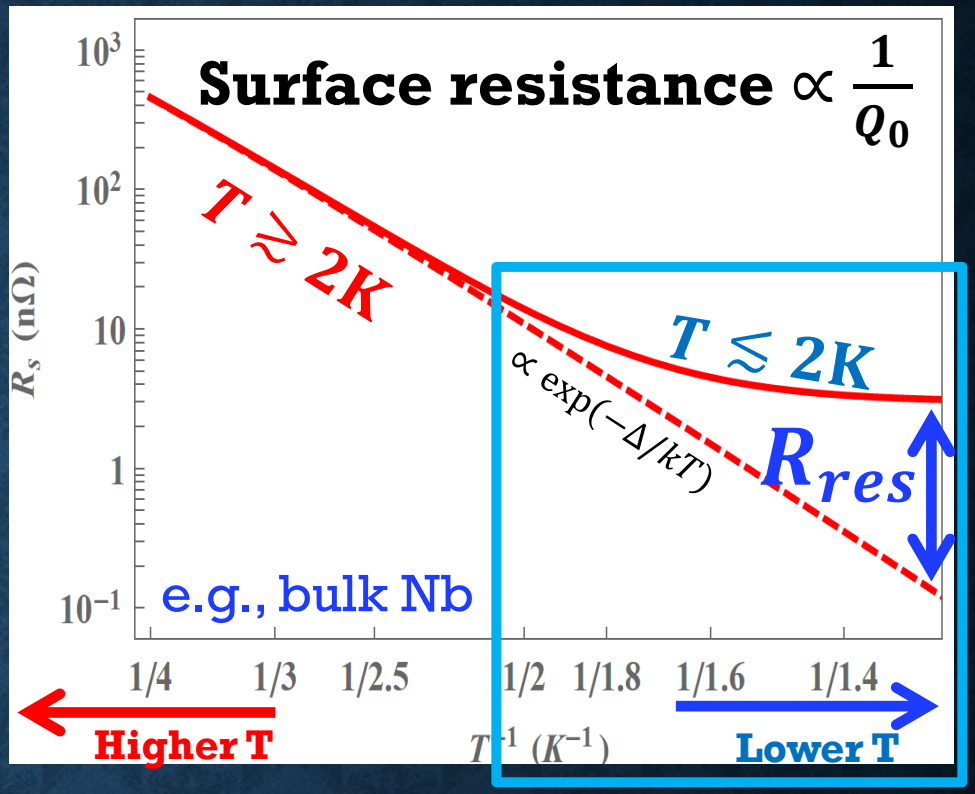
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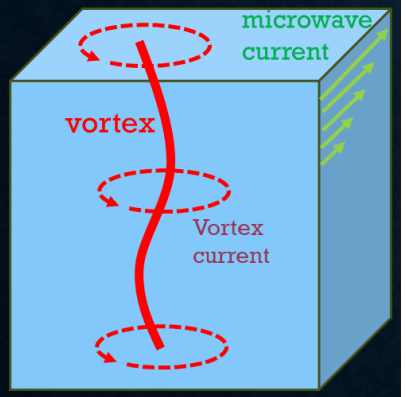
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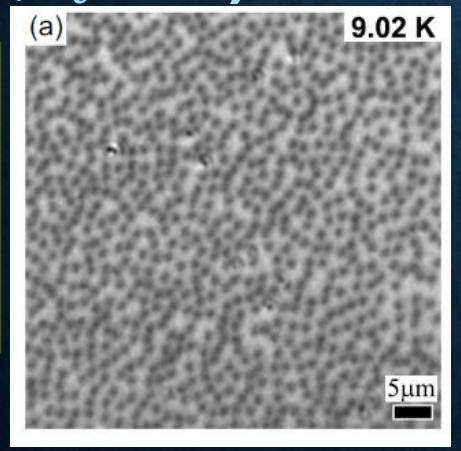




There are two different temperature regimes

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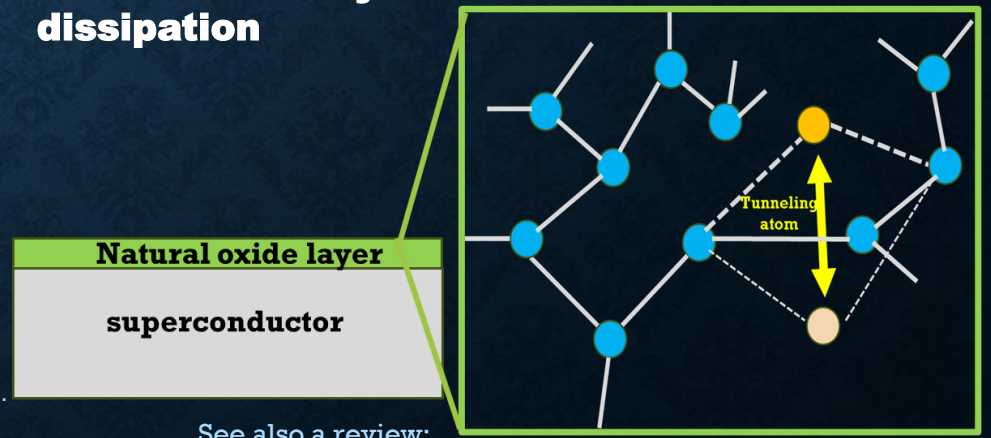
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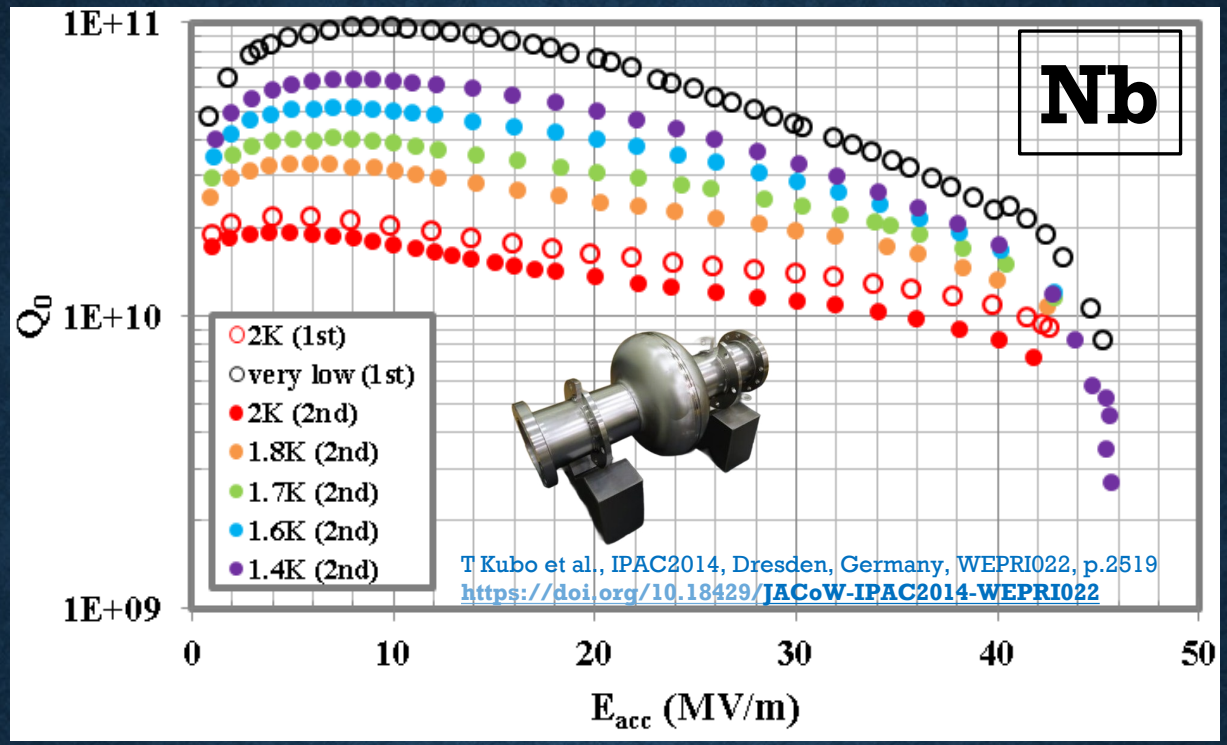
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Two-level system (TLS) defects on the natural oxide layer also contributes to the dissipation



See also a review:
Müller et al Rep. Prog. Phys. **82**, 124501 (2019)

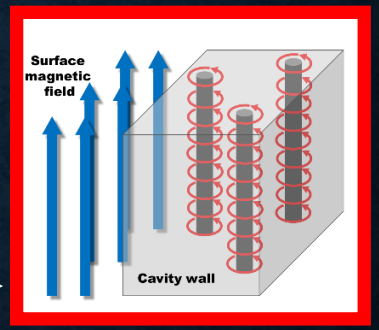
Summary of this part



T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

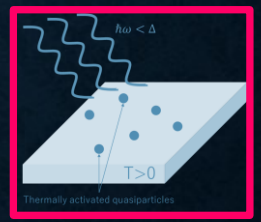
There are two key performance metrics

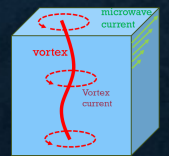
(1) Maximum field (stability of the vortex-free state)

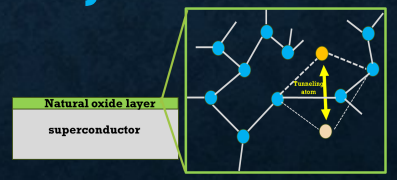


(2) Quality factor Q_0 (smallness of dissipation)

- **Quasiparticles (at $T \gtrsim 2$ K for Nb)**
- **Trapped vortex, TLS etc. (at $T \lesssim 2$ K for Nb)**







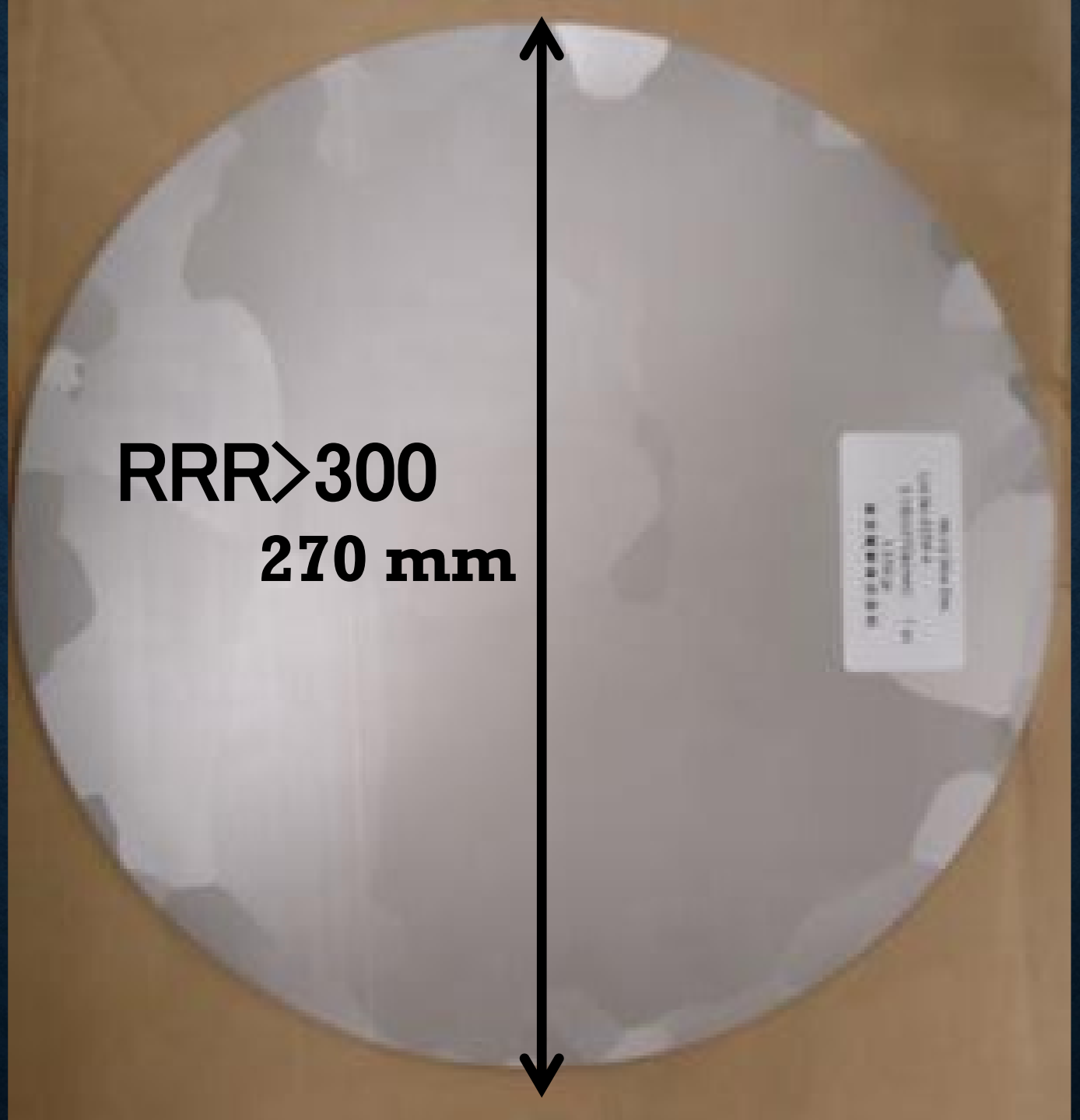


Nb cavity technology for accelerators:

Classical ILC recipe as a representative example

ILC recipe

- Materials
- Fabrication
- Electropolishing(1) (100-200μm)
- Annealing (750-900°C, 2-3hours)
- Electropolishing(2) (several μm)
- High pressure rinsing
- Assembling
- Low-temperature baking (100-140°C, 48hours)
- Test cavity



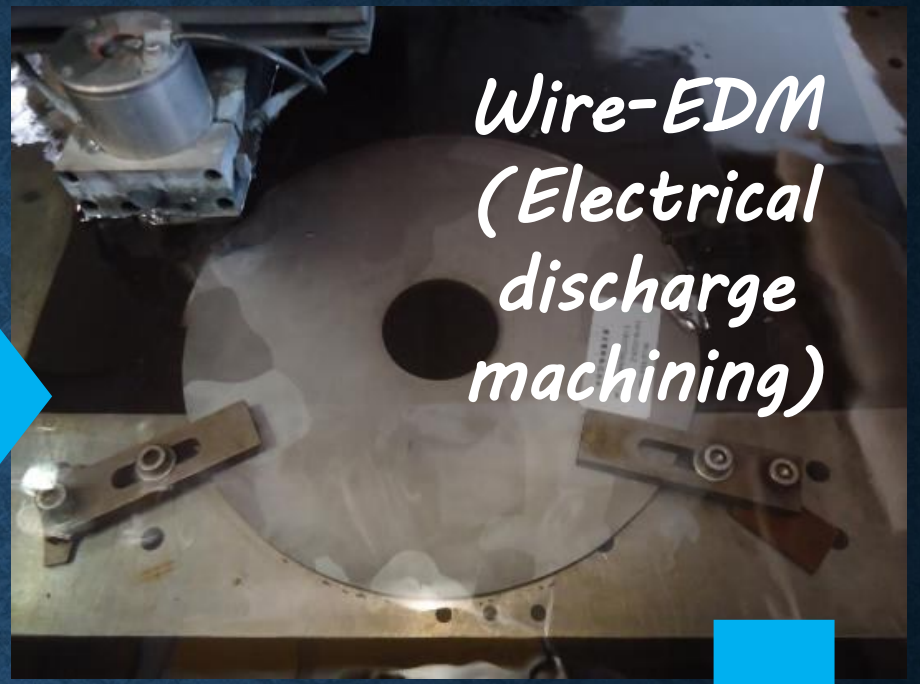
High purity Nb

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Let's start fabrication.





Wire-EDM (Electrical discharge machining)





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Electron beam welding

Cavity material quality is maintained

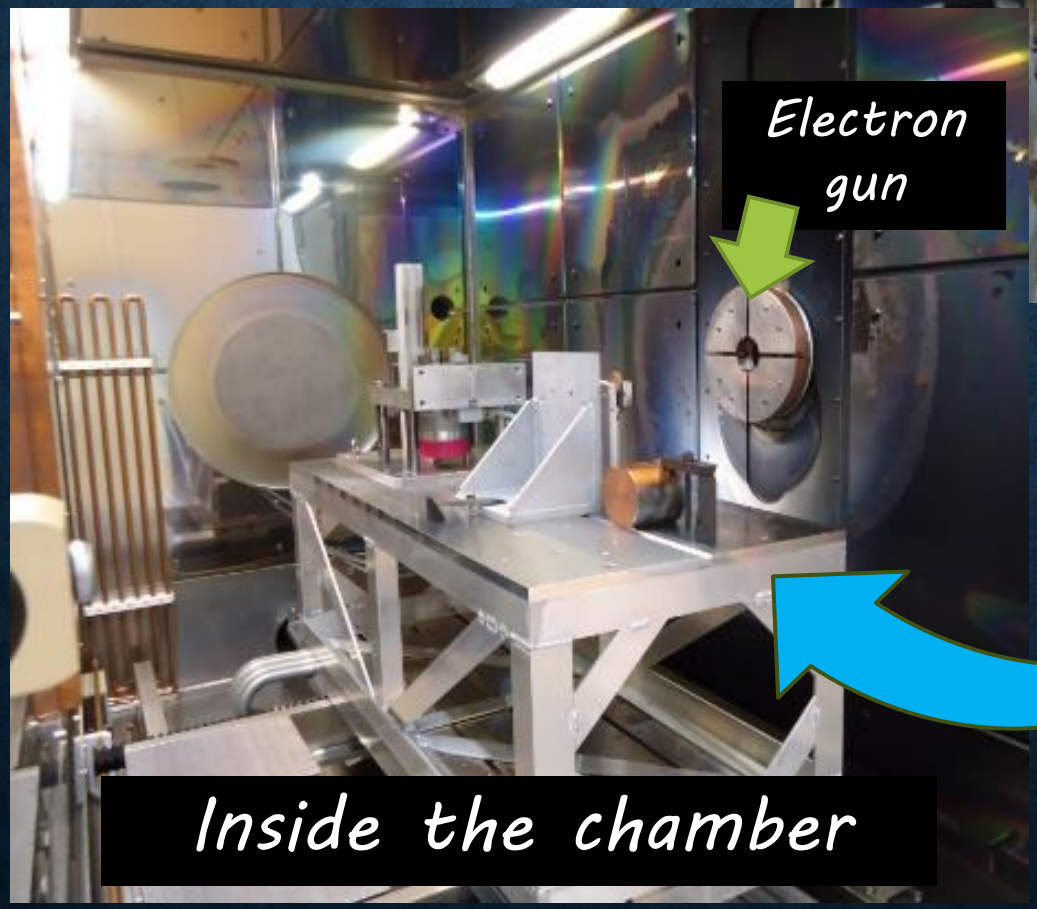


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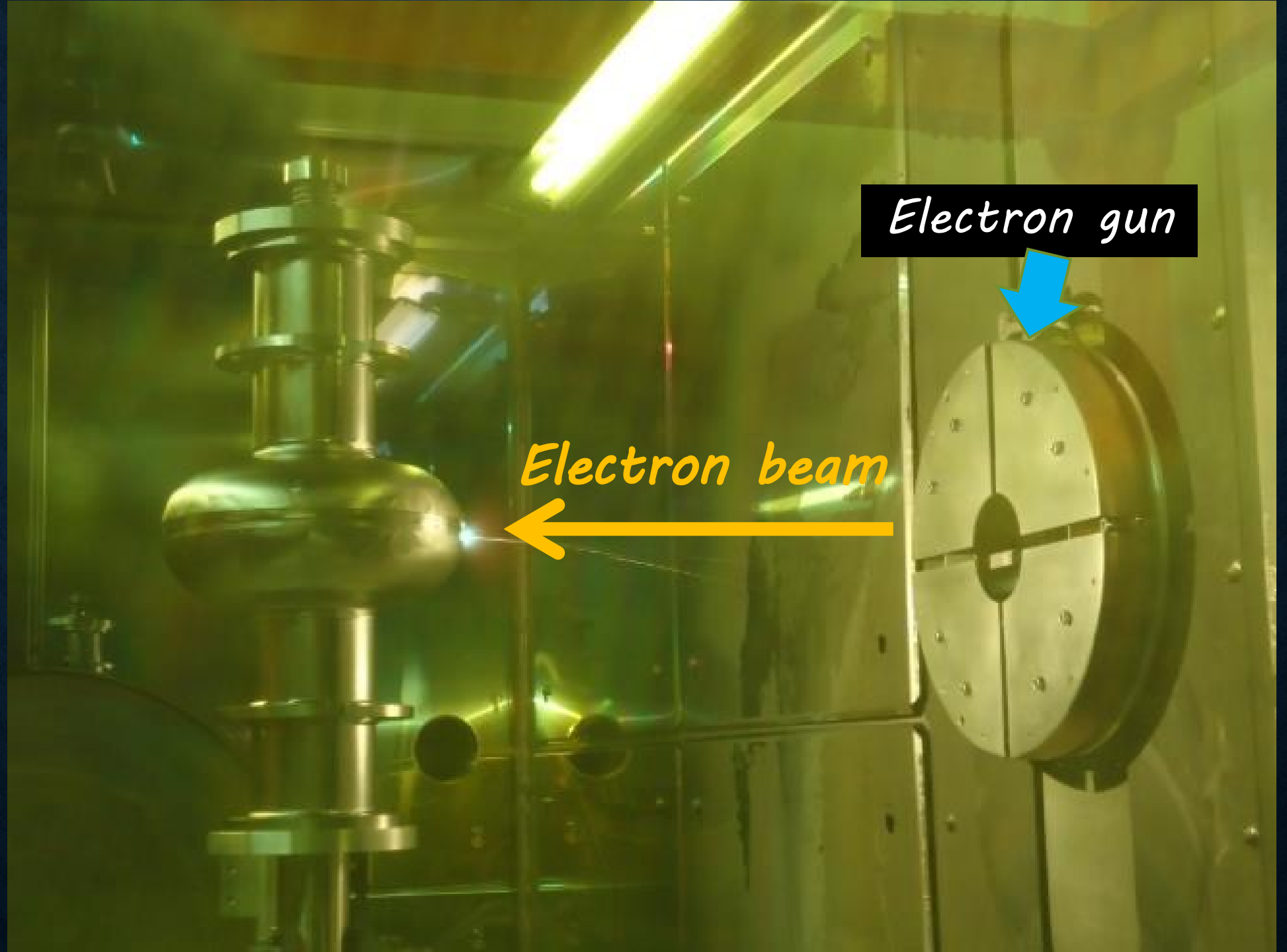






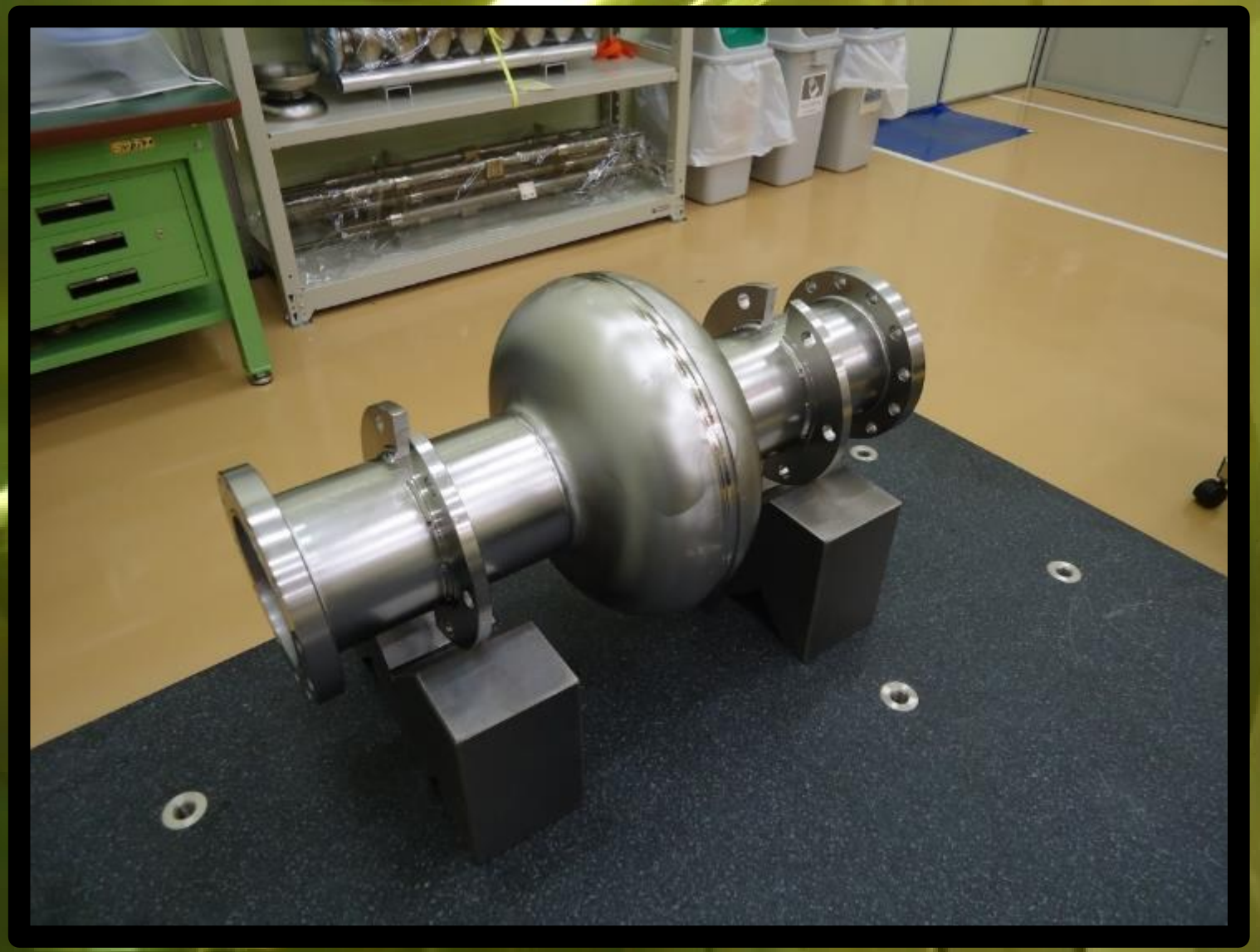
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Electropolishing (EP)

- The electrolyte is a mixture of (9:1) sulfuric acid and hydrofluoric acid (HF).
- A DC voltage is applied with Nb as the anode and aluminum as the cathode.
- Niobium is oxidized to form Nb_2O_5.
- This oxide layer is dissolved and polished away by HF present in the electrolyte.











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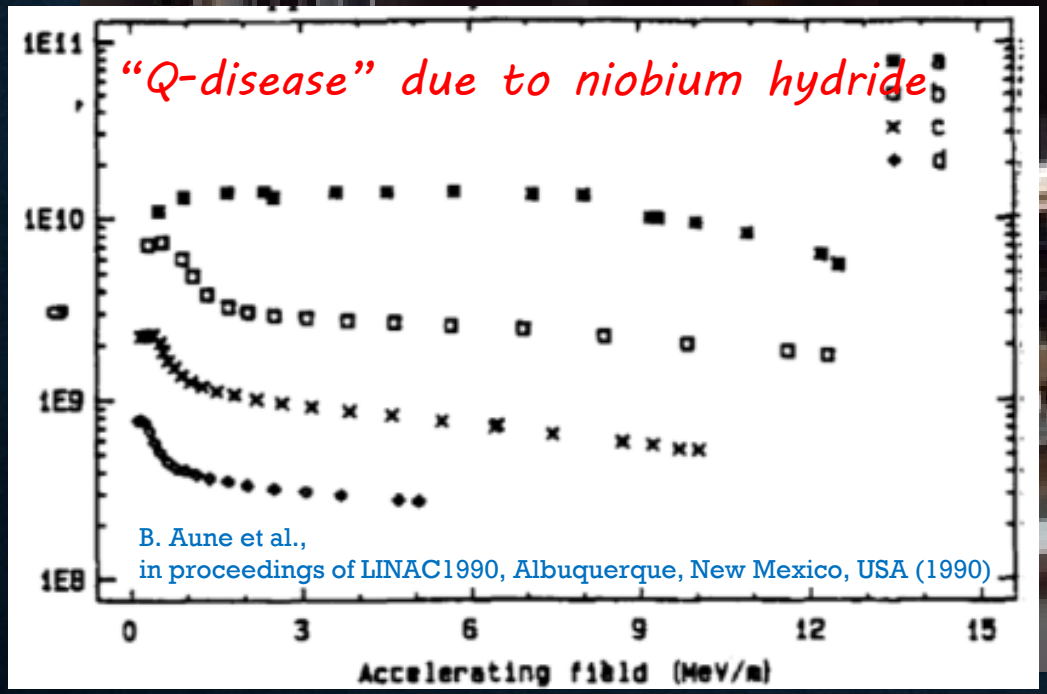


Annealing

- To degas hydrogen — prevents hydrogen Q-disease (requires >700 ℃)

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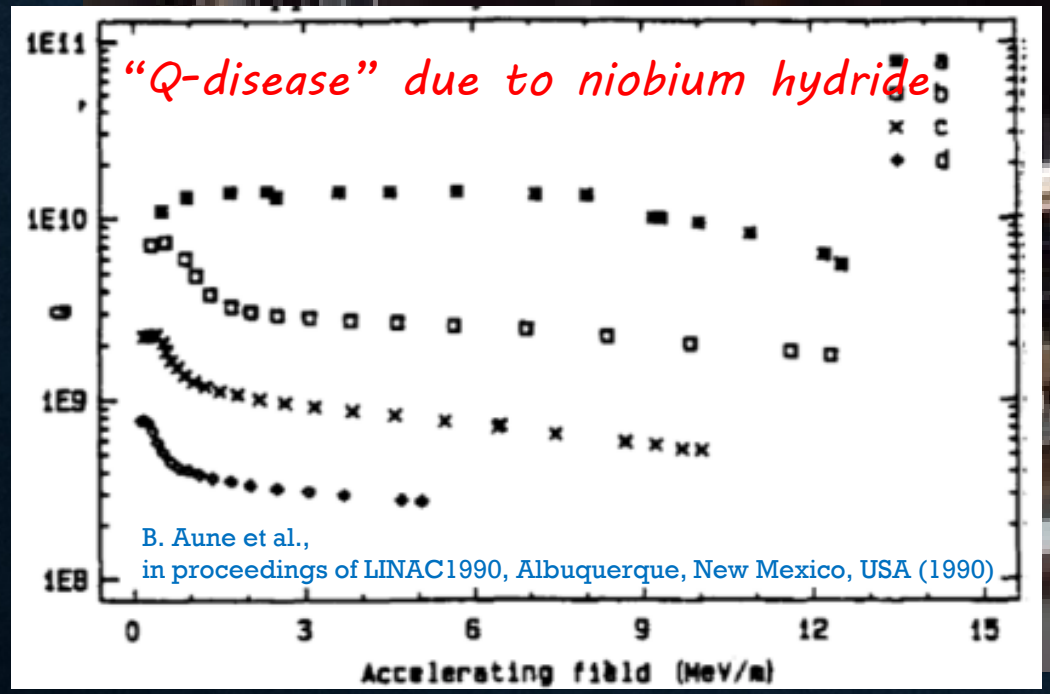


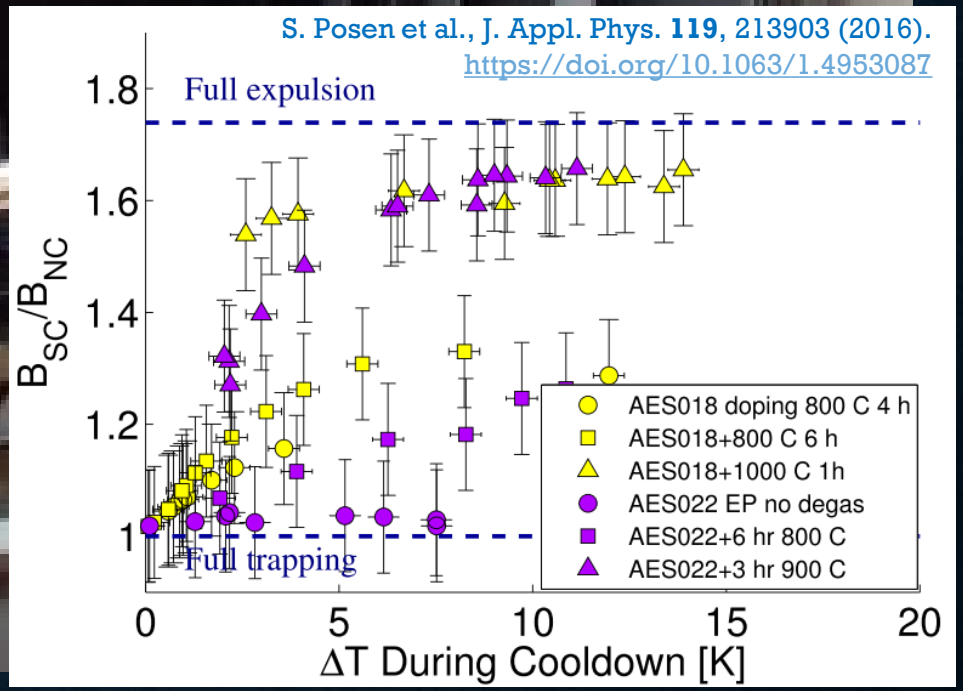
Annealing

- To degas hydrogen — prevents hydrogen Q-disease (requires >700 ℃)
- To reduce pinning centers — minimizes trapped flux (requires ~900 ℃)

ILC recipe

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- Test cavity





A second EP step, removing only a few microns.

ILC recipe

- Materials
- Fabrication
- Electropolishing(1) (100-200µm)
- Annealing (750-900°C, 2-3hours)
- **Electropolishing(2) (several µm)**
- High pressure rinsing
- Assembling
- Low-temperature baking (100-140°C, 48hours)
- Test cavity

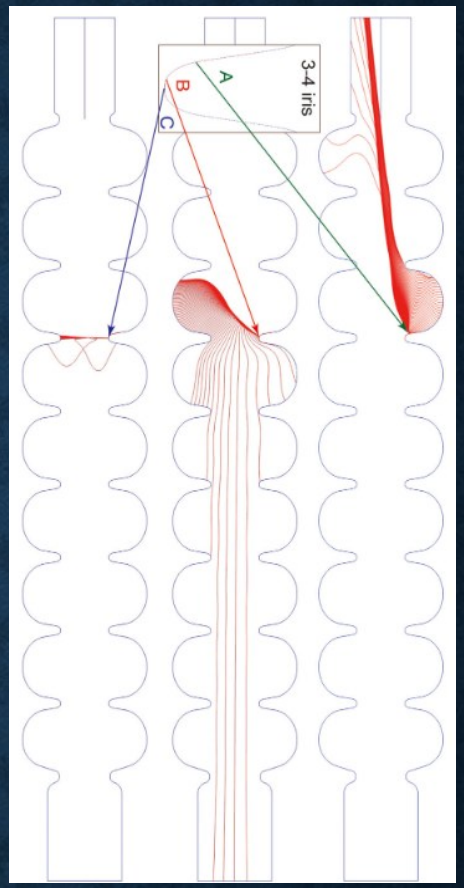


ILC recipe

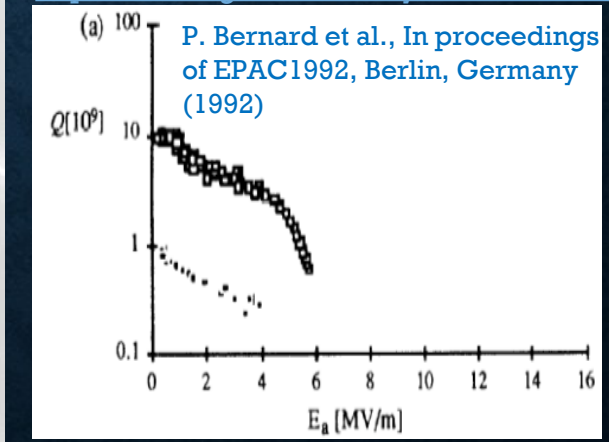
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- Test cavity



Courtesy of Mathieu Omet (KEK)



H. Sakai et al.,
Phys. Rev. Accel. Beams **22**,022002 (2019)
https://doi.org/10.1103/PhysRevAccelBeams.22.022002



P. Bernard et al., In proceedings of EPAC1992, Berlin, Germany (1992)

ILC recipe

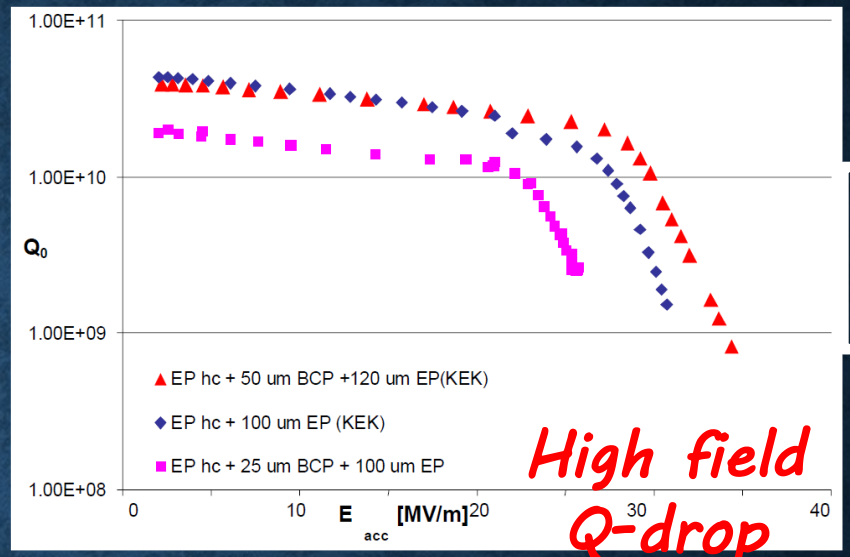
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120°C—48h bake

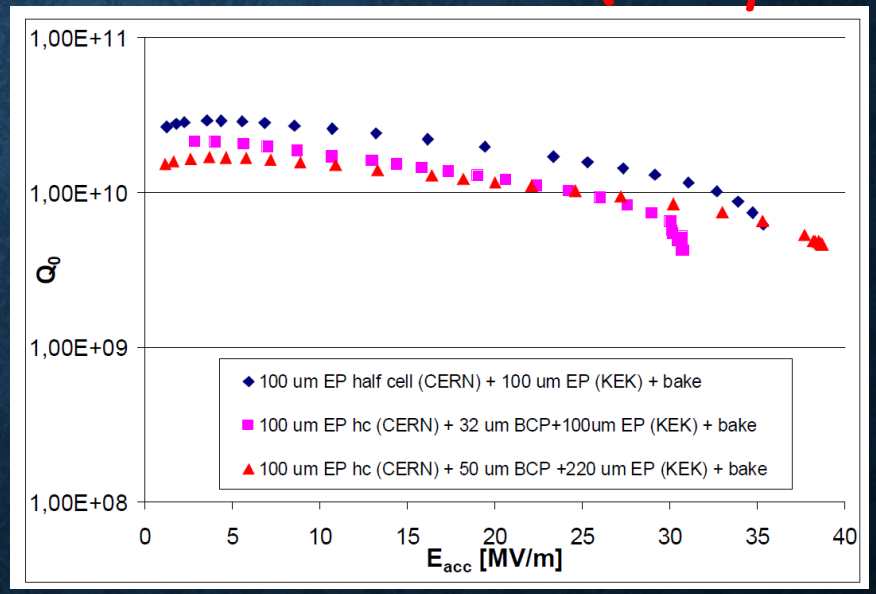


Courtesy of Mathieu Omet (KEK)

L. Lilje et al., in Proceedings of SRF1999, Santa Fe, New Mexico, USA (1999), p. 74, TUA001.



Without baking



With baking

Spatial variation of the mean free path on the scale of the penetration depth suppresses the surface current, thereby increasing the maximum achievable field.

T. Kubo, Supercond. Sci. and Technol. **30**, 023001 (2017); Supercond Sci. and Technol. **34**, 045006 (2021)
V. Ngampruetikorn and J. A. Sauls, Phys. Rev. Research **1**, 012015(R) (2019)

ILC recipe

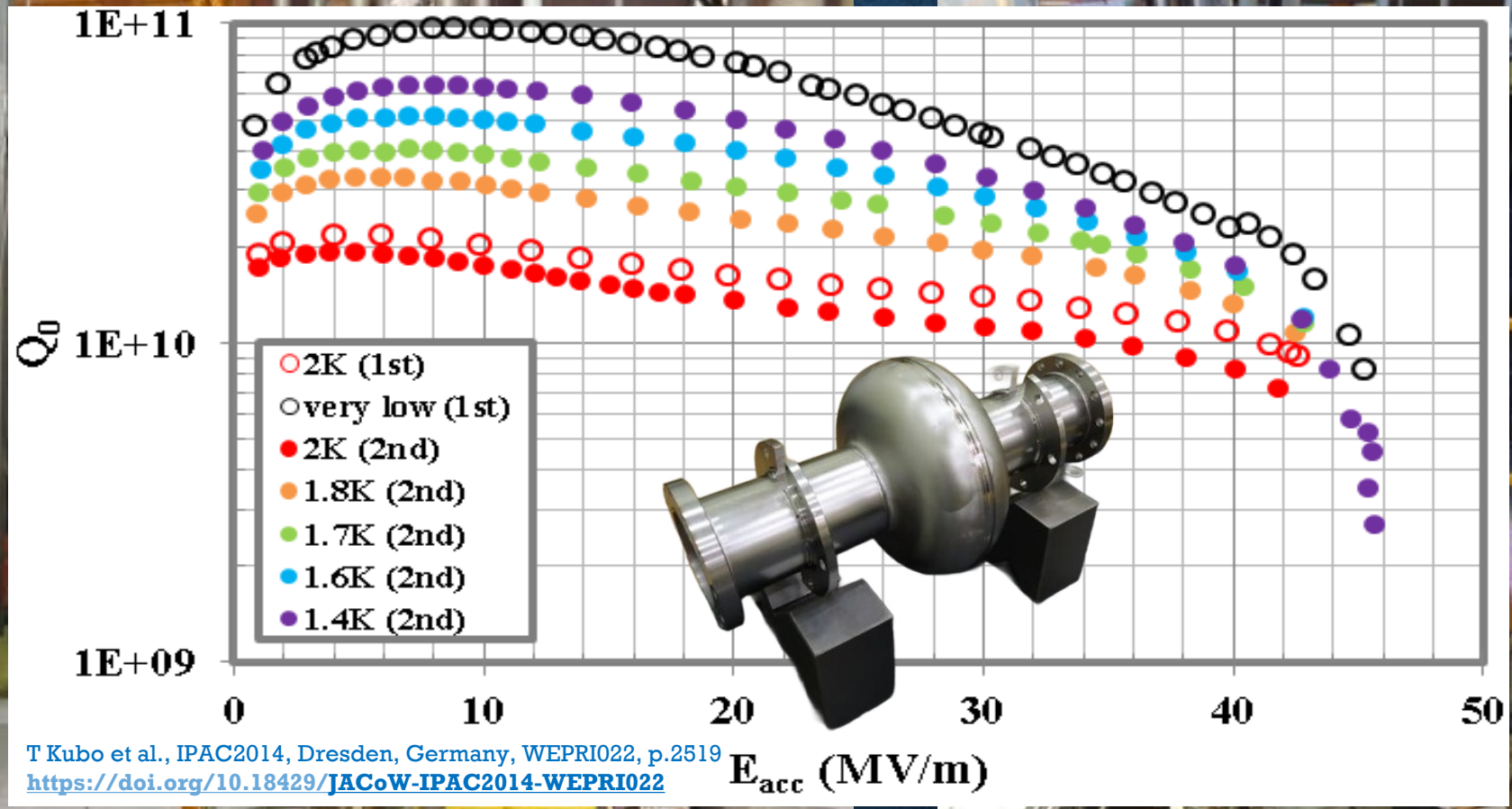
- Materials
- Fabrication
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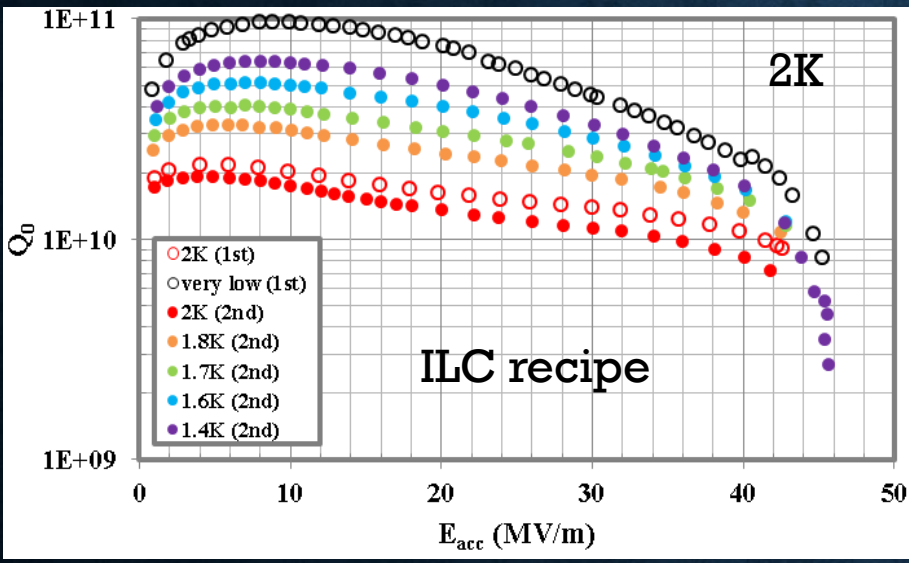
ILC recipe

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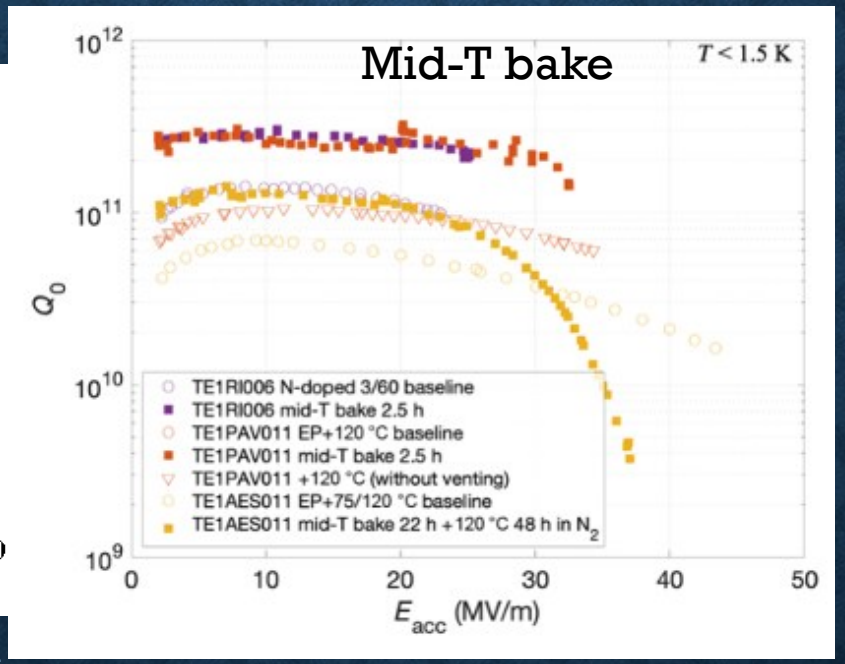


T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

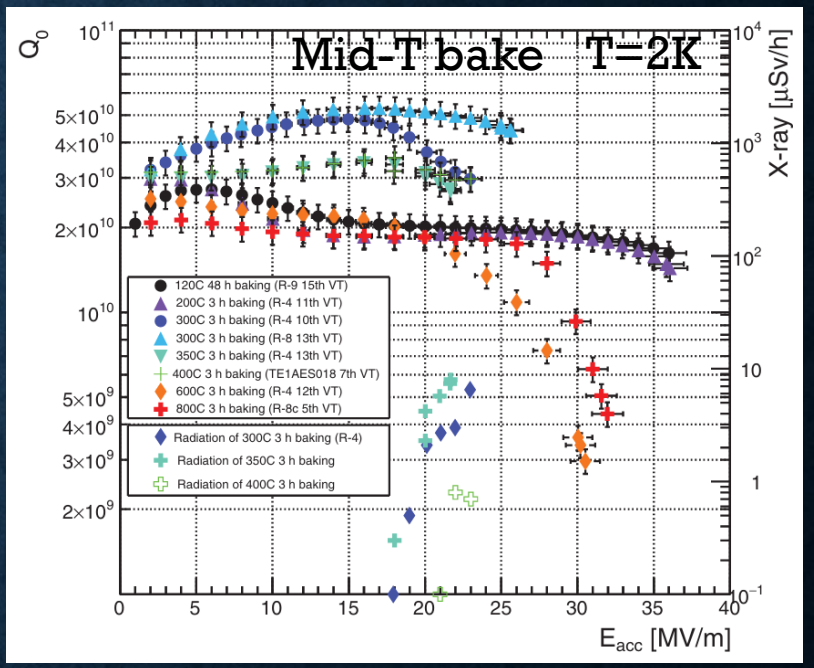
Many modern recipes exist: N-doping, N-infusion, two-step bake, mid-T bakes, etc. **They are all variants of the classical recipe**, with some steps replaced or modified.



T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

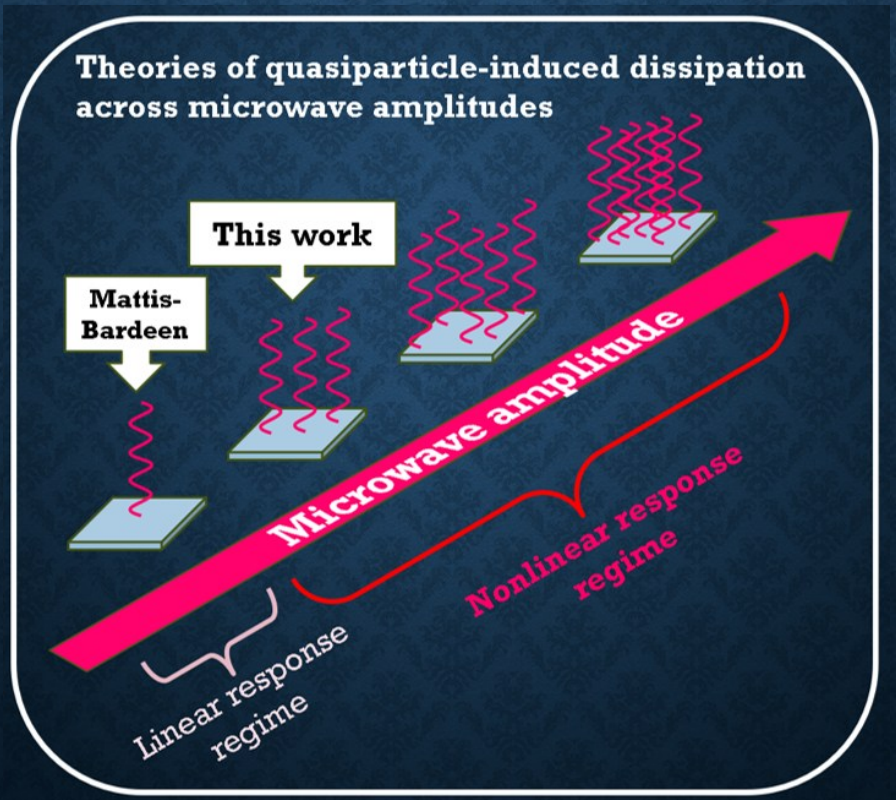


S. Posen et al., Ultralow Surface Resistance via Vacuum Heat Treatment of Superconducting Radio-Frequency Cavities,
Phys. Rev. Applied **13**, 014024 (2020)
https://doi.org/10.1103/PhysRevApplied.13.014024



H. Ito et al., Influence of furnace baking on Q–E behavior of superconducting accelerating cavities,
Prog. Theor. Exp. Phys. **2021**, 071G01 (2021)
https://doi.org/10.1093/ptep/ptab056

Understanding more about physics of Q factor



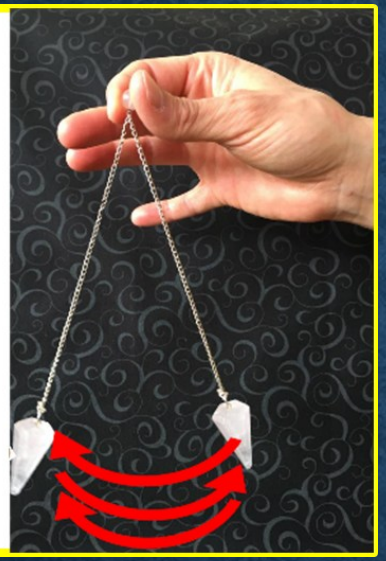
In the accelerator regime (T~2K), the quality factor is largely determined by quasiparticle dissipation.

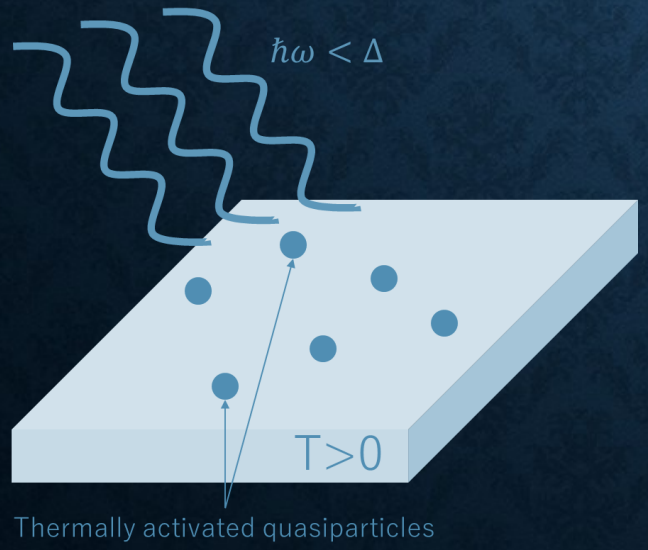
$$Q_0 = \frac{\omega U}{P} \sim \frac{U}{P_1} \propto \frac{1}{R_s}$$

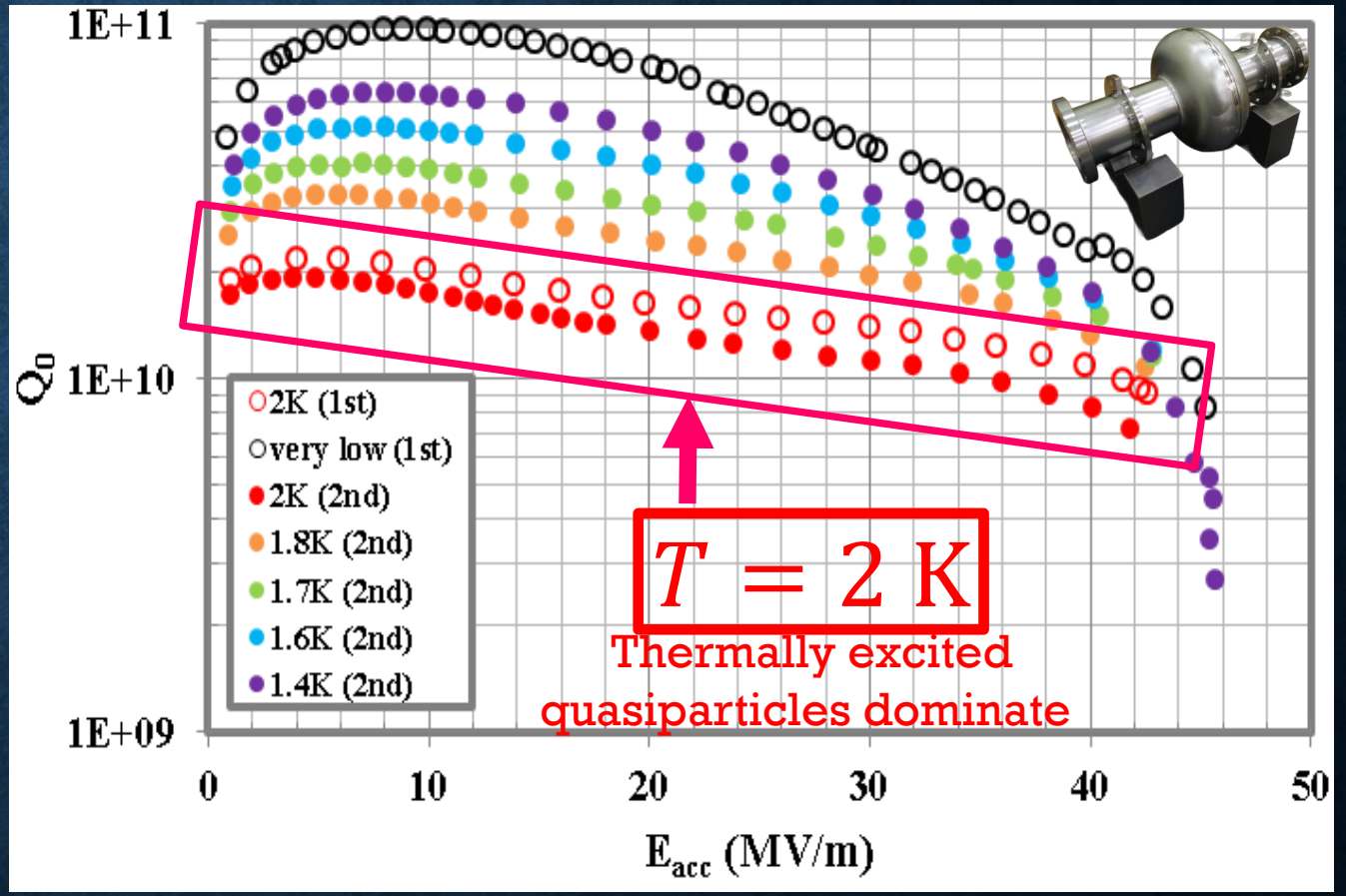
P_1 : dissipation per RF cycle

Q_0 : how many cycles it takes for the stored energy to completely dissipate

What are the causes of dissipation?







T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

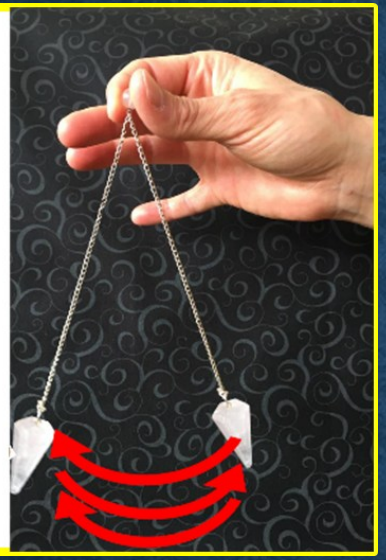
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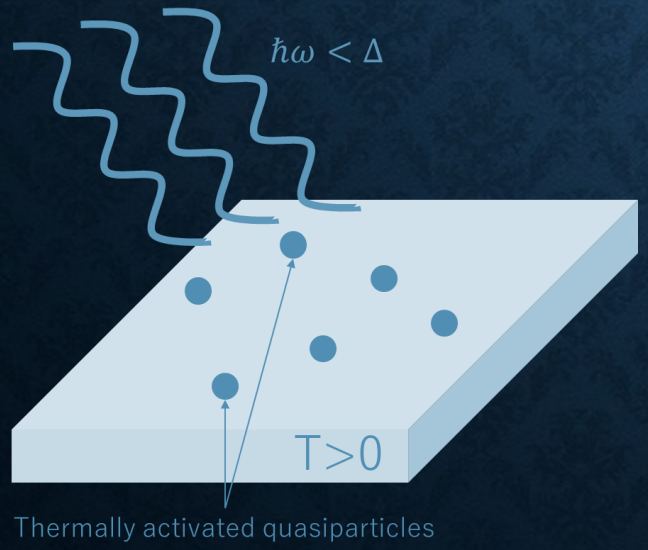
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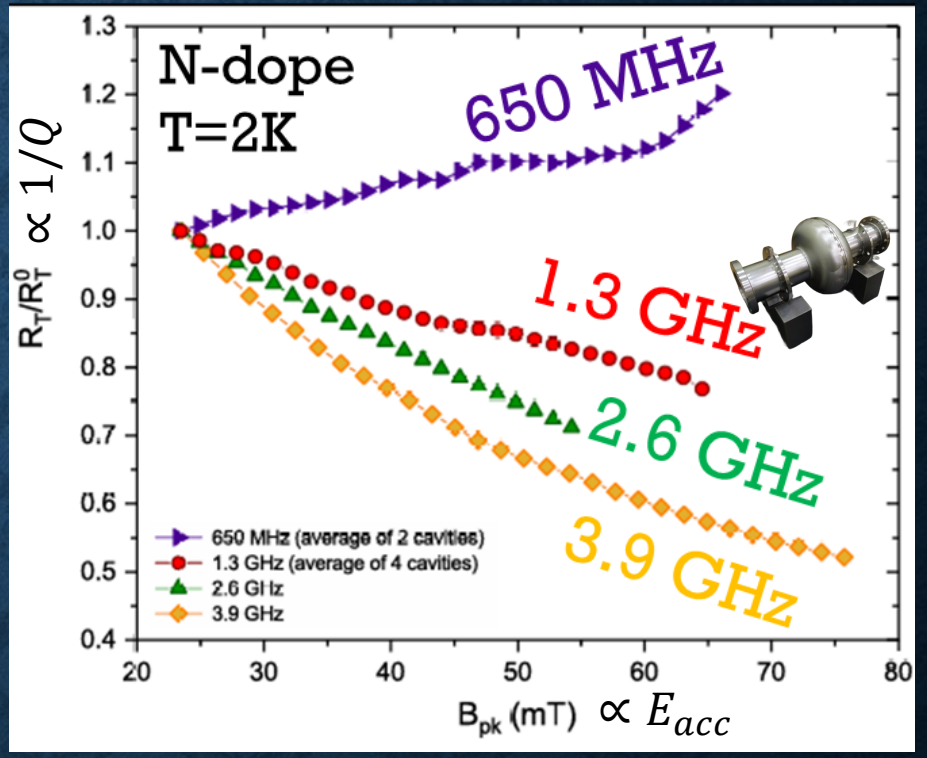
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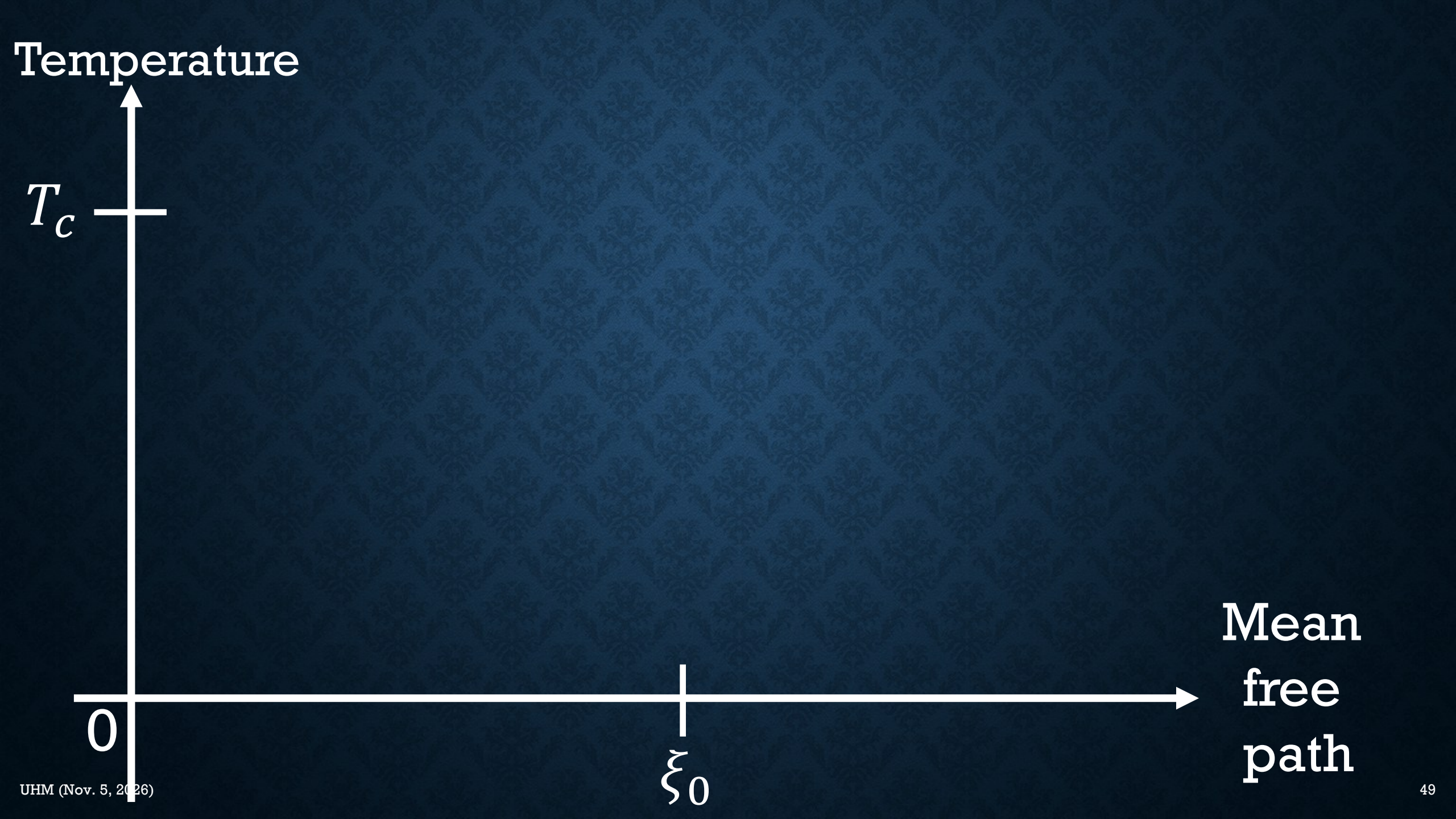


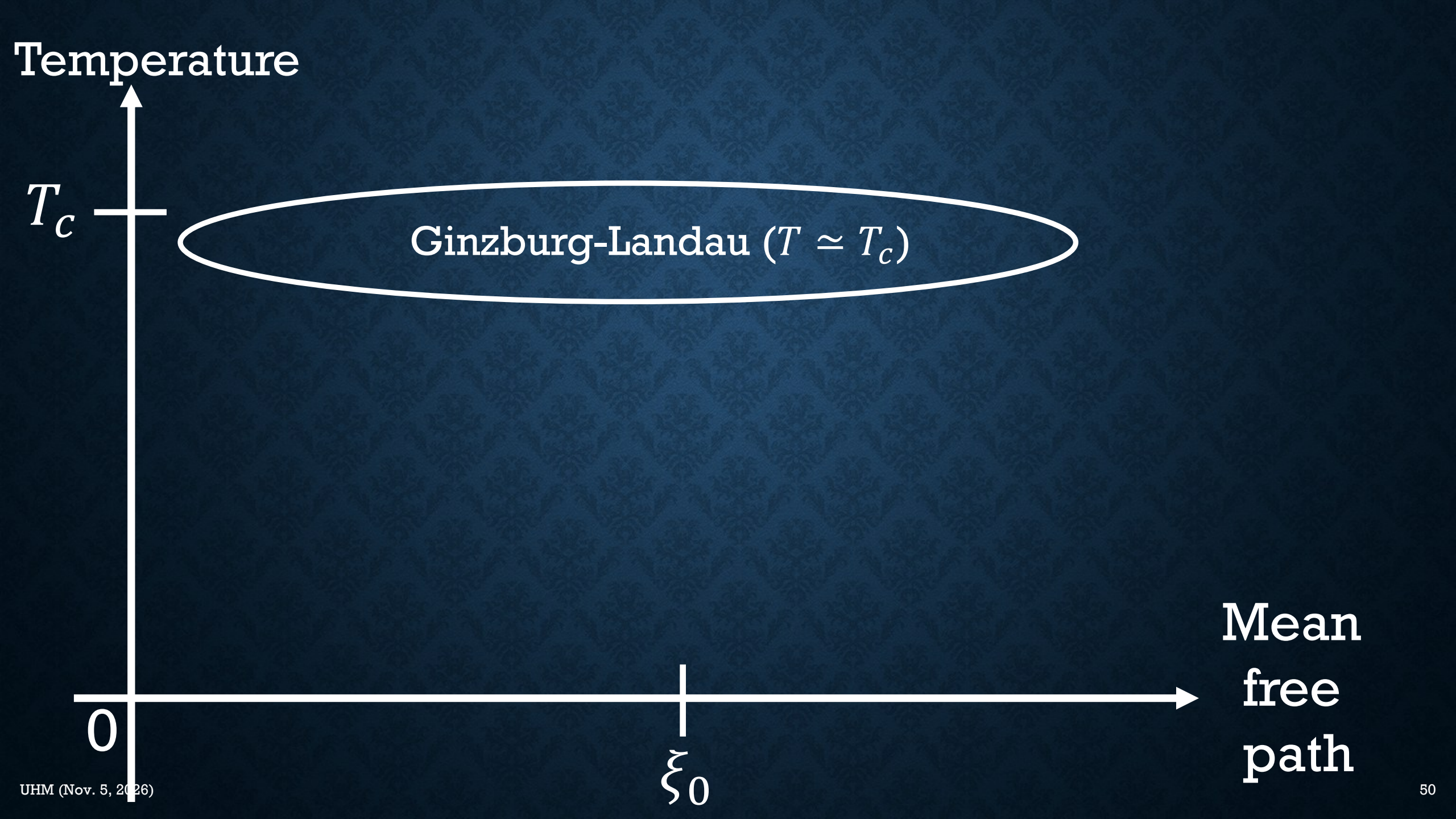


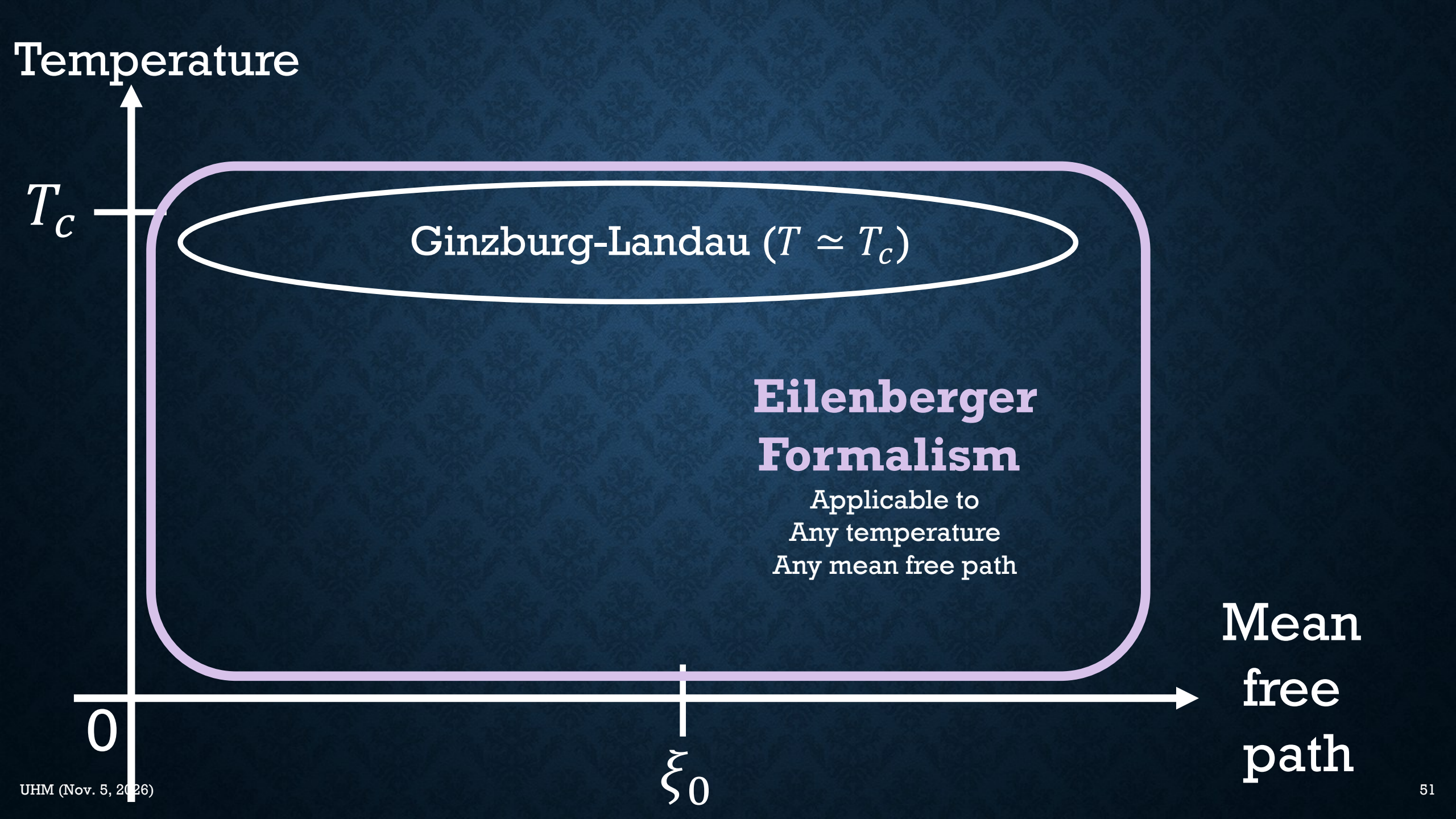


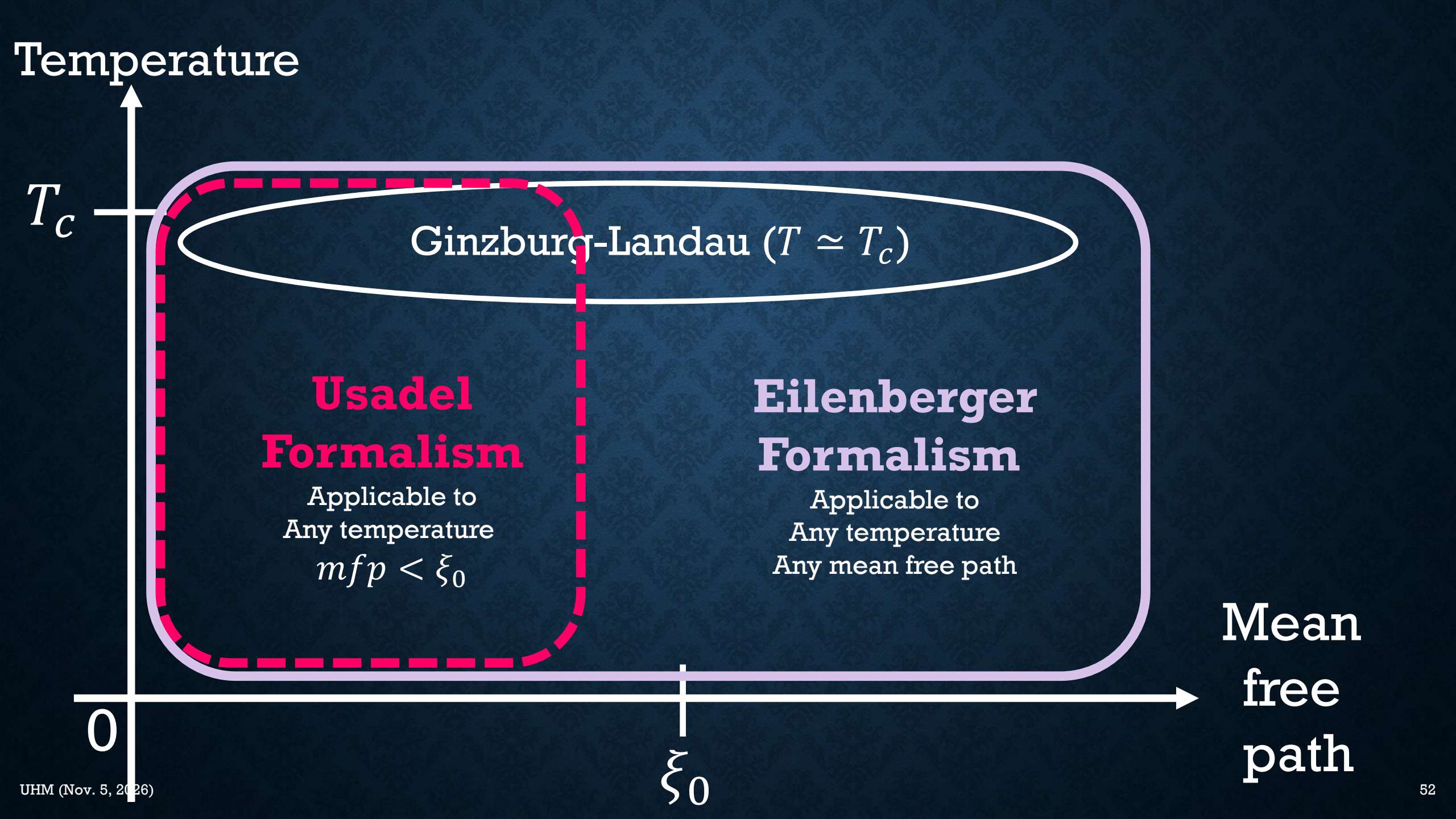
M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
DOI: https://doi.org/10.1103/PhysRevLett.121.224801

Actually, this is a very challenging problem

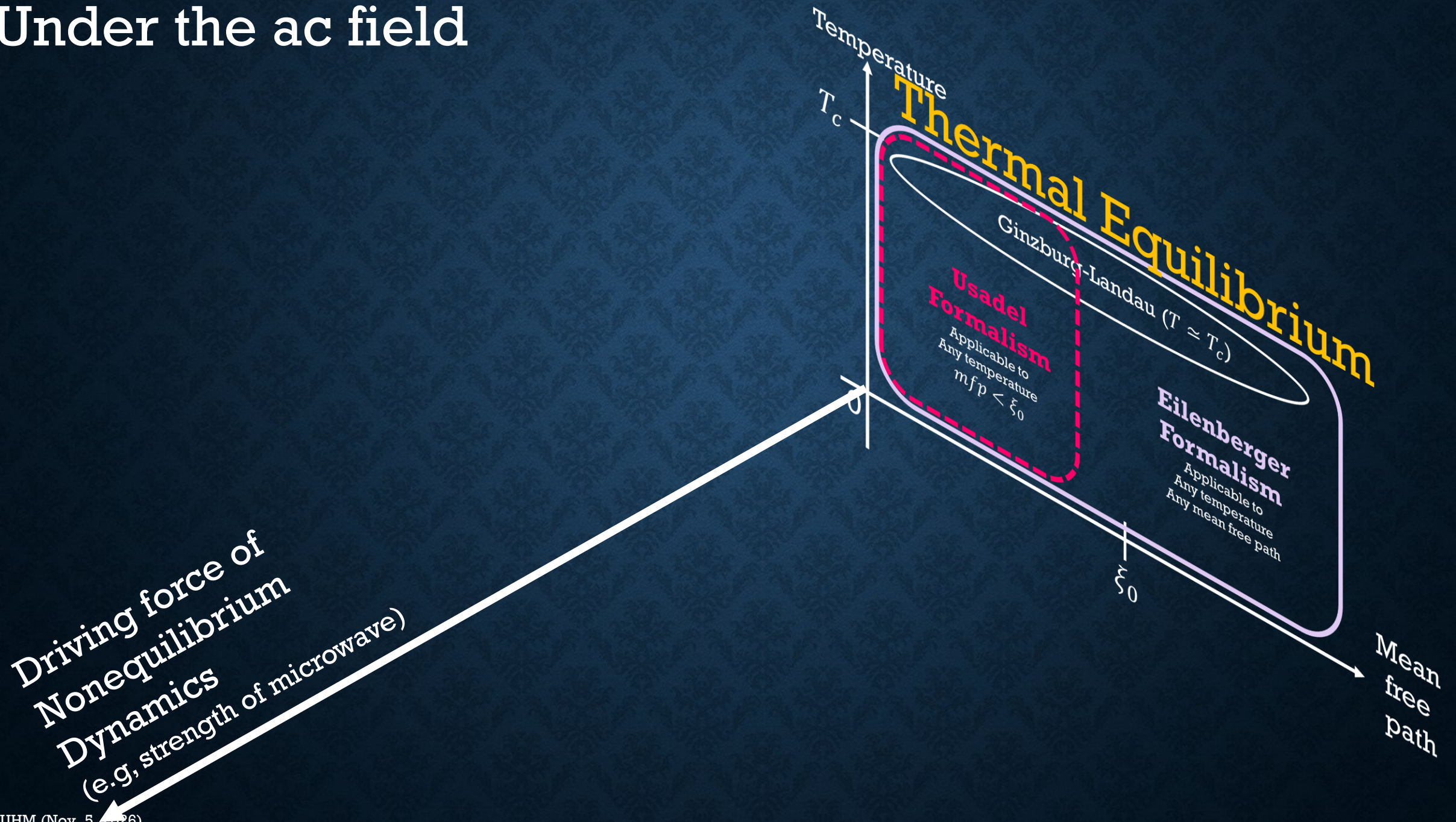

Temperature
T_c
0
ξ_0
Mean free path


Temperature
T_c
Ginzburg-Landau ($T \simeq T_c$)
0
ξ_0
Mean free path

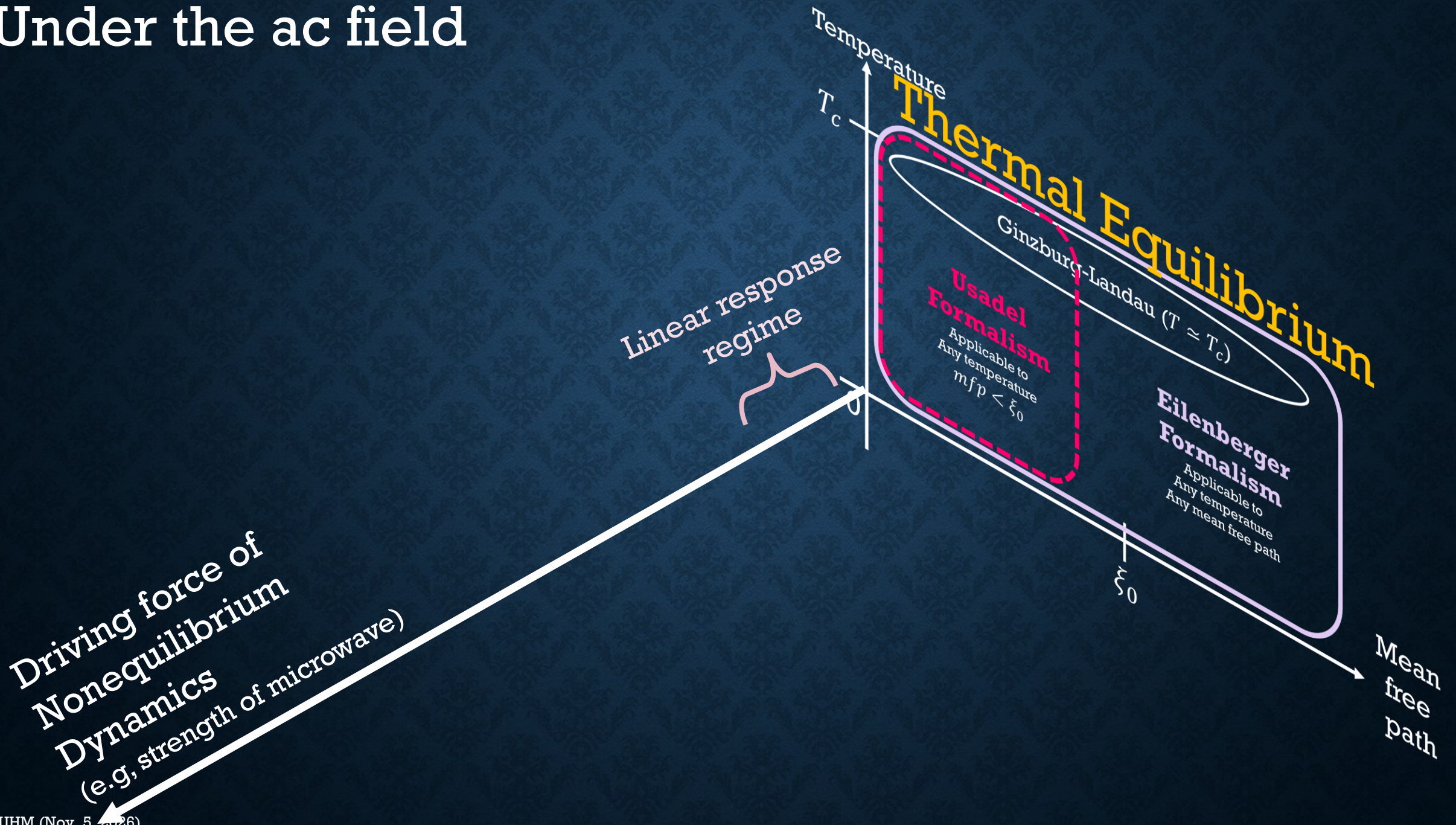

Temperature
T_c
Ginzburg-Landau ($T \simeq T_c$)
Eilenberger Formalism
Applicable to
Any temperature
Any mean free path
Mean free path
0
ξ_0


Temperature
T_c
Ginzburg-Landau ($T \simeq T_c$)
Usadel Formalism
Applicable to
Any temperature
$mfp < \xi_0$
Eilenberger Formalism
Applicable to
Any temperature
Any mean free path
0
ξ_0
Mean free path

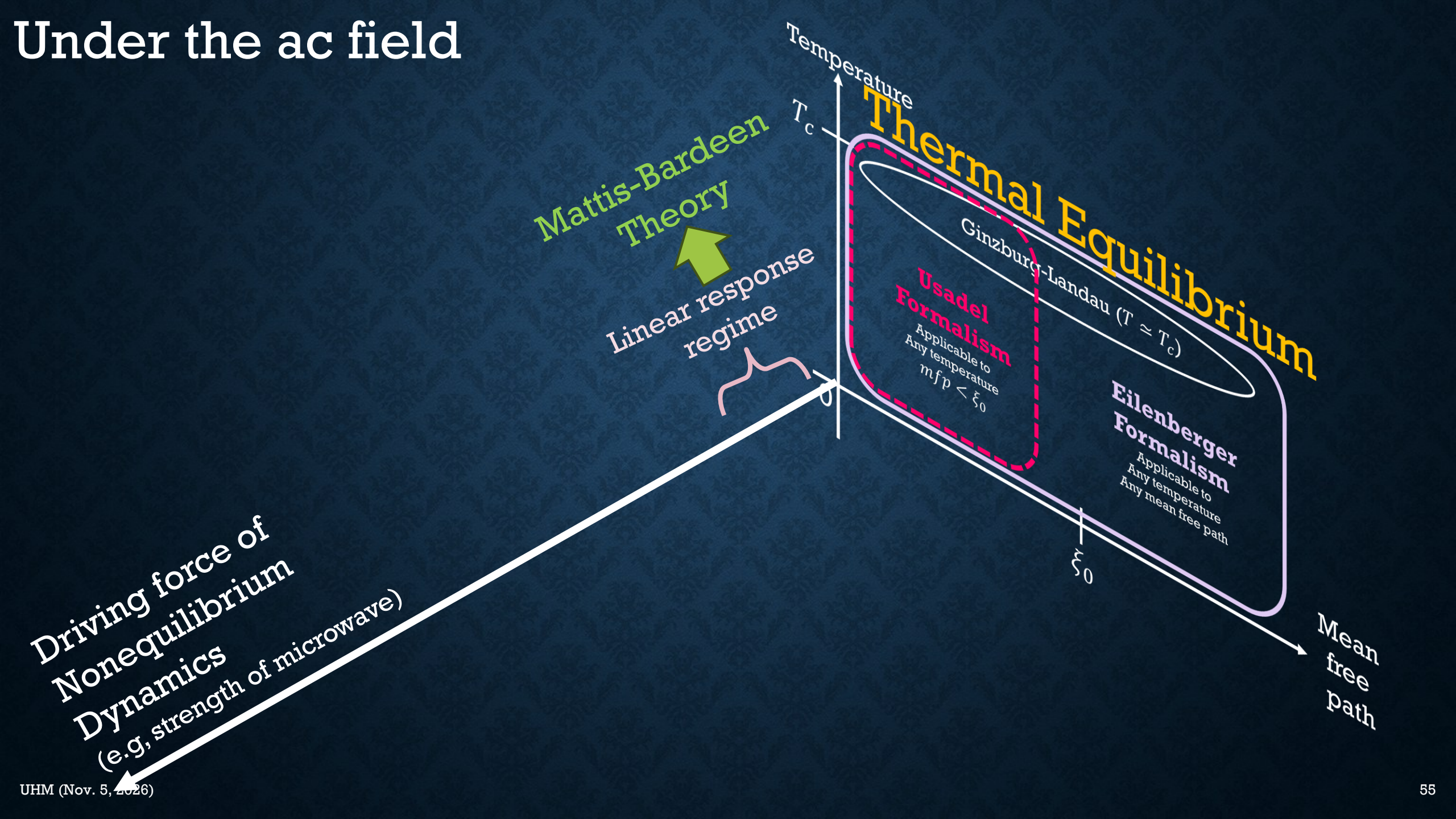
Under the ac field



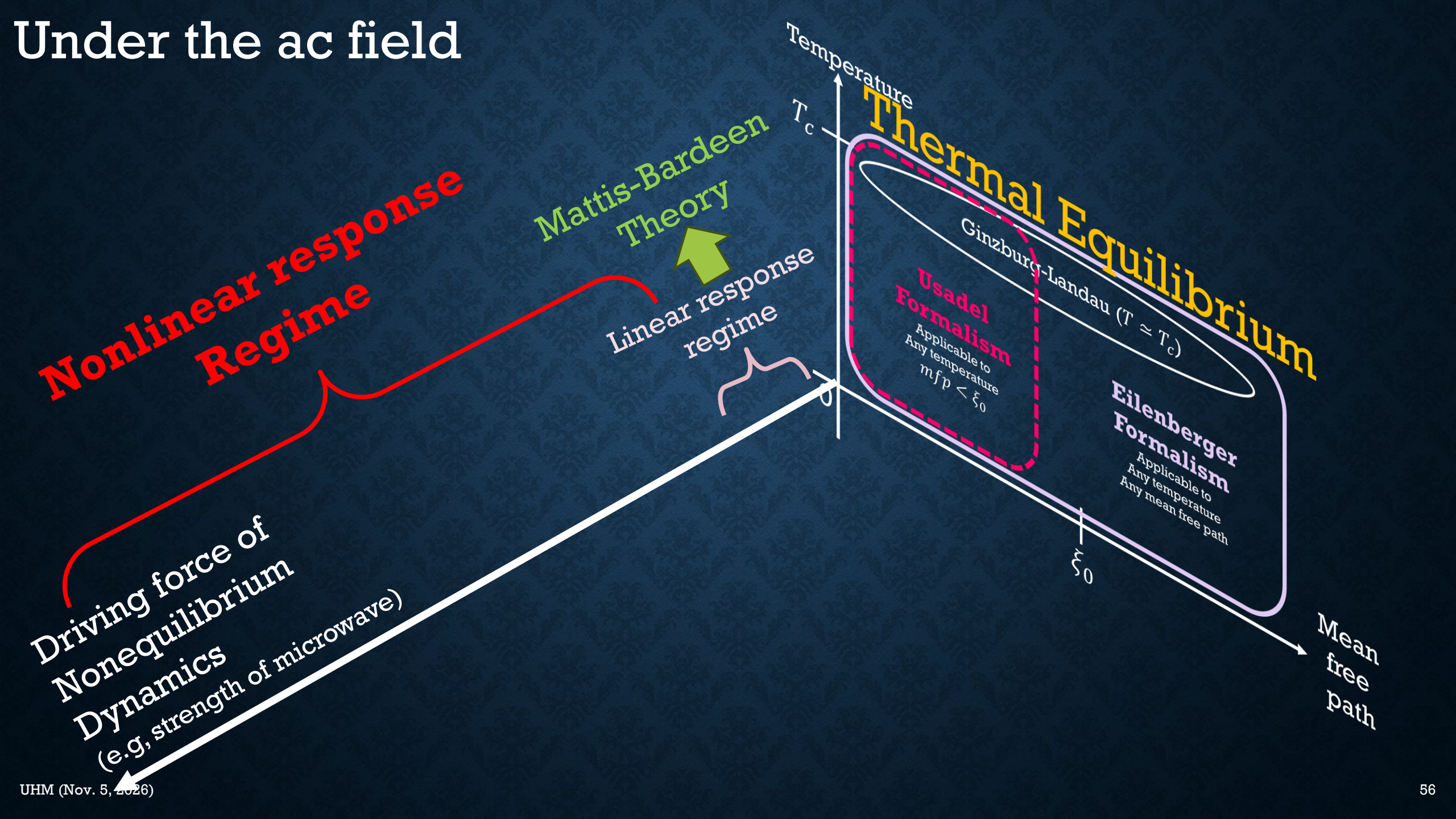
Under the ac field



Under the ac field



Under the ac field



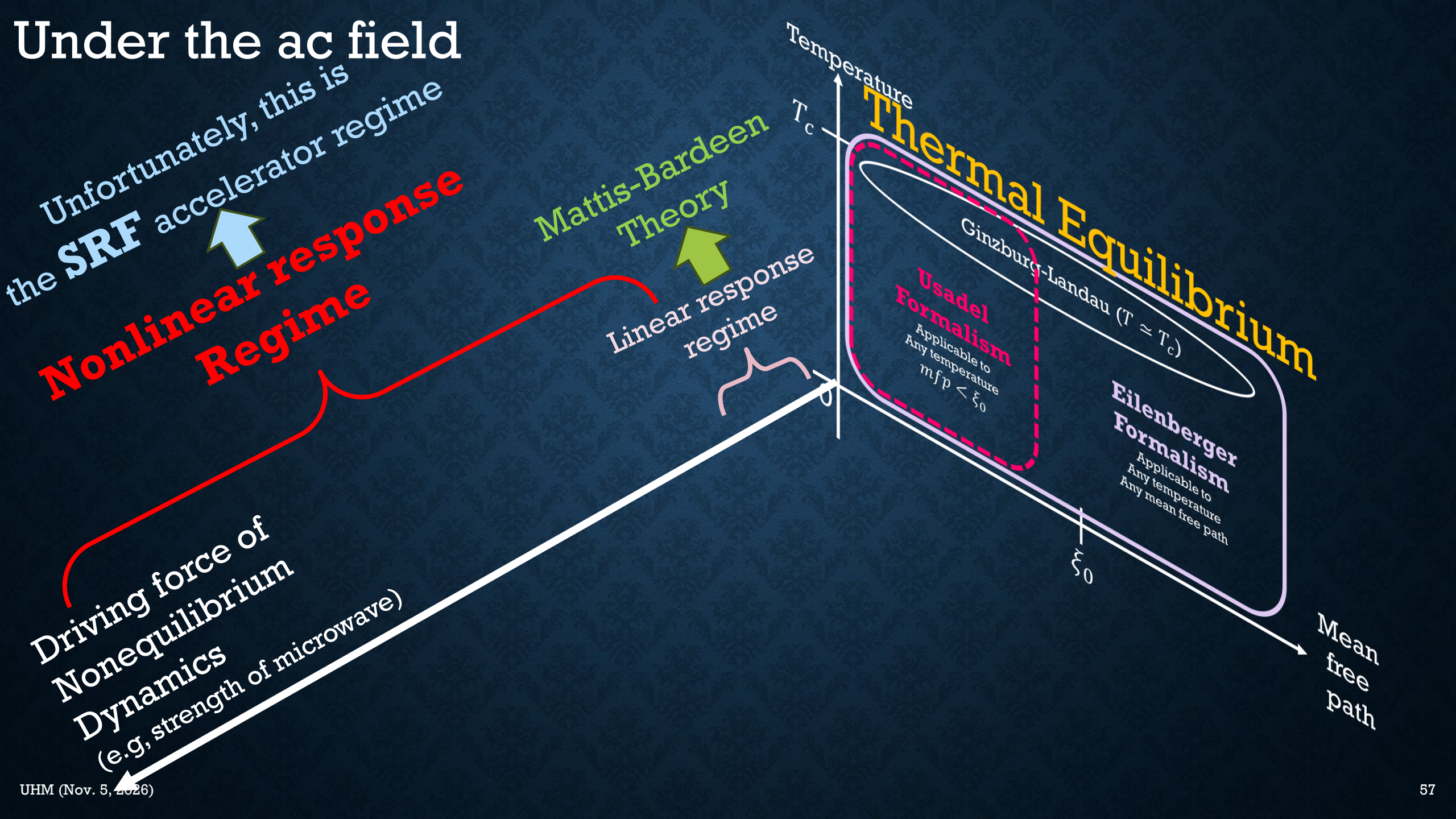
Under the ac field

Unfortunately, this is the **SRF** accelerator regime

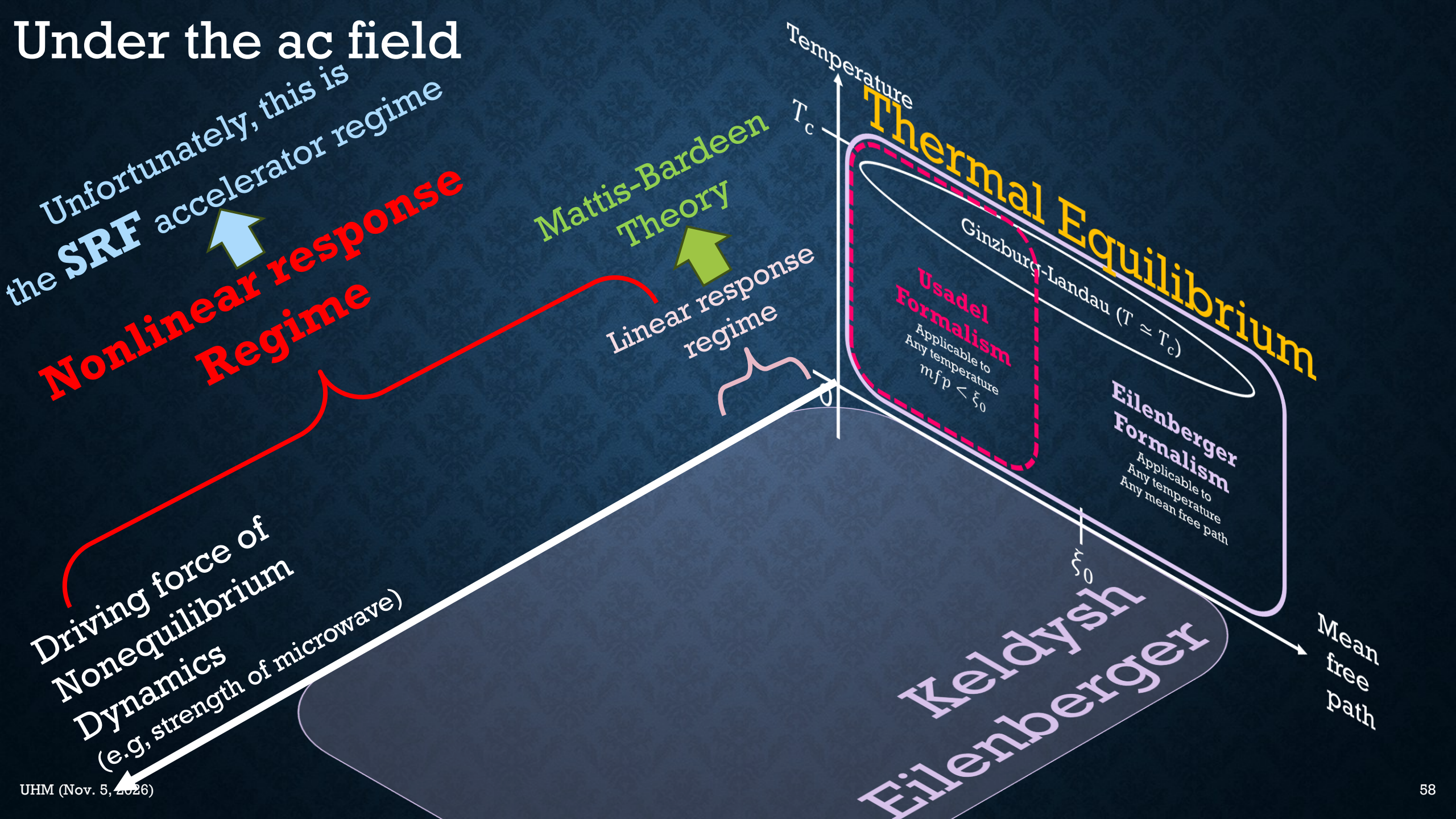
Nonlinear response Regime

Mattis-Bardeen Theory

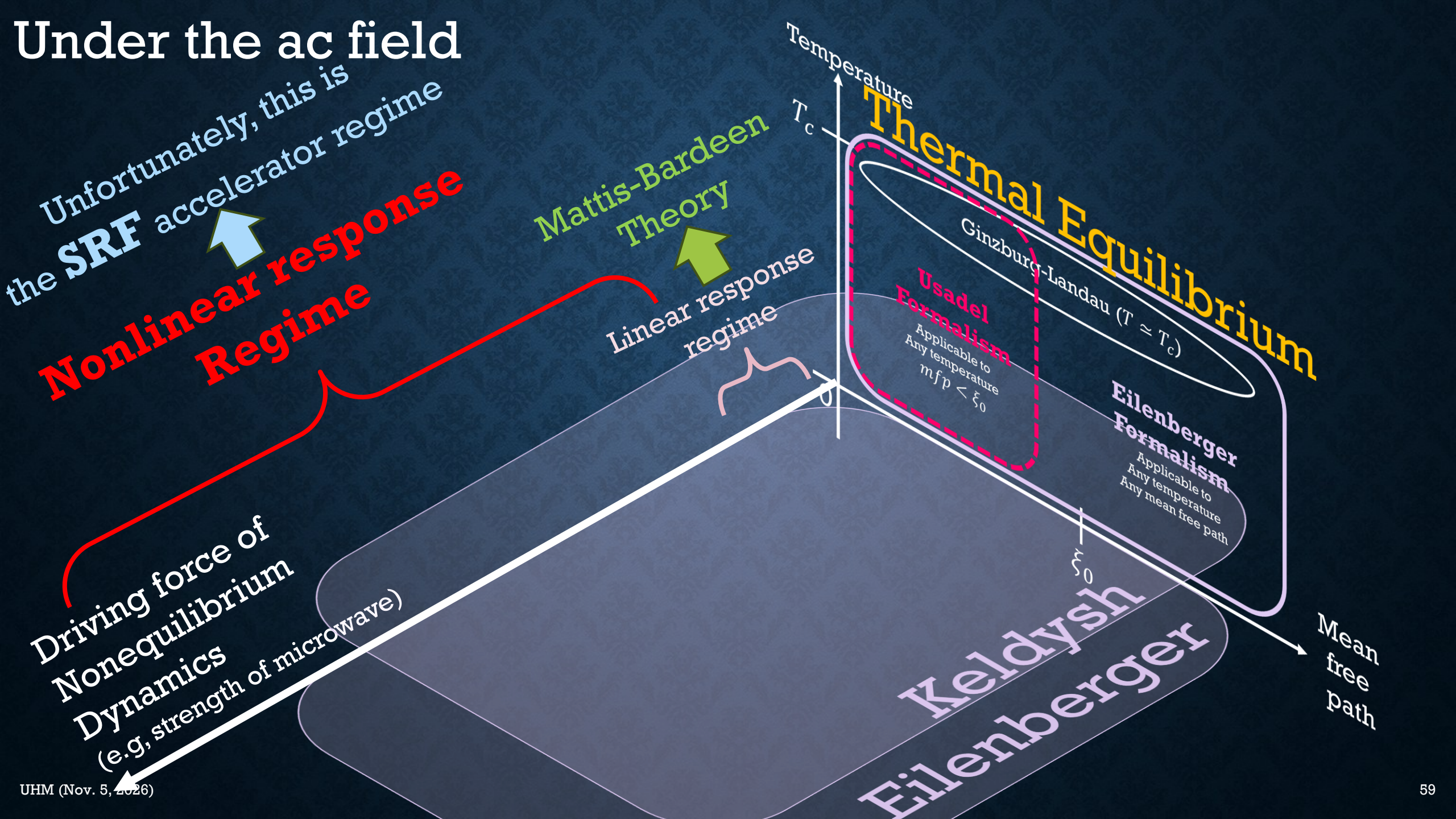
Linear response regime



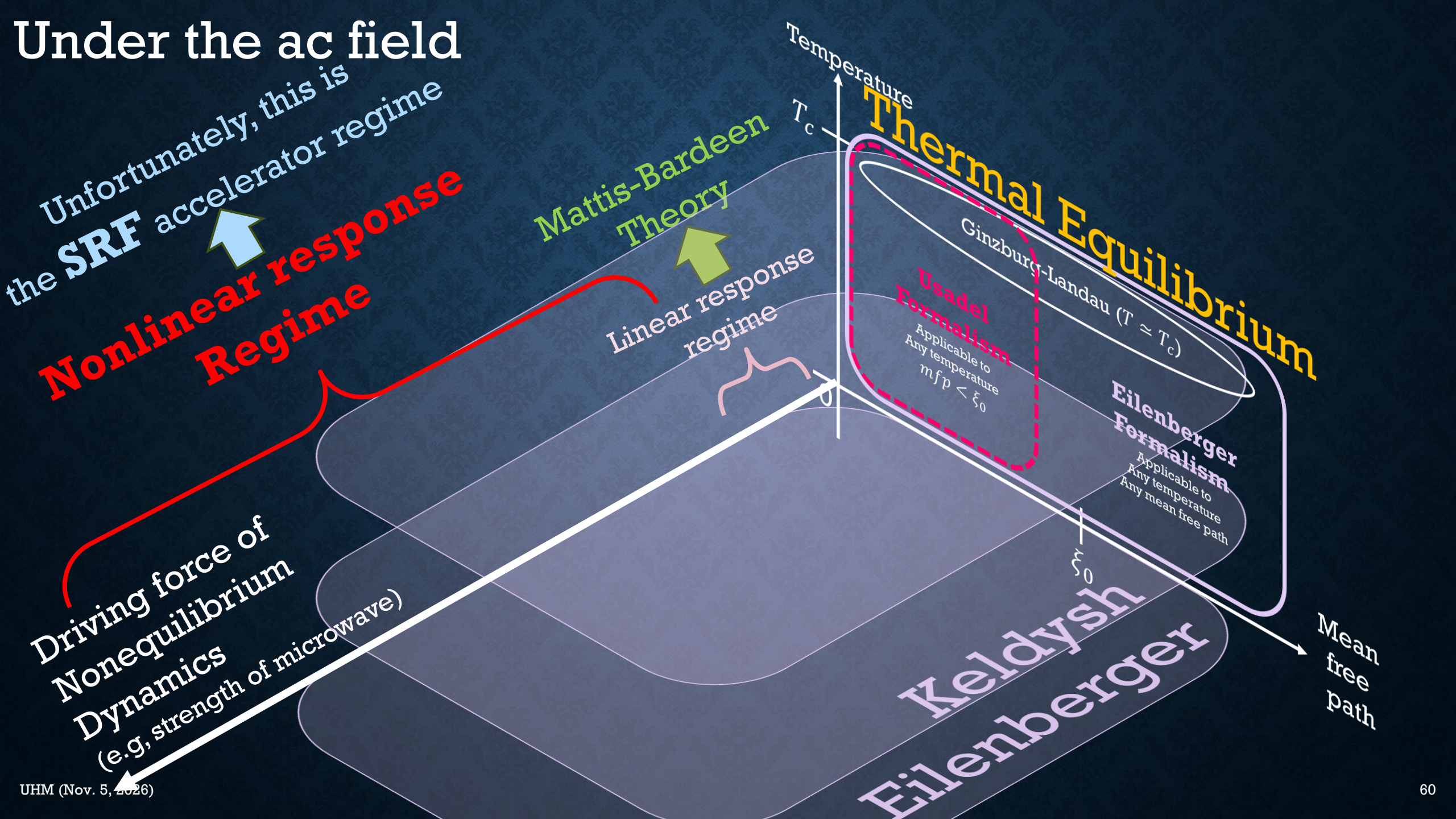
Under the ac field



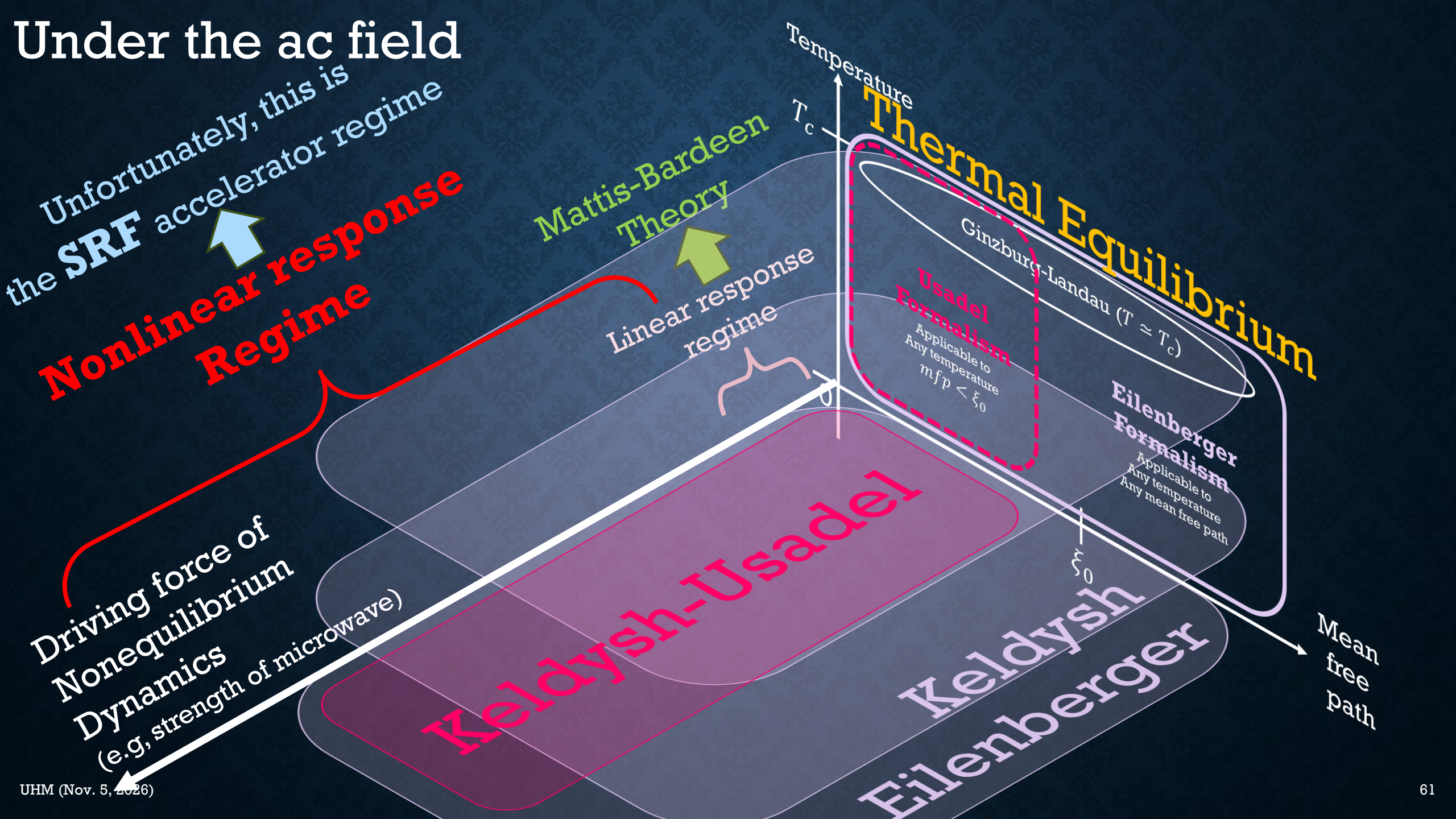
Under the ac field



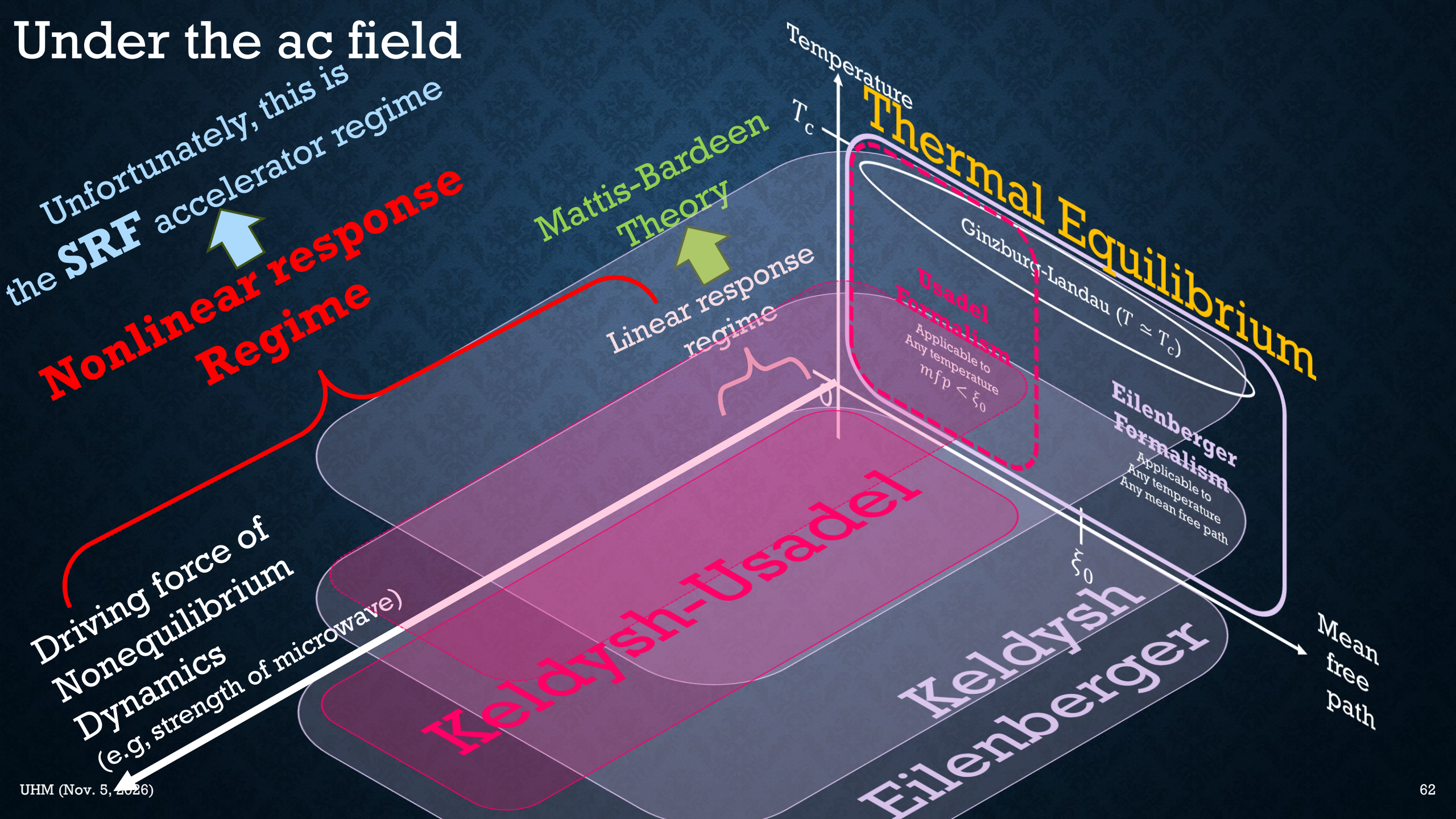
Under the ac field



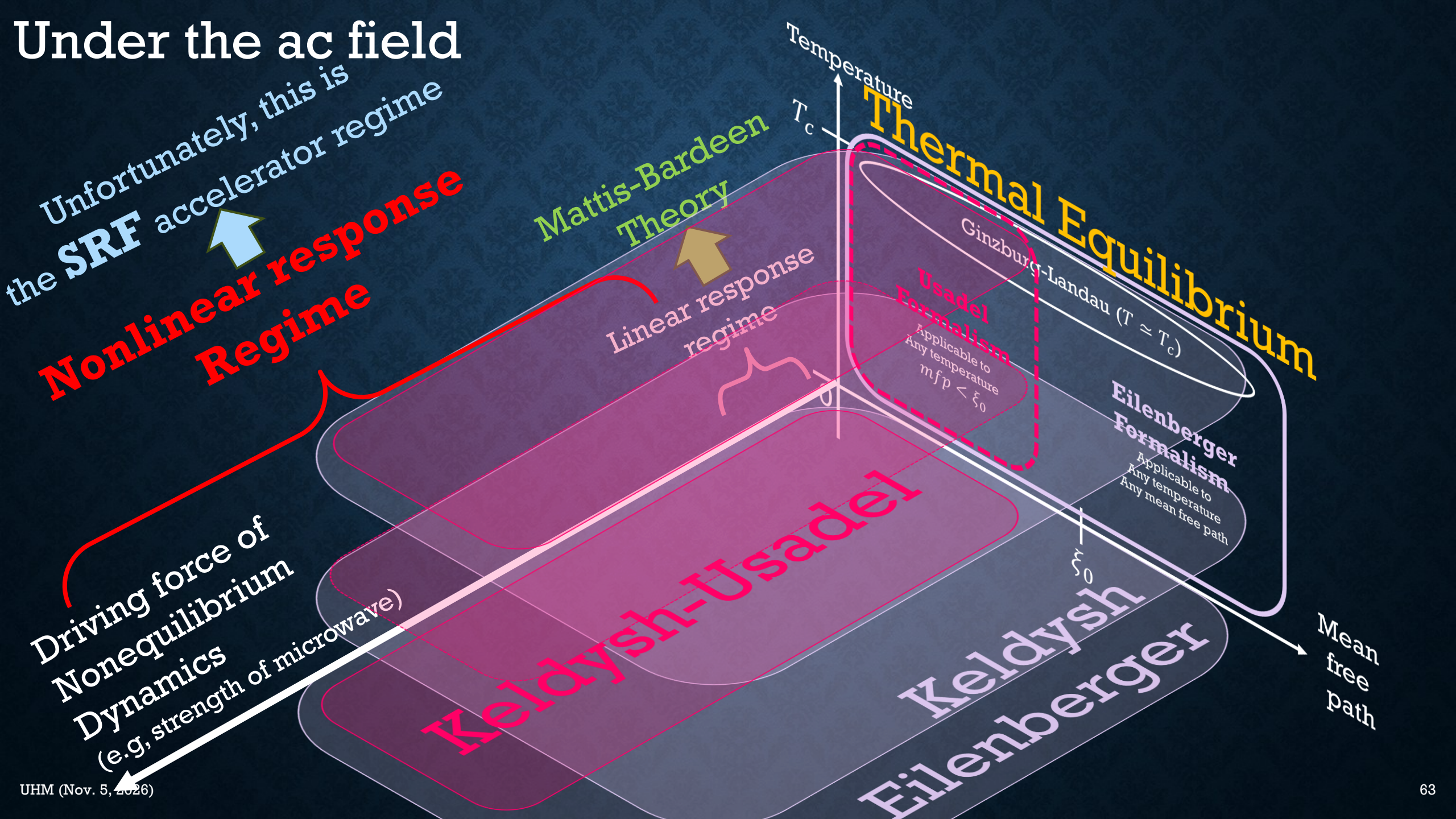
Under the ac field



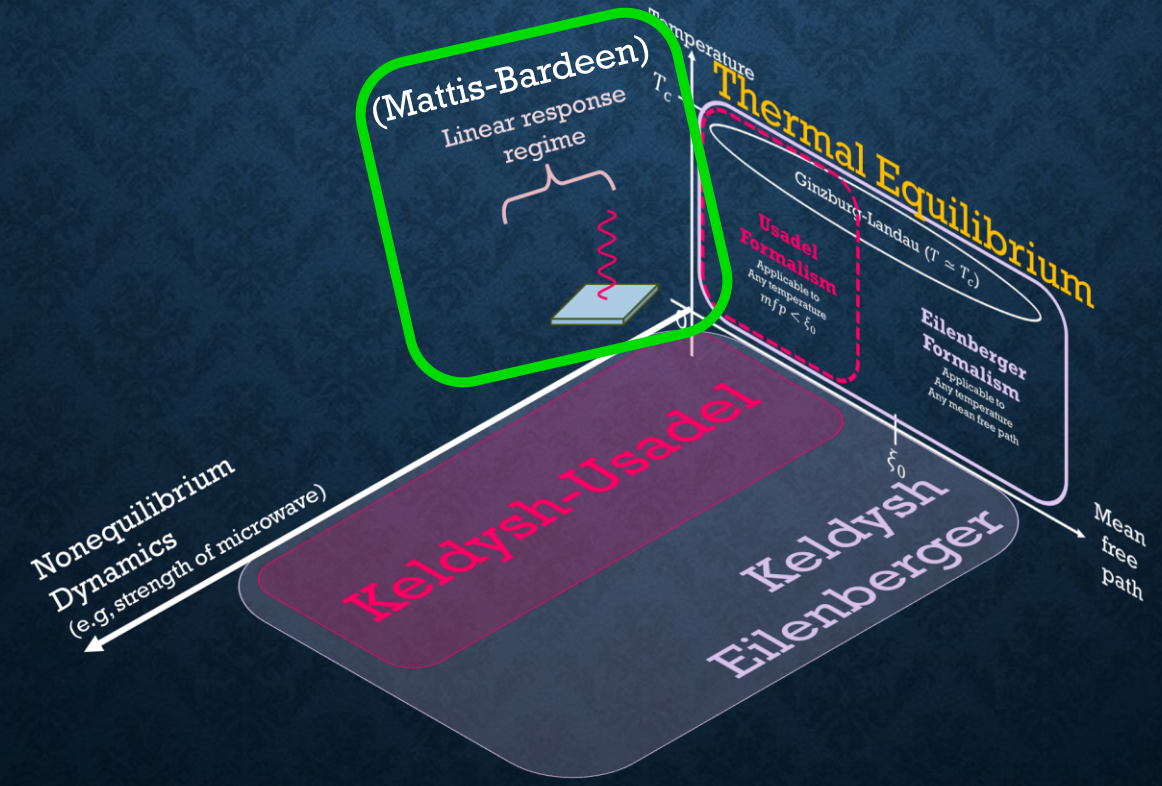
Under the ac field



Under the ac field



Let us start from the linear response



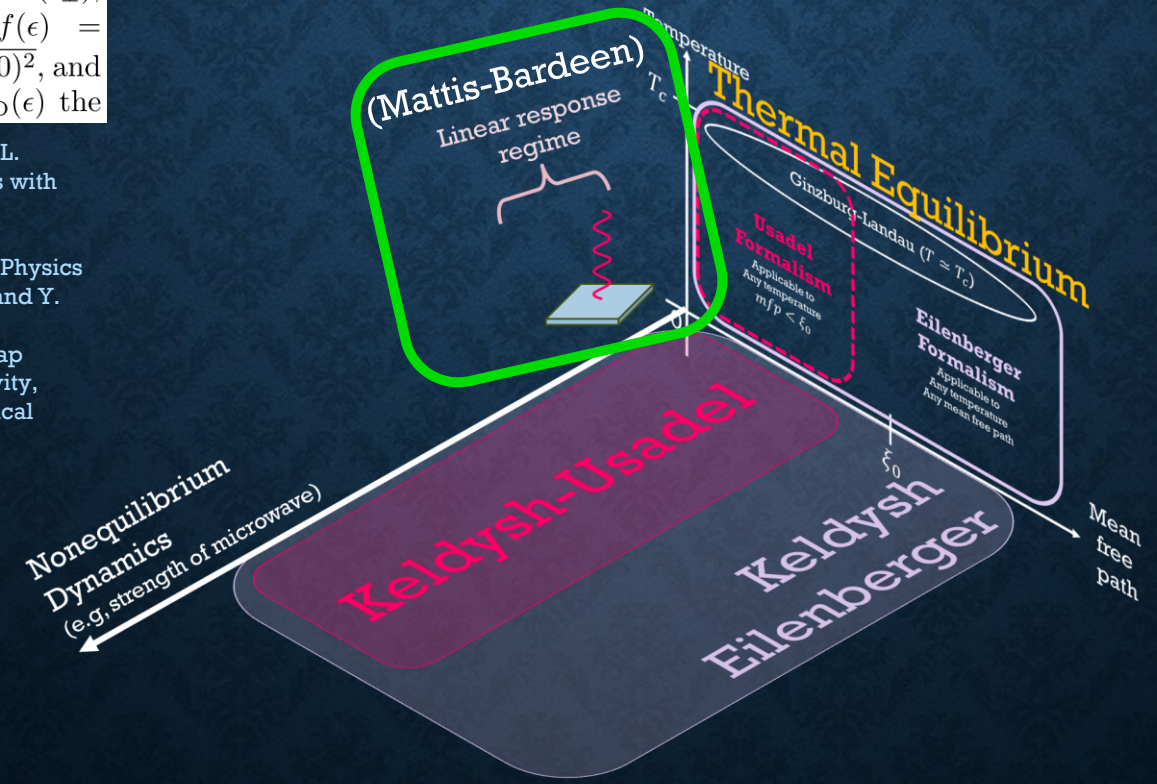
Let us start from the linear response

$$\sigma(\ell, T, \omega) = \sigma_1 + i\sigma_2$$
$$= \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon \left[\kappa \mathcal{T}_- + \kappa^* \mathcal{T}_+ + \kappa^a (\mathcal{T}_+ - \mathcal{T}_-)\right],$$
$$\kappa = \frac{1}{3} \frac{g_+ g_- + f_+ f_- - 1}{d_+ + d_-},$$
$$\kappa^a = \frac{1}{3} \frac{-g_+ g_-^* - f_+ f_-^* - 1}{d_+ - d_-^*}.$$

$\tau = \ell/v_f$ is the electron relaxation time. We define $\epsilon_\pm := \epsilon \pm \hbar\omega/2$ and set $g_\pm := g(\epsilon_\pm)$, $f_\pm := f(\epsilon_\pm)$, $d_\pm := d(\epsilon_\pm)$, and $\mathcal{T}_\pm := \mathcal{T}(\epsilon_\pm)$, where $g(\epsilon) = -i(\epsilon + i0)/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $f(\epsilon) = -i\Delta/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $d(\epsilon) = i\gamma + i\sqrt{\Delta^2 - (\epsilon + i0)^2}$, and $\mathcal{T}(\epsilon) := \tanh(\epsilon/2k_BT) = 1 - 2f_{\mathrm{FD}}(\epsilon)$, with $f_{\mathrm{FD}}(\epsilon)$ the

- W. Zimmermann, E. Brandt, M. Bauer, E. Seider, and L. Genzel, Optical conductivity of BCS superconductors with arbitrary purity, Physica C **183**, 99 (1991).
- D. Rainer and J. A. Sauls, Strong coupling theory of superconductivity, in Superconductivity: From Basic Physics to the Latest Developments, edited by P. N. Butcher and Y. Lu (World Scientic, Singapore, 1995), p. 45.
- T. Kubo, Effects of Nonmagnetic Impurities and Subgap States on the Kinetic Inductance, Complex Conductivity, Quality Factor, and Depairing Current Density, Physical Review Applied **17**, 014018 (2022)

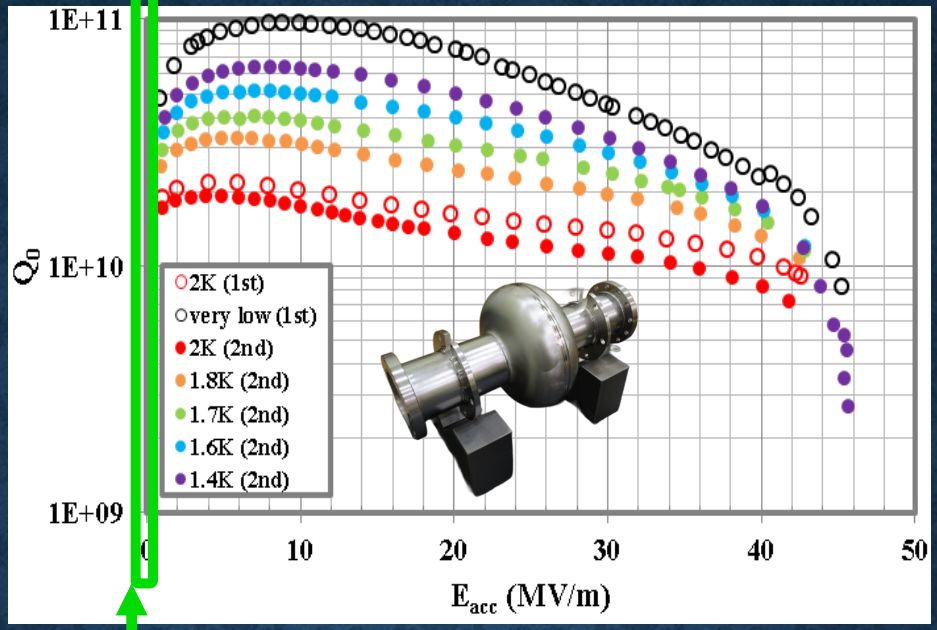
The theory was established decades ago.
We just need to plug in parameters such as temperature.



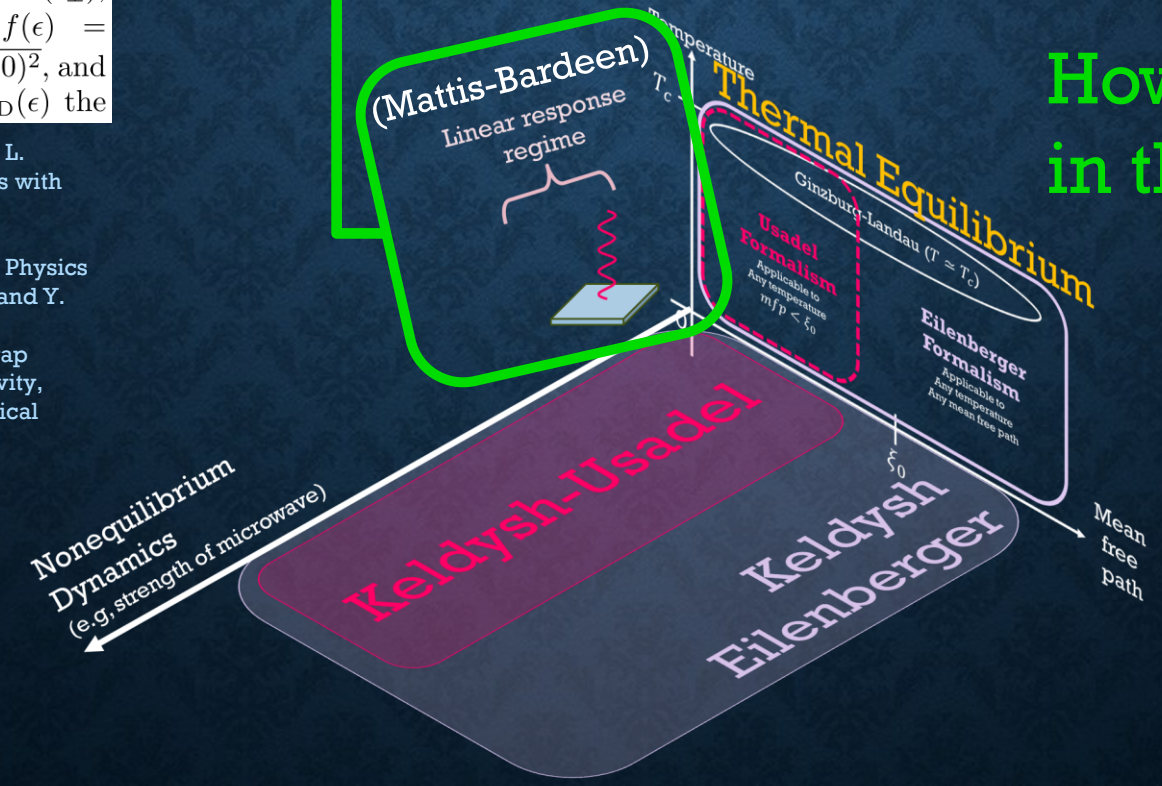
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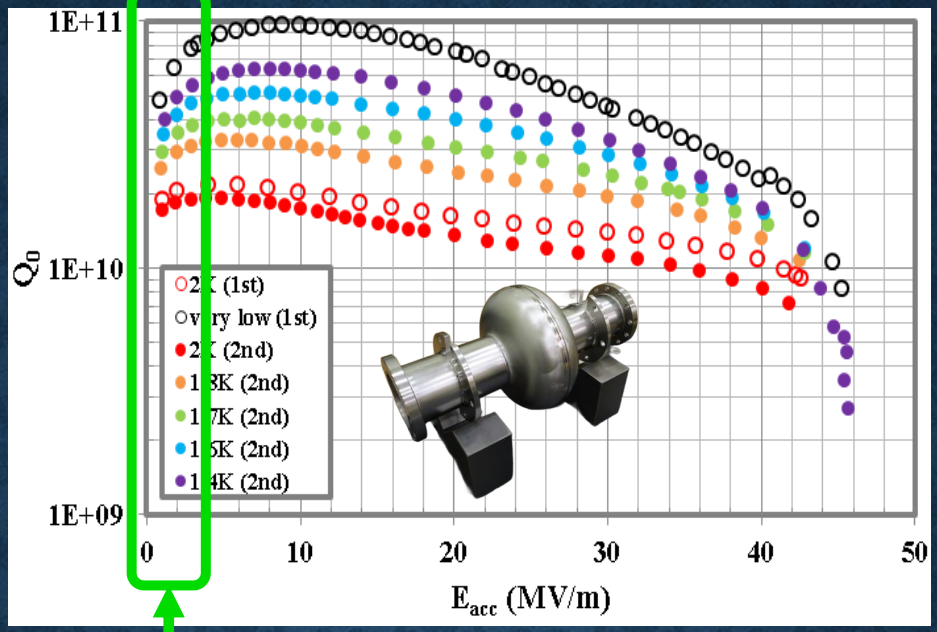


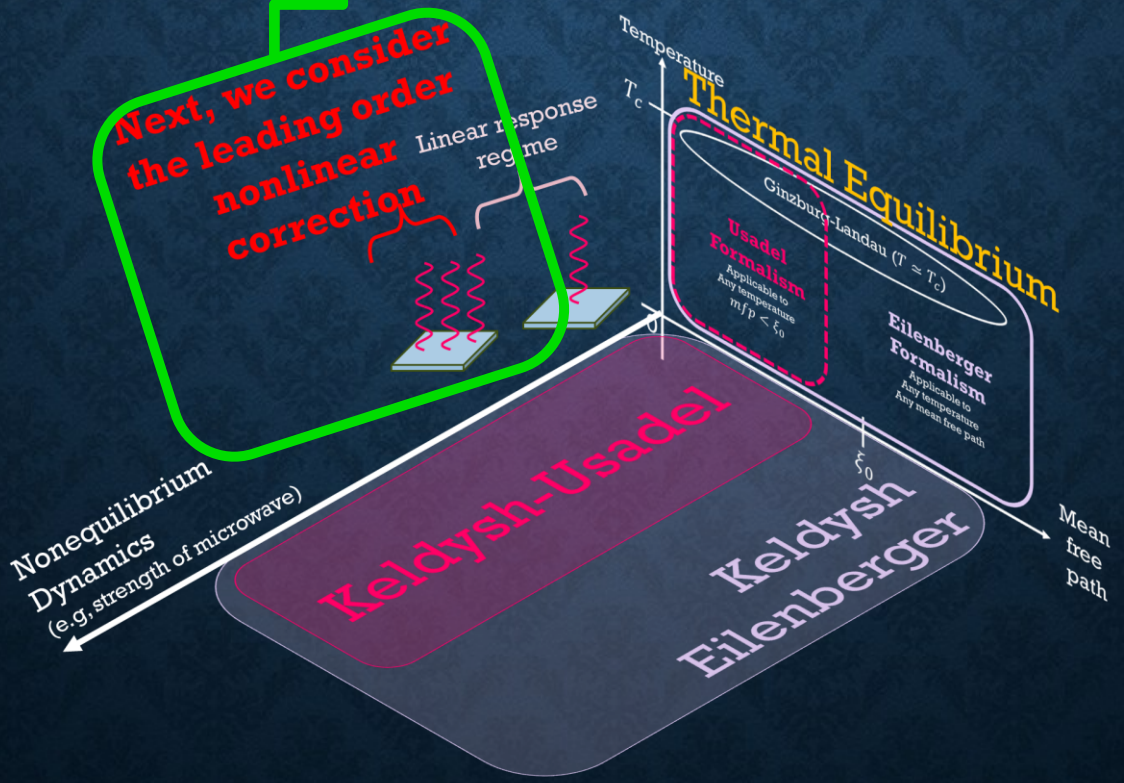
T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519 https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022



However, it is only valid in the limit $E_{acc} \to 0$

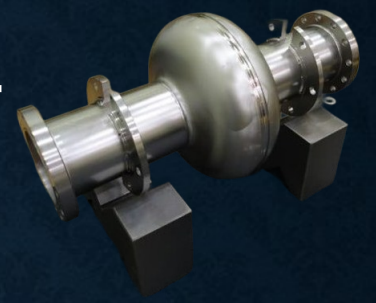
The leading-order nonlinear correction allows us to calculate the field dependence in the weak-field regime.





Keldysh–Usadel Equation for SC under a strong ac field

T. Kubo,
Phys. Rev. Applied
24, 064061 (2025)
https://doi.org/10.1103/8fh8-nknj



$$
\begin{aligned}
&-i\frac{s}{8}\sum_{\eta=\pm\omega_{\rm ac}}\sum_{\eta'=\pm\omega_{\rm ac}}2\pi\delta(\omega+\eta+\eta')\times\\
&\Big[\hat{\tau}_3\hat{g}_e^r\Big(\epsilon-\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^r\Big(\epsilon+\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\\
&-\hat{g}_e^r\Big(\epsilon-\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^r\Big(\epsilon+\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\Big]\\
=\;&\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)-\Big(\epsilon-\frac{\hbar\omega}{2}\Big)\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\\
&+\delta\hat{\Delta}(\omega)\hat{g}_e^r\Big(\epsilon-\frac{\hbar\omega}{2}\Big)-\hat{g}_e^r\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\delta\hat{\Delta}(\omega)\\
&+[\hat{\Delta}_e,\delta\hat{g}^r(\epsilon,\omega)]. \qquad (11)
\end{aligned}
$$

$$
\delta\Delta(\omega)=-\frac{\mathscr{G}}{8}\int d\epsilon\,\mathrm{Tr}[(-i\tau_2)\delta\hat{g}^K(\epsilon,\omega)]. \qquad (15)
$$

$$
\begin{aligned}
&-i\frac{s}{8}\sum_{\eta=\pm\omega_{\rm ac}}\sum_{\eta'=\pm\omega_{\rm ac}}2\pi\delta(\omega+\eta+\eta')\times\\
&\Big[\hat{\tau}_3\hat{g}_e^R\Big(\epsilon-\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^K\Big(\epsilon+\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\\
&-\hat{g}_e^R\Big(\epsilon-\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^K\Big(\epsilon+\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\\
&+\hat{\tau}_3\hat{g}_e^K\Big(\epsilon-\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^A\Big(\epsilon+\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\\
&-\hat{g}_e^K\Big(\epsilon-\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^A\Big(\epsilon+\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\Big]\\
=\;&\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)-\Big(\epsilon-\frac{\hbar\omega}{2}\Big)\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\\
&+\delta\hat{\Delta}(\omega)\hat{g}_e^K\Big(\epsilon-\frac{\hbar\omega}{2}\Big)-\hat{g}_e^K\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\delta\hat{\Delta}(\omega)\\
&+[\hat{\Delta}_e,\delta\hat{g}^K(\epsilon,\omega)]. \qquad (13)
\end{aligned}
$$

We solve these equations to obtain the **nonequilibrium corrections** ($\delta\hat{g}^{R,A,K}$ and $\delta\Delta$).

To obtain the current response, we substitute the solutions ($\delta\hat{g}^{R,A,K}$, $\delta\Delta$) into

$$
\begin{aligned}
\mathbf{J}(t)&=\mathbf{J}^{(1)}(t)+\mathbf{J}^{(3)}(t)\\
&=\frac{\sigma_n}{8e}\int\Big\{\mathbf{S}^{(1)}(\epsilon,t)+\mathbf{S}^{(3)}(\epsilon,t)\Big\}d\epsilon, \qquad (49)
\end{aligned}
$$

$$
\mathbf{S}^{(1)}(\epsilon,t)=\mathrm{Tr}\big[\hat{\tau}_3\big\{\hat{g}_e^R\circ(\hat{\partial}\circ\hat{g}_e^K)+\hat{g}_e^K\circ(\hat{\partial}\circ\hat{g}_e^A)\big\}\big] \qquad (50)
$$

← Linear response

$$
\begin{aligned}
\mathbf{S}^{(3)}(\epsilon,t)=\mathrm{Tr}\big[\hat{\tau}_3\big\{&\hat{g}_e^R\circ(\hat{\partial}\circ\delta\hat{g}^K)+\hat{g}_e^K\circ(\hat{\partial}\circ\delta\hat{g}^A)\\
&+\delta\hat{g}^R\circ(\hat{\partial}\circ\hat{g}_e^K)+\delta\hat{g}^K\circ(\hat{\partial}\circ\hat{g}_e^A)\big\}\big]. \qquad (51)
\end{aligned}
$$

← **Non**linear correction of $\mathcal{O}(A^3)$

Schematically, the results can be illustrated as

Linear response

Nonlinear corrections

Current $J \sim \boxed{Agg} + \boxed{Ag\delta g}$ $\delta g \sim A^2 + \delta\Delta$ and $\delta\Delta \sim \Psi A^2$

Schematically, the results can be illustrated as

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Nonlinear corrections

Current $J \sim \boxed{Agg} + \boxed{Ag\delta g}$ $\delta g \sim A^2 + \delta\Delta$ and $\delta\Delta \sim \Psi A^2$

$$\sim A(A^2 + \delta\Delta)$$

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Current $J \sim \boxed{Agg} + \boxed{Ag\delta g}$ $\delta g \sim A^2 + \delta\Delta$ and $\delta\Delta \sim \Psi A^2$

$\sim A(A^2 + \delta\Delta)$

Time averaged $\delta\Delta$: **Eliashberg effect**

Time dependent $\delta\Delta$: **Higgs mode**

Note: This theory automatically includes the **Eliashberg effect** (microwave-induced gap enhancement) as well as the **Higgs-mode** contribution.

Eliashberg effect? Higgs mode?

Eliashberg effect? Higgs mode?

The variation of order parameter $\delta\Delta$ does not appear in standard linear-response theories such as Mattis–Bardeen theory because $\delta\Delta$ is an $\mathcal{O}(A^2)$ effect, but it does emerge in nonlinear response.

$$\delta\Delta \propto A_0^2 \cos^2 \omega t$$

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$$\delta\Delta \propto A_0^2 \cos^2 \omega t \propto 1 + \cos 2\omega t$$

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Time-averaged nonequilibrium gap variation

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Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation

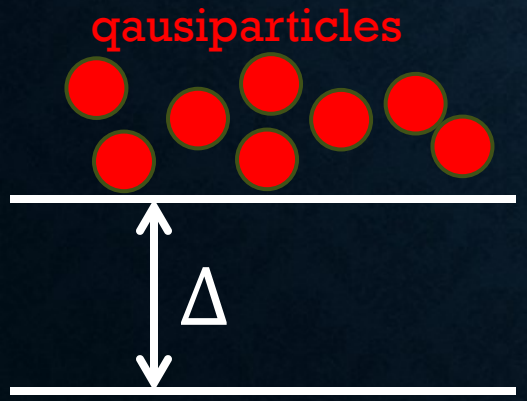
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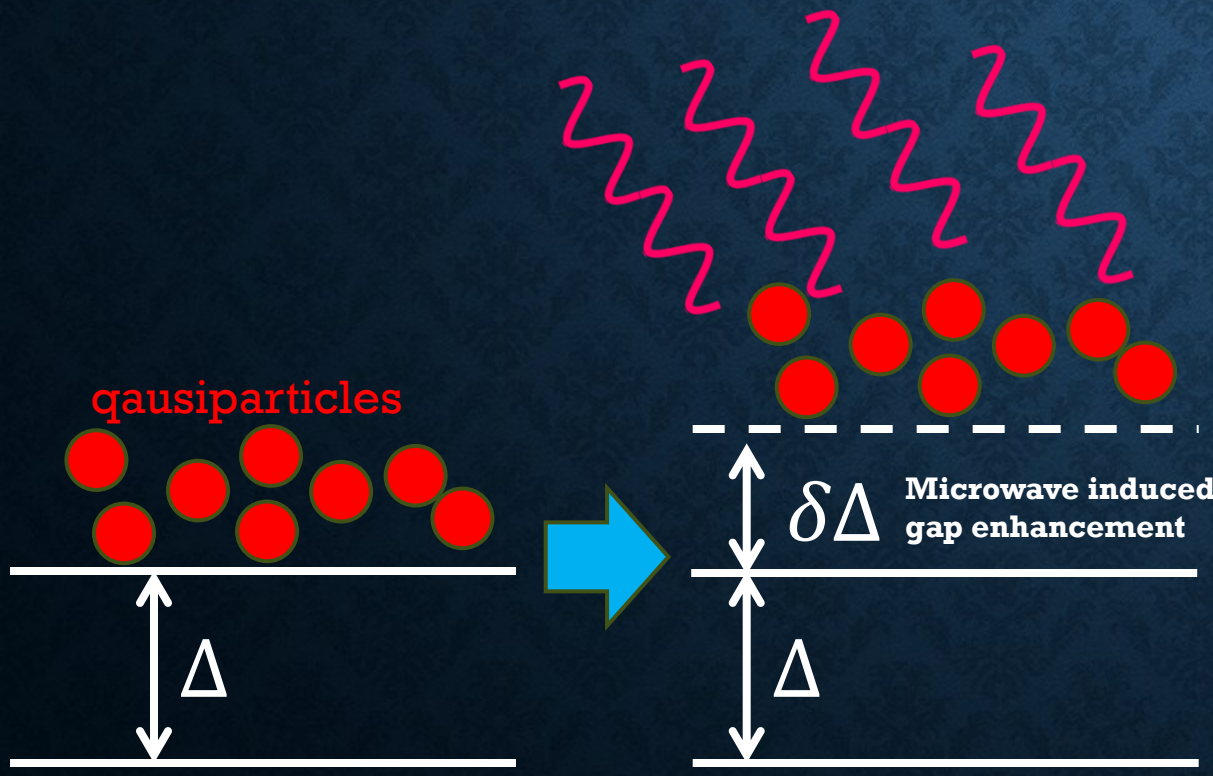
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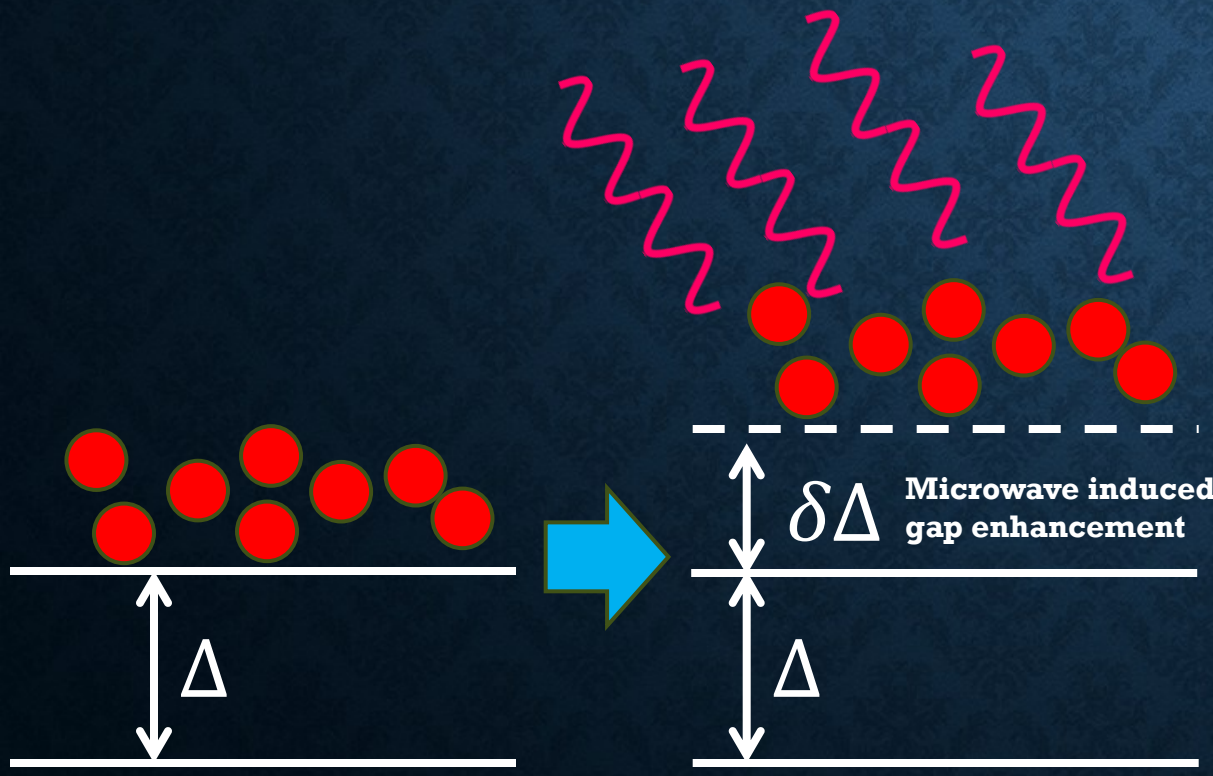


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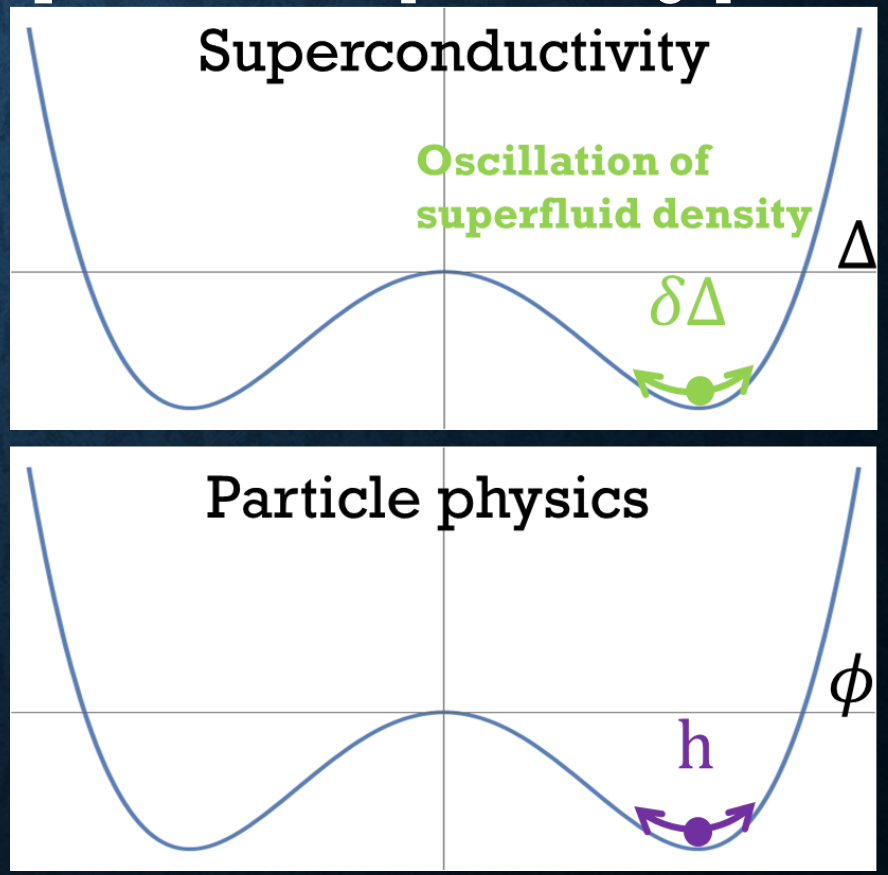
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Time-averaged nonequilibrium gap variation



Time dependent nonequilibrium gap variation



Nonlinear correction

$$\delta J \sim A(A^2 + \delta\Delta)$$

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Nonlinear correction

$$\delta J \sim A(A^2 + \delta\Delta) \Rightarrow \delta\sigma \sim \frac{\delta J}{A} \sim A^2 + \delta\Delta$$

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$$\delta\sigma_1 = \delta\sigma_1^{qqq} + \delta\sigma_1^{\mathrm{Higgs}} + \delta\sigma_1^{\mathrm{Eliash}}$$

$$\delta\sigma_1^{qqq} = 2\sqrt{\pi}\sigma_n \mathrm{Re}[I_{1\mathrm{H}}^{qqq}]\frac{s}{\hbar\omega_{\mathrm{ac}}} \qquad I_{1\mathrm{H}}^{\mathrm{qqq}} = \frac{-1}{16\sqrt{\pi}}\int K_{1\mathrm{H}} d\epsilon,$$

$$K_{1\mathrm{H}}(\epsilon,\omega_{\mathrm{ac}}) = \sum_{i=1}^{6} K_{1\mathrm{H},i}, \tag{61}$$

$$K_{1\mathrm{H},1} = i\frac{(F_1 - F_{-3})F_{-1} + (G_1 - G_{-3})G_{-1}}{4\hbar\omega_{\mathrm{ac}}(F_1 + F_{-3})}$$
$$\times\Big[\Big\{(F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^*\Big\}(\mathcal{T}_{-1} - \mathcal{T}_{-3})$$
$$+\Big\{(F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1}\Big\}\mathcal{T}_{-1}\Big], \tag{62}$$

$$K_{1\mathrm{H},2} = i\frac{(G_3 - G_{-1})G_1 + (F_3 - F_{-1})F_1}{4\hbar\omega_{\mathrm{ac}}(F_3 + F_{-1})}$$
$$\times\{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1\}\mathcal{T}_{-1}, \tag{63}$$

$$K_{1\mathrm{H},3} = -i\frac{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1}{4\hbar\omega_{\mathrm{ac}}(F_3 - F_{-1}^*)}$$
$$\times\Big[\Big\{(G_3 + G_{-1}^*)G_1^* + (F_3 + F_{-1}^*)F_1^*\Big\}(\mathcal{T}_3 - \mathcal{T}_1)$$
$$+\Big\{(G_3 + G_{-1}^*)G_1 + (F_3 + F_{-1}^*)F_1\Big\}(\mathcal{T}_{-1} - \mathcal{T}_1)\Big], \tag{64}$$

$$K_{1\mathrm{H},4} = -i\frac{F_1G_{-1} + G_1F_{-1}}{2(\epsilon_{-1}G_{-1} - F_{-1}\Delta)}$$
$$\times\{(F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1}\}\mathcal{T}_{-1}, \tag{65}$$

$$K_{1\mathrm{H},5} = -i\frac{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1}{2(\epsilon_1G_1 - F_1\Delta)}$$
$$\times\Big\{(G_1F_{-1}^* + F_1G_{-1}^*)(\mathcal{T}_{-1} - \mathcal{T}_1)$$
$$+(G_1F_{-1} + G_{-1}F_1)\mathcal{T}_{-1}\Big\}, \tag{66}$$

$$K_{1\mathrm{H},6} = -\frac{G_{-1}^*\mathrm{Im}F_1 + F_{-1}^*\mathrm{Im}G_1}{\epsilon_1\mathrm{Im}G_1 - \mathrm{Im}F_1\Delta}$$
$$\times\Big\{(\mathrm{Re}G_3\mathrm{Im}F_1 + \mathrm{Im}G_1\mathrm{Re}F_3)(T_3 - T_1)$$
$$+(\mathrm{Re}G_{-1}\mathrm{Im}F_1 + \mathrm{Im}G_1\mathrm{Re}F_{-1})(T_{-1} - T_1)\Big\}, \tag{67}$$

Higgs Contribution: Time dependent $\delta\Delta$

$$\delta\sigma_1^{\mathrm{Higgs}} = 2\sqrt{\pi}\sigma_n \mathrm{Re}[I_{1\mathrm{H}}^{\mathrm{Higgs}}]\frac{s}{\hbar\omega_{\mathrm{ac}}}$$

$$I_{1\mathrm{H}}^{\mathrm{Higgs}} = \frac{-1}{16\sqrt{\pi}}\Psi_{2\mathrm{H}}\int Z_{1\mathrm{H}}^{\mathrm{Higgs}} d\epsilon$$

$$Z_{1\mathrm{H}}^{\mathrm{Higgs}}(\epsilon,\omega_{\mathrm{ac}}) = \sum_{i=1,2,3} Z_{1\mathrm{H},i}^{\mathrm{Higgs}}, \tag{71}$$

$$Z_{1\mathrm{H},1}^{\mathrm{Higgs}} = -\frac{2(F_1 - F_{-3})}{\hbar\omega_{\mathrm{ac}}(F_1 + F_{-3})}$$
$$\times\Big[\Big\{(F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^*\Big\}(\mathcal{T}_{-1} - \mathcal{T}_{-3})$$
$$+\Big\{(F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1}\Big\}\mathcal{T}_{-1}\Big], \tag{72}$$

$$Z_{1\mathrm{H},2}^{\mathrm{Higgs}} = -\frac{2(F_3 - F_{-1})}{\hbar\omega_{\mathrm{ac}}(F_3 + F_{-1})}$$
$$\times\Big\{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1\Big\}\mathcal{T}_{-1}, \tag{73}$$

$$Z_{1\mathrm{H},3}^{\mathrm{Higgs}} = -\frac{2(F_3 + F_{-1}^*)}{\hbar\omega_{\mathrm{ac}}(F_3 - F_{-1}^*)}$$
$$\times\Big\{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1\Big\}(\mathcal{T}_3 - \mathcal{T}_{-1}) \tag{74}$$

Eliashberg Contribution: Time averaged $\delta\Delta$

$$\delta\sigma_1^{\mathrm{Eliash}} = 2\sqrt{\pi}\sigma_n \mathrm{Re}[I_{1\mathrm{H}}^{\mathrm{Eliash}}]\frac{s}{\hbar\omega_{\mathrm{ac}}}$$

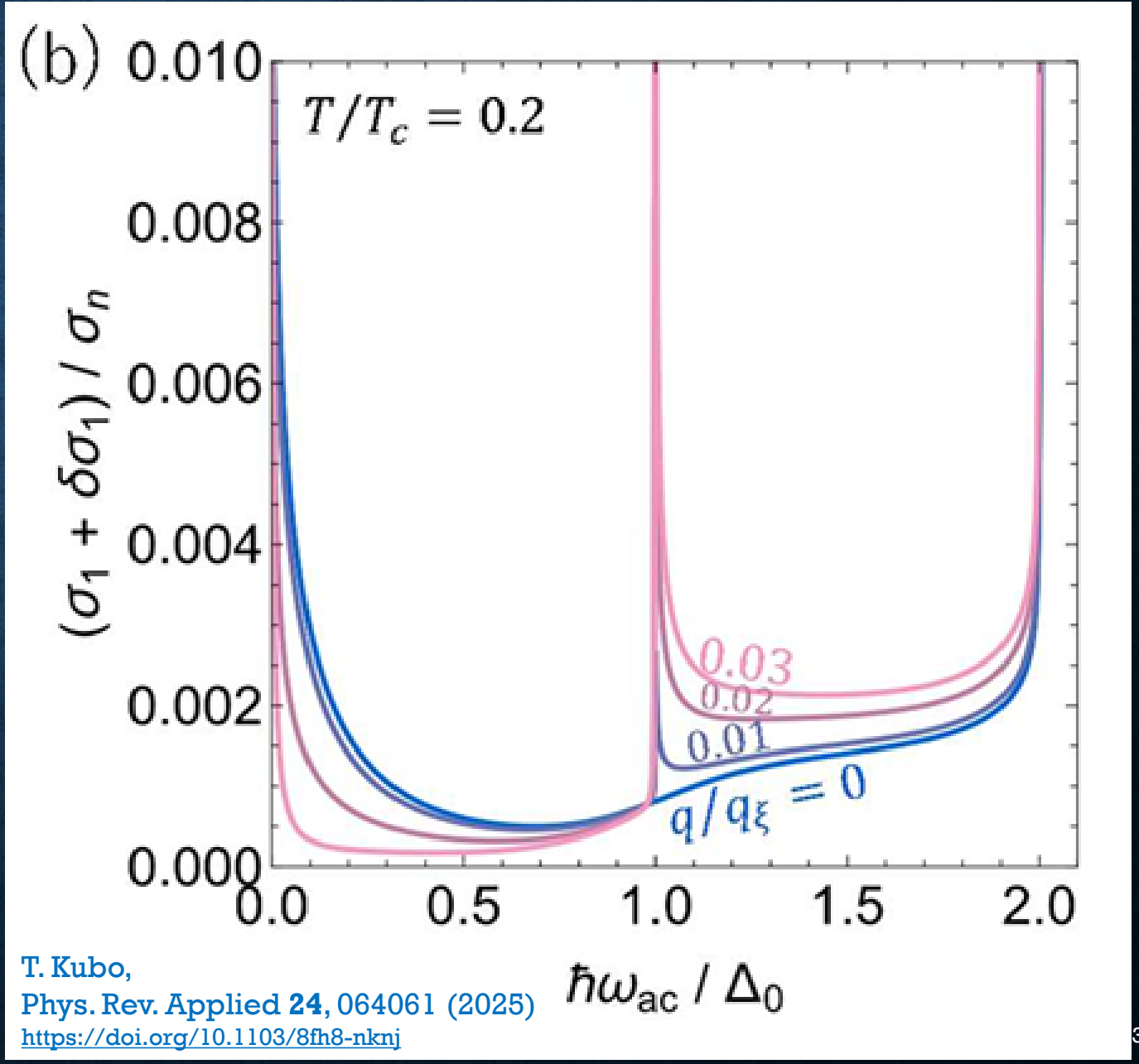
$$I_{1\mathrm{H}}^{\mathrm{Eliash}} = \frac{-1}{16\sqrt{\pi}}\Psi_{0\mathrm{H}}\int Z_{1\mathrm{H}}^{\mathrm{Eliash}} d\epsilon$$

$$Z_{1\mathrm{H}}^{\mathrm{Eliash}}(\epsilon,\omega_{\mathrm{ac}}) = \sum_{i=1,2} Z_{1\mathrm{H},i}^{\mathrm{Eliash}}, \tag{68}$$

$$Z_{1\mathrm{H},1}^{\mathrm{Eliash}} = \frac{4G_{-1}(G_1F_{-1} + F_1G_{-1})}{\epsilon_{-1}G_{-1} - F_{-1}\Delta}\mathcal{T}_{-1}, \tag{69}$$

$$Z_{1\mathrm{H},2}^{\mathrm{Eliash}} = \frac{4G_1}{\epsilon_1G_1 - F_1\Delta}\Big\{(G_1F_{-1} + G_{-1}F_1)\mathcal{T}_{-1}$$
$$+(G_1F_{-1}^* + F_1G_{-1}^*)(\mathcal{T}_{-1} - \mathcal{T}_1)\Big\}, \tag{70}$$

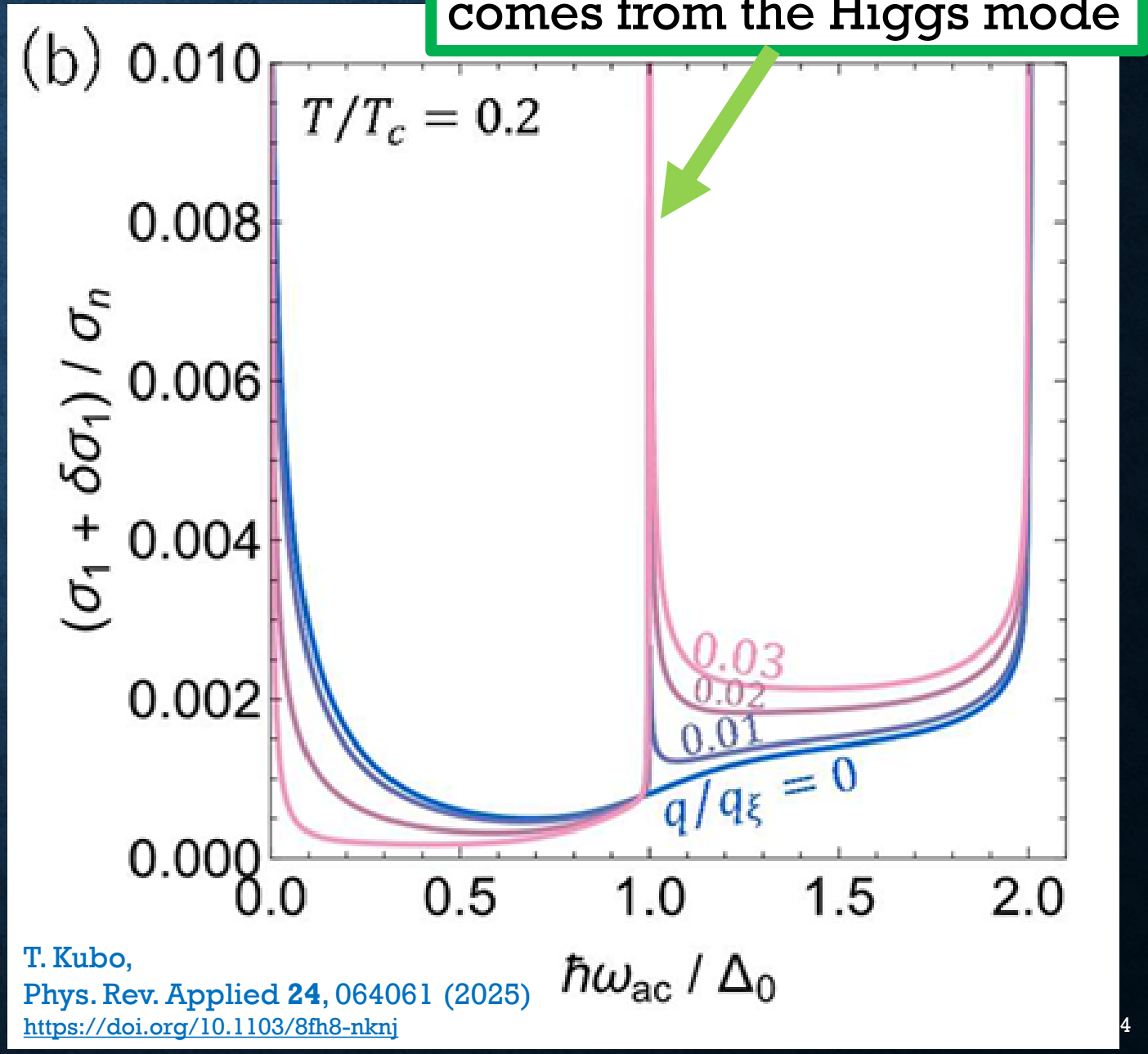
Results



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https://doi.org/10.1103/8fh8-nknj

Results

The resonance peak at Δ comes from the Higgs mode

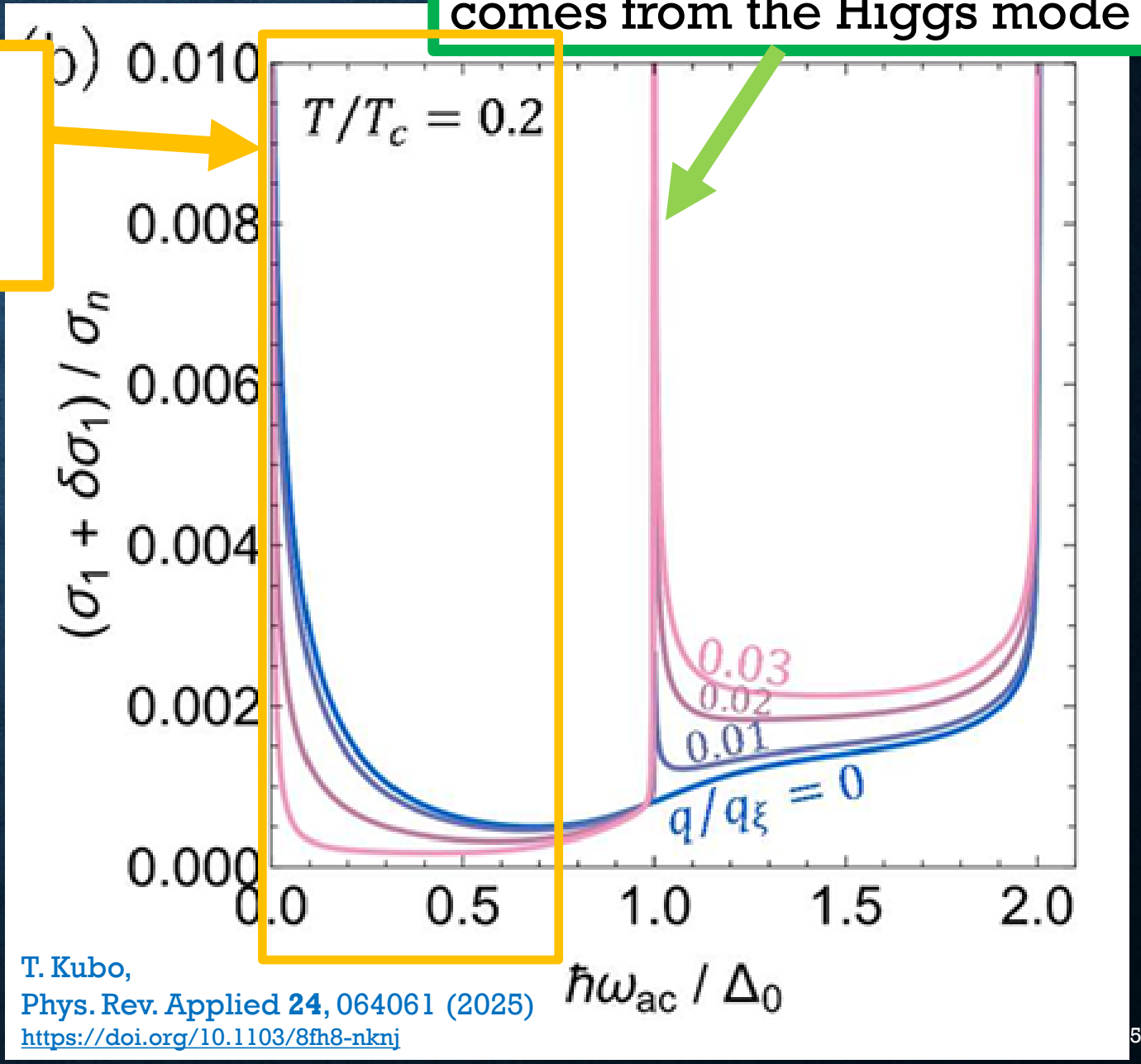


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Results

At lower frequencies,
$\sigma_1 + \delta\sigma_1$ can be a decreasing function of the ac amplitude (q).

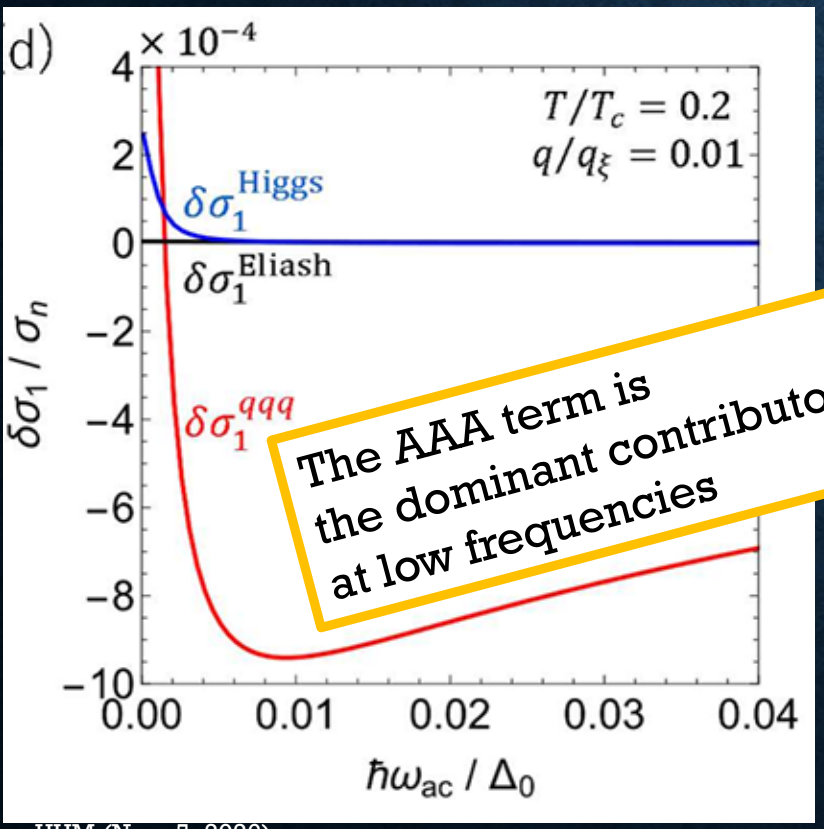
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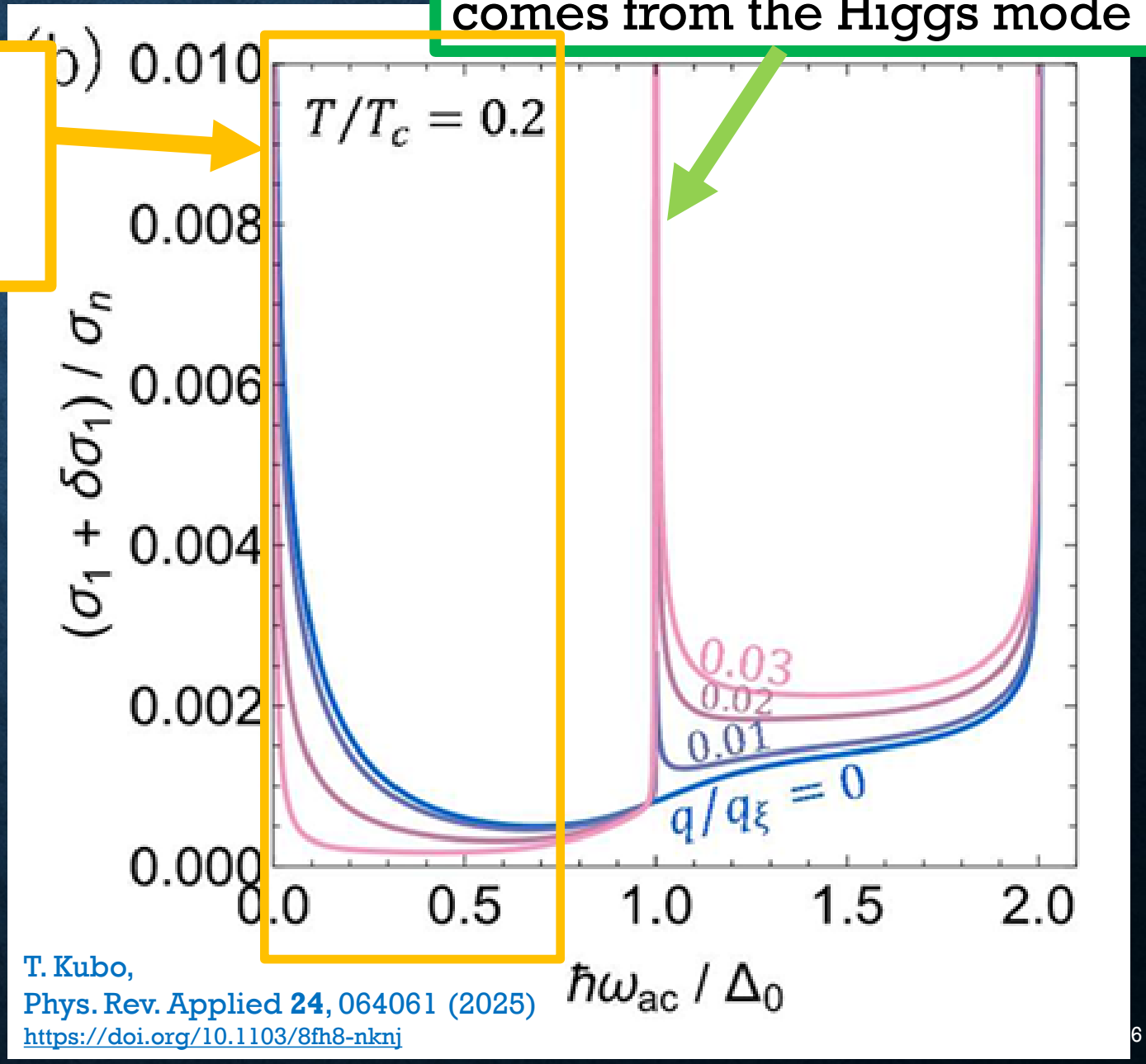
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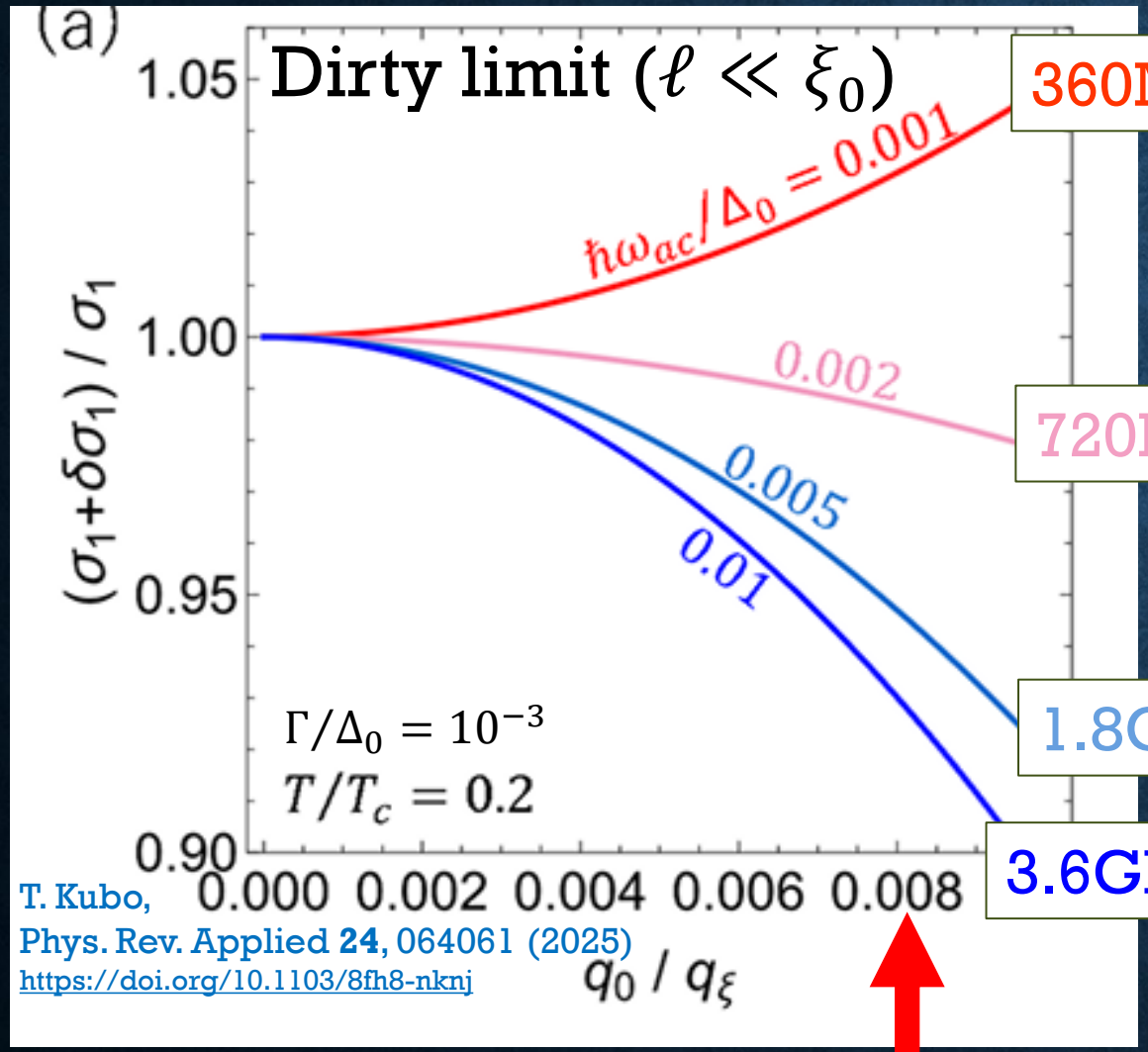
The AAA term is the dominant contributor at low frequencies

The resonance peak at Δ comes from the Higgs mode



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ac amplitude dependent dissipative conductivity



360MHz for Nb

720MHz for Nb

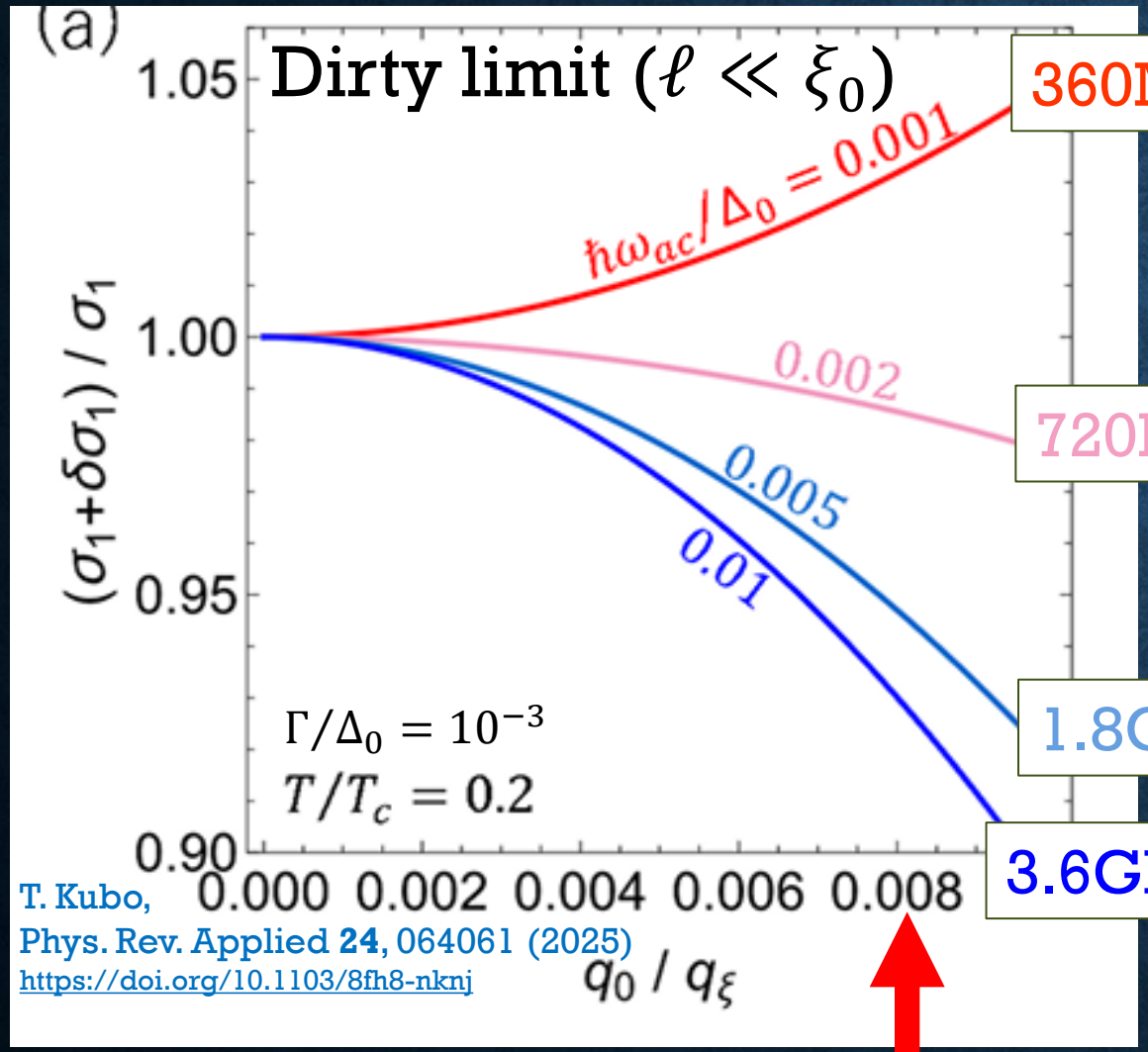
1.8GHz for Nb

3.6GHz for Nb

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Note: this corresponds to **a few (mT)**

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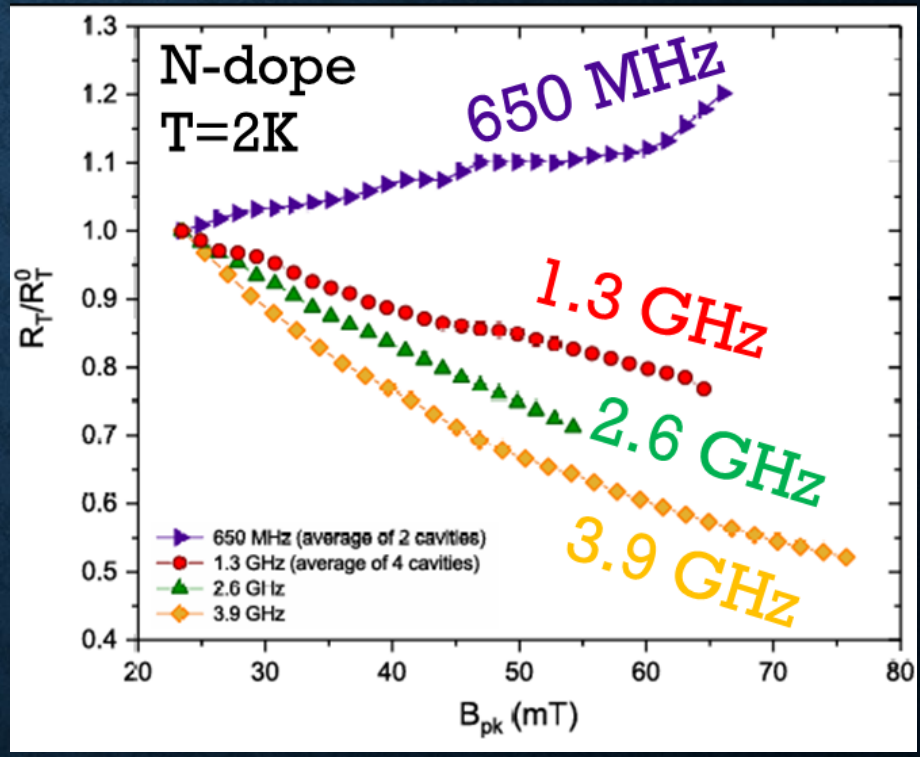
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Experiments: Moderately dirty ($\ell \sim \xi_0$)

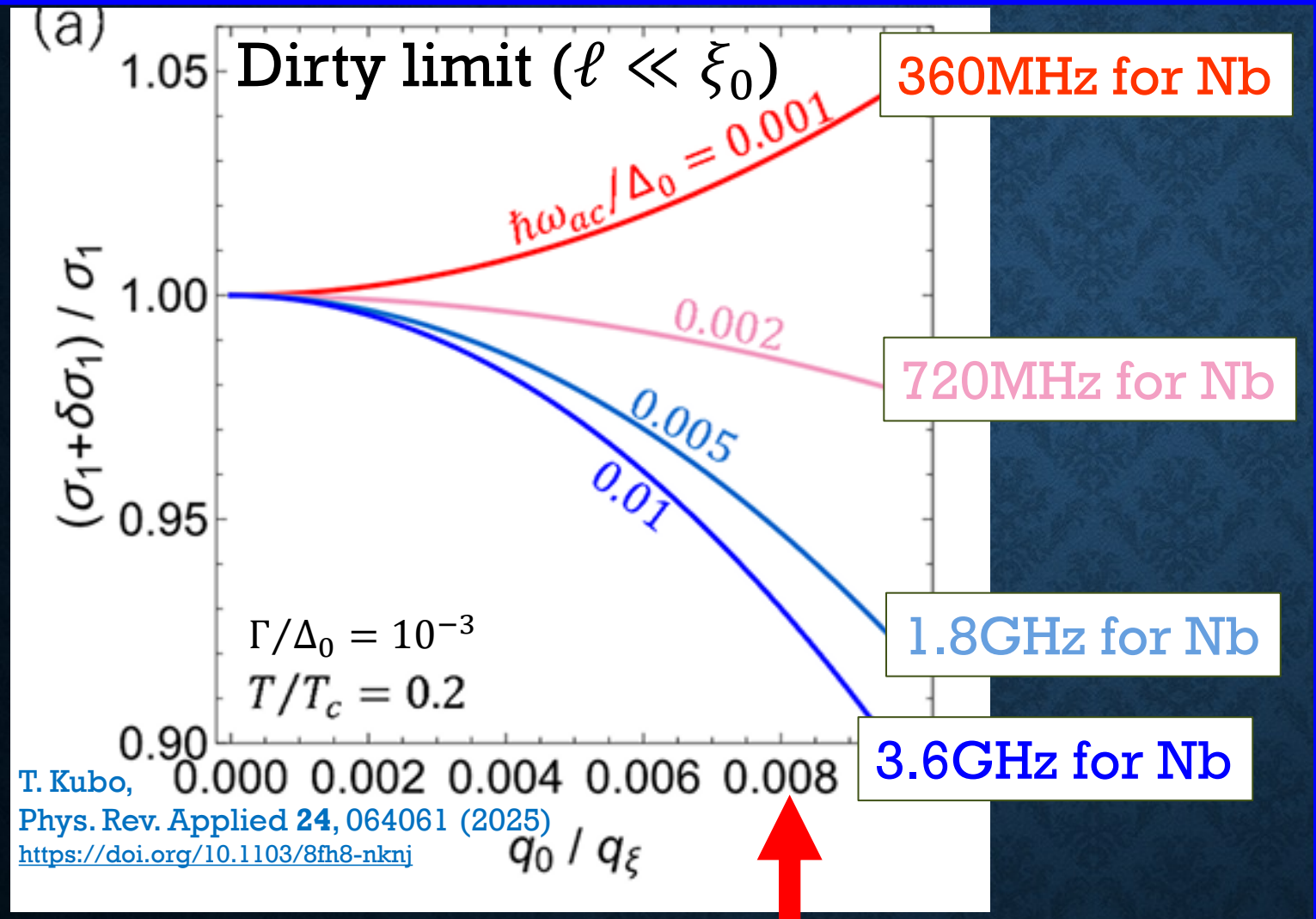
- M. Martinello et al., PRL. **121**, 224801 (2018)
- P. Dhakal et al., PRAB **27**, 062001 (2024)



M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
DOI: https://doi.org/10.1103/PhysRevLett.121.224801

Note: this corresponds to a few (mT)

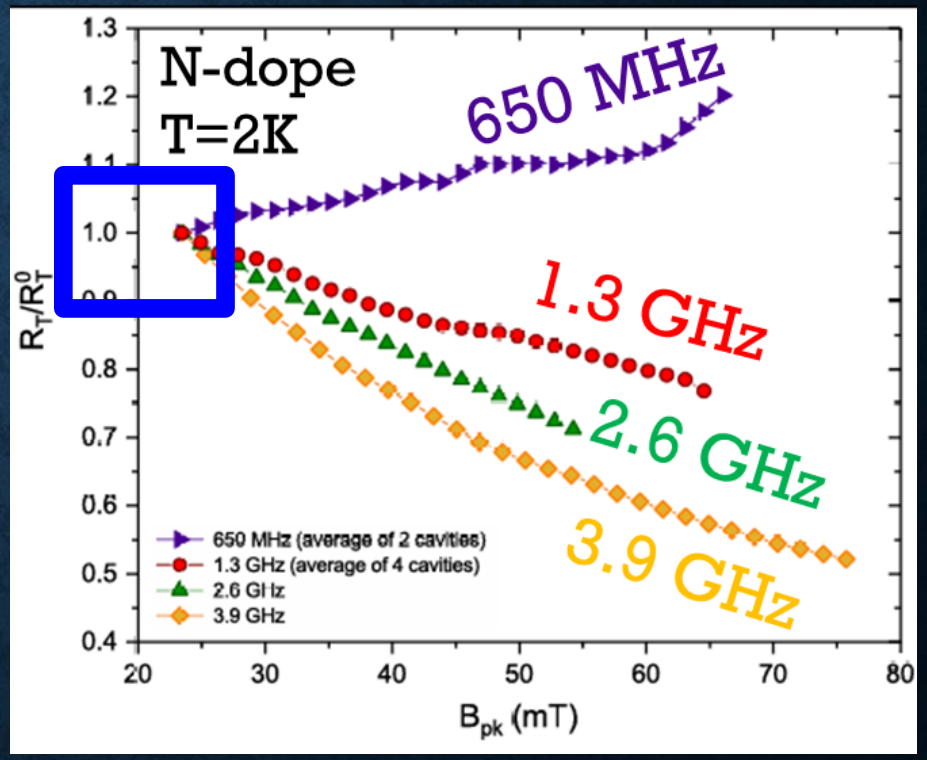
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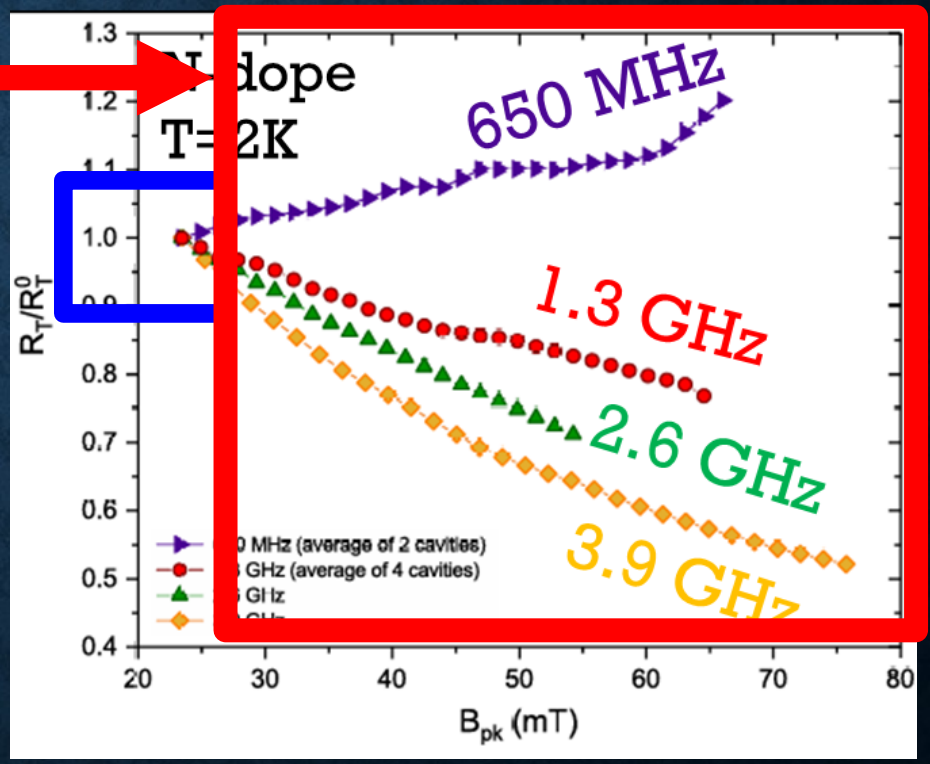
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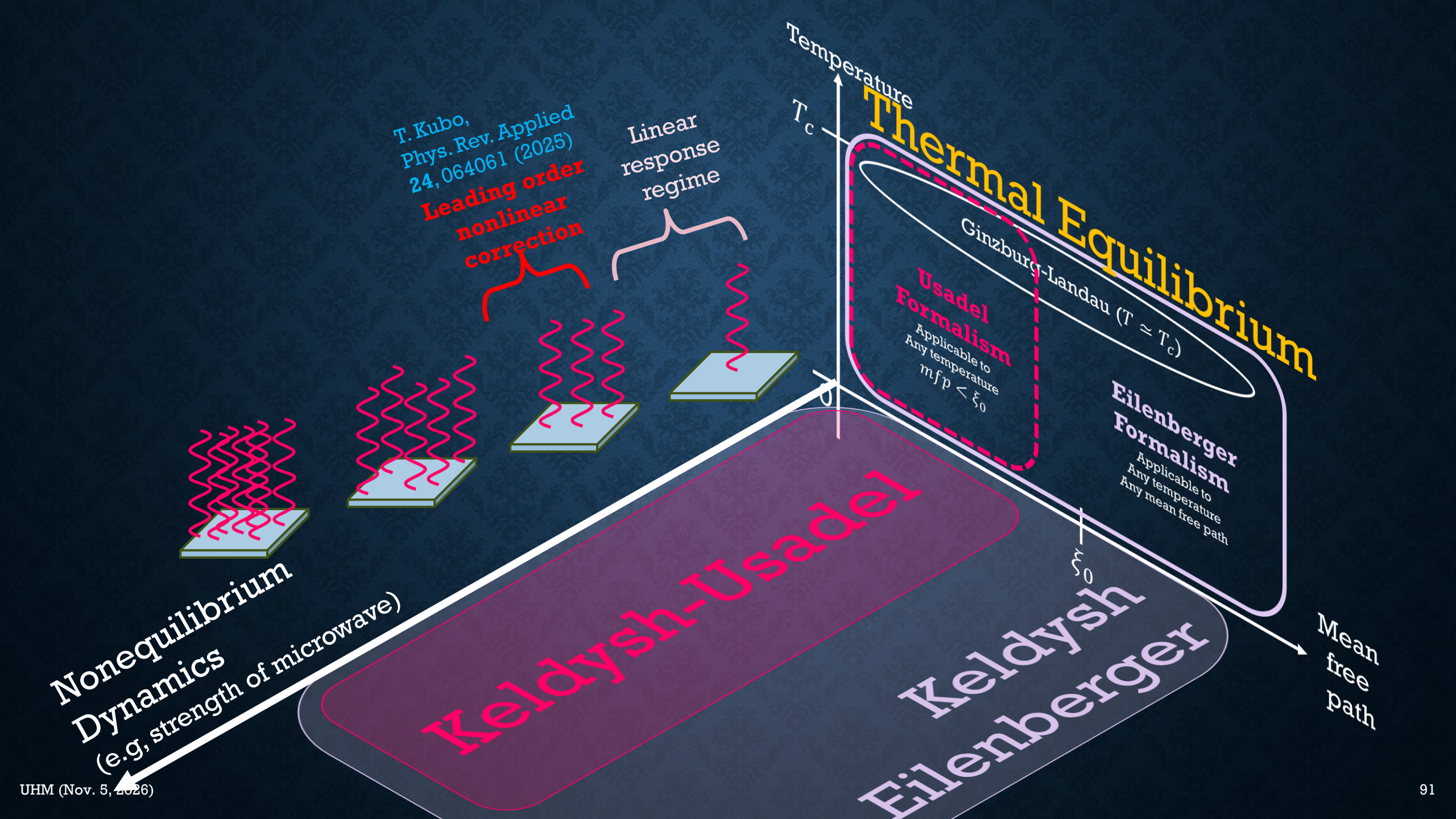
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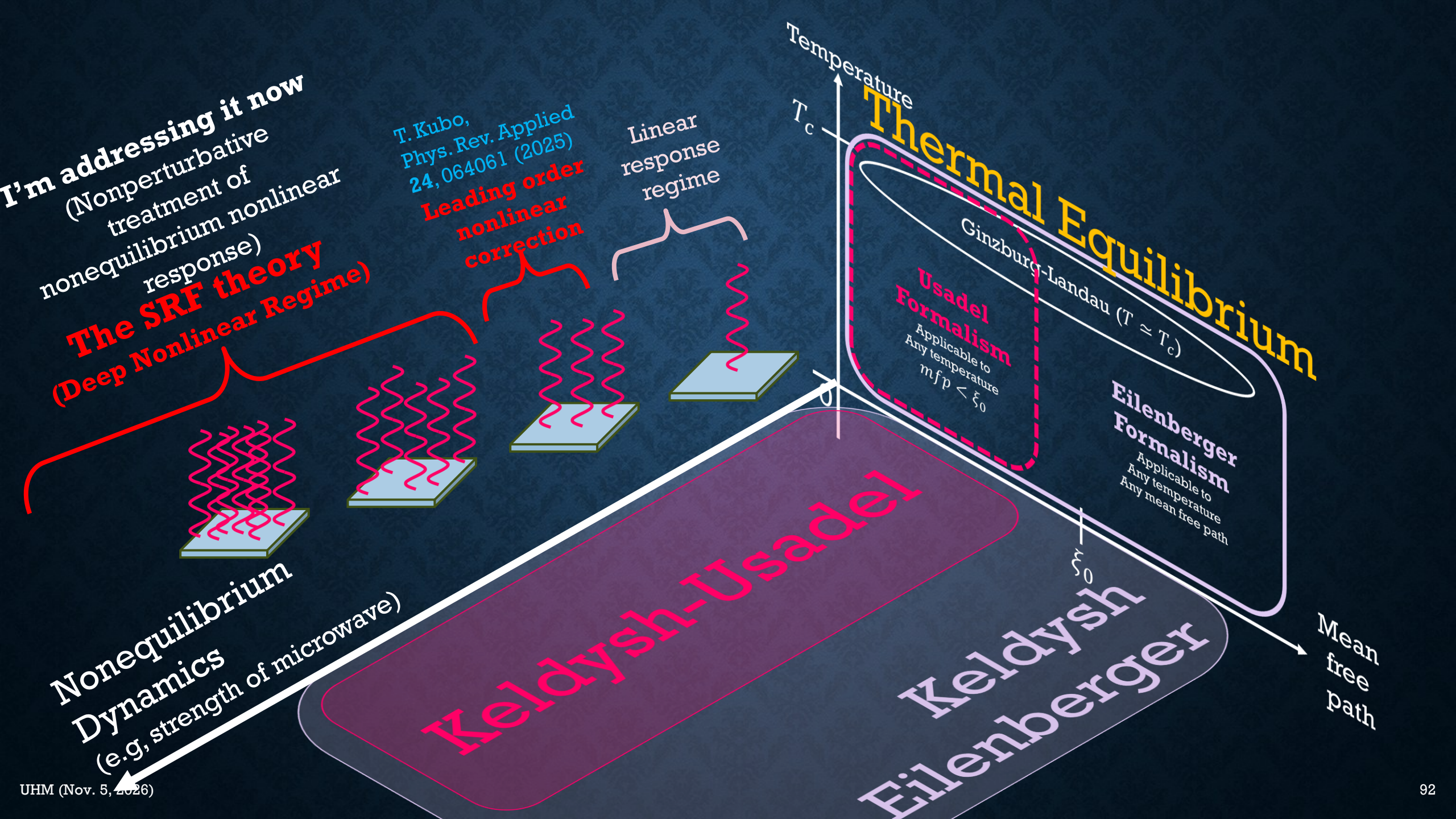
To deal with **the red-framed region**, the theory must be extended **nonperturbatively** to capture the quasiparticle spectrum modified by strong AC currents (**DOS smearing**).

Gurevich, PRL **113**, 087001 (2014)
Kubo-Gurevich, PRB **100**, 064522 (2019)

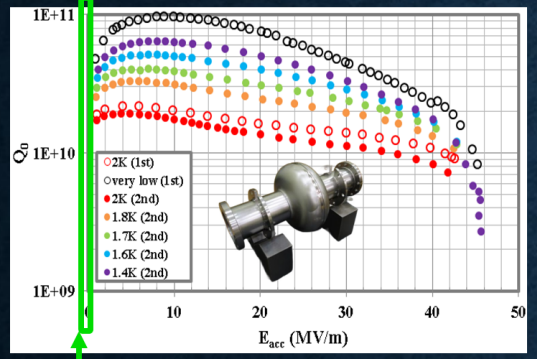


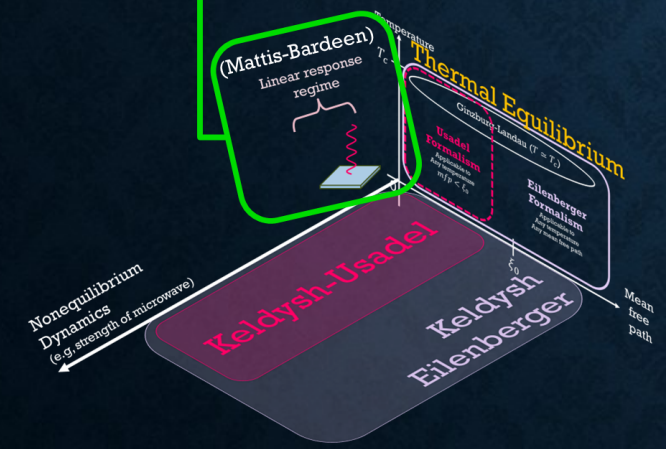
M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
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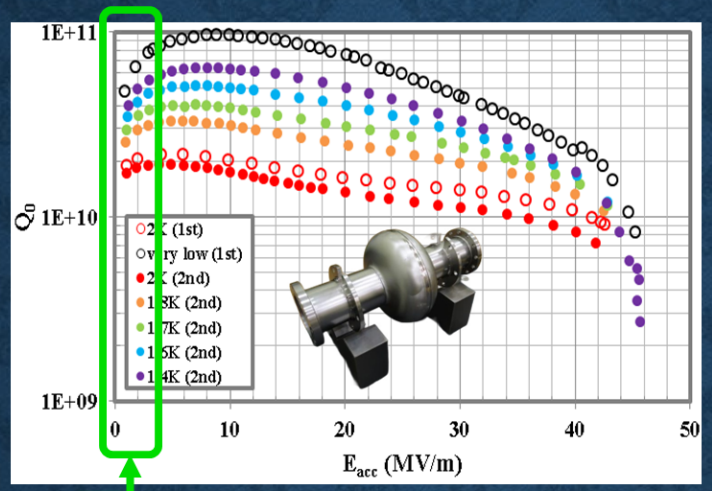

T. Kubo,
Phys. Rev. Applied
24, 064061 (2025)
Leading order
nonlinear
correction
Linear
response
regime
Temperature
T_c
Thermal Equilibrium
Ginzburg-Landau ($T \simeq T_c$)
Usadel
Formalism
Applicable to
Any temperature
$mfp < \xi_0$
Eilenberger
Formalism
Applicable to
Any temperature
Any mean free path
0
ξ_0
Nonequilibrium
Dynamics
(e.g, strength of microwave)
Keldysh-Usadel
Keldysh
Eilenberger
Mean
free
path

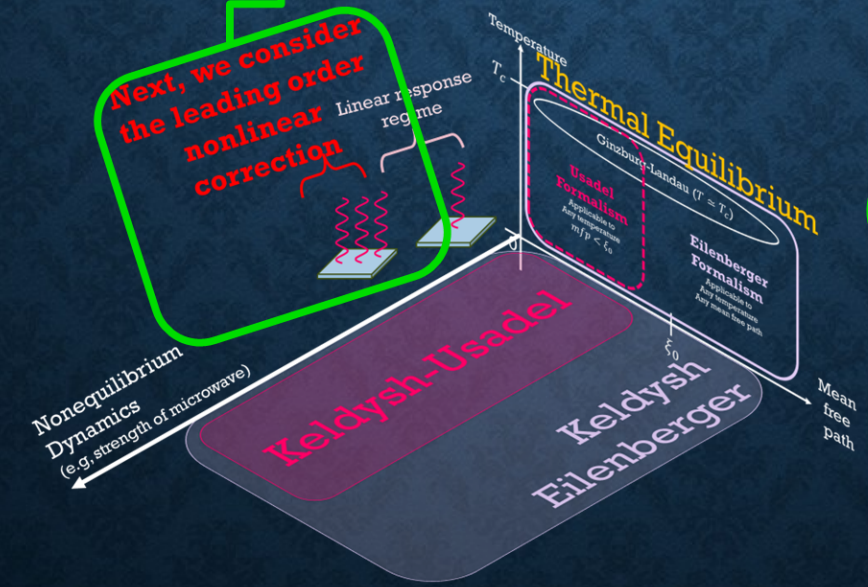

I'm addressing it now
(Nonperturbative treatment of nonequilibrium nonlinear response)
The SRF theory
(Deep Nonlinear Regime)
T. Kubo, Phys. Rev. Applied 24, 064061 (2025)
Leading order nonlinear correction
Linear response regime
Temperature
T_c
Thermal Equilibrium
Ginzburg-Landau ($T \simeq T_c$)
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Nonequilibrium Dynamics
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Mean free path

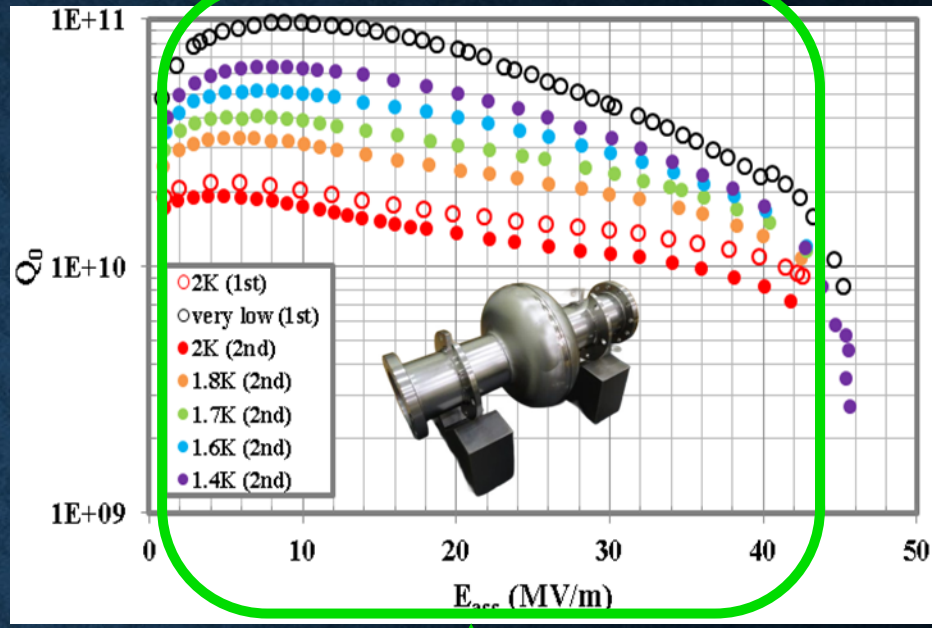
Summary of this part

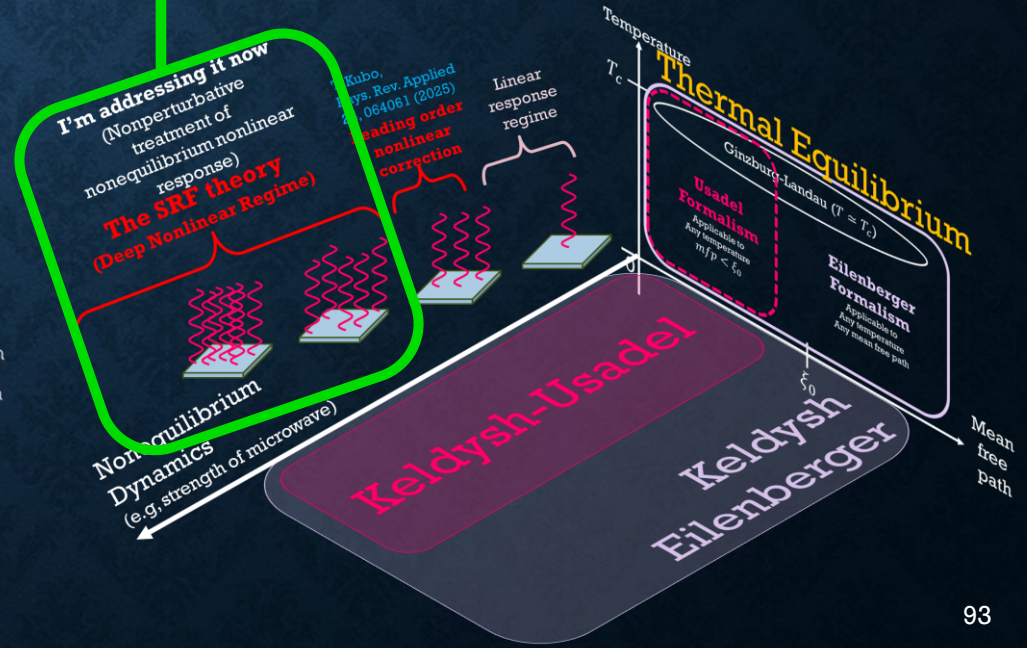


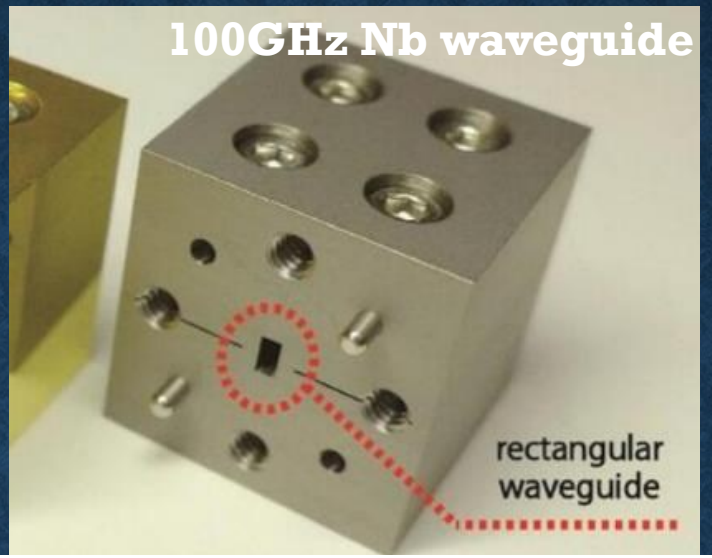












T. Nakajima et al., Propagation in Superconducting Niobium Rectangular Waveguide in the 100 GHz band, J. Phys.: Conf. Ser. **2545**, 012021 (2023)
DOI : 10.1088/1742-6596/2545/1/012021

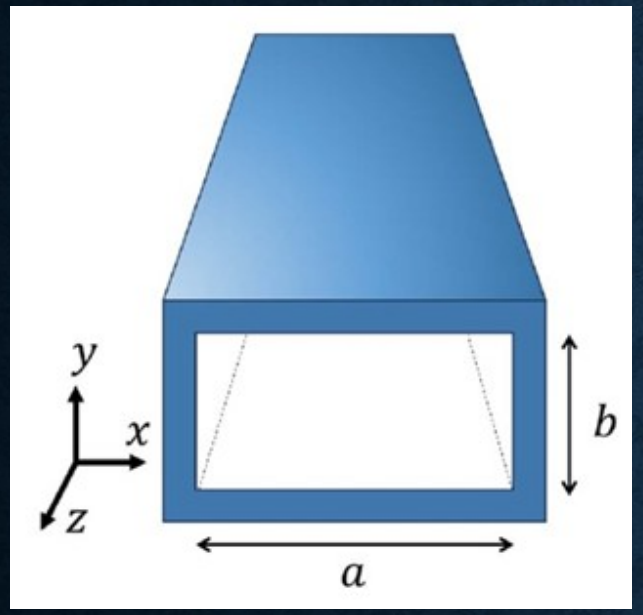
1.27 mm × 2.54 mm
(the so-called WR10)

Non-accelerator application (1)
Millimeter wave and THz superconducting waveguide

Why superconducting waveguides?
Ultra-low-loss transmission at millimetre-wave and sub-terahertz frequencies
→ Higher-sensitivity receiver systems for astronomy (Waveguide multiplexer) and
Improved detection of extremely weak astronomical signals
→ Potential applications to atmospheric remote sensing
→ Possible applications to high-frequency communication systems above 100 GHz

Attenuation at different frequencies and for different material purities

Linear response



$$\alpha = \frac{P_{\text{loss}}}{2P_{\text{flow}}} = \frac{R_s}{Z_0 b} \frac{1 + (2b/a)(\omega_c/\omega)^2}{\sqrt{1 - (\omega_c/\omega)^2}},$$

$$R_s(\ell, T, \omega) = \sqrt{\frac{\mu_0 \omega}{2}} \frac{\sqrt{|\sigma| - \sigma_2}}{|\sigma|}.$$

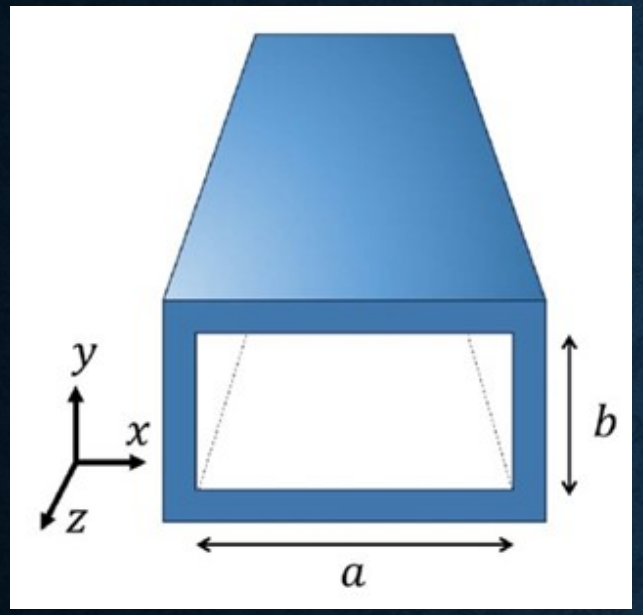
$$\begin{aligned}
\sigma(\ell, T, \omega) &= \sigma_1 + i\sigma_2 \\
&= \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon [\kappa \mathcal{T}_- + \kappa^* \mathcal{T}_+ + \kappa^a (\mathcal{T}_+ - \mathcal{T}_-)], \\
\kappa &= \frac{1}{3} \frac{g_+ g_- + f_+ f_- - 1}{d_+ + d_-}, \\
\kappa^a &= \frac{1}{3} \frac{-g_+ g_-^* - f_+ f_-^* - 1}{d_+ - d_-^*}.
\end{aligned}$$

Table 2. Dimensions and frequency ranges of standard rectangular waveguides. Here a and b are the broad and narrow wall dimensions, respectively.

Waveguide name	a [mm]	b [mm]	Cutoff of TE_{10} [GHz]	Cutoff of next mode [GHz]
WR28	7.112	3.556	21.1	42.2
WR15	3.759	1.880	39.9	79.7
WR10	2.54	1.27	59.0	118.0
WR8	2.032	1.016	73.8	147.5
WR5.1	1.295	0.648	115.7	231.4
WR3.4	0.864	0.432	173.6	347.1
WR2	0.508	0.254	295.1	590.1
WR1	0.254	0.127	590.1	1180.3

T. Kubo, Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity, Supercond. Sci. Technol. **39**, 075036 (2026)

Attenuation at different frequencies and for different material purities



Linear response

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$$R_s(\ell, T, \omega) = \sqrt{\frac{\mu_0 \omega}{2}} \frac{\sqrt{|\sigma| - \sigma_2}}{|\sigma|}.$$

$$\sigma(\ell, T, \omega) = \sigma_1 + i\sigma_2 = \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon [\kappa \mathcal{T}_- + \kappa^* \mathcal{T}_+ + \kappa^a (\mathcal{T}_+ - \mathcal{T}_-)],$$

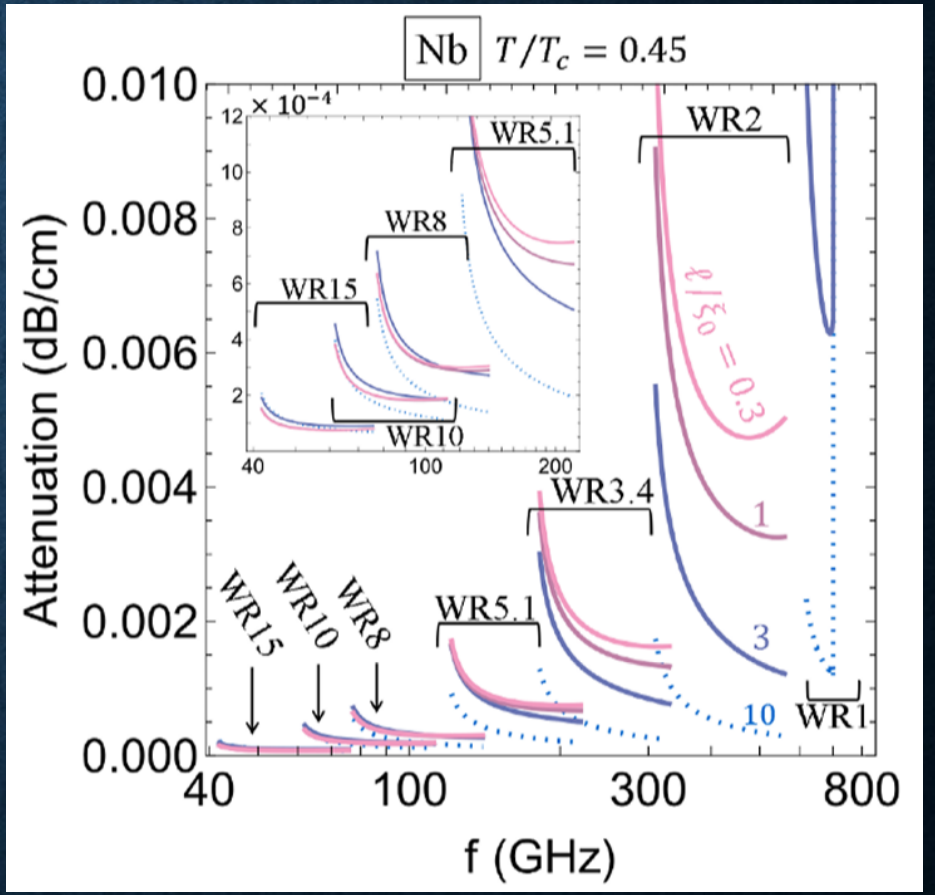
$$\kappa = \frac{1}{3} \frac{g_+ g_- + f_+ f_- - 1}{d_+ + d_-},$$

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High frequency: higher purity is better
Low frequency: higher purity is not always better

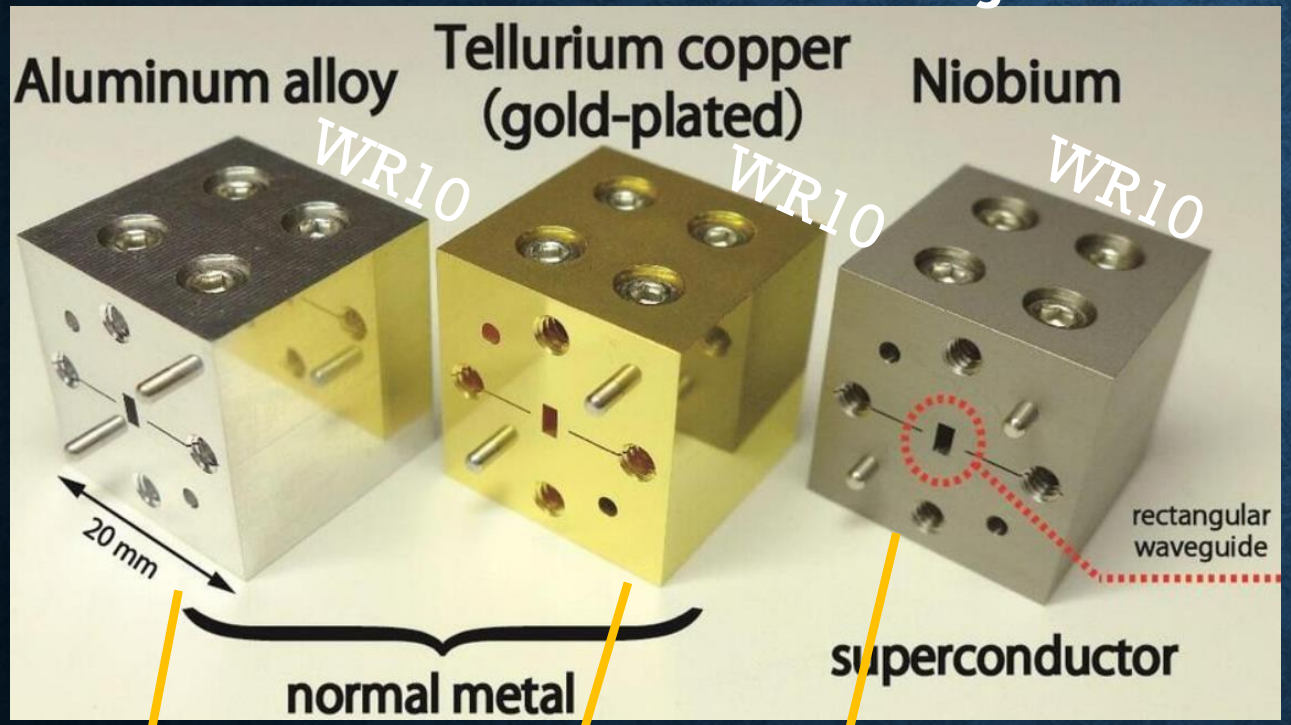


T. Kubo, Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity, Supercond. Sci. Technol. **39**, 075036 (2026)

Attenuation at different frequencies and for different material purities

T. Nakajima et al., Propagation in Superconducting Niobium Rectangular Waveguide in the 100 GHz band, J. Phys.: Conf. Ser. **2545**, 012021 (2023);
DOI : 10.1088/1742-6596/2545/1/012021

100GHz waveguides

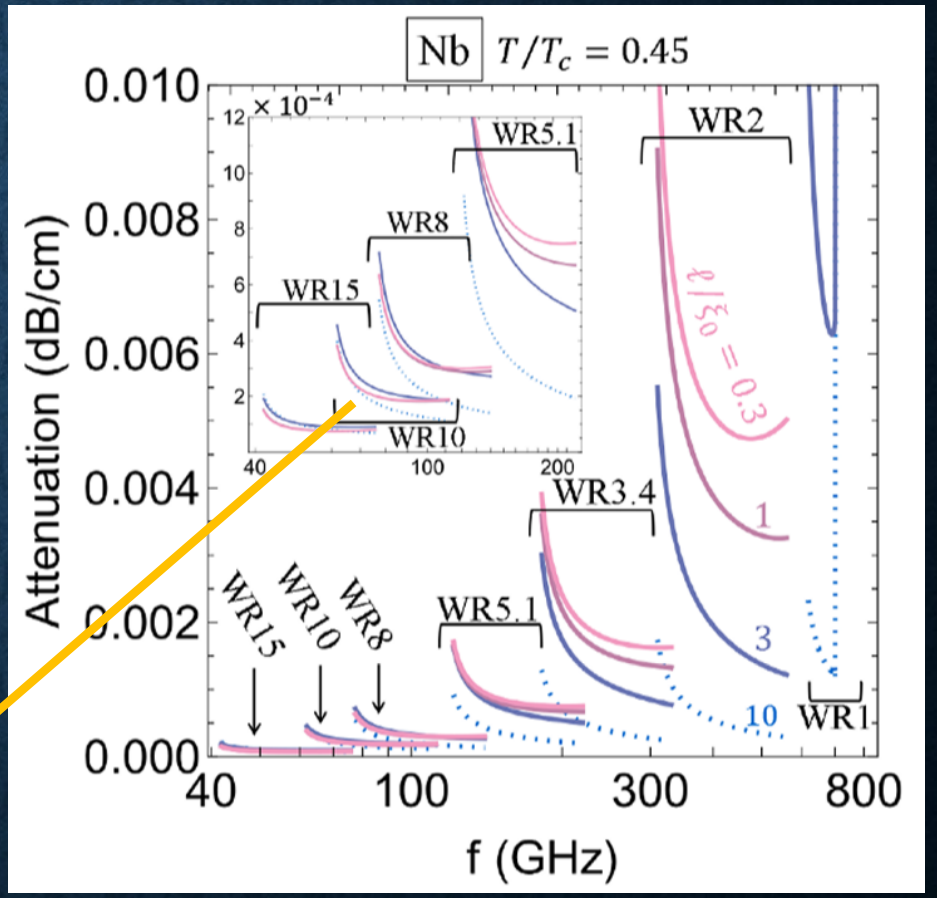


2.6×10^{-2} dB/cm

1.1×10^{-2} dB/cm

5×10^{-4} dB/cm
(Same order of magnitude as theory $\simeq 2 \times 10^{-4}$ dB/cm)

High frequency: higher purity is better
Low frequency: higher purity is not always better



T. Kubo, Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity, Supercond. Sci. Technol. **39**, 075036 (2026)

Attenuation at different frequencies and for different material purities

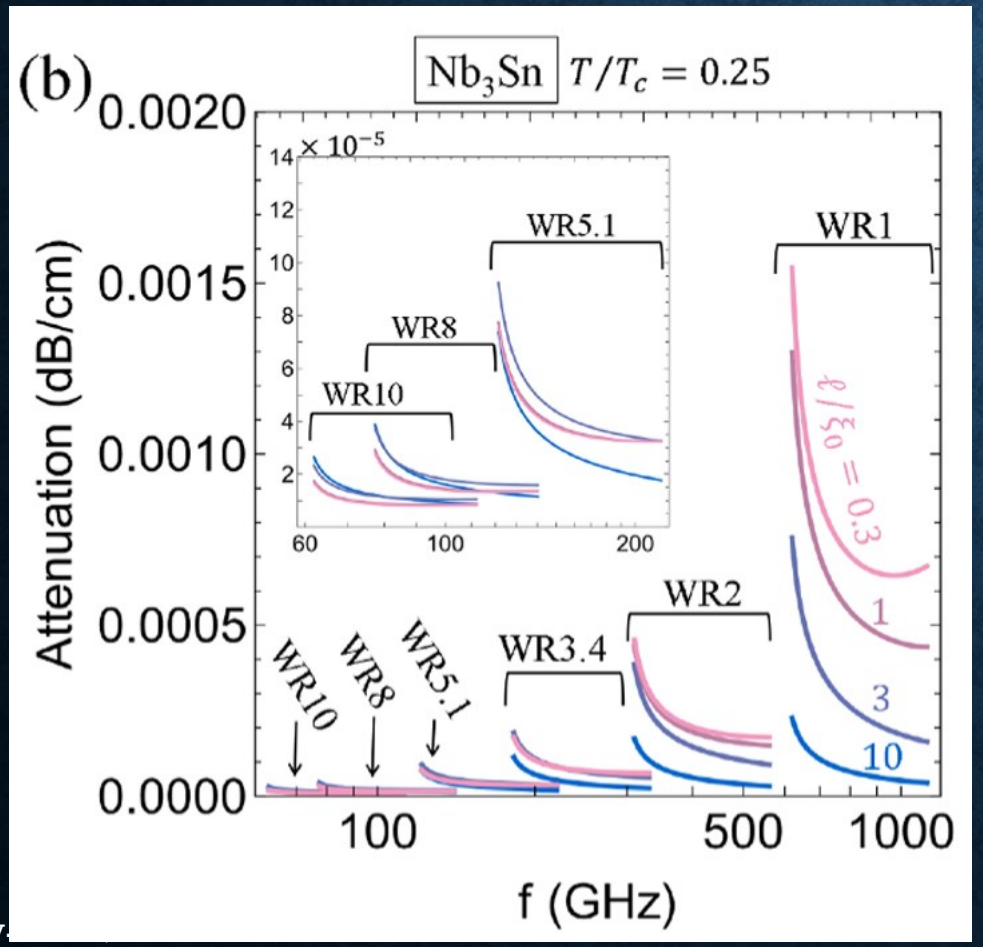
Higher-gap materials are needed toward the THz range.

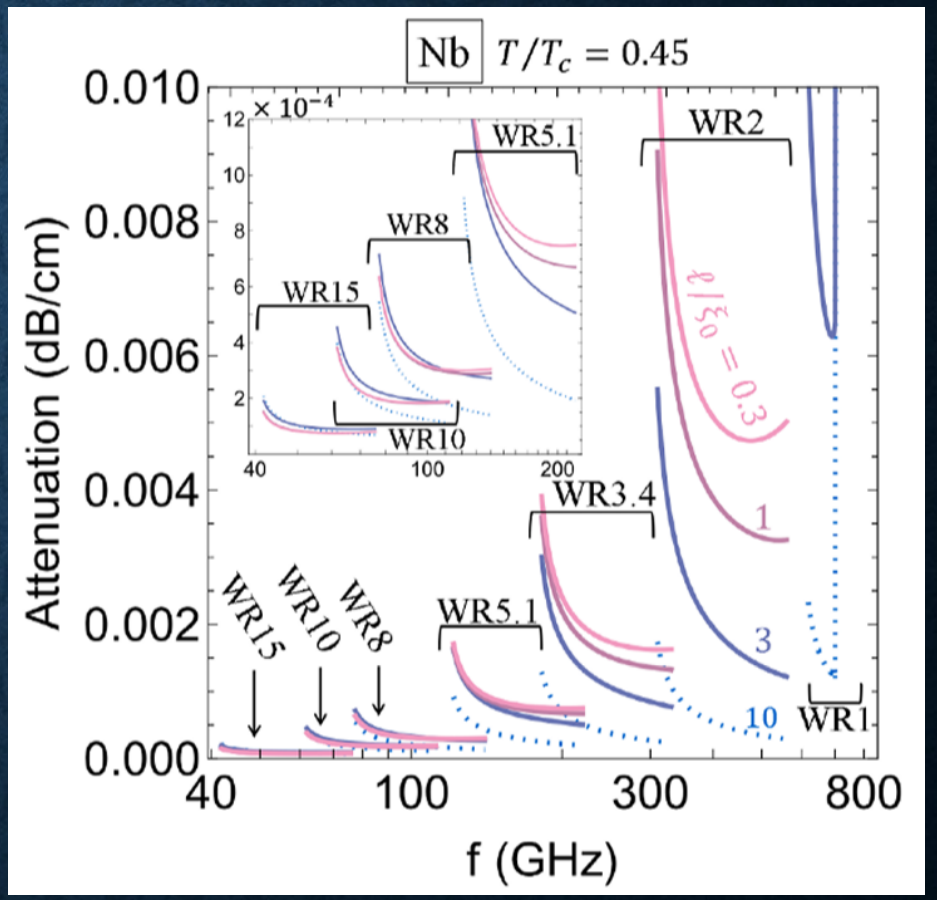
Nb3Sn: gap frequency

$f_g = 2\Delta/h \simeq 1.4$ THz

Nb: gap frequency

$f_g = 2\Delta/h \simeq 720$ GHz





T. Kubo,
Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity,
Supercond. Sci. Technol. **39**, 075036 (2026)

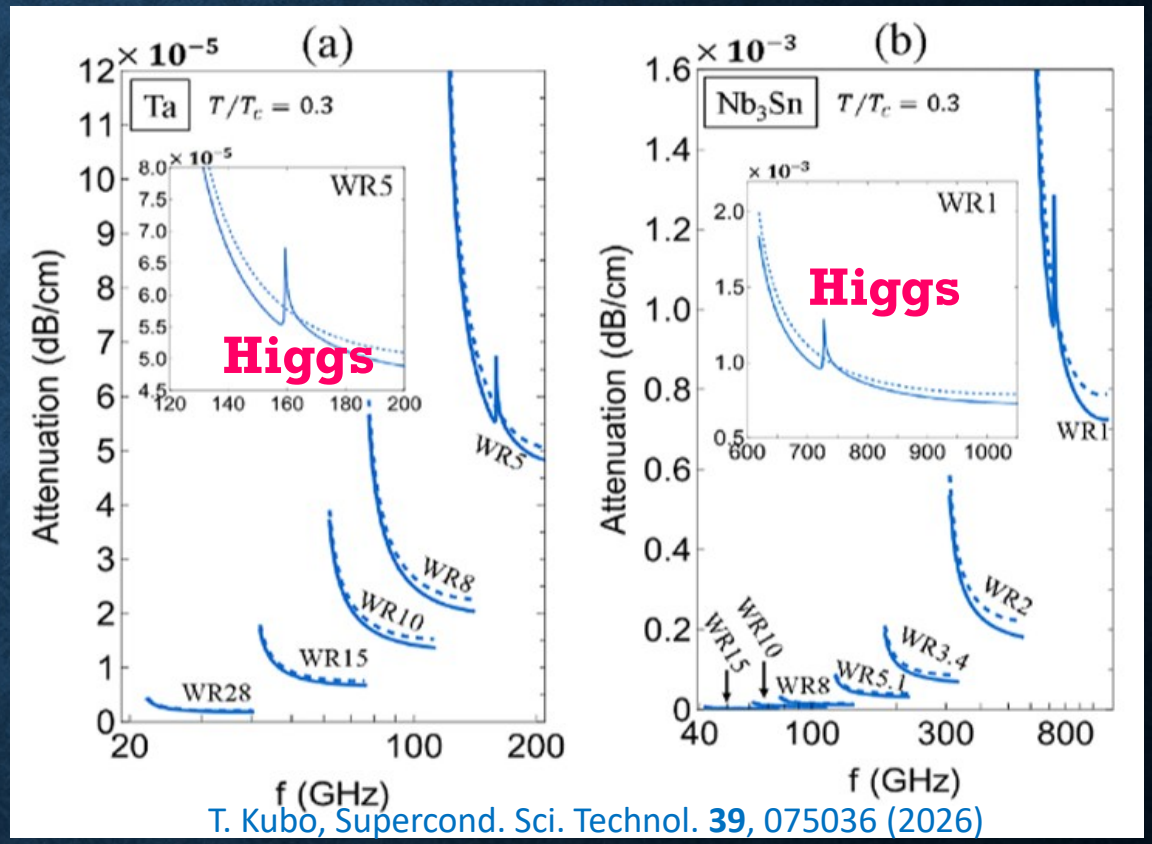
Strong-signal regime: nonlinear electrodynamics and Higgs-mode signatures

$$\alpha = \frac{P_{\mathrm{loss}}}{2P_{\mathrm{flow}}} = \frac{R_s}{Z_0 b}\frac{1+(2b/a)(\omega_c/\omega)^2}{\sqrt{1-(\omega_c/\omega)^2}},$$

$$\frac{R_s}{R_0} = \frac{1}{2\pi\sqrt{\ell/\xi_0}}\left(\frac{\hbar\omega}{\Delta_0}\right)^2\left(\frac{\lambda}{\lambda_{0,\mathrm{dirty}}}\right)^3 \times \left[\frac{\sigma_1}{\sigma_n} + \frac{u(T,\omega)}{2\pi}\left(\frac{B_{\mathrm{ac}}}{B_c}\right)^2\right]$$

$$u(T,\omega) = \frac{2\sqrt{\pi}}{\hbar\omega/\Delta_0}\mathrm{Re}\left(I_{1\mathrm{H}}^{qqq} + I_{1\mathrm{H}}^{\mathrm{Higgs}} + I_{1\mathrm{H}}^{\mathrm{Eliash}}\right)$$

T. Kubo, Phys. Rev. Applied **24**, 064061 (2025)



T. Kubo, Supercond. Sci. Technol. **39**, 075036 (2026)

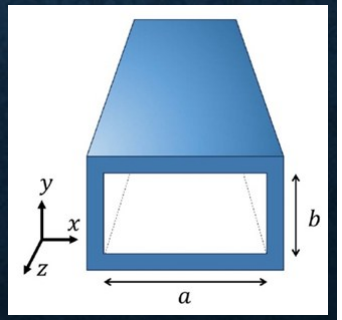


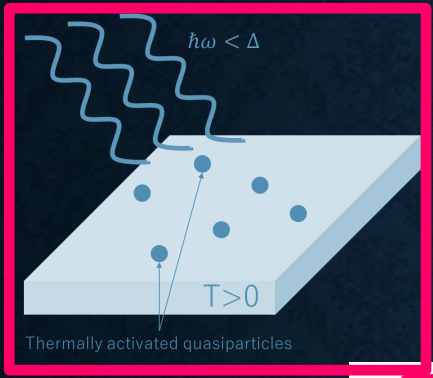
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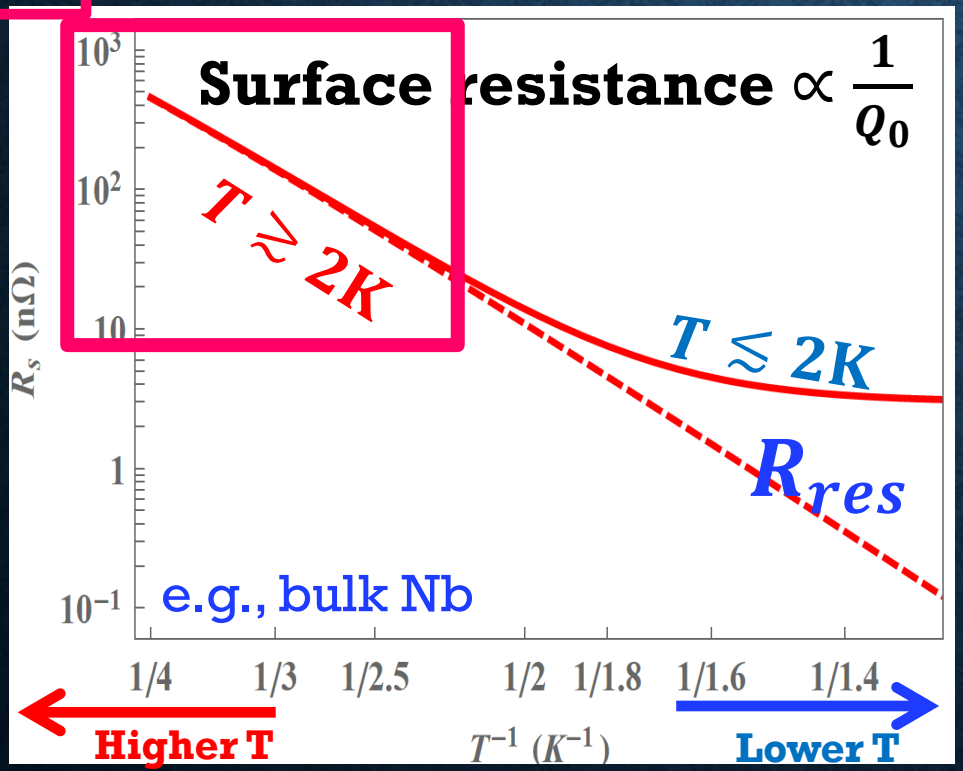
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Low-temperature Q factor

Accelerator regime

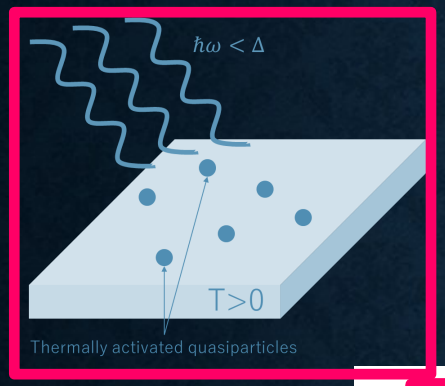
$T \gtrsim 2\mathrm{K}$



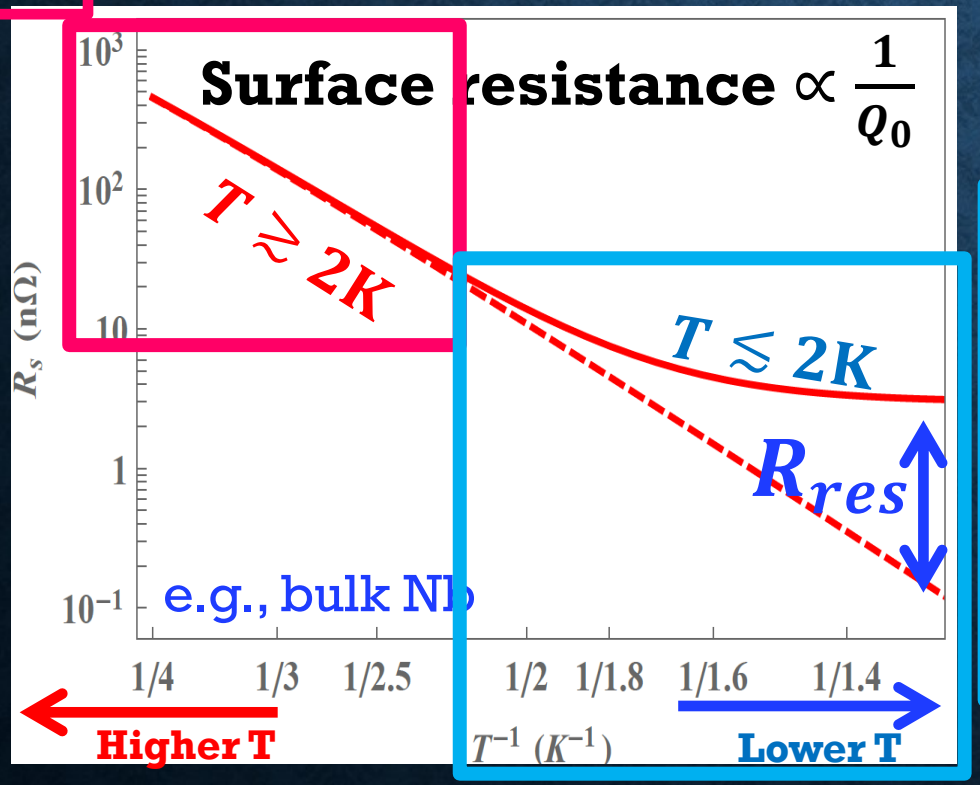


Accelerator regime

$T \gtrsim 2\mathrm{K}$



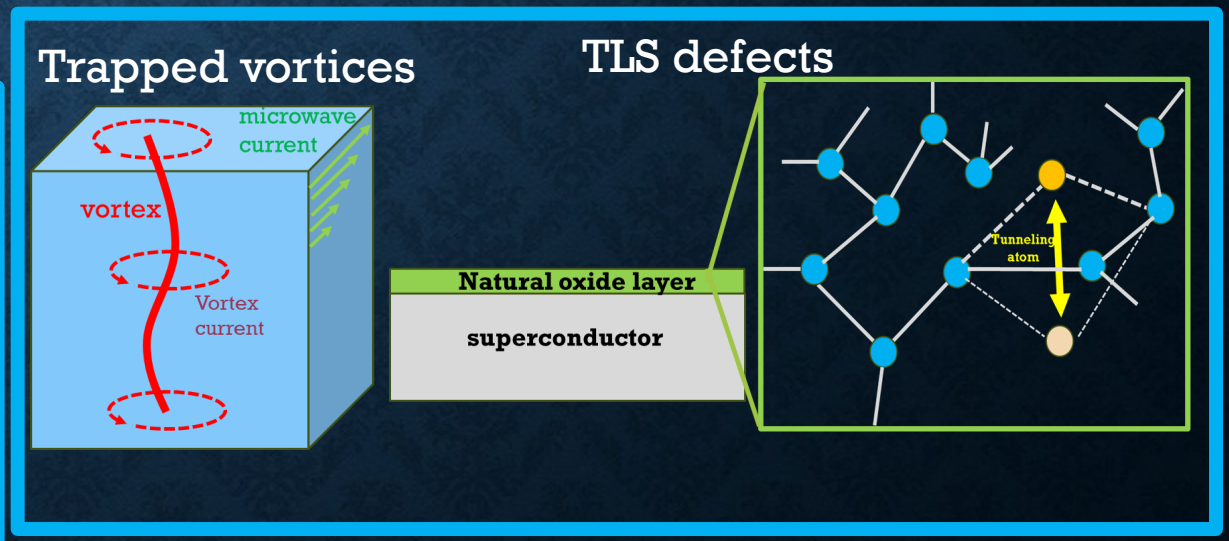
Can accelerator technologies be useful beyond the accelerator regime?



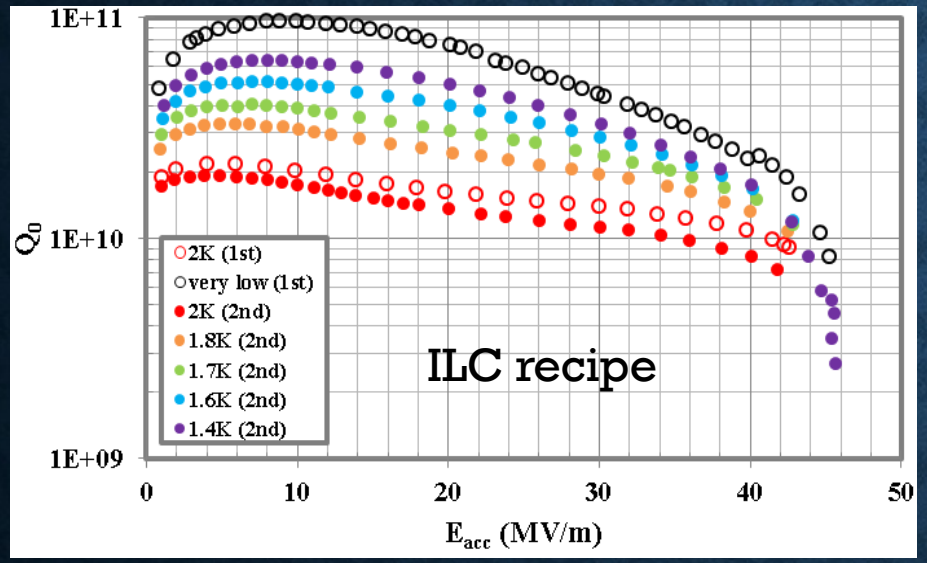
$T \lesssim 2\mathrm{K}$

Non-accelerator regime

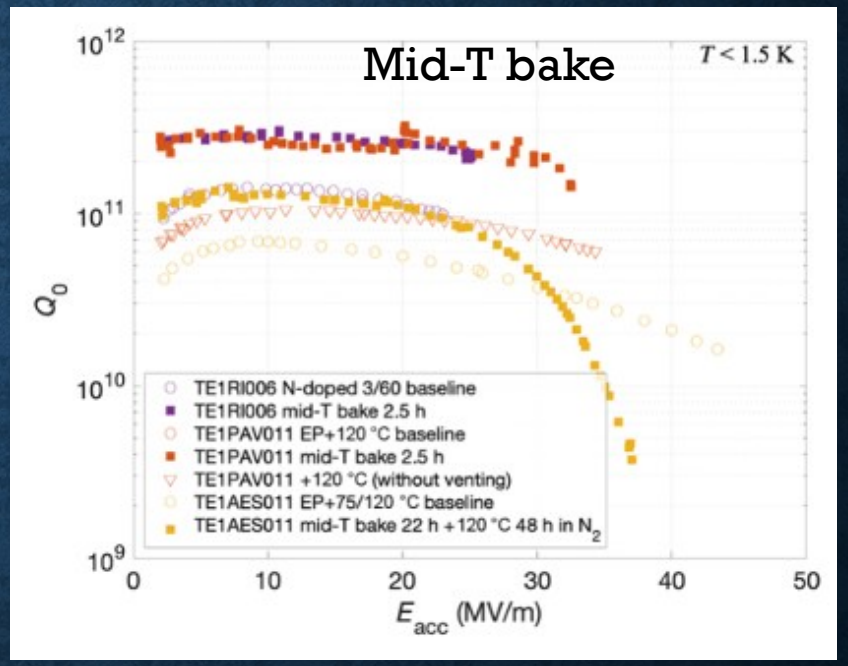
Including quantum regime (~10mK)



Acutually, the accelerator community can build **the best cavities on the planet** — achieving $Q \gtrsim 10^{11}$ at $T = 1$-2 K.



T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022



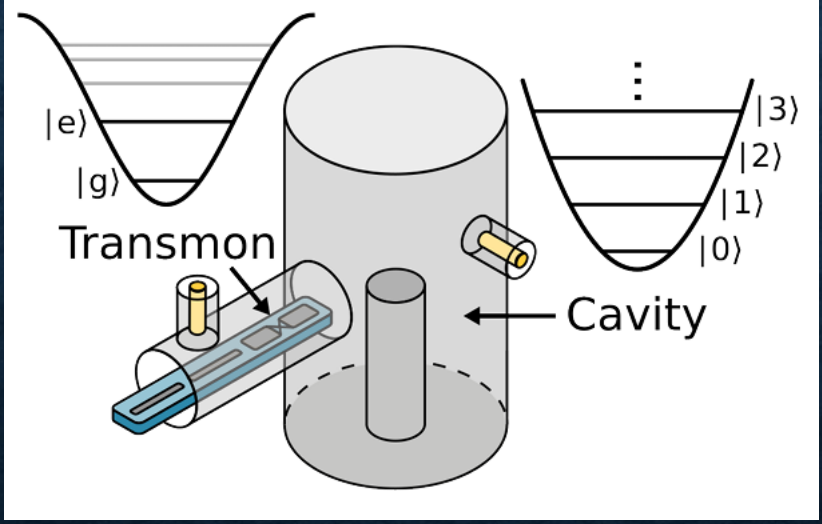
S. Posen et al., Ultralow Surface Resistance via Vacuum Heat Treatment of Superconducting Radio-Frequency Cavities,
Phys. Rev. Applied **13**, 014024 (2020)
https://doi.org/10.1103/PhysRevApplied.13.014024

But they do not assure the performace at T < 1 K.

Non-accelerator application (2)

Quantum applications

Cavities for quantum computing

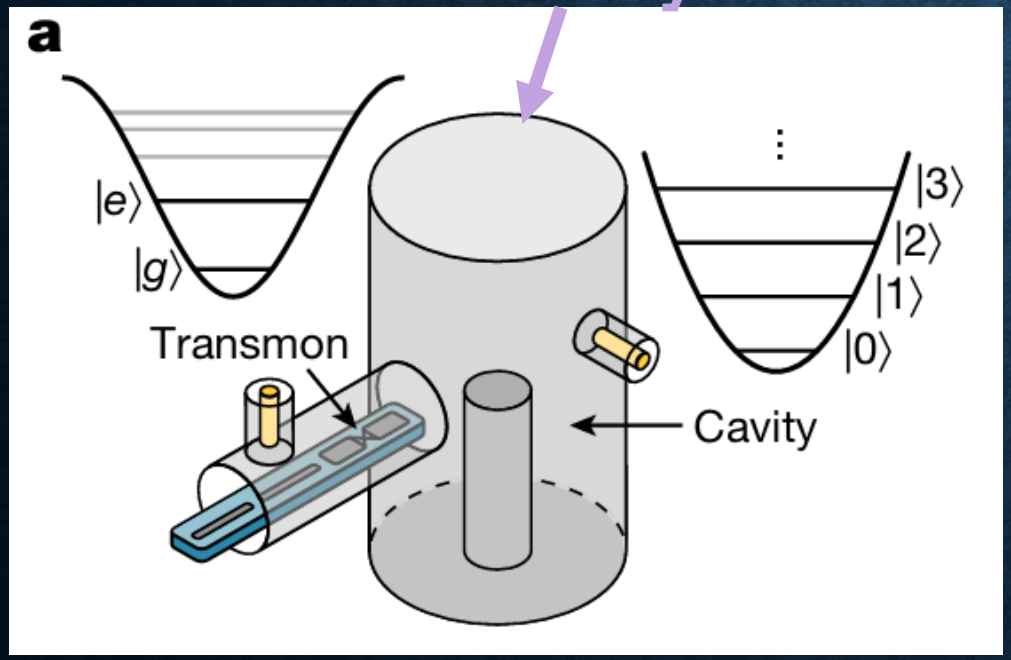


Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y

Superconducting Cavities in Quantum Technologies

- Superconducting cavities offer a promising approach to **hardware-efficient quantum information processing,** potentially reducing the overhead of quantum error correction.
- Various bosonic encodings are being explored, including cat codes, GKP codes, and dual-rail encoding.
- Startups are actively developing cavity-based quantum computing architectures.

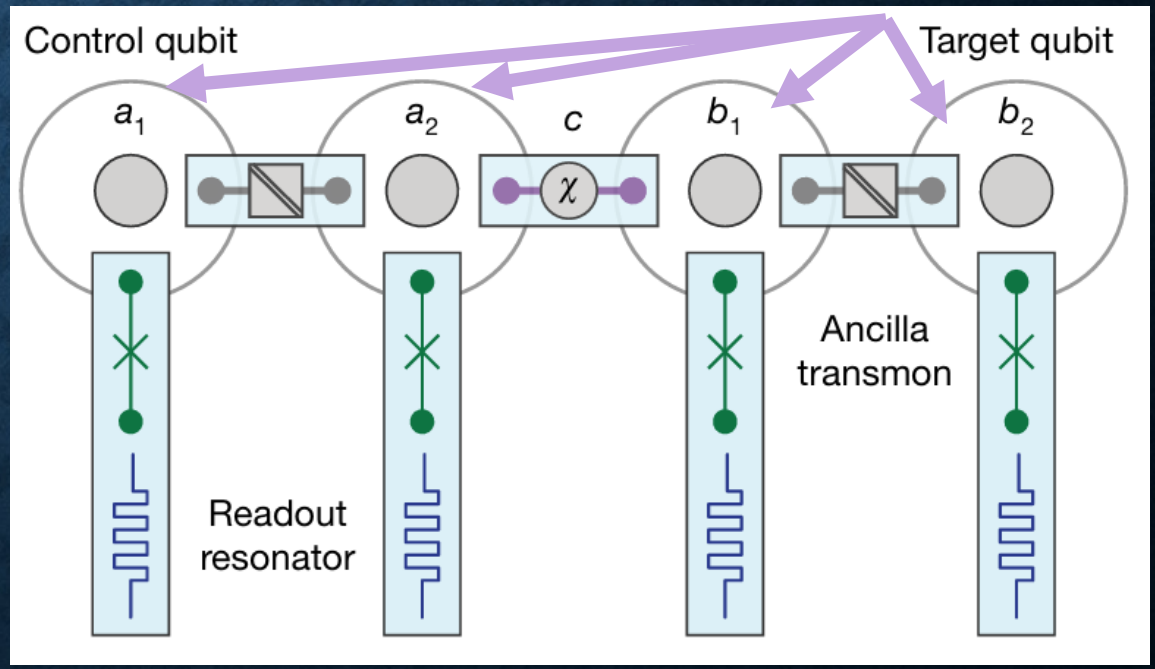
cavity



Brock et al., Nature **641**, 612 (2025)

Yale University

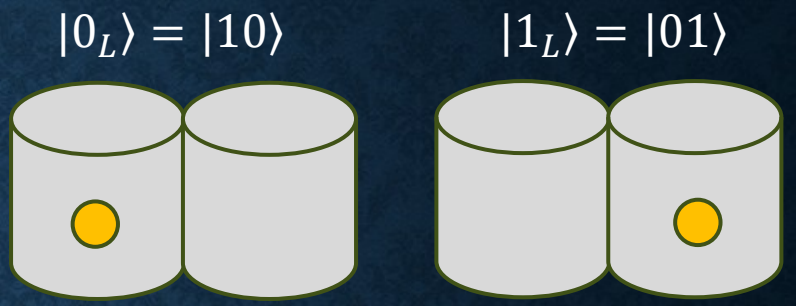
cavities



D-Wave Quantum Inc., Nature **656**, 47 (2026)

Quantum Circuits, Inc.

e.g., Dual-Rail Cavity Qubit

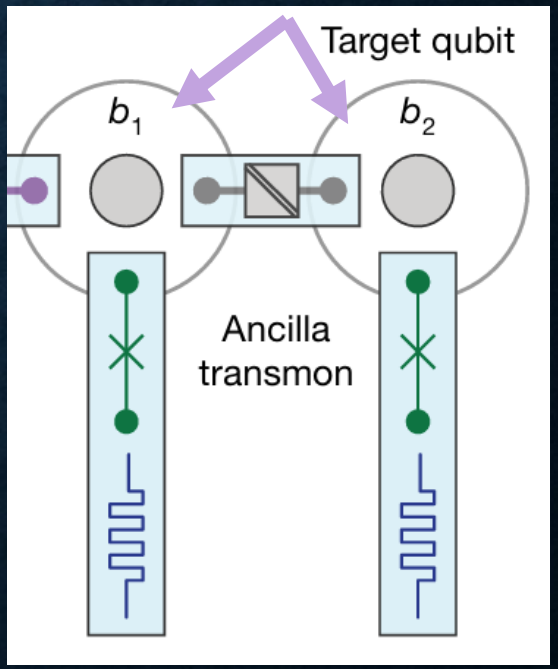


- **Two superconducting cavities:** Store a single photon encoding one qubit, $|0_L\rangle = |10\rangle$, $|1_L\rangle = |01\rangle$.

- **Tunable cavity-cavity coupler:** Enables coherent photon exchange and single-qubit gates.

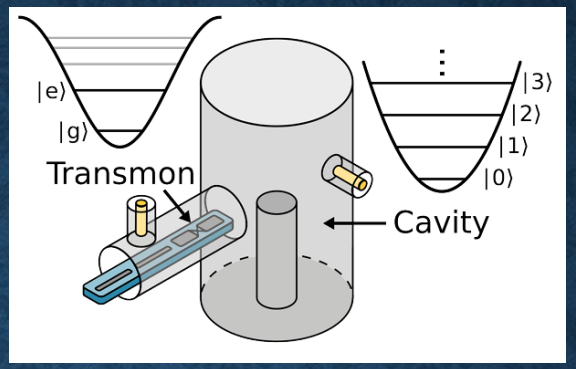
- **Ancilla transmon:** Enables state preparation, logical readout, and detection of photon-loss errors.

Key advantage: Photon loss can be detected and converted into a *located erasure error*, making quantum error correction more efficient.



D-Wave Quantum Inc., Nature **656**, 47 (2026)
Quantum Circuits, Inc.

e.g. binomial code



Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y

$$|0\rangle_L = \frac{|0\rangle + \sqrt{3}\,|6\rangle}{2}$$

$$|1\rangle_L = \frac{\sqrt{3}|3\rangle + |9\rangle}{2}$$

1 photon loss

$|5\rangle$

$$\frac{|2\rangle + |8\rangle}{\sqrt{2}}$$

1 photon loss

$|4\rangle$

$$\frac{|1\rangle + 2|7\rangle}{\sqrt{3}}$$

correct

a remainder of 2 when divided by 3

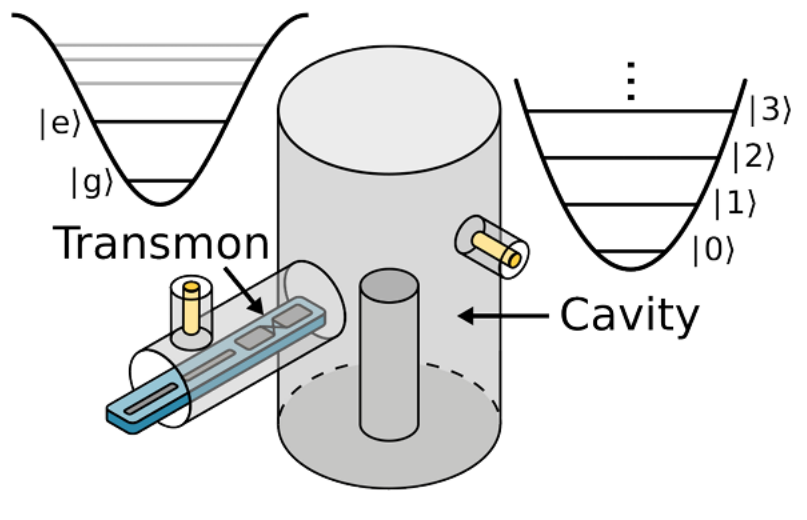
a remainder of 1 when divided by 3

correct

3D coaxial cavity progress

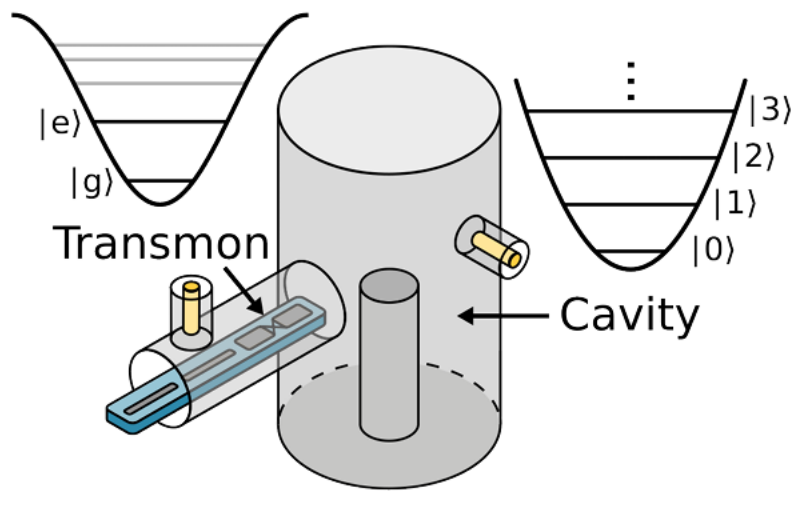
- Reagor et al., Phys. Rev. B **94**, 014506 (2016) Yale University
  Aluminum,
  $Q = 7 \times 10^7$ (single photon level)

Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y



3D coaxial cavity progress

Brock et al., Nature **641**, 612 (2025)
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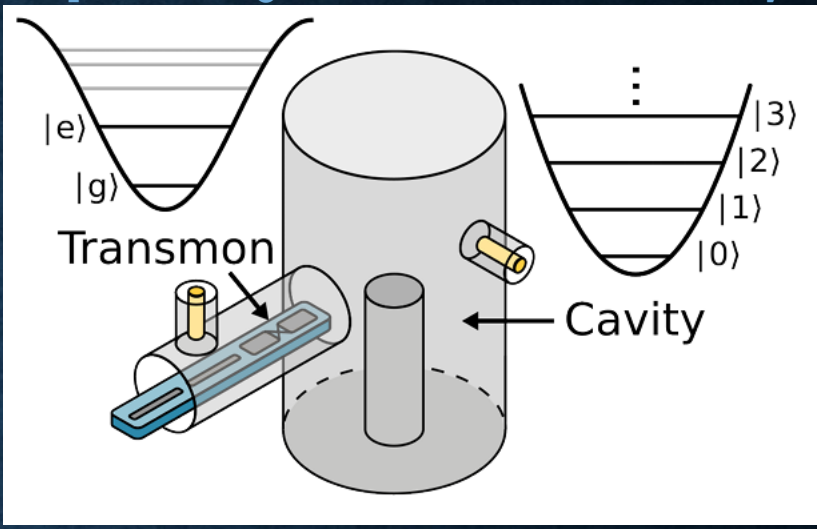


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- Heidler et al., Phys. Rev. Applied **16**, 034024 (2021) University of Innsbruck
  Niobium (1:1:1 buffered chemical polishing at 0-5℃),
  $Q = 5 \times 10^8$ (single photon level)

3D coaxial cavity progress

Brock et al., Nature **641**, 612 (2025)
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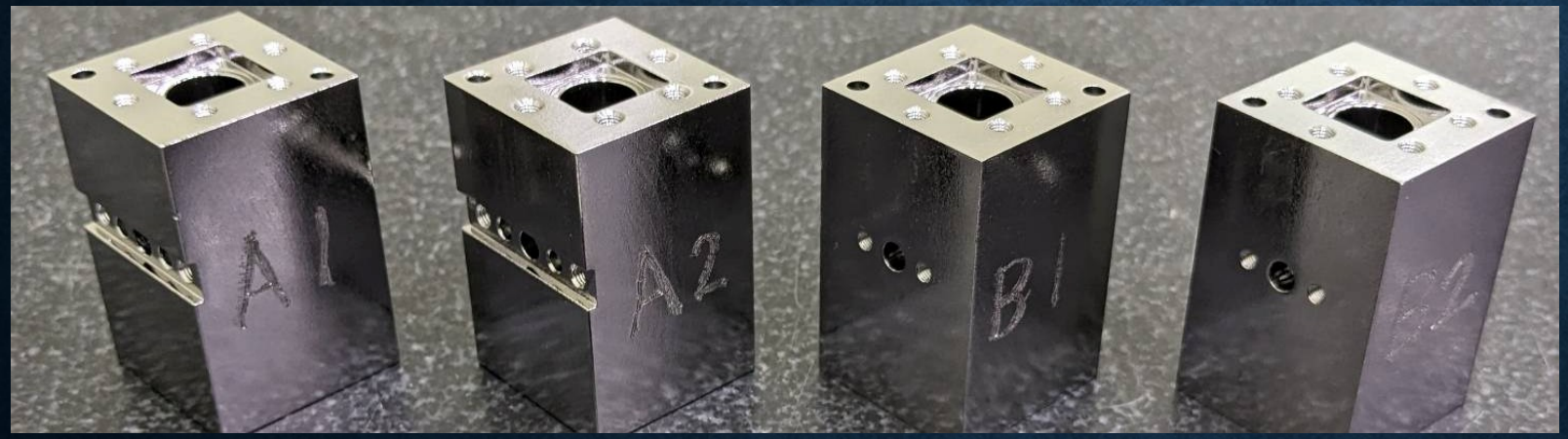
- Oriani et al., Phys. Rev. Applied **24**, 044080 (2025) University of Chicago
  Niobium (1:1:2 water buffered chemical polishing below 10℃),
  $Q = 1.4 \times 10^9$ (single photon level)

↑

The world record (as of 2024)

Let's break the world record

Nb coaxial cavity of KEK-NTT collaboration

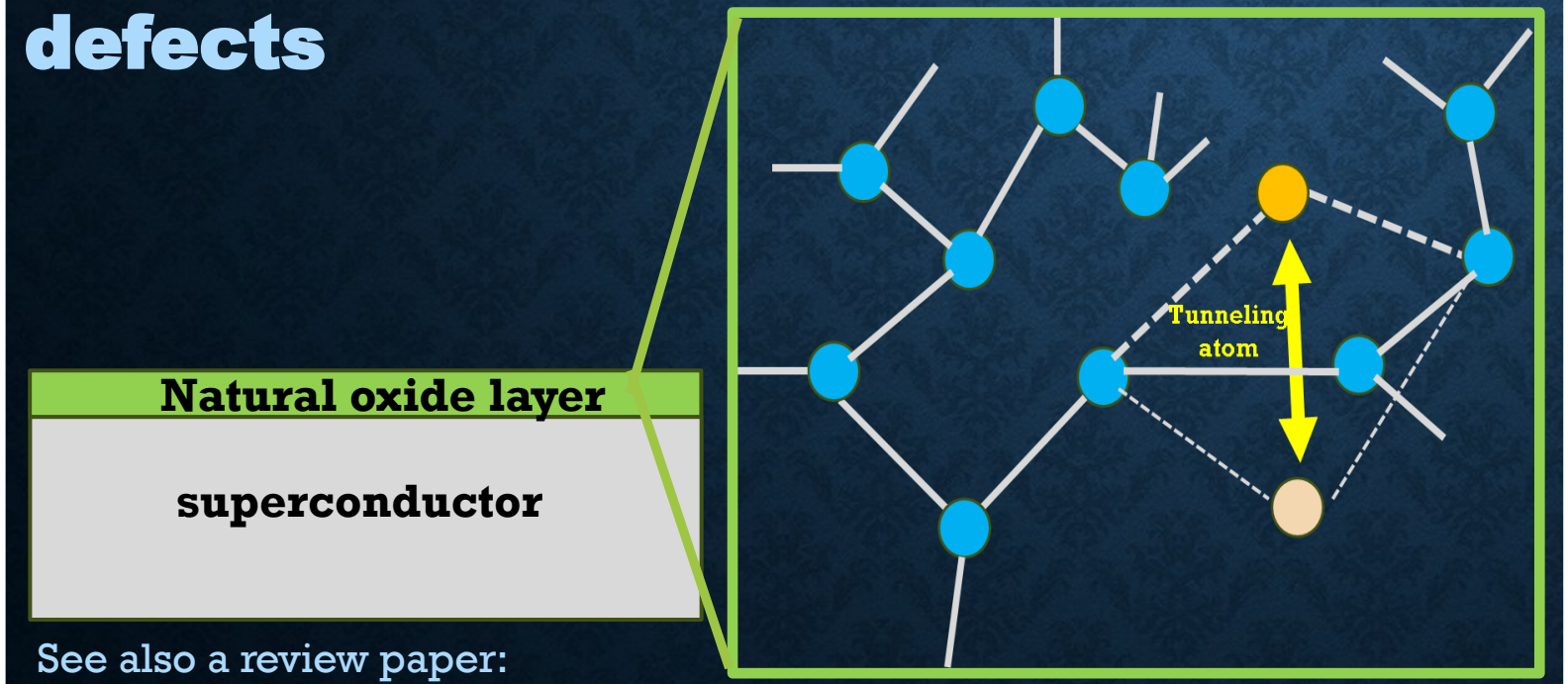




The greatest enemy of SRF technology in quantum applications is **TLS** defects—that is, a problem originating outside the superconductor itself.

Most amorphous solids (e.g. glass) show universal behavior at temperatures below a few kelvin. This behavior can be explained by the well-established "*standard tunneling model*".

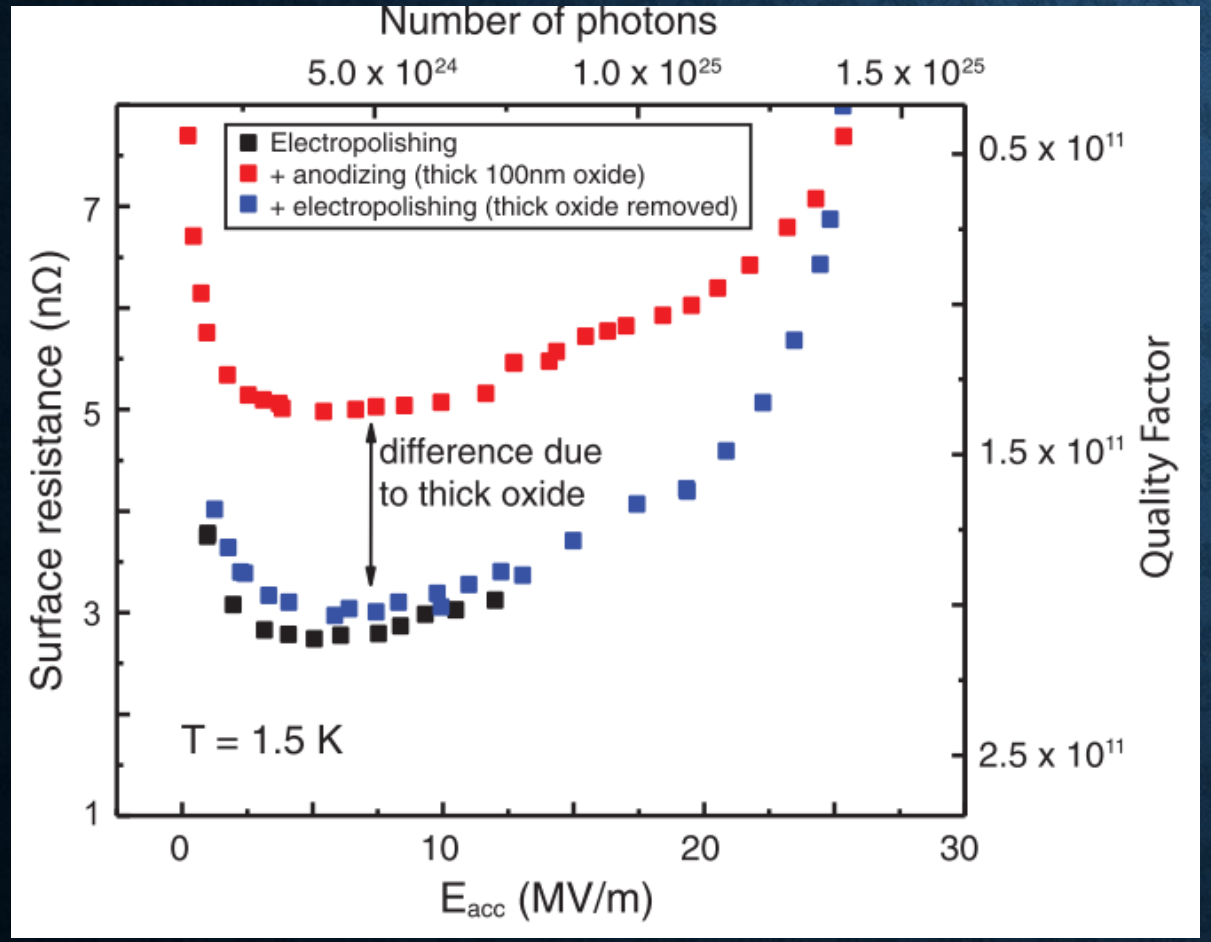
Two-level system (TLS) defects



See also a review paper:
Müller et al Rep. Prog. Phys. **82**, 124501 (2019)

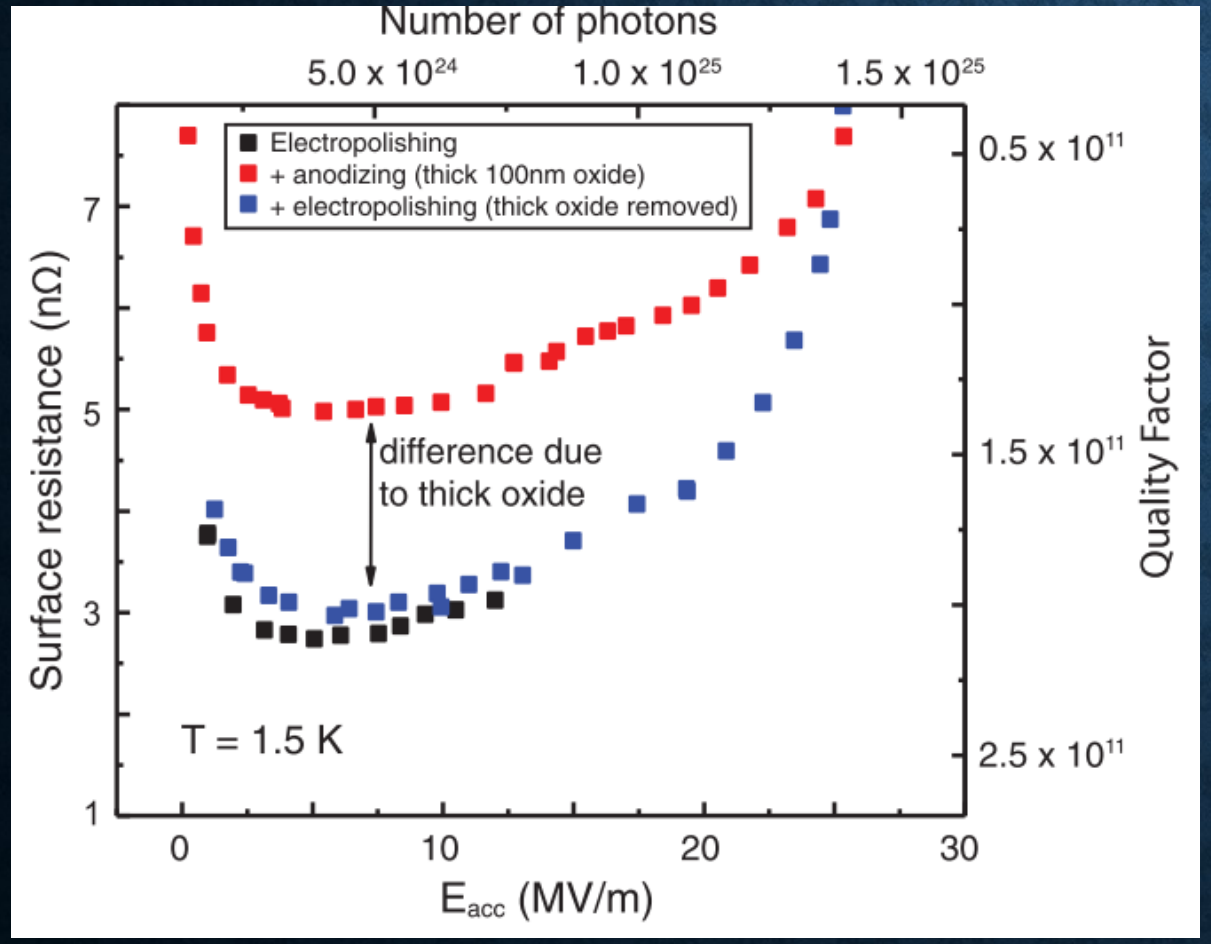
$$\frac{1}{Q_{\mathrm{int}}(\bar{n})} = F\delta^{0}_{\mathrm{TLS}} \frac{\tanh\left(\frac{hf_r}{2kT}\right)}{\sqrt{1+\left(\frac{\bar{n}}{n_c}\right)^{\beta}}} + \frac{1}{Q_{\mathrm{res}}},$$

TLSs live in the oxide layer on the inner surface of the niobium cavity.

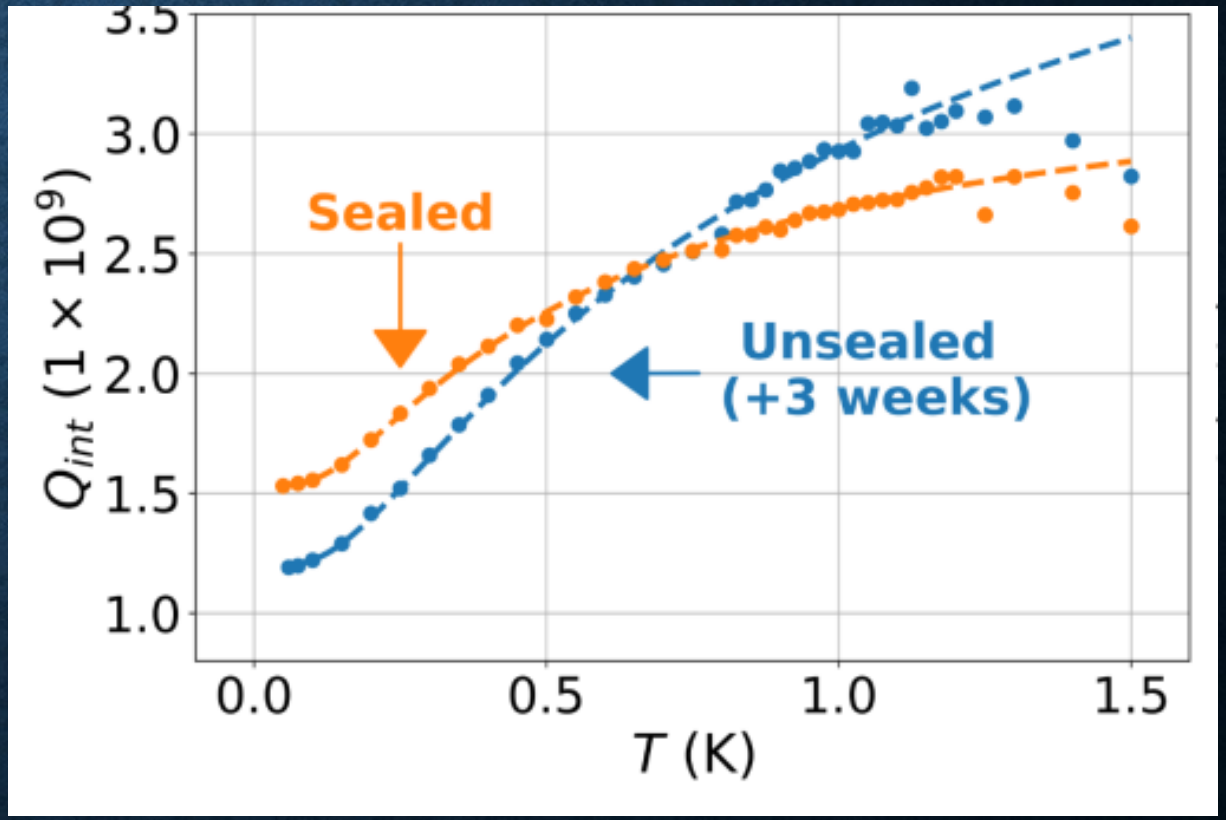


Romanenko, PRL **119**, 264801 (2017)

TLSs live in the oxide layer on the inner surface of the niobium cavity.
The oxide inevitably regrows after any surface treatment.
Whether TLSs can be reduced by surface treatment is not obvious.



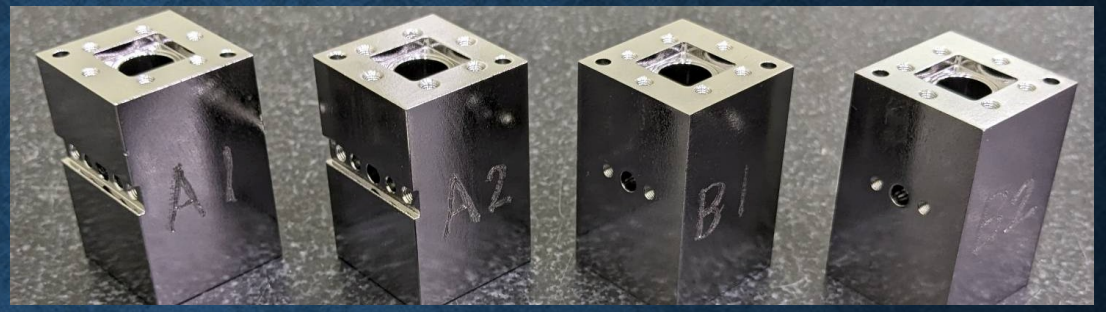
Romanenko, PRL **119**, 264801 (2017)



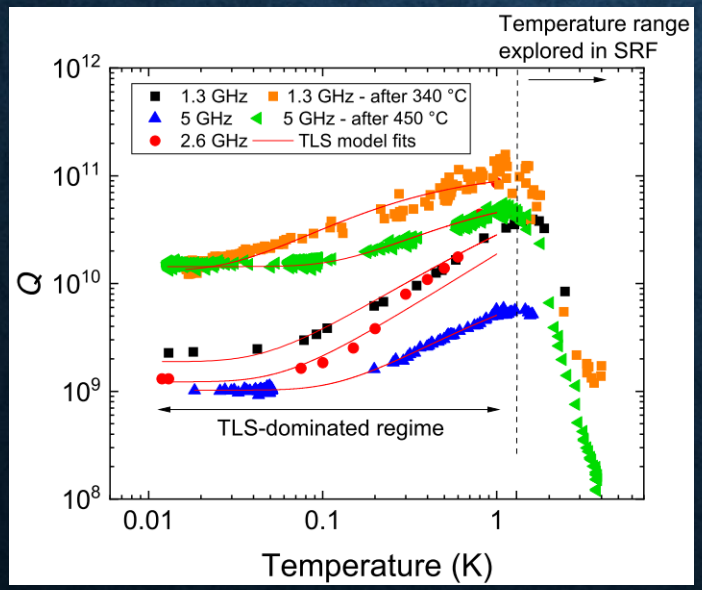
Oriani et al., arXiv:2403.00286

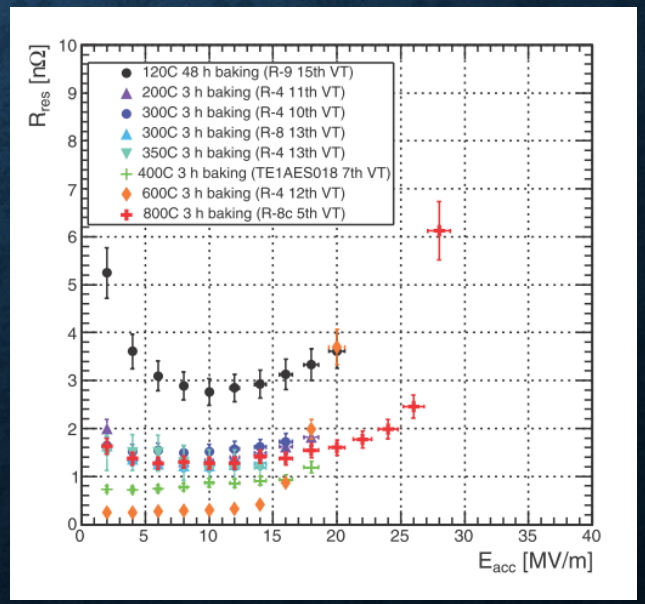
We applied accelerator-type surface treatments to the quantum cavity, while omitting processes intended for high-gradient operation.

1. Removal of approximately 100 μm of material from the inner surface using buffered chemical polishing (BCP) at room temperature ($\gtrsim 20°C$).
2. Baking the cavity in a vacuum furnace at 900 °C for 3 hours (high-T bake).
3. Light BCP ("BCP flush") to remove a few additional micrometers of material.
4. Additional treatment for surface optimization



Prior work suggests a possible route: despite inevitable regrowth, a **mid-temperature bake** that disrupts the oxide may help.





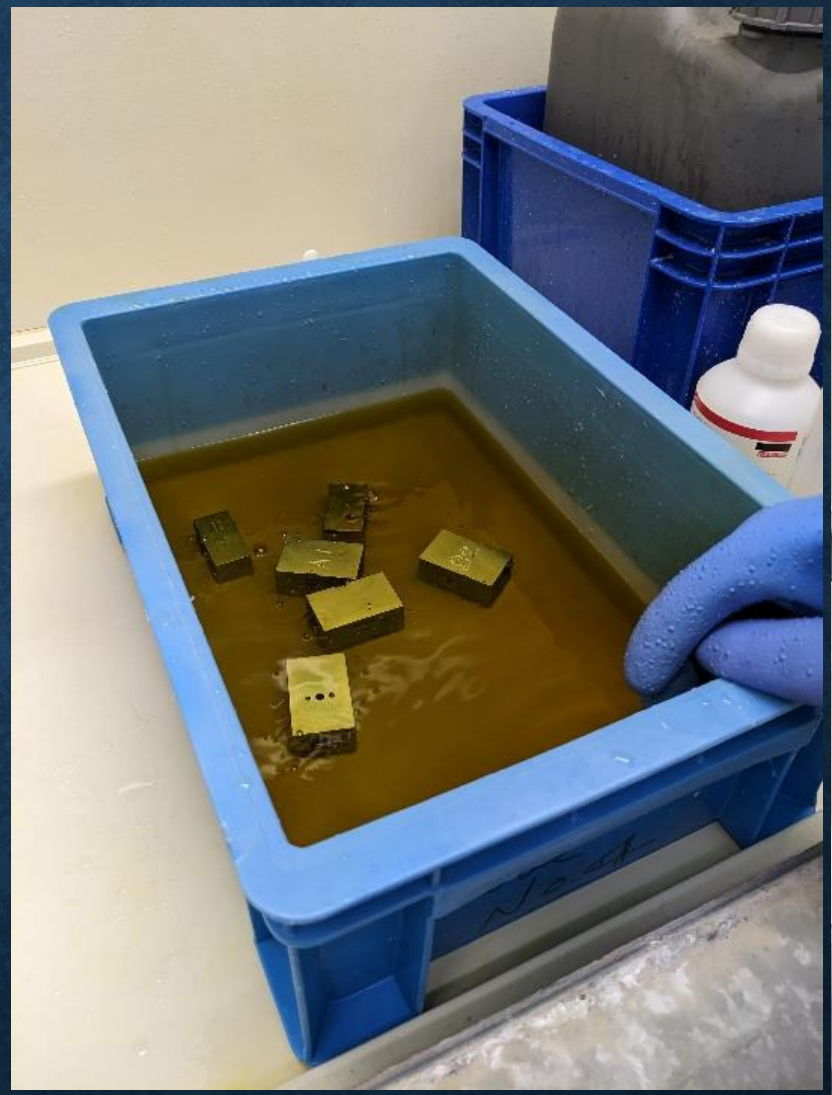
A. Romanenko et al., Phys. Rev. Applied **13**, 034032 (2020)

H. Ito et al., PTEP **2021**, 071G01 (2021)

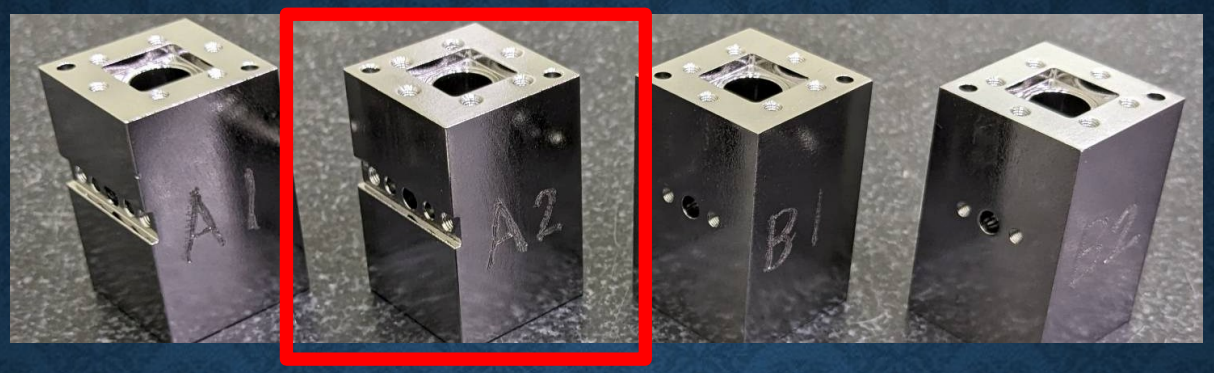
Ultrasonic cleansing



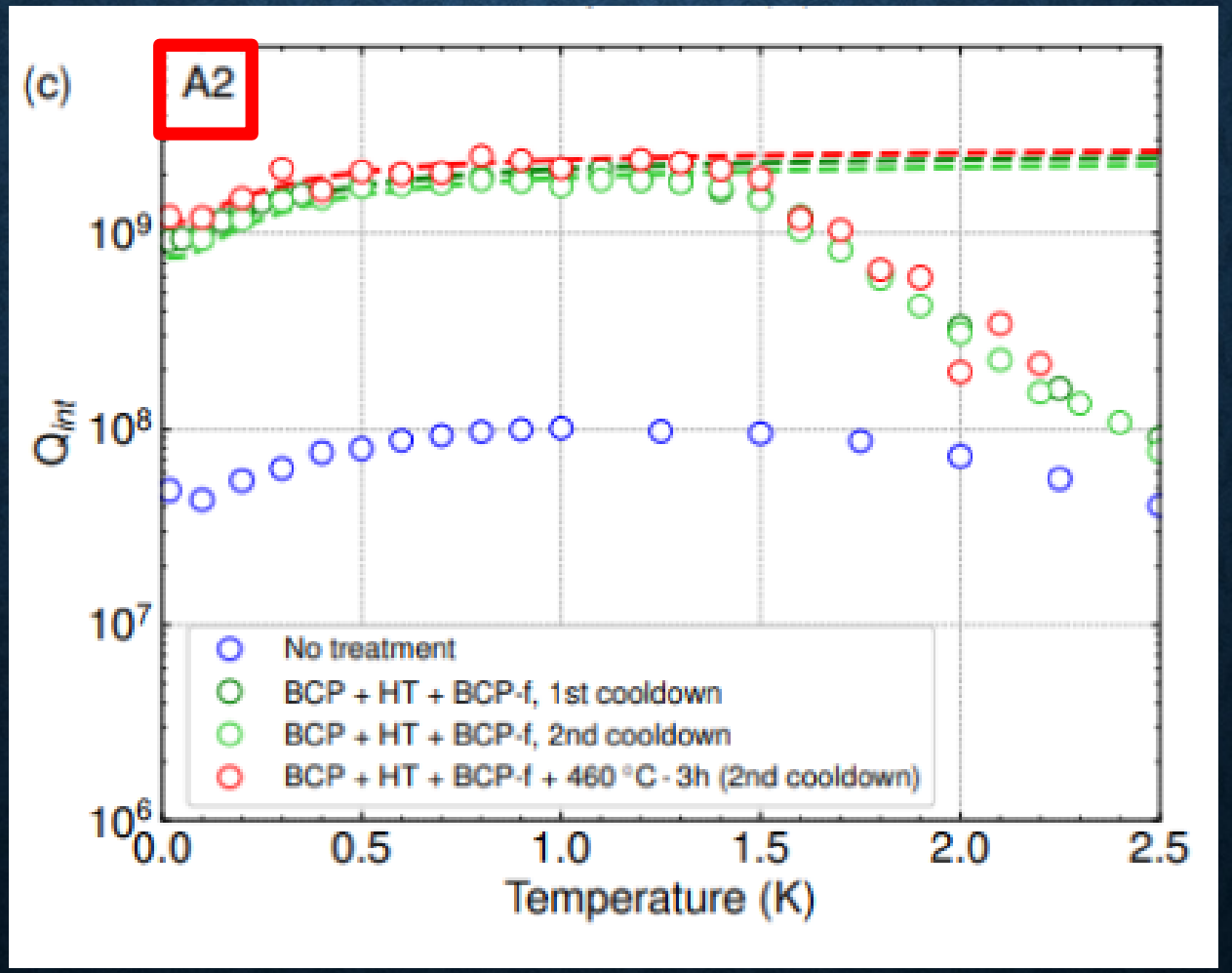
Buffered Chemical Polishing (BCP)



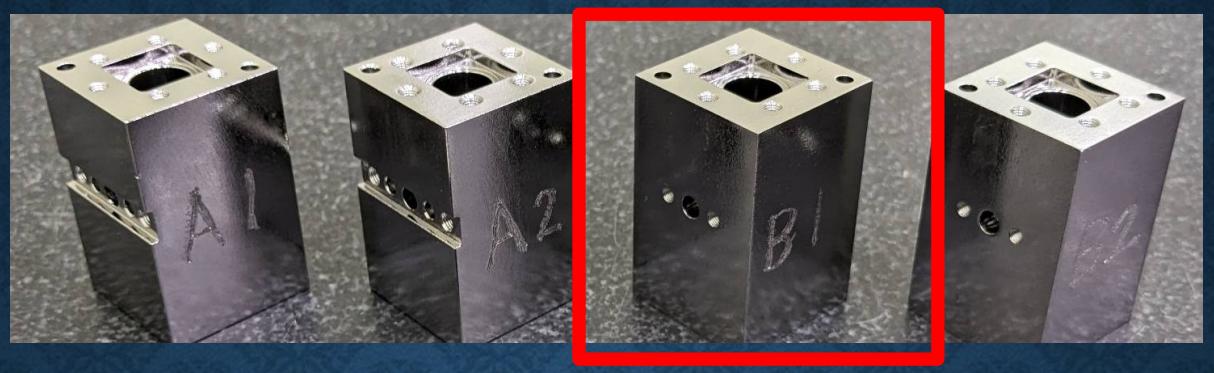




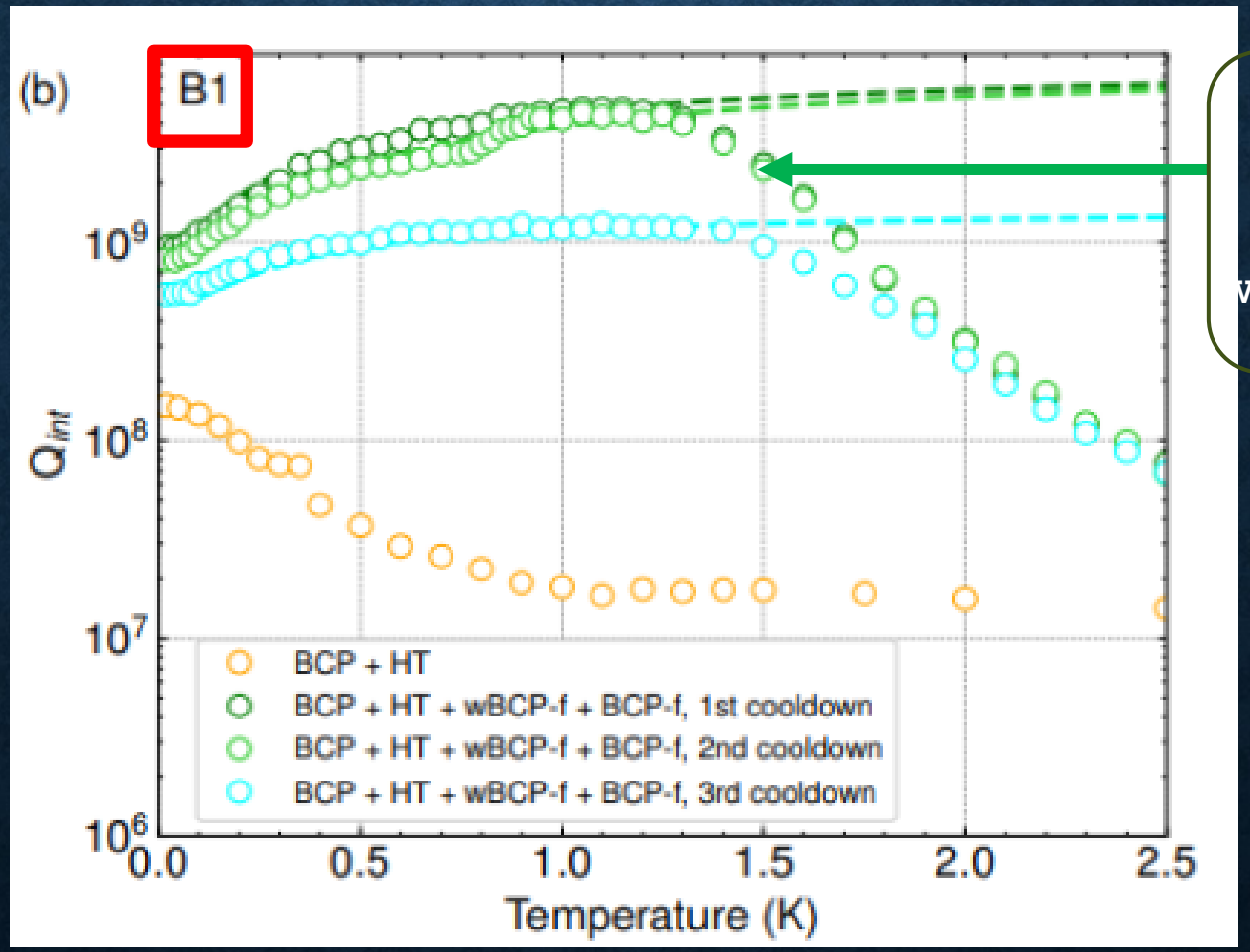
comparable to
the world record→



T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).



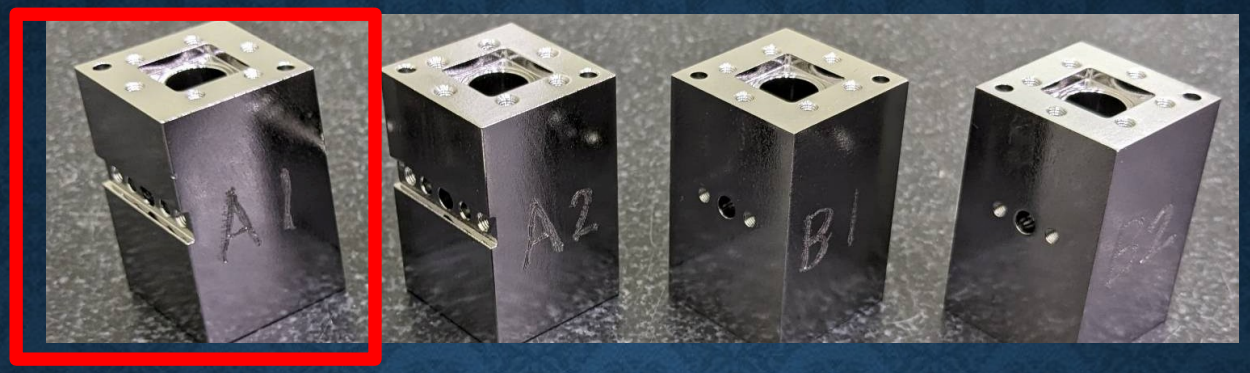
comparable to
the world record→



Room temperature
wBCP (1:1:1) flush

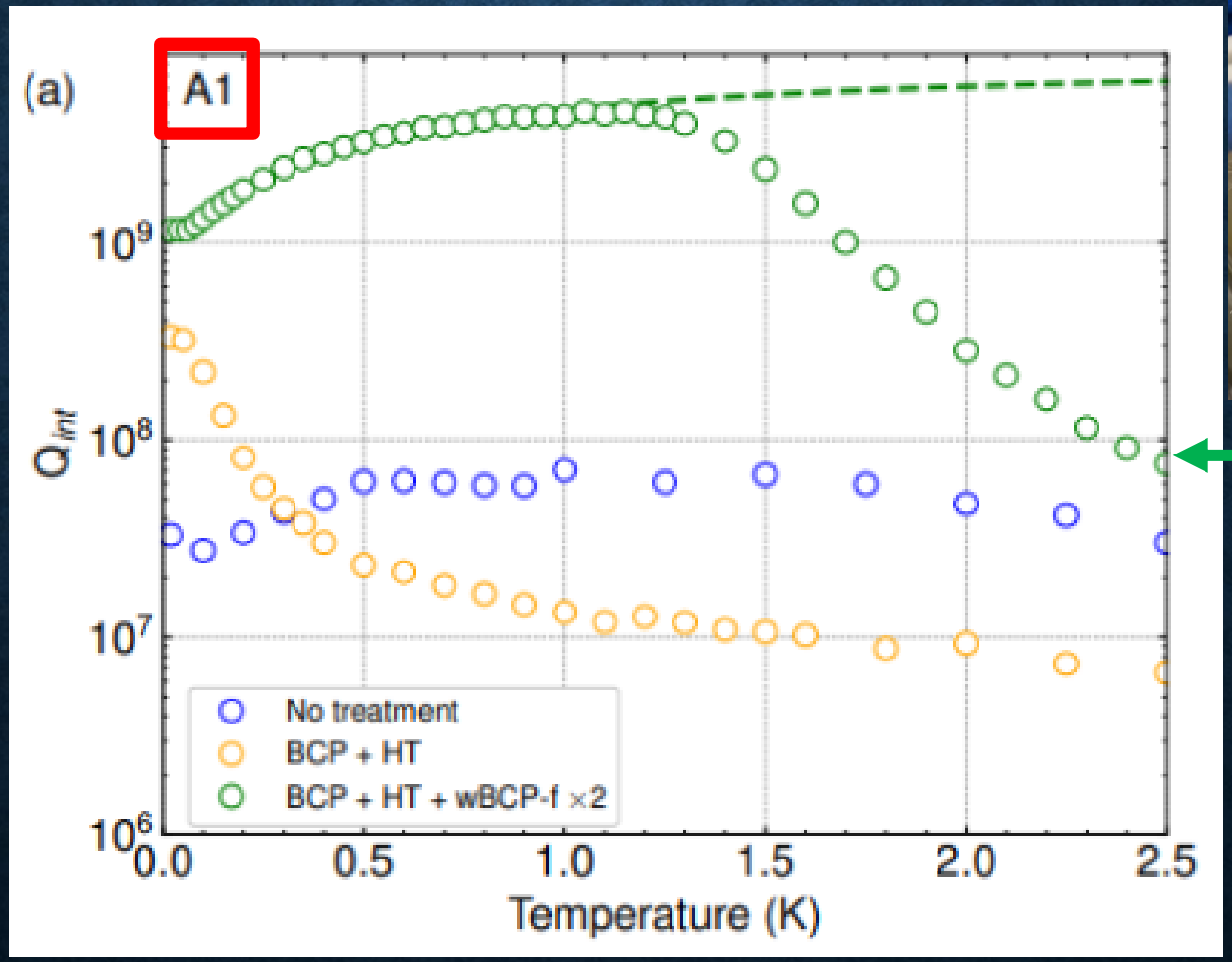
wBCP=water buffered CP

T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).





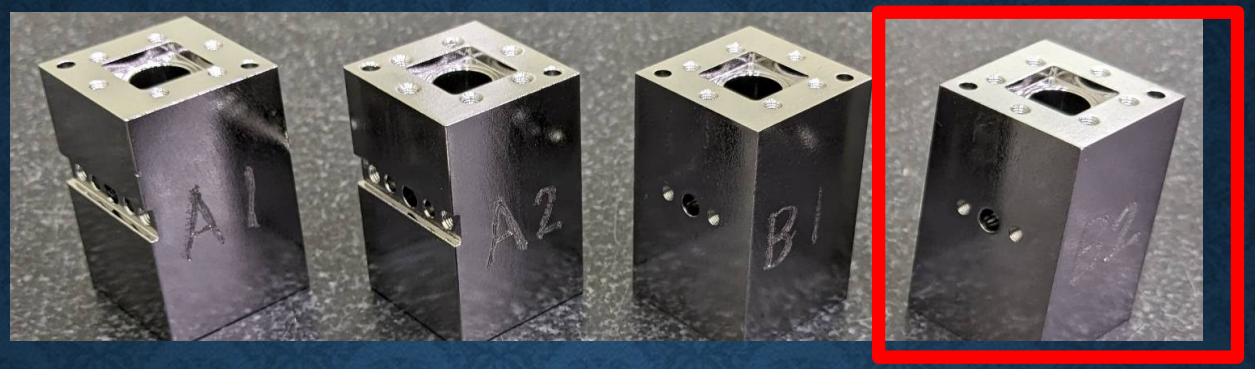
comparable to
the world record→



Cold (5℃)
wBCP (1:1:2) flush

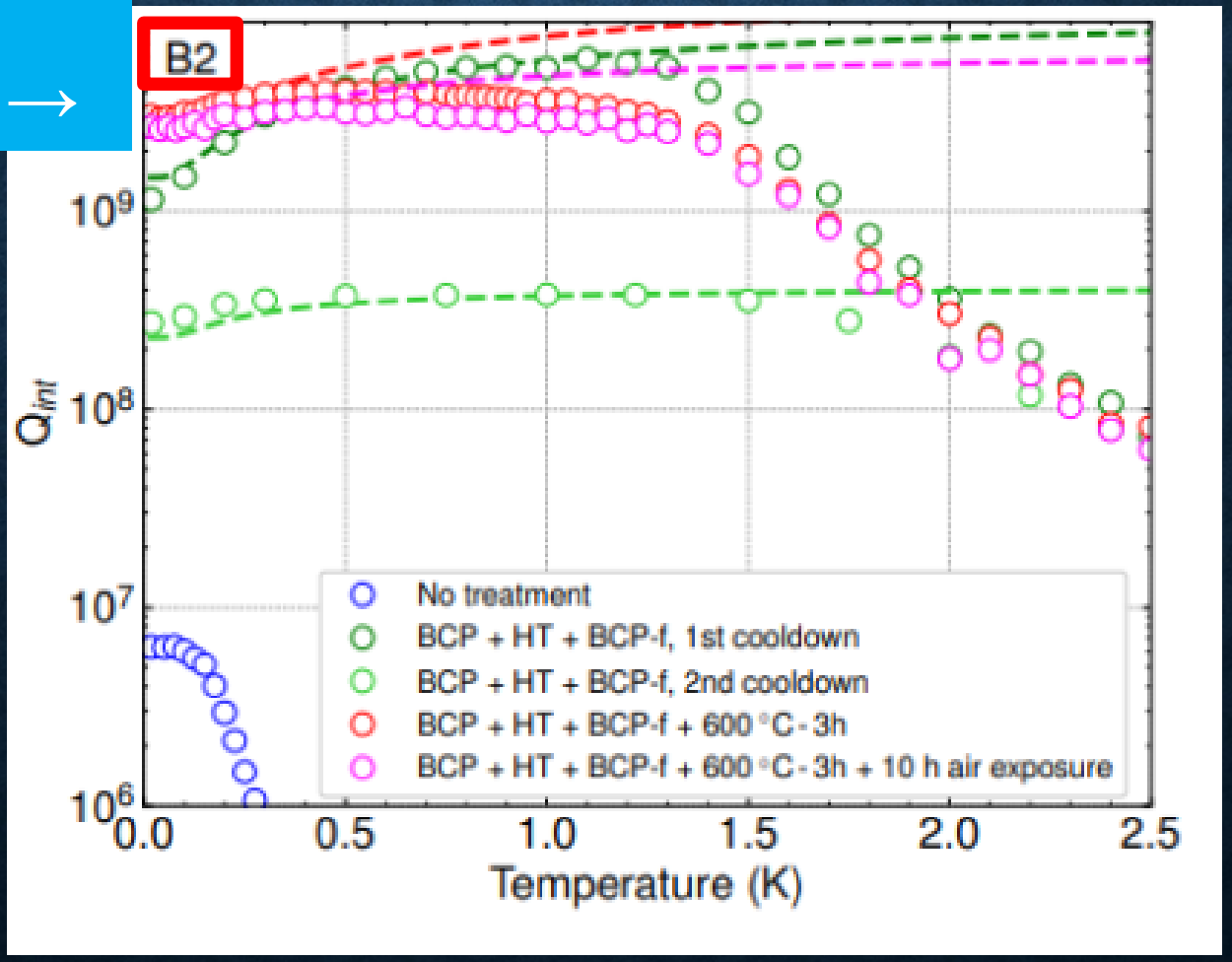
wBCP=water buffered CP

T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).



We broke the world record!!!→

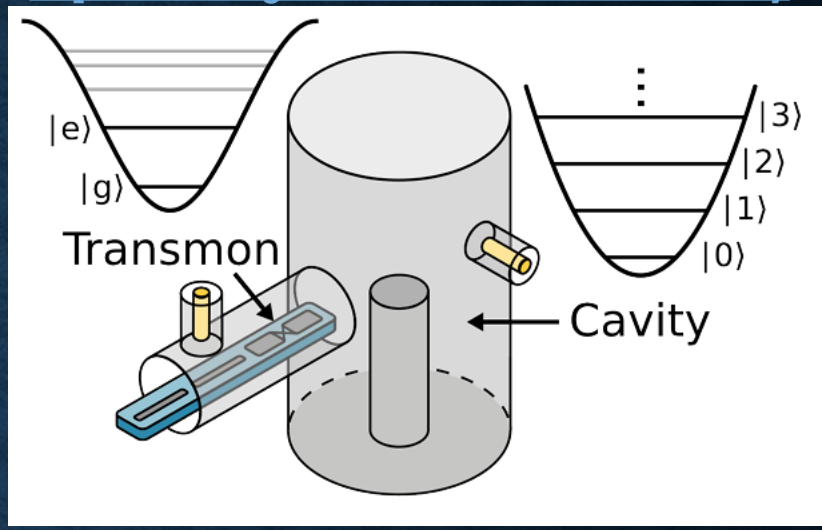
Corresponding to photon lifetime of ~0.1 sec.



T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).

3D coaxial cavity progress

Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y

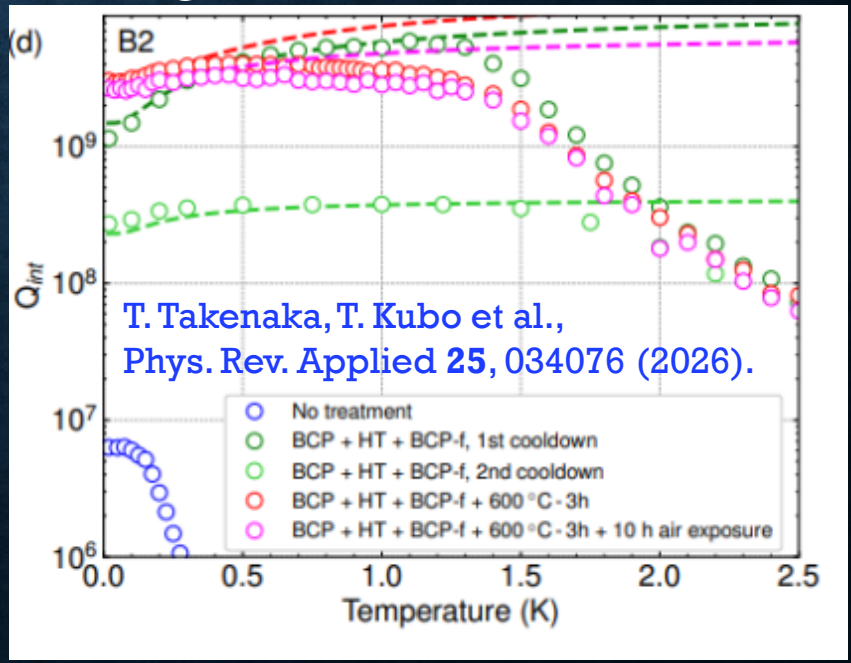


- Reagor et al., Phys. Rev. B **94**, 014506 (2016) Yale University
  Aluminum,
  $Q = 7 \times 10^7$ (single photon level)

- Heidler et al., Phys. Rev. Applied **16**, 034024 (2021) University of Innsbruck
  Niobium (1:1:1 buffered chemical polishing at 0-5℃),
  $Q = 5 \times 10^8$ (single photon level)

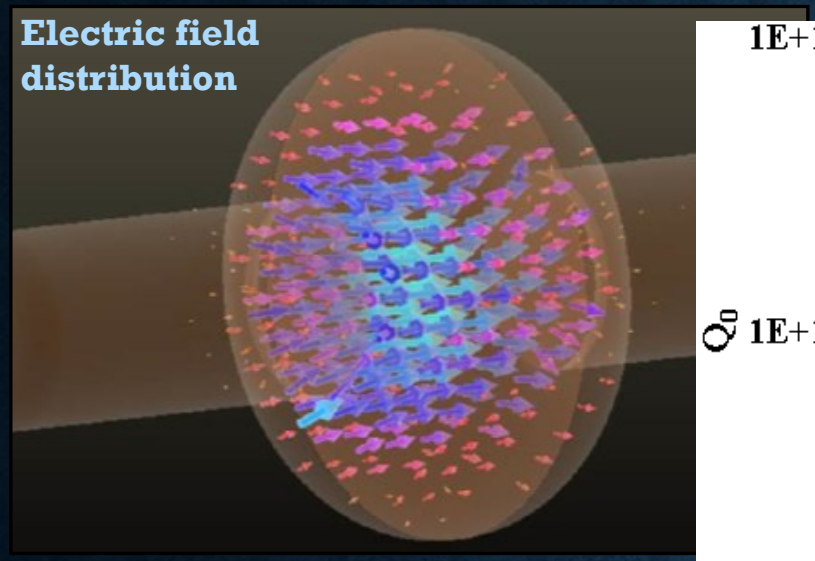
- Oriani et al., Phys. Rev. Applied **24**, 044080 (2025) University of Chicago
  Niobium (1:1:2 water buffered chemical polishing below 10℃),
  $Q = 1.4 \times 10^9$ (single photon level)

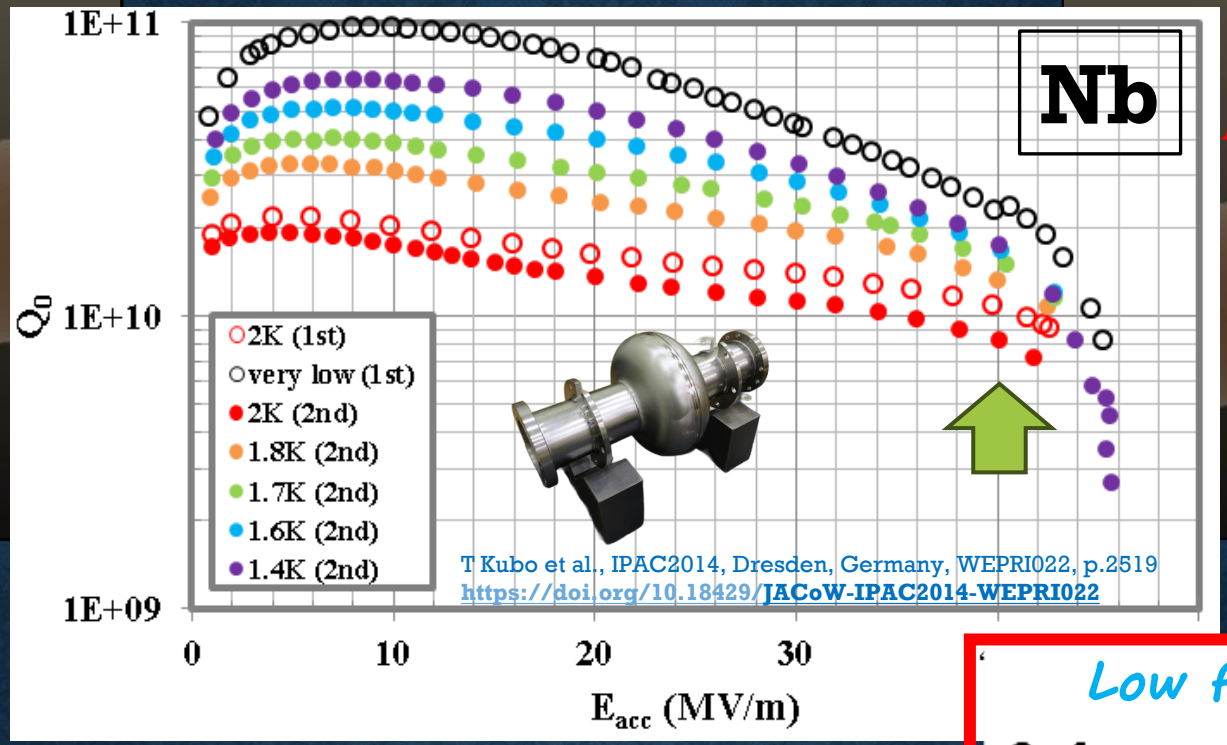
- T. Takenaka, T. Kubo et al., Phys. Rev. Applied 25, 034076 (2026). KEK&NTT
  Niobium (1:1:1 BCP at 25℃ + 900℃ anneal + 1:1:1 BCP flush at 25℃ + 600℃ anneal)
  $Q=3 \times 10^9$ (single photon level)



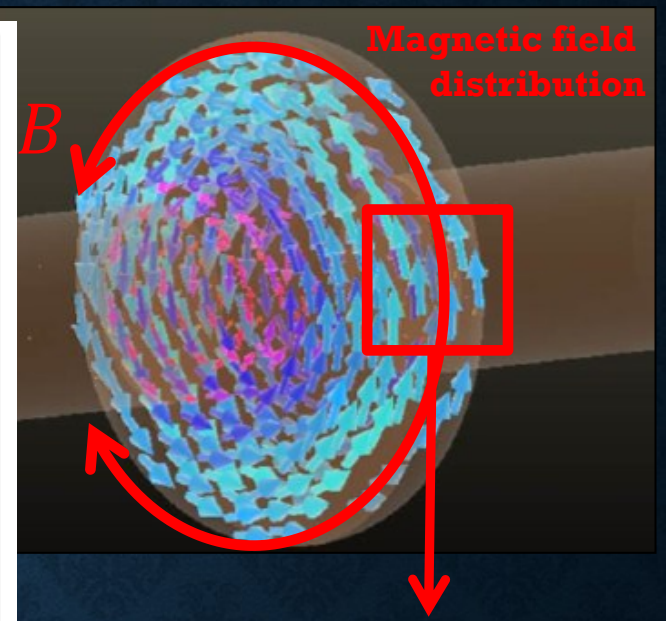
The world record

Stability of the vortex-free state
(very short introduction)

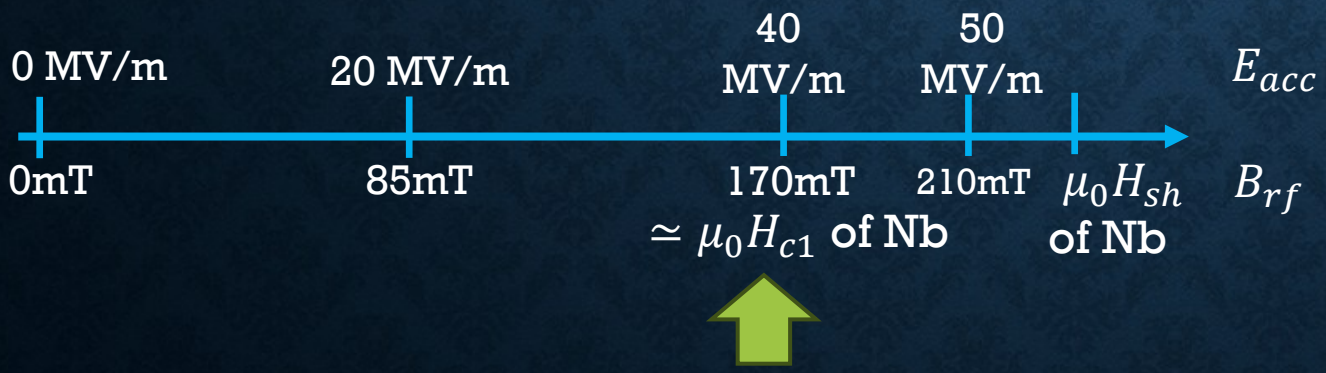




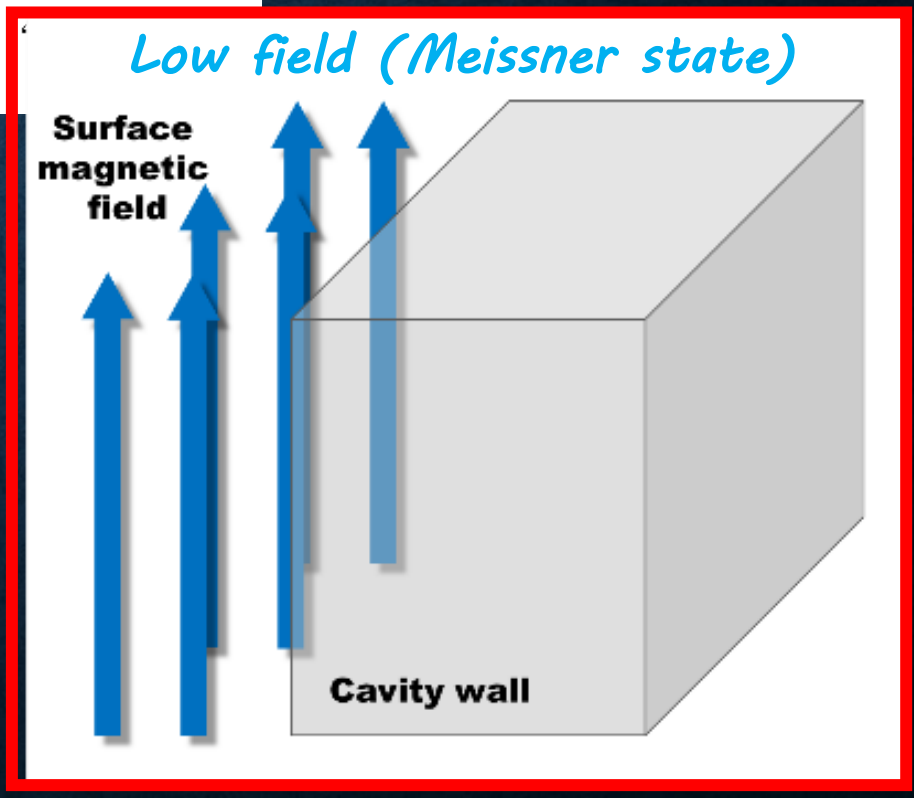
T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

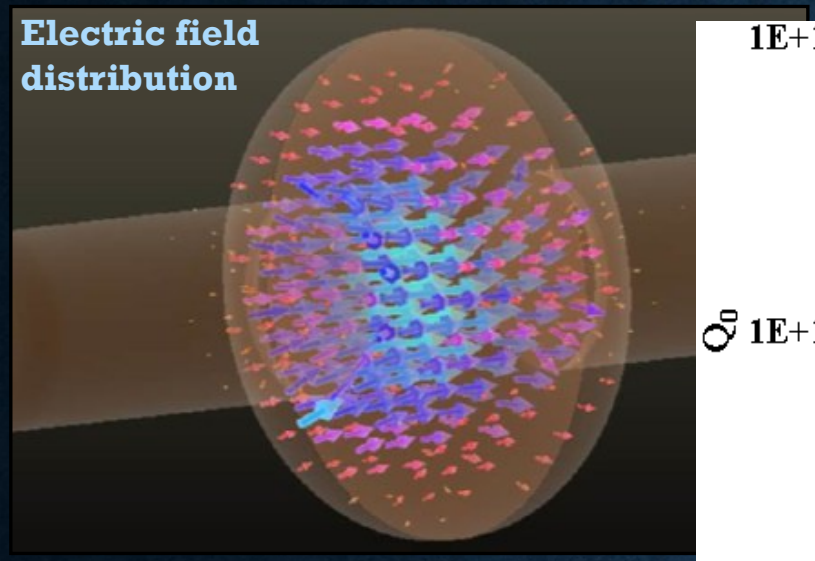


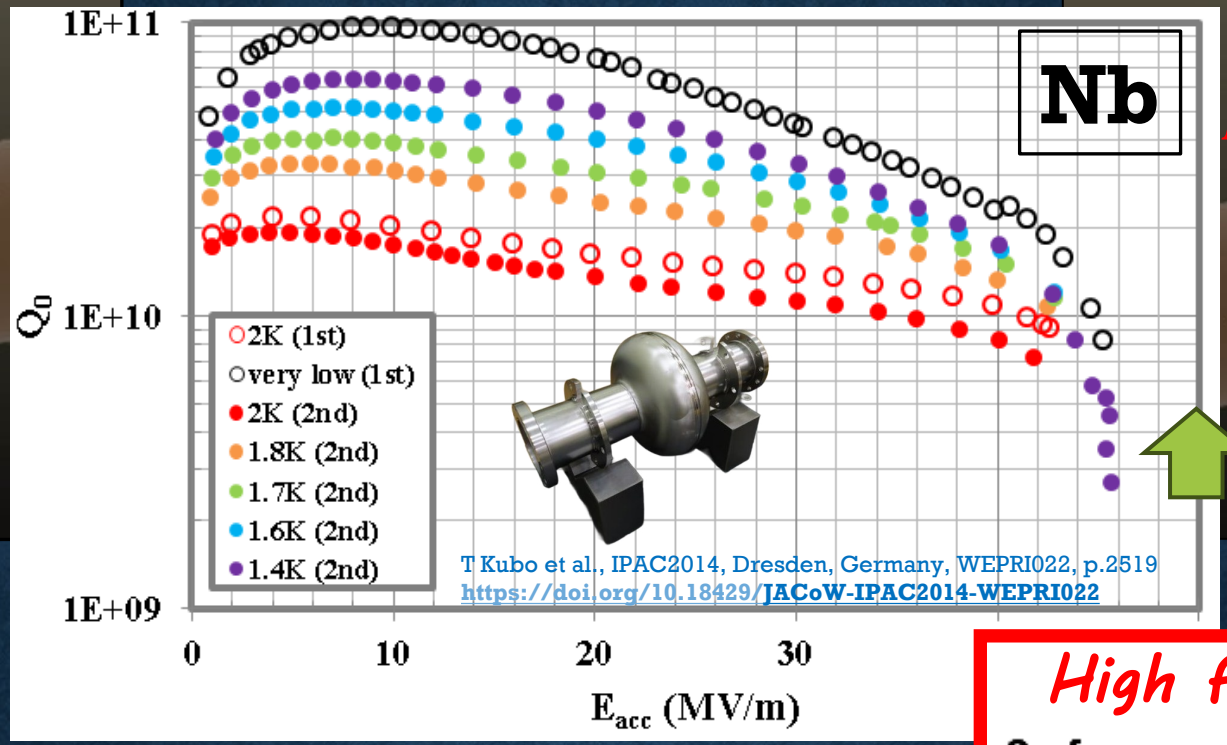
Note: Cavities are operated under the **Meissner state**



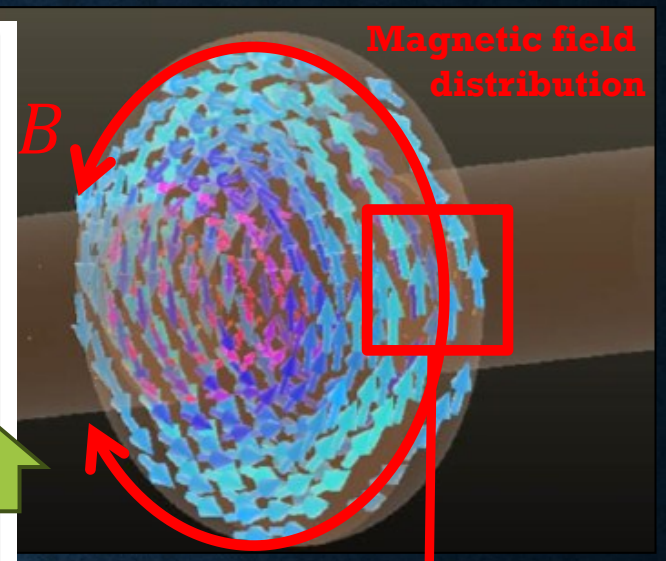
(⌐■_■) good



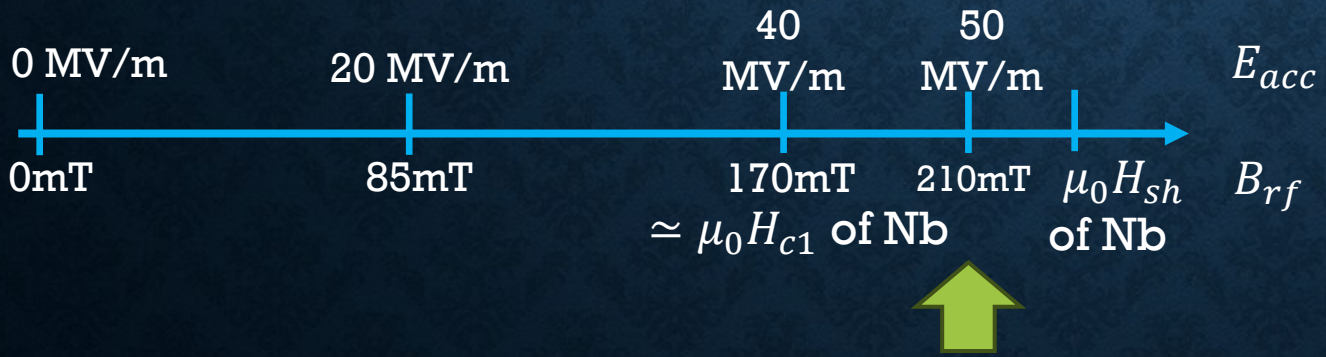




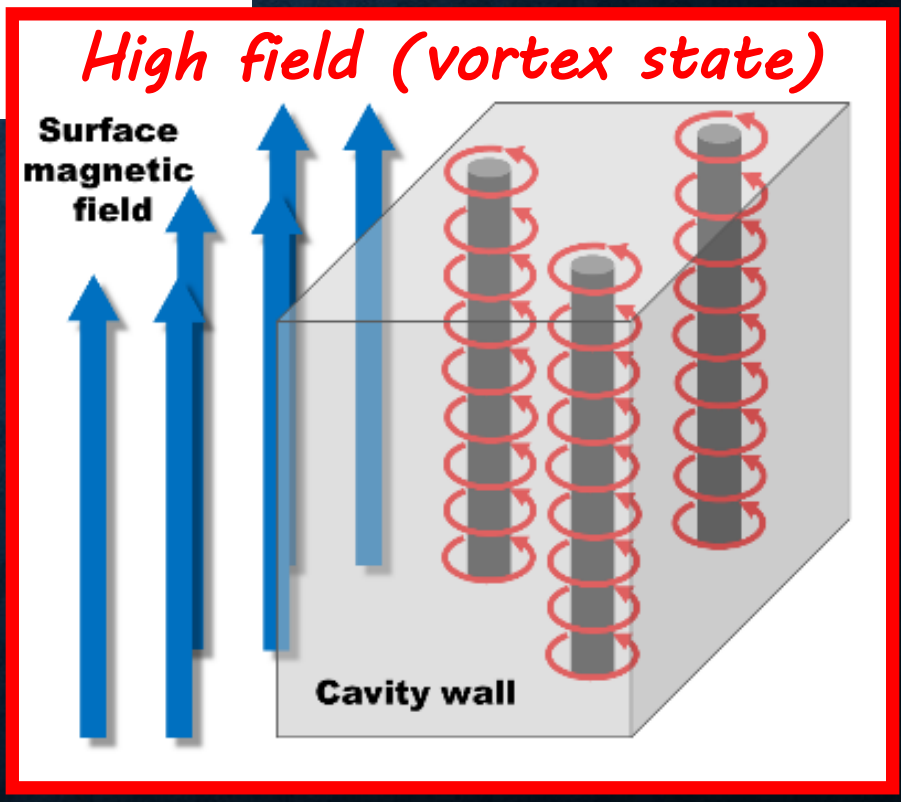
T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022



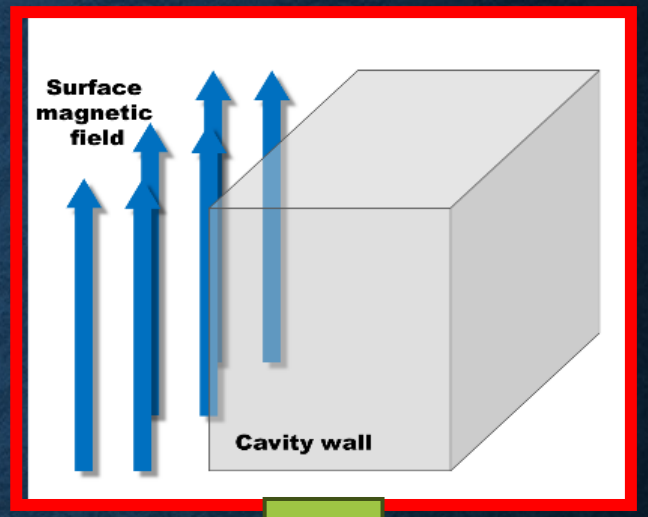
Note: Cavities are operated under the **Meissner state**



＼(^o^)／オワタ
I'm doomed

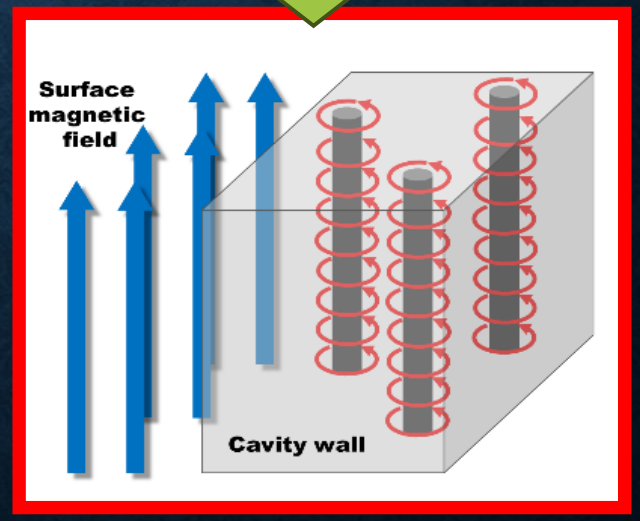


What is the fundamental theoretical limit?



Bulk

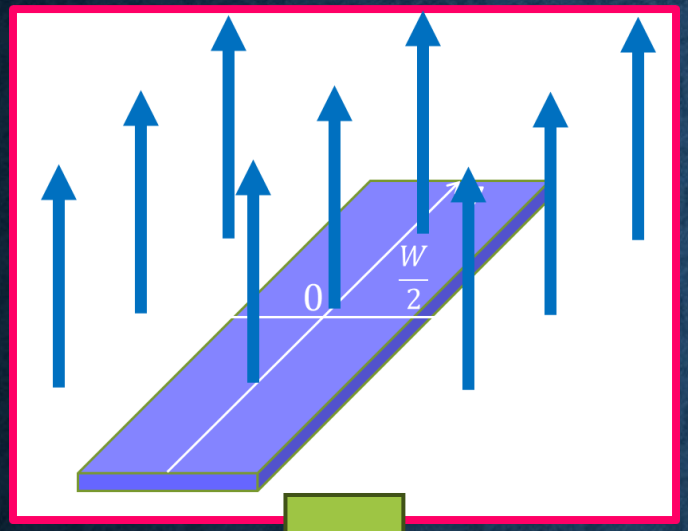
$B > B_{sh} > B_{c1}$



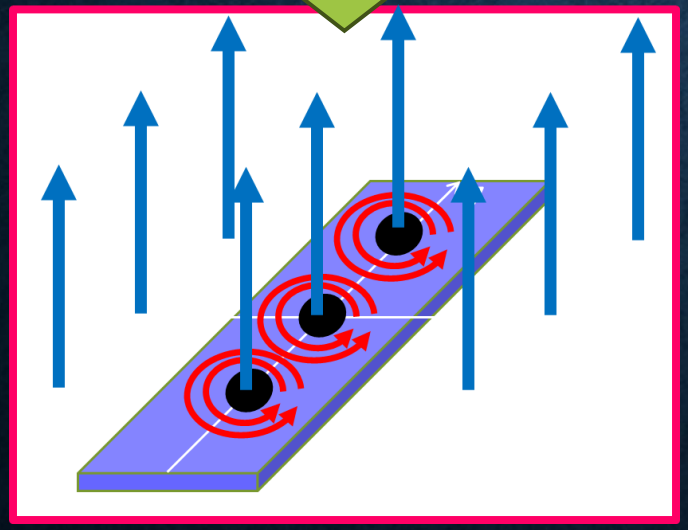
What is the fundamental theoretical limit?

Actually, this is a common fundamental problem in superconducting devices.

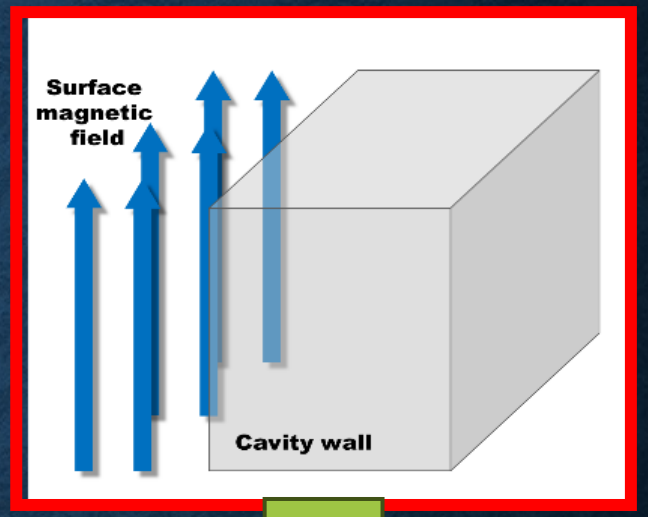
Narrow thin film



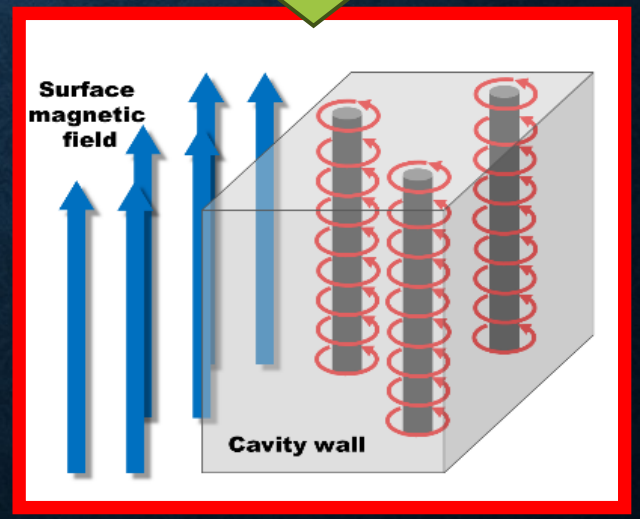
$B > B_s > B_{c1}$



Bulk



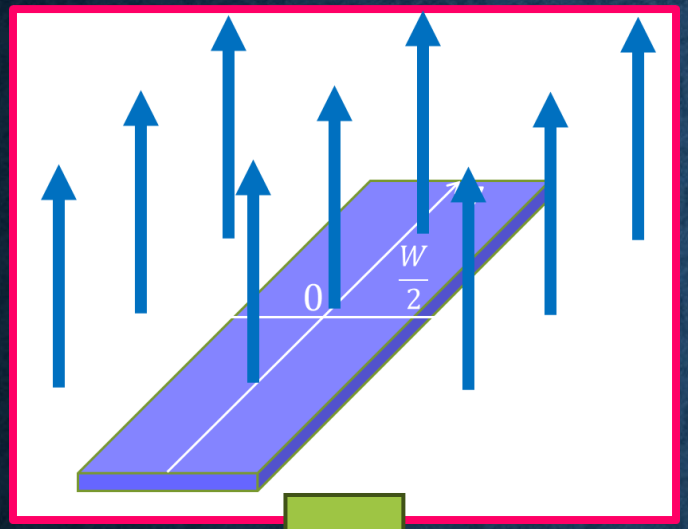
$B > B_{sh} > B_{c1}$



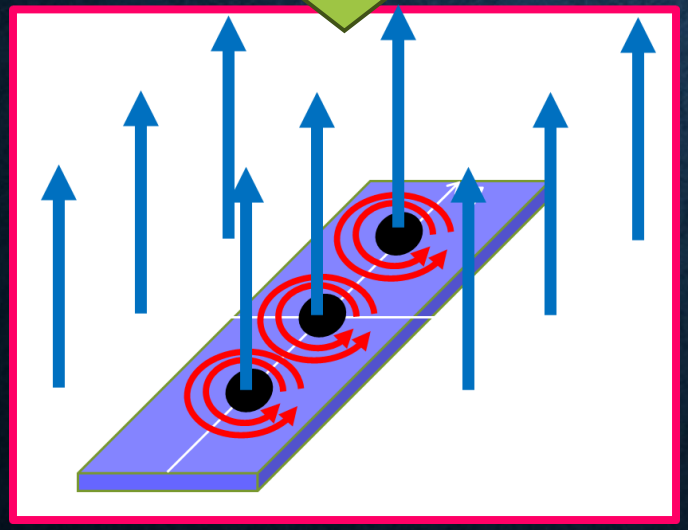
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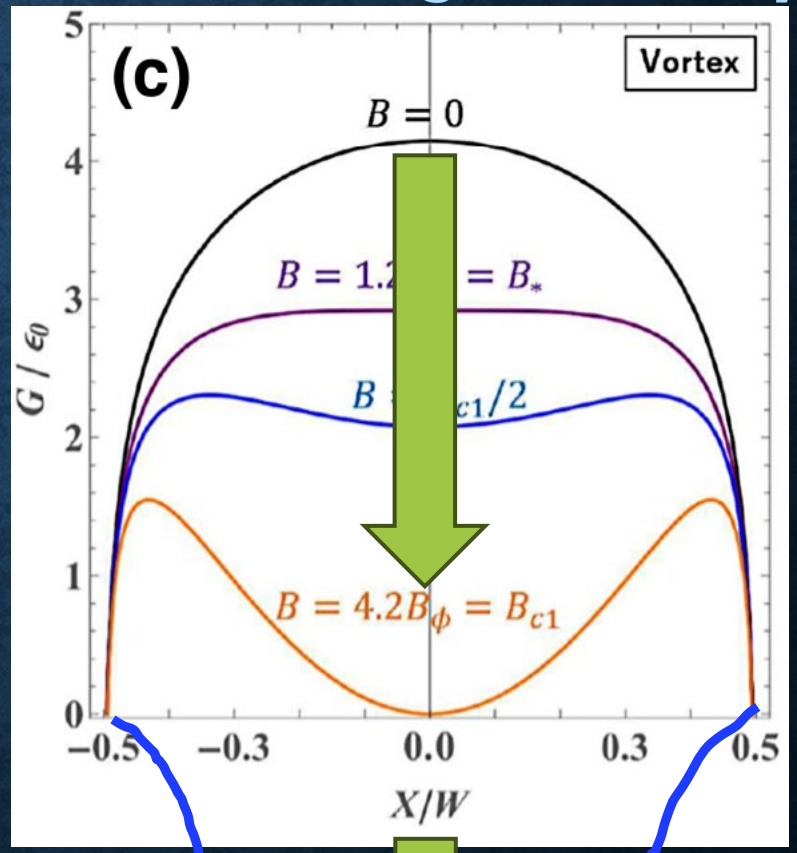
Narrow thin film



$B > B_s > B_{c1}$



Phenomenological-theory



$B = B_s$

Barrier disappears

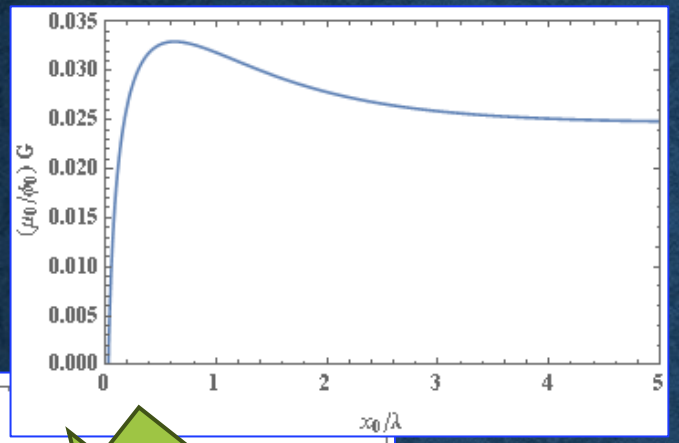
See, e.g.,
T. Kubo, Phys. Rev. Applied **20**, 034033 (2023)

What is the fundamental theoretical limit?

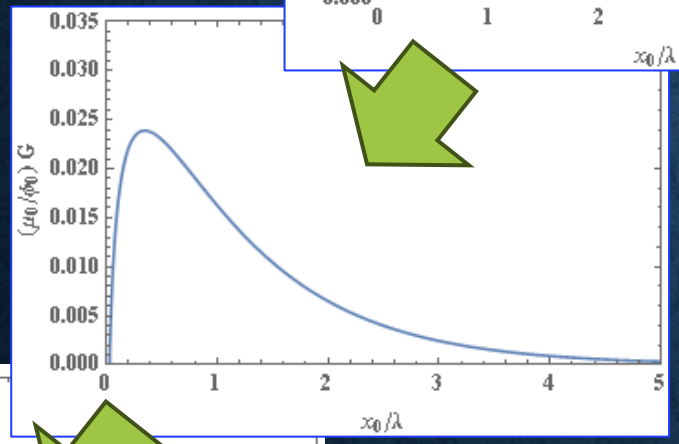
Actually, this is a common fundamental problem in superconducting devices.

Phenomenological-theory

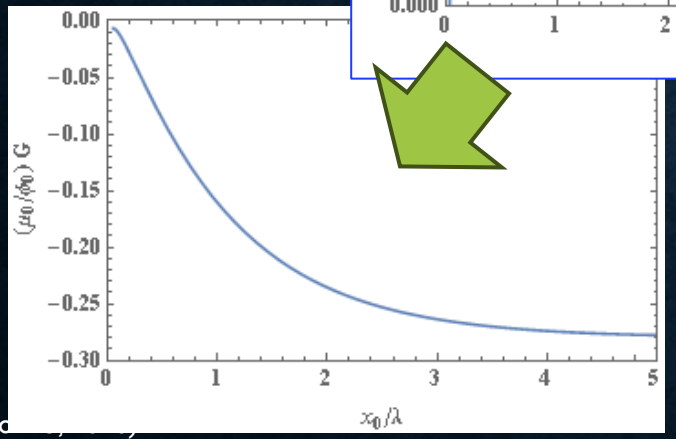
$B < B_{c1}$



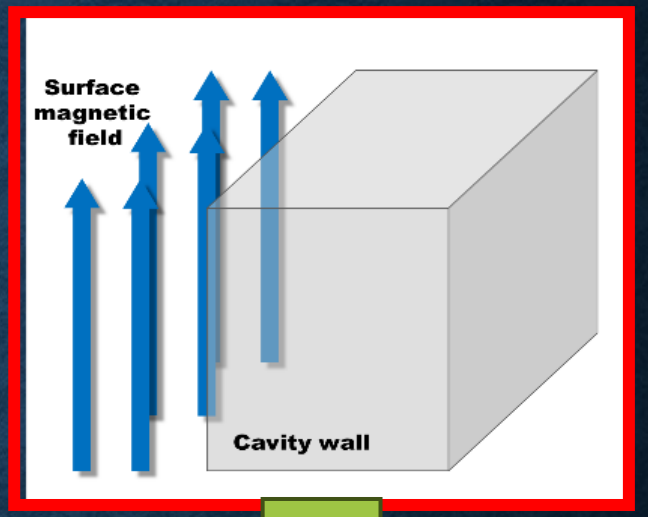
$B = B_{c1}$



$B = B_{sh}$

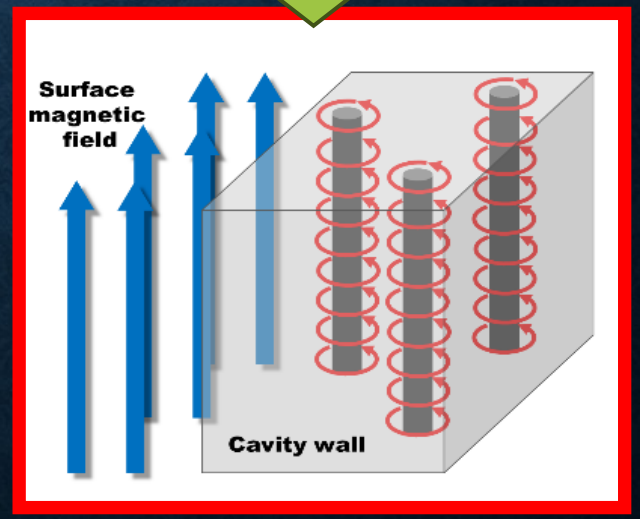


Barrier disappears

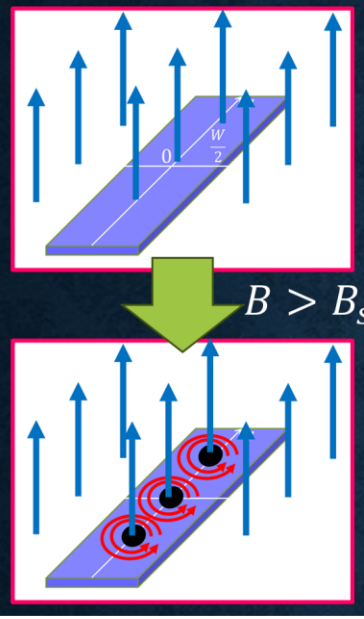


Bulk

$B > B_{sh} > B_{c1}$

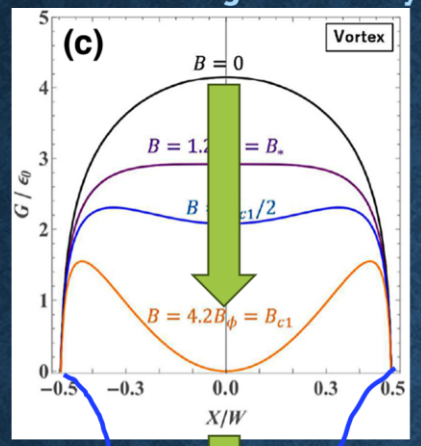


Narrow thin film

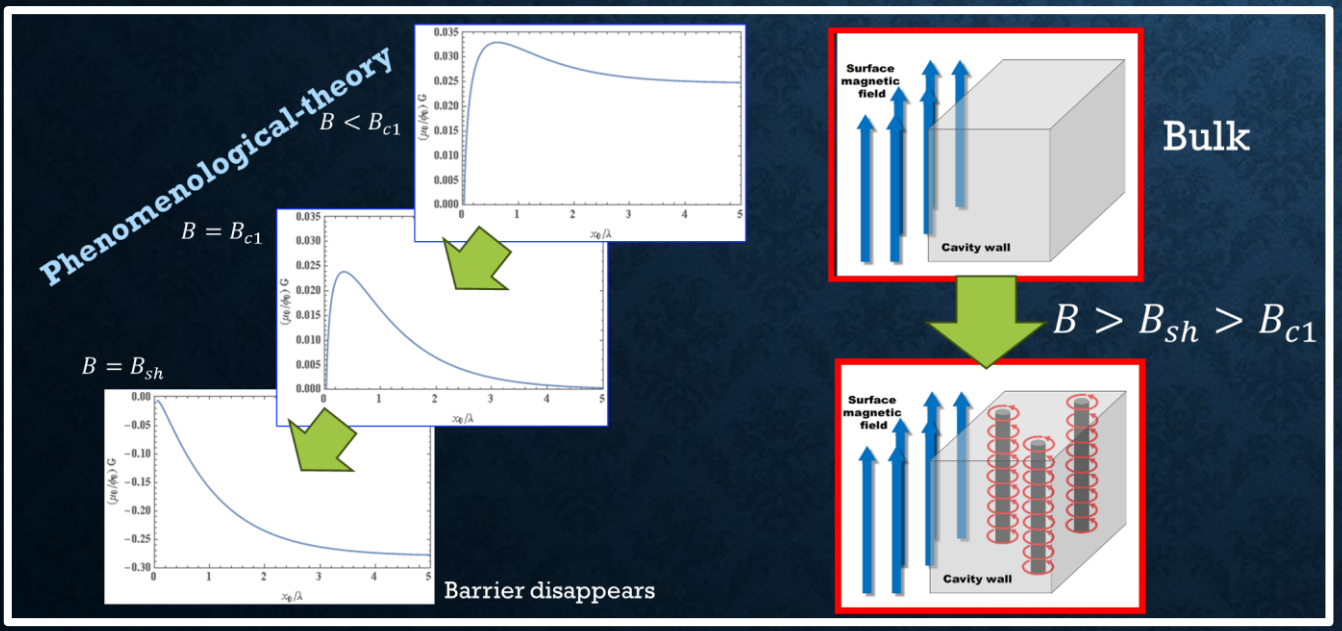


$B > B_s > B_{c1}$

Phenomenological-theory



$B = B_s$ Barrier disappears



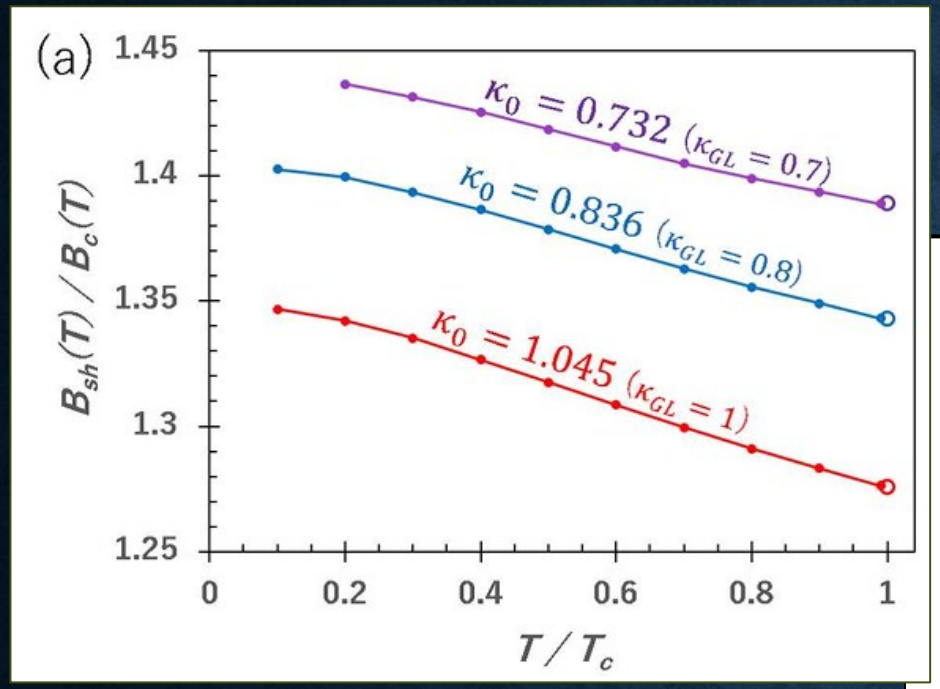
These discussions based on the "edge barrier" and "Bean–Livingston barrier" are phenomenological and are not always quantitatively—or even qualitatively—correct.

We need a stability analysis based on the microscopic theory of superconductivity

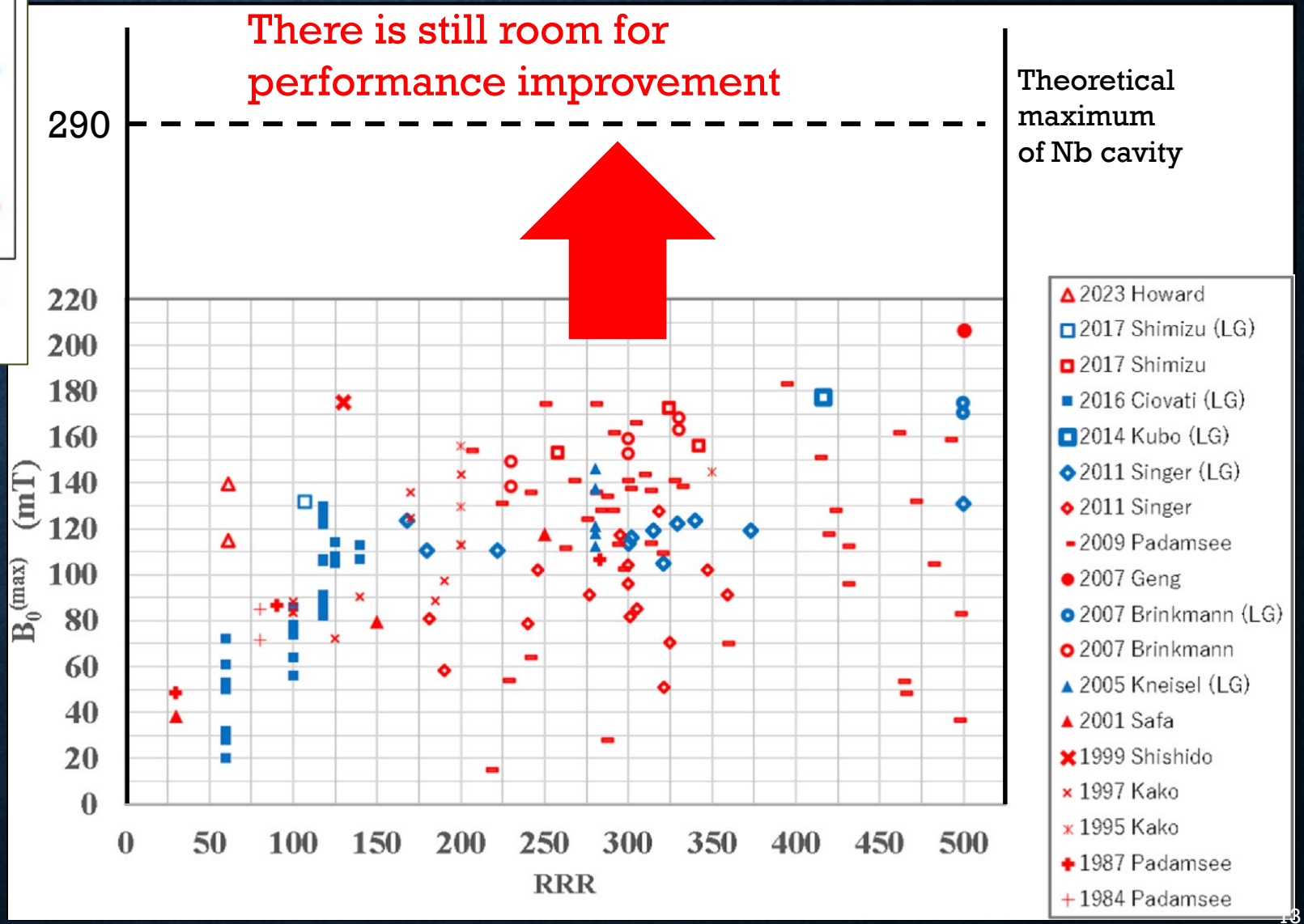
T. Kubo, arXiv:2608.04508 (2026)
and
T. Kubo, Phys. Rev. Accel. Beams (accepted as a **Letter** article) arXiv:2606.10420 (2026)

Microscopic theory of the superheating field

Intrinsic $\frac{\lambda_0}{\xi_0}$	Purity (mfp ℓ)	Theory (applicable to SRF temperature $T \ll T_c$)	
$\frac{\lambda_0}{\xi_0} \gg 1$ (e.g., Nb_3Sn, NbN, NbTiN etc.)	$\ell \gg \xi_0$ (clean)	$H_{sh}(0) = 0.84 H_c(0)$ $H_{sh}(\mathrm{T})$ for $0 \le T \le T_c$	Analytical result for T=0. V. P. Galaiko, Sov. Phys. JETP **23**, 475 (1966) • G. Catelani and J. P. Sethna, Phys. Rev. B **78**, 224509 (2008) • F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)
	any ℓ (from clean limit to dirty limit)	$H_{sh}(\mathrm{T})$ for $0 \le T \le T_c$	F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)
	$\ell \ll \xi_0$ (dirty)	$H_{sh}(\mathrm{T})$ for $0 \le T \le T_c$ $H_{sh}(0) = 0.795 H_c(0)$	• F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012) • T. Kubo, Phys. Rev. Research **2**, 033203 (2020) Analytical result for T=0. • T. Kubo, Phys. Rev. Research **2**, 033203 (2020) • T. Kubo, Supercond.Sci. Tecnol. **34**, 045006 (2021)
$\frac{\lambda_0}{\xi_0} \lesssim 1$ (e.g., Nb)	any ℓ	N/A	
	$\ell \gg \xi_0$ (clean)	$H_{sh}(\mathrm{T})$ for $0.1 \le T \le T_c$	T. Kubo, Phys. Rev. Accel. Beams (accepted as a **Letter** article) arXiv:2606.10420 (2026)

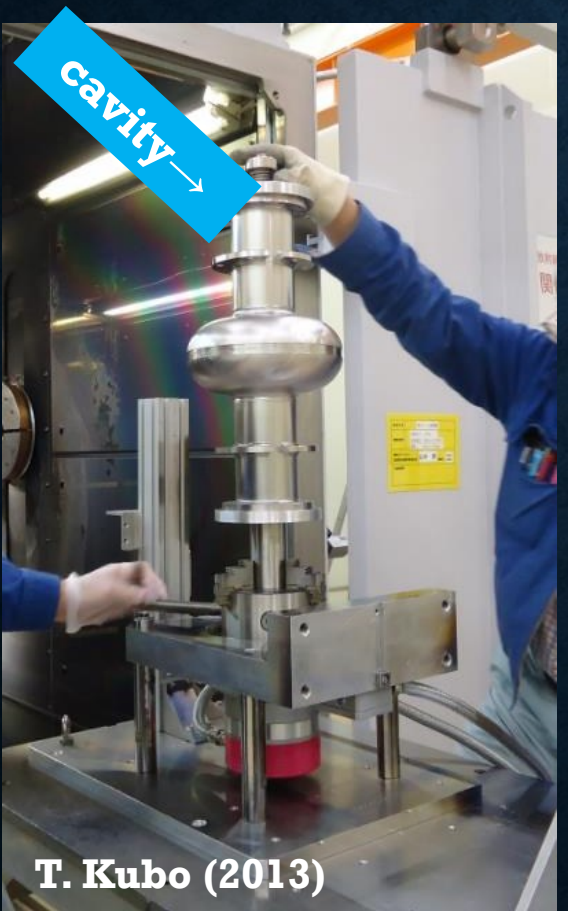


T. Kubo, Phys. Rev. Accel. Beams
(accepted as a **Letter** article)
arXiv:2606.10420 (2026)

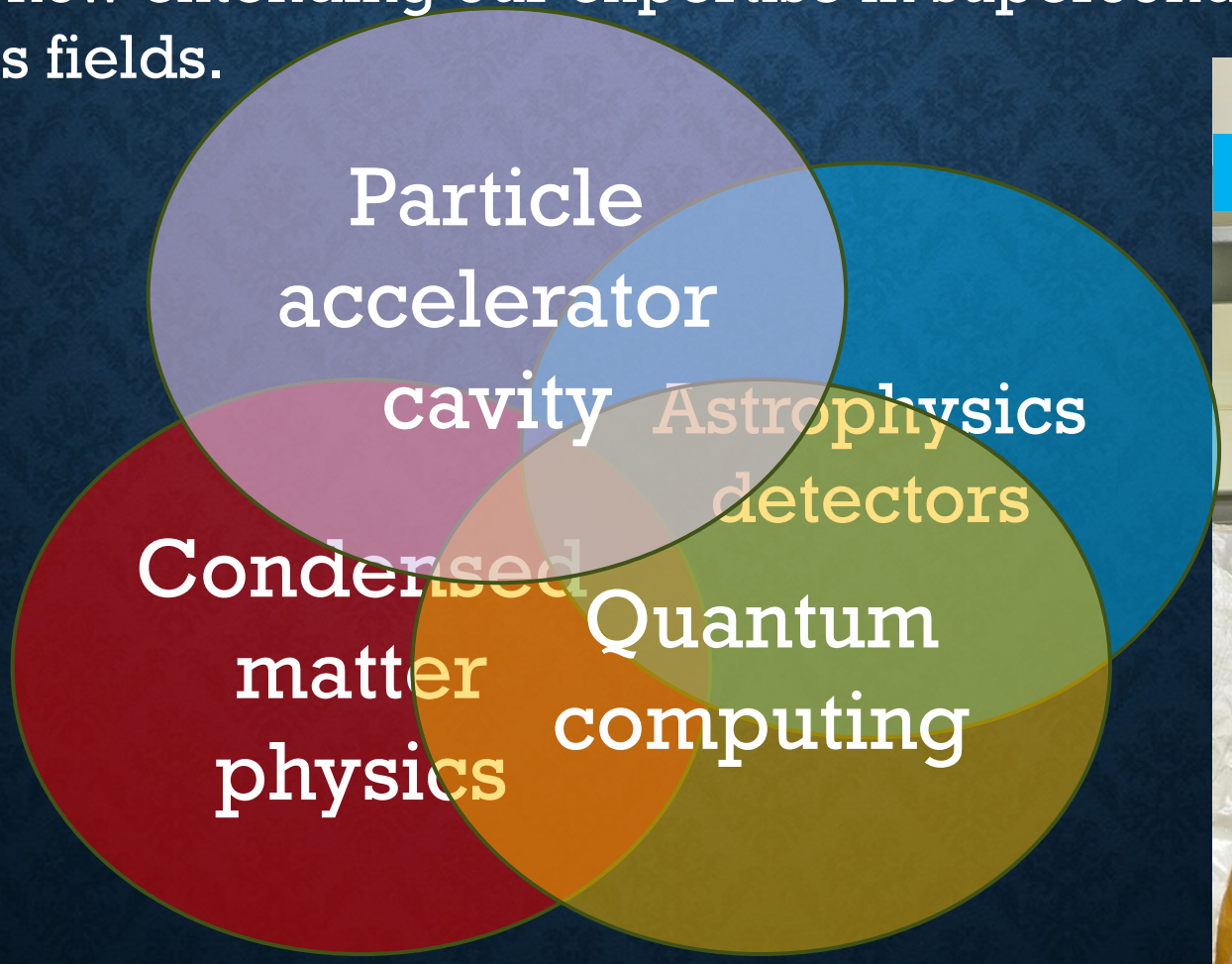


Summary

Superconducting resonators lie at the intersection of accelerator physics, astronomy, and quantum technologies, opening new opportunities for scientific discovery. We have been at the forefront of research on superconducting RF cavities for particle accelerators. We are now extending our expertise in superconducting physics and technology to various fields.



T. Kubo (2013)





T. Kubo (2024)